Functional Analysis

— ***EYNTKA*** **Series** —

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Contents

Part I

Functional Analysis

The transition from Calculus and Linear Algebra to Functional Analysis marks a fundamental shift in mathematical perspective. In classical analysis, we study individual functions, their derivatives, and their integrals. In linear algebra, we study vectors in finite dimensions. *Functional Analysis* we treat the function f itself as a point and look at transformations between spaces of these functions.

Note the term functional historically meant a function F that takes a function and returns a number. We shall show that most linear functionals are given by integrating against a function (section 2.2 and section 2.3), and many important exceptions will be shown to still be given by integrating against distributions.

The Historical Imperative: Taming the Infinite

Historically, the field grew out of a crisis in physics and differential equations. By the early 20th century, physicists dealing with Quantum Mechanics and classical wave equations realized that their problems could not be solved by looking at individual coordinates.

- In Quantum Mechanics, the state of a particle is not a position in $\mathbb{R}^3$, but a wave function ψ. The superposition principle—the fact that if ψ_1 and ψ_2 are states, then $\alpha\psi_1 + \beta\psi_2$ is a state—suggests that these wave functions live in a linear vector space.
- In Partial Differential Equations, viewing the heat or wave equation as an operation on an infinite-dimensional space allows us to decouple the "geometry" of the operator from the specific details of the function.

However, unlike the finite-dimensional world of $\mathbb{R}^n$, where linear algebra is rigid and predictable, infinite-dimensional spaces are wild. The intuitions of Euclidean geometry breaks down: bounded sets are no longer compact, linear maps are not necessarily continuous, and the notion of "convergence" splinters into multiple, distinct topologies.

Functional Analysis is the rigorous attempt to tame this infinite wilderness. It builds a hierarchy of structures to recover the tools of calculus and algebra in the infinite setting.

The Three Pillars of the Theory

The narrative of this course is driven by the tension between three mathematical forces:

1. The Algebraic Structure (Linearity): At its core, we are doing linear algebra. We want to solve equations like $Lx = y$. If L is a matrix, we invert it. If L is a differential operator, we want to do the same. The dream of Functional Analysis is to treat differential operators as "infinite matrices" and apply the Spectral Theorem to diagonalize them. When this works (as in Hilbert Spaces), we can solve complex PDEs with the same ease as a 2×2 matrix equation.
2. The Topological Structure (Continuity): Algebra alone is insufficient in infinite dimensions. In $\mathbb{R}^n$, all norms are equivalent, and continuity is automatic for linear maps. In infinite dimensions, this rigidity evaporates. We must introduce *Topology* via norms and metrics to distinguish between operators that are "well-behaved" (bounded) and those that are explosive (unbounded, like differentiation). The interplay between the algebraic dual space (all linear functionals) and the topological dual space (continuous functionals) becomes a central theme of the theory.

3. The Geometric Structure (Distance and Angle): This is where the "Zoo" of spaces emerges.
   (a) *Hilbert Spaces (L^2):* These are "correct" generalization of finite-dimensional euclidean space. They possess an inner product, allowing us to measure angles and define orthogonality.
   (b) *Banach Spaces ($L^p, C(K)$):* They have a notion of size (norm) but lack a notion of angle (i.e. a linear notion of relatedness). The geometry here can be strange - unit balls can be "pointy" or "flat," and the shortest path to a subspace might not be unique.
   (c) *Fréchet and Topological Vector Spaces:* necessary for distributions and generalized functions, where the concept of a "norm" is abandoned for families of semi-norms. They are equivalent to invariant metric spaces

There are more general spaces, but they are beyond an introductory text.

Ultimately, Functional Analysis is the study of *Dualities*. It is the duality between a space and the (linear) functions acting on it. We will see that by lifting concrete problems into these abstract spaces, we gain a new vantage point. The goal of this Part is to map this universe, understanding how the shape of the space dictates the physics of the operators within it.

1

Banach and Fréchet Spaces

(these are all spaces key spaces to consider with normed

(alternative title: Fréchet, Banach, and Hilbert Spaces)

1.1 Banach Spaces

Let k be either $\mathbb{R}$ or $\mathbb{C}$, let V or X denote a vector-space over k, with 0 representing the zero vector. Let kx denote the subspace of X spanned by x. If $M, N \subseteq X$, then $M + N = \{m + n \mid m \in M, n \in N\}$.

Definition 1.1.1: Semi-Norm And Norm

Let V be a vector-space. A *semi-norm* on V is a function $x \mapsto \|x$
from V to $[0, \infty)$ satisfying the propeties:

1. $\|x + y\| \leq \|x\| + \|y\|$ (the triangle inequality)
2. $\|\lambda x\| = |\lambda| \|x\|$ (homogeneity of scalars)

If furthermore:

(3) $\|x\| = 0$ if and only if $x = 0$

then the function is called a *norm*

Definition 1.1.2: Normed Vector-Space

Let V be a vector space. Then if there exists a function $\|\cdot\|$ that is defined on V, then $(V, \|\cdot\|)$ is called a *normed vector-space.*

If V is a normed vector-space, we can induce a metric $\rho(x, y) = \|x - y\|$, namely since:

$$\|x - z \leq \|x - y\| + \|y - z\| \qquad \|x - y\| = \|(-1)(x - y)\| = \|y - x\|$$

Since any metric induces a topology, the induced topology by the metric induced by a norm $\|\cdot\|$ is called the *norm topology* on V. A nice consequence of the induced norm topology is that addition and scalar multiplication is continuous. From the axioms of a semi-norm, $\|0\| = 0$. There are norms for which $x \neq 0$ but $\|x\| = 0$, as we'll soon show. Note that the norm $\|\cdot\| : V \to \mathbb{F}$ is evidently continuous in the topology it induces[1]. If $x_n \to x$ in X, then for all $\epsilon > 0$, there is an $N \in \mathbb{N}$ such that for all $n \geq N$, $\|x_n - x\| < \epsilon$. By the reverse triangle inequality:

$$|\|x_n\| - \|x\|| \leq \|x_n - x\| < \epsilon$$

showing that $|\|x_n\| - \|x\||$ get's arbitrarily close, and so $\|x_n\| \to \|x\|$, i.e. $\|\cdot\|$ is continuous. This means that the pre-image of $\{0\}$ of $\|\cdot\|$ is closed. Furthermore, it is a vector-space. Denoting

$$K = \{x \in V \mid \|x\| = 0\}$$

Then if $x, y \in K$, $0 \leq \|x+y\| \leq \|x\|+\|y\| = 0$, and $\|cx\| = |c|\|x\| = |c|0 = 0$. Hence, we may consider V/K. What norm to put on V/K that is natural with respect to the projection map $\pi : V \to V/K$ will be addressed soon.

Two norms $\|\cdot\|_1$ and $\|\cdot\|_1$ are said to be *equivalent* if there exits some $C_1, C_2 > 0$ st

$$C_1\|\cdot\|_1 \leq \|\cdot\|_2 \leq C_2\|\cdot\|_1$$

Equivalent norms define equivalent metrics and the same topology, and so the same Cauchy sequences. Recall that in finite dimensions, all norms are equivalent, meaning there is only one notion of distance. As we'll see, this is not the case in infinite dimensions.

Since we have a metric, we have a notion of completeness. As is the case when working with analysis, we will require our spaces to be complete:

Definition 1.1.3: Banach Space

Let V be a normed vector space. Then V is a *Banach space* if it is complete with respect to it's induced metric.

Example 1.1.4: Banach Spaces

1. Every finite dimensional vector space over $\mathbb{R}$ or $\mathbb{C}$ is a Banach space. In this space, all linear map are continuous.

2. Let K be a compact space, and take the set of all continuous function, $C_0(K)$. Then $C(K)$ is a Banach space with the supremum norm $\|f\|_\infty = \sup_{x\in K} |f(x)|$ (also called uniform

[1] Note that in the metric space topology, continuous if and only if sequentially continuous

norm). We will later show (Banach–Mazur theorem) that all real separable Banach spaces are isometric (the isomorphisms in the category of Banach spaces) to a subspace of $C([0,1])$, giving us a theoretical universal way of thinking about Banach spaces. We shall later see that $C([0,1])$ has the Approximation property and has a Schauder basis, but there exists Banach spaces without these properties. This shows that for Banach spaces, it is possible to have nice total spaces with "bad" subspaces. In particular, if X is an isometric embedding into $C([0,1])$ there are not always nl nice projection $P : C([0,1]) \to X$.

If the space is not separable, if the smallest dense set D of a Banach space X has density of cardinality α, then X is isometric to $C_0([0,1]^\alpha, \mathbb{R})$[a]

3. If X is a topological space, then $B(X,k)$ and $BC(X,k)$ (all bounded functionals and bounded continuous functionals) form a Banach space with the uniform norm $\|f\|_u = \sup_{x\in X} |f(x)|$.
4. For p where $1 \le p \le \infty$, the space $L^p(X)$ for some measurable space X is a Banach space with the L^p-norm (recall that $L^p(\mu)$ is the set of equivalence classes functions that agree a.e.; if $L^p(\mu)$ represented only functions, then $\|\cdot\|_p$ would be a semi-norm)
5. Two other common examples of Banach spaces we will encounter later will be special subspaces spaces of differentiable functions: Banach Algebras, and Hilbert Spaces.

There are also examples of spaces that are not Banach Spaces:

1. Let X be a locally compact Hausdorff space and consider $C_c(X)$ with the ∞-norm. Then it is not a Banach space, not being complete. It is easy to see that $C_c(X)$, the set of continuous bounded function, is a Banach space and contains $C_c(X)$, and so it suffices to find the closure of $C_c(X)$ in $C_b(X)$. The closure turns out to be $C_0(X)$.
2. Let $U \subseteq \mathbb{R}^n$ be open and take $C_0(U,\mathbb{C})$. Then This space is *not* a Banach space
3. The space $H(U,\mathbb{C})$ of all holomorphic functions from U to $\mathbb{C}$ is *not* a Banach Space
4. The space $C_c^\infty(\mathbb{R}^n)$ is *not* a Banach space
5. (If you know some PDE's) The space of distributions $\mathcal{D}'$ is *not* a Banach Space

These are all examples of a weaker structure known as a *topological vector space* which we shall cover in section 1.4.

[a]An interesting result using Banach-Mazur's theorem by Luis Rodríguez-Piazza shows that the space of smooth functions with the uniform norm is isometric to the space of nowhere differentiable functions

There is a nice way to check whether a vector space V is a Banach space which requires the following definition (this essentially follows from the fact that continuity if and only if sequential continuity in a metric space and that $\|\cdot\|$ is continuous)

Definition 1.1.5: Absolutely Convergent

Let V be a normed vector space. and $\{x_k\}_k^\infty \subseteq V$ a sequence. Then the sequence is said to *converge absolutely* if

$$\sum_k^\infty \|x_k\| < \infty$$

Lemma 1.1.6: Cauchy and Absolute Convergence

Let V be a normed vector space. Then V is complete if and only if every absolutely convergent series in V converges.

The idea is that $\mathbb{R}$ has an order, making it easier to produce many arguments about sequences.

***Proof*:**
Let's first say that X is complete and that $\sum_{n=1}^\infty \|x\| < \infty$. To show it converges, we take advantage of completeness and show there is a Cauchy sequence. In particular, let $S_n = \sum_{k=1}^n x_k$ be the partial sum. Then since the series is absolutely convergent, there for all $n > m$,

$$\|S_n - S_m\| \leq \sum_{m+1}^n \|x_k\| \to 0 \quad \text{as } m, n \to \infty$$

so the sequence $\{S_n\}$ is Cauchy, and so converges, so $\{x_k\}$ converges.

Conversely, let's say an absolutely convergent series converges, and choose any Cauchy sequence $\{x_k\}$. Since it's a Cauchy sequence, choose a subsequence such that

$$\left\|x_{k_{i+1}} - x_{k_i}\right\| < 2^{-i}$$

Take $y_1 = x_{k_1}$ and $y_i = x_{i+1} - x_i$ for $i > 1$ so that $\sum_{i=1}^n y_i = x_{k_n}$. Then notice that $\{y_i\}$ is in fact absolutely convergent:

$$\sum_{i=1}^\infty \|y_i\| \leq \|y_1\| + \sum_{i=1}^\infty 2^{-i} = \|y_1\| + 1 < \infty$$

So $\sum_{i=1}^\infty y_i = \lim x_{k_n}$ exists. Since the subsequence of every Cauchy sequence approaches the same "point" (just like convergent sequences), we see that $\{x_n\}$ also converges to the same point, showing that Cauchy sequences are in fact convergent sequences, as we sought to show.

Next, we would want to study which linear functions between two vector-spaces are *continuous*. It will *not* be the case that all linear functions will be continuous. The following builds up to the classification of continuous linear functions:

Definition 1.1.7: Bounded Linear Map

Let $T : X \to Y$ be a linear map between two normed vector space. Then if there exists some C such that for all $x \in X$

$$\|T(x)\| \leq C\|x\|$$

Then T is said to be *bounded*, or is a *bounded linear operator*.

Note how this is different then boundedness of a set (ex. the range is bounded if $\|T(x)\| \leq C$). This definition is would be useless to us since no nonzero linear map would satisfy it. All linear map over finite dimensional vector spaces clearly bounded (can you find the C?), so this concept gets interesting in infinite dimesion

Proposition 1.1.8: Bounded $\Leftrightarrow$ Continuous

Let $T : X \to Y$ be a linear map between two normed vector spaces. Then the following are equivalent:

1. T is continuous
2. T is continuous at 0
3. T is bounded

***Proof*:**
If T is the zero map, then we're done, so assume $T \neq 0$

(1) implies (2) by definition. For (2) implies (3), Since T is continuous at 0, for for all $\epsilon > 0$, there exists a $\delta > 0$ such that if $\|x - 0\| < \|x\| < \delta$, then $\|T(x) - T(0)\| = \|T(x) - 0\| = \|T(x)\| < \epsilon$. Choosing $\epsilon = 1$, we get that there exists a δ_1 such that

$$\|x\| < \delta_1 \to \|T(x)\| < 1$$

we know want to show there exists a C such that $\|T(y)\| \leq C\|y\|$ for all y in the domain. We will do this by taking advantage of linearity and shrinking element so that we can take advantage of continuity.

Define $x(y) = \frac{\delta_1}{2\|y\|}y$ so that

$$\|x(y)\| = \left\|\frac{\delta_1}{2\|y\|}y\right\| = \frac{\delta_1}{2}\frac{1}{\|y\|}\|y\| = \frac{\delta_1}{2} < \delta_1$$

Therefore, by continuity and linearity:

$$\|T(x(y))\| = \left\|T\left(\frac{\delta_1}{2\|y\|}y\right)\right\| = \frac{\delta_1}{2\|y\|}\|T(y)\| < 1$$

since this is true for all y, we if we re-arrange we have:

$$\|T(y)\| < \frac{2}{\delta_1}\|y\|$$

showing that T is indeed bounded.

Finally, if $\|T(x)\| \leq C\|x\|$, then for all $\epsilon > 0$, choose $\delta = C^{-1}\epsilon$ so that

$$\|T(y) - T(x)\| = \|T(y - x)\| \leq \epsilon$$

completing the proof.

Note that for $T \in B(X, Y)$ to be an isomorphism, we require that T^{-1} also be bounded[2]. If $\|T(x)\| = \|x\|$ for all x, then T is called an *isometry.*

Since the topology is that of norms, then continuous linear functions between normed vector-spaces are the homomorphisms, meaning they preserve the norm (to the degree of boundedness mentioned in the above proposition) and the vector-space structure (to the degree the map is injective). We will denote the set of bounded linear functions (which we will call linear operators) as $B(X, Y)$, $\mathcal{L}(X, Y)$, or $\mathrm{Hom}_k(X, Y)$, where we understand that since X and Y are normed spaces $\mathrm{Hom}_k(X, Y)$ refers to continuous lienar maps.

Like with most fields of mathematics, we are more often more interested in the functions between spaces rather than the spaces itself, for it is function that ultimatly give us information about the structure we are working with. Thus, if we can use the technics we will be developing on $B(X, Y)$, we may gain many interesting results. For this to be the case, we require to introduce a norm on $B(X, Y)$. Naturally, we want that the norm on $B(X, Y)$ to be compatible with X and Y in some way, especially the codomain Y (as we usually investigate the structure of the image of a map). Sicne we are working with bounded operators, we have that $\|T(x)\| \leq C\|x\|$. Furthermore, since we are working with linear operators, by some scaling we see that we only need to consider when $\|x\| = 1$. This moivates the following norm on $B(X, Y)$:

Definition 1.1.9: Operator Norm

$B(X, Y)$ is a normed vector space with norm[a]

$$\begin{aligned}\|T\| &= \sup_{\|x\|=1} \|T(x)\| \\ &= \inf\{C \mid \|T(x)\| \leq C\|x\| \text{ for all } x\} \\ &= \sup_{\|x\|\neq 0} \frac{\|T(x)\|}{\|x\|}\end{aligned}$$

If $\|T(x)\| = \|x\|$, then T is called an *isometry*

[a] the third equality holds if $V \neq \{0\}$

The operator norm can be thought of as the smallest consant that will satisfy $\|T(x)\| \leq C\|x\|$ when T is bounded, that is $\|T(x)\| \leq \|T\|\|x\|$. In the spacial case where we have $B(X, k)$, we write V^* and call it the dual space with the *norm topology.* It is useful to establish when is $B(X, Y)$ a Banach space

[2] We will show using the open mapping theorem that it suffices to show T is continuous and bijective

Proposition 1.1.10: Completness Of $B(X, Y)$

Let X, Y be normed vector-space and $B(X, Y)$ the normed vector-space of all bounded linear functions from X to Y. If Y is complete, so is $B(X, Y)$.

***Proof*:**
Let (T_n) be a cauchy sequence in $B(X, Y)$. For each $x \in X$, we have a cauchy sequence $(T_n(x)) \subseteq Y$, and thus a limit vector which we will denote $T(x)$. This defines a function $T : X \to Y$ which is clearly linear. The inequality

$$\begin{aligned} \|T(x) - T_n(x)\| &= \lim_m \|T_m(x) - T_n(x)\| \\ &\leq (\limsup_m \|T_m - T_n\|)\|x\| \end{aligned}$$

shows that T is bounded and that $T_n \to T$.

Hence, V^* is always a banach space, since $\mathbb{R}$ and $\mathbb{C}$ are complete. Next, we see what happens with common vector-space constructions like products and quotients.

Subspaces, Product spaces, Quotient Spaces

There are 3 equivalent norms we can put on the product of two normed vector-spaces:

1. $\|(x, y)\| = \max(\|x\|, \|y\|)$
2. $\|(x, y)\| = \|x\| + \|y\|$
3. $\|(x, y)\| = (\|x\|^2 + \|y\|^2)^{1/2}$

The key to these being equivalent is that we are only multiplying by a finite number of spaces, a different strategy would have to be implemented for norms on infinite dimensional vector spaces (recall the different ℓ^p-norms put on the set of all sequences). For subspace, it is clear that if $W \subseteq V$ is a subspace of a normed vector space V, then W has an induced norm with respect to V. What is more interesting is if V is a banach space and we want W to be a Banach space. We then require all sequences to converge. Since V is Hausdorff (even metrizable since there is a norm on V), we see that W is complete if W is a *closed subspace* with respect to the topology of V. For quotients V/W, we would like to define a new norm on V/W so that the natural projeciton map $\pi : V \to V/W$ is continuous. This leads us to the following definition:

Proposition 1.1.11: Banach Quotient Space

Let X be a normed vector space and $Y \subseteq X$ a subspace of X. Then $\pi : X \to X/Y$ is the quotient map. This induces the topology on X/Y by defining:

$$\|\pi(x)\| = \|x + Y\| = \inf_{y \in Y} \|x - y\|$$

which is a *seminorm* on X/Y. If Y is closed, then this is in fact a norm, and if furthermer X is Banach, then X/Y is a banach space.

The intuition in my mind for this definition of quotient norm is to "eliminate" any effect of the quotienting space Y on any element of $x+Y$. Take for example $\mathbb{R}^2/\mathbb{R}$. Then the the norm of $\|(x,y)+\mathbb{R}\|$ should really be $|x|$. Furthermore, π is naturally a continuous map since it is norm-decreasing, meaning $\|\pi(x)\| \leq \|x\|$ (simply choose $C=1$).

***Proof*:**
For subadditivity, we take advantage of the definition of infimum: for every $x_1, x_2 \in X$, there exists an $\epsilon > 0$ and elements $y_1, y_2 \in Y$ such that

$$\begin{aligned}\|\pi(x_1)\| + \|\pi(x_2)\| + \epsilon &\geq \|x_1 - y_2\| + \|x_2 - y_2\| \\ &\geq \|x_1 + x_2 - (y_1 + y_2)\| \\ &\geq \|\pi(x_1 + x_2)\|\end{aligned}$$

Since ϵ was arbitrary, we have subaditivity. Homogeneity is immediate since π is linear, and hence we have a seminorm.

If $\|\pi(x)\| = \|x+Y\| = 0$, there is a sequence $(y_n) \subseteq Y$ such that $\|x - y_n\| \to 0$. If Y is closed, $\lim_n(y_n) = x \in Y$, meaning $\pi(x) = 0$. It follows that X/Y is a normed space if and only if Y is closed in X.

Now, say $(z_n) \subseteq X/Y$ is a cauchy sequence. Then we can find a subsequence (z'_n) were

$$\|z'_{n+1} - z'_n\| < 2^{-n}$$

By induction we can choose $x_n \in X$ such that $\pi(x_n) = z'_n$ and $\|x_{n+1} - x_n\| < 2^{-n}$. If X is banach, (x_n) converges to an element $x \in X$, and since π is "norm decreasing" (the norm the same or smaller), then (z'_n) converges to $\pi(x)$. Hence, (z_n) converges to $\pi(x)$, and so X/Y is complete, as we sought to show.

Proposition 1.1.12: Universal Property of Quotient Maps

If $T \in B(X,Y)$ where X and Y are normed spaces, and if $W \subseteq X$ is a closed subspace of X containing $\ker(T)$, then then there exists an operator $\tilde{T} \in B(X/W, Y)$ defined pointwise as $\tilde{T}(\pi(x)) := T(x)$ such that

$$\|\tilde{T}\| = \|T\|$$

in particular, the following diagram commutes in **Ban**:

$$\begin{tikzcd} X \arrow[r, "T"] \arrow[d, "\pi"'] & Y \\ X/W \arrow[ur, "\tilde{T}"'] & \end{tikzcd}$$

***Proof*:**
First, by elementary linear algebra we have that $\tilde{T}$ is an operator on X/W, which is a normed space by the earlier proposition. Since

$$\|\tilde{T}(\pi(x))\| = \|T(x)\| = \|T(x-z)\| \leq \|T\|\|x-z\|$$

for every $z \in W$, it follows form the definition of the quotient norm that

$$\|\tilde{T}(\pi(x)) \leq \|T\|\|\pi(x)\|$$

hence, $\|\tilde{T}\| \leq \|T\|$. The reverse inequality $\|\tilde{T}\| \geq \|T\|$ is clear since Q is norm decreasing.

Proposition 1.1.13: Extending Banach Spaces

Let X be a normed space and Y a closed subspace. Then if Y and X/Y are Banach spaces, then X is a Banach space.

***Proof*:**
Let $(x_n) \subseteq X$ be a cauchy sequence. Since $\pi : X \to X/Y$ is norm-decreasing, $(\pi(x_n)) = (y_n) \subseteq X/Y$ is a cauchy sequence. Since X/Y is complete, it converges to a point $\overline{x} \in X/Y$. Next, by the definition of the quotient norm, for every x_n, there exists an n st

$$\|(x - x_n) - y_n\| \leq \frac{1}{n} + \|\pi(x - x_n)\|$$

Then we get:

$$\begin{aligned} \|y_n - y_m\| &= \|(y_n + x - x_n) - (y_m + x - x_m) + x_n - x_m\| \\ &\geq \frac{1}{n}\|\pi(x - x_n)\| + \frac{1}{m}\|\pi(x - x_m)\| + \|x_n + x_m\| \end{aligned}$$

which is clearly cauchy, and hence since Y is banach it is complete and so this sequence converges to y. Finally, it is easy to see that

$$\|x_n - (x + y)\| \leq \|x_n - (x - y_n)\| + \|y - y_n\|$$

showing that $x - n \to x + y$, as we sought to show.

Finite space naturally have none of the problems introduced in infinite dimensions and preserve all of their nice properties even when subspace of infinite dimensinoal vectorspace:

Proposition 1.1.14: Finite Dimensional Normed Space

Every finite-dimensional subspace Y of a normed space X is a Banach space, and consequently is closed in X. Moreover, if $\dim(Y) = n$, every linear isomrophism of $\mathbb{F}^n$ onto Y is a homeomorphism

***Proof*:**
p. 46 Analysis now.

Working with infinite dimensional space dense subsets become of key interest. As a simple example, the smooth bump functions form a dense subset of L^p for all $1 \leq p < \infty$. The following simplifies for us how we may define functions from spaces given a known dense set:

Proposition 1.1.15: Defining Operator using Dense Sets

Let X and Y be banach spaces and X_0 is a dense subspace of X. Then every operator $T_0 \in B(X_0, Y)$ has a unique extension to an operator $T \in B(X, Y)$ and $\|T\| = \|T_0\|$

***Proof*:**
For each $x \in X$, there is a sequence $(x_0) \subseteq X_0$ that converges to x: $x_n \to x$. Since T_0 is a bounded operator ($\|T_0(x_n)\| \leq C\|x_n\|$), it is clear that $(T_0(x_0))$ is a convergent sequence in Y; let's say it converges to $T(x)$. It should be clear that if we choose another sequence converging to x and do this procedure, then we will get the same value of $T(x)$. Thus, the map $x \to T(x)$ is a operator that is linear, extends T_0, and by construciton

$$\|T\| = \|T_0\|$$

an excellent example would be polynomial over a compact set are dense over continuous functions over compact sets, which themselves are dense over all L^p spaces. Hence in the compact case, for real or complex functions, it suffices to define our function over a the polynomials. For the case of locally compact spaces, Note that smooth bump functions are dense in L^p, and since they have compact support a polynomial sequnce uniformally converges to them. Hence, we may define a lot of operators on different real/complex functions spaces over locally compact spaces using polynomals.

For future reference, it will be important to remember that all normed spaces can be completed and that the norm between a normed space X and its completion $\overline{X}$ is isometric:

Proposition 1.1.16: Unique Extension To Banach Space

For each normed space X, there is a Banach space $\overline{X}$, uniquely determiend up to isometric isomoprhism, such that $\overline{X}$ contains X as a dense subspace

***Proof*:**
p. 47 Analysis now

An important concept in any infinite dimensional space is to figure out which sets are compact. For example, you may recall in real analysis that we proved the Stone-Weistrass theorem, or even simpler that in $\mathbb{R}^n$ a set is compact if it is closed and bounded. The following shows that subsets that are "almost finite-dimensional" are compact.

Proposition 1.1.17: Compact Sets In Banach Spaces

Let $K \subseteq V$ be a subset of a Banach Space V. Then the following are equivalent:

1. K is compact
2. K is sequentially compact
3. K is closed and bounded, and for every $\epsilon > 0$, K lies in the ϵ-neighborhood
$$\{x \in V \mid \|x - y\| < \epsilon \text{ for some } y \in W\}$$
of a finite dimensional subspace W of V.

Suppose further that there is a nested sequence $V_1 \subseteq V_2 \subseteq \cdots$ of finite dimensional subspaces of V such that $\bigcup_{n=1}^{\infty} V_n$ is dense. Then the following statement is equivalent to the first three:

4. K is closed and bounded, and for every $\epsilon > 0$, there exits an n such that K lies in the ϵ-neighborhood of V_n

***Proof*:**
exercise

Example 1.1.18: Compact Sets In Banach Spaces

Let $1 \leq p < \infty$ and take a set $K \subseteq \ell^p(\mathbb{N})$. In order for it to be compact in the norm topology, it needs to be closed, bounded, and also uniformly p^{th}-power integrable at spatial infinity, that is for every $\epsilon > 0$, there exists an $n > 0$ such that

$$\left(\sum_{m>n} |f(m)|^p \right)^{1/p} \leq \epsilon$$

for all $f \in K$. This shows that the moving bump set is not uniformally p^{th} power integrale and thus not a compact subset, even though it's closed and bounded.

This condition doesn't translate to the continuous equivalent $L^p(\mathbb{R})$. We also need some uniform integrability at a "finte scale", which would be described using harmonic analysis tools like Fourier transforms, and hence we simply mention it here without developing it further.

Given the Banach space $\mathrm{End}_k(V)$, it is natural have a second binary operation of composition and ask if it is also continuous (hence bounded). More generally, if $T \in B(X, Y)$, and $S \in L(Y, Z)$, then

$$\|ST(x)\| \leq \|S\|\|T(x)\| \leq \|S\|\|T\|\|x\|$$

and so $ST \in B(X, Z)$. In particular, that makes $B(X, X) = \mathrm{End}_k(X)$ into an algebra, and when X is complete, $B(X, X)$ is a *Banach Algebra*. More generally:

Definition 1.1.19: Banach Algebra

Let X be a normed k-algebra where addition, multiplication, and scalar multiplication are continuous with respect to the induced topology. Then if X is complete, X is a *Banach Algebra*.

For the multiplication to be continuous, it suffices to check that $\|xy\| \leq \|x\|\|y\|$ (can you see why?). If X has a multiplicative unit, it is called *unital*. Not all Banach algebra's have a multiplicative unit, $C_0(X)$ where X is a locally compact (hausdroff) space and $C_0(X)$ is the set of functions from X to $\mathbb{R}$ (or $\mathbb{C}$) that vanish at infinity is *not* unital. On the otherhand, $C_b(X)$, the set of bounded functions with pointwise operations with the supremum norm *is* unital. We will work more with Banach Algebras later.

The Local Geometry of Banach Spaces

While infinite-dimensional Banach spaces can be geometrically wild (as we will see with L^1 and L^∞), their finite-dimensional subspaces exhibit a surprising regularity. We previously noted that all norms on a finite-dimensional space are equivalent. The following theorem, a cornerstone of the "Local Theory of Banach Spaces," makes a much stronger claim: Euclidean geometry is "locally universal".

Theorem 1.1.20: Dvoretzky's Theorem

Let $(X, \|\cdot\|)$ be an infinite-dimensional Banach space. For every integer $n \geq 1$ and every $\epsilon > 0$, there exists a subspace $E_n \subset X$ with $\dim(E_n) = n$ such that E_n is $(1+\epsilon)$-isomorphic to the Euclidean space ℓ_2^n.
In terms of the Banach-Mazur distance, this means:

$$d(E_n, \ell_2^n) = \inf \|T\|\|T^{-1}\| \leq 1 + \epsilon$$

where the infimum is taken over all isomorphism $T : E_n \to \ell_2^n$.

Geometrically, this is often described as the "slicing problem." If you visualize the unit ball of a Banach space X as a high-dimensional convex body K, Dvoretzky's theorem states that there is always a slice through the center of K (a subspace) that is nearly spherical.

This theorem highlights a tension between local and global structure:

1. *Globally*, X might be very far from a Hilbert space (like ℓ_∞ or the Enflo space).
2. *Locally*, X contains "copies" of Hilbert spaces of all finite dimensions.

Thus, while the "global" geometry of Banach spaces is a vast zoo of structures, the "atomic" local geometry is always Euclidean.

Exercise 1.1.1

1. Let $A, B \subseteq V$ are two closed subspaces. Show that $A + B$ need not be a closed subspace! However, if one of A or B is finite dimensional, then $A + B$ is closed.

2. Show that the open unit ball in X maps onto (by π) the open unit ball of X/Y, but a similar statement about closed unit balls holds only in special cases.
3. For the categorically inclined, it should be verified that the topology on X/Y is the quotient topology corresponding to the equivalent relation $\sim$ where $x_1 \sim x_2$ if and only if $x_1 - x_2 \in Y$.

1.2 Baire Category Theorem

In this section, we introduce some of the basic tools that allow us to work with functions between Banach spaces. Of keen interest to us is the *open mapping theorem*, the *closed graph theorem*, and the *uniform boundedness theorem*. The open mapping theorem makes it sufficient that a surjective continuous linear operator is open, which in particular means that if it is also bijective it is an isomorphism. The closed graph theorem allows us to show that a linear operator $T : X \to Y$ is continuous if the graph $\Gamma(T)$ is closed, i.e. if all sequences in the graph converge. This will lead to great simplifications in showing a function is continuous. Finally, the uniform boundedness theorem will give a powerful criterion for upgrading pointwise boundedness of a family of operators to uniform boundedness.

The underpinning results we need are about dense sets. Let's say we have a countable family of dense sets, which can be thought of as our "generalized basis". Then it is natural that their intersection may be empty, for example taking linearly independent transcendental elements $r_1, r_2, ... \in \mathbb{R}$, then $\bigcap_r \mathbb{Q} + r = \emptyset$. What we need is to ensure that the intersection is non-empty. For that, we require that the limit of an infinite intersection exists (ex. our space is LCH or a complete metric space), and that our sets have a nice property that ensures non-emptiness. The following deals with the case of the space being a complete metric space:

Proposition 1.2.1: Baire Category Theorem I

Let (A_n) be a sequence of open dense subset of a complete metric space (X, d). Then $\bigcap_n A_n$ is dense in X.

***Proof*:**
We want to show that $\bigcap_n A_n$ is dense, so for every ball $B_0 \subseteq X$, $B_0 \cap \bigcap_n A_n \neq \emptyset$.

Let B_0 be any closed ball in X with radius $r > 0$. Then since A_1 is dense in X and is open, the set $A_1 \cap \operatorname{int}(B_0)$ is a nonempty open set, and hence contains a closed ball B_1 with radius at most $r/2$. Since A_2 is dense and open, we can repeat this with $A_2 \cap \operatorname{int}(B_1)$ and find a closed ball B_2 with radius at most $r/2^2$. By induction, we get a sequence of closed balls (B_k) in X such that $B_k \subseteq A_k \cap \operatorname{int}(B_{k-1})$, each with radius at most $r/2^k$. The sequence of centers is Cauchy (since the radii tend to zero and the balls are nested), and since X is complete, there exists a unique point $\{x\}$ such that

$$\{x\} = \bigcap_k B_k \subseteq B_0 \cap \left(\bigcap_k A_k\right)$$

This shows that $\bigcap_k A_k$ intersects every nontrivial ball in X, which implies density, completing the proof.

Note that this is a fully topological proof, and hence is invariant under homeomorphism. In particular, completeness is *not* a topological property (it is a uniform property), however $(0,1) \cong_{\mathbf{Top}} \mathbb{R}$. Furthermore, since each U_n is open, its complement must be nowhere dense. For if some non-empty open set O satisfies $O \subseteq \overline{U_n^c} = U_n^c$ (since U_n^c is closed), then $U_n \subseteq O^c$, and so

$$X = \overline{U_n} \subseteq O^c \subsetneq X$$

a contradiction. This leads to the following very useful corollary:

Corollary 1.2.2: Countable union of Empty Interior Sets

The countable union of closed sets with empty interior has empty interior. If X is a complete metric space, $X \neq \bigcup_i \overline{X_i}$ for nowhere dense sets X_i.

As a consequence, if $X = \bigcup_n X_n$, then there exists some n_0 such that X_{n_0} has nonempty interior.

***Proof*:**
Let (F_n) be a countable collection of closed sets with empty interior. Then each F_n^c is open and dense (open since F_n is closed; dense since F_n has empty interior, meaning F_n^c is dense). By the Baire Category Theorem, $\bigcap_n F_n^c$ is dense in X, and in particular nonempty. Hence $\bigcup_n F_n \neq X$, and since $\bigcup_n F_n$ is contained in a countable union of closed sets with empty interior, it itself has empty interior.

If each X_i is nowhere dense, then each $\overline{X_i}$ is closed with empty interior, so $X \neq \bigcup_i \overline{X_i} \supseteq \bigcup_i X_i$. Contrapositively, if $X = \bigcup_n X_n$, then some X_{n_0} must have nonempty interior, for otherwise each $\overline{X_n}$ would be closed with empty interior, contradicting the above.

This consequence is very interesting, since it shows that through a countable process, we require a set with nonempty interior to form our entire space. This motivates the following definition:

Definition 1.2.3: Meager and Generic Set

Let X be a topological space. Then $A \subseteq X$ is said to be *meager* (or of first-category) if it is a countable union of nowhere dense sets. The complement of a meager set is a *residual* (or co-meager, or generic, or of second-category).

Example 1.2.4: Interesting Meager Sets

1. The Cantor set $\mathcal{C}$ is meager in $\mathbb{R}$. Indeed, $\mathcal{C}$ is closed (as the intersection of closed sets in its construction) and has empty interior (it contains no intervals, since at each stage of the construction every remaining interval is split and its middle third removed). Hence $\mathcal{C}$ is nowhere dense, and in particular meager (as a single-term union of a nowhere dense set).

2. Take $C([0,1])$ with the uniform convergence topology. Let E be the set of all continuous functions that are differentiable somewhere. Then E is in fact a *meager* set in $C([0,1])$. To see this, take:

$$E_n = \{f \mid \exists x_0 \in [0,1],\ |f(x) - f(x_0)| < n|x - x_0| \ \forall x \in [0,1]\}$$

Then all differentiable functions are contained in $\bigcup_n E_n$, and it should be shown that this is a closed set and that there exists functions g such that $\|f-g\|_u < \epsilon$ for any $f \in E_n$ but $g \notin E_n$. Hence, we satisfied the conditions of the Baire Category theorem. As a consequence, the set of continuous nowhere differentiable functions is residual! Thus more generally, smooth and analytic functions are meager.

3. $\mathbb{Q}$ is not a G_δ subset of $\mathbb{R}$. Suppose for contradiction that $\mathbb{Q} = \bigcap_n U_n$ for open sets U_n. Since $\mathbb{Q}$ is dense, each U_n is open and dense. The irrationals $\mathbb{R} \setminus \mathbb{Q}$ can also be written as a countable intersection of open dense sets: $\mathbb{R} \setminus \mathbb{Q} = \bigcap_q (\mathbb{R} \setminus \{q\})$ where q ranges over $\mathbb{Q}$. Then

$$\emptyset = \mathbb{Q} \cap (\mathbb{R} \setminus \mathbb{Q}) = \left(\bigcap_n U_n\right) \cap \left(\bigcap_q (\mathbb{R} \setminus \{q\})\right)$$

would be a countable intersection of open dense subsets of the complete metric space $\mathbb{R}$. By the Baire Category Theorem, this intersection must be dense, contradicting the fact that it is empty. Hence $\mathbb{Q}$ is *not* a G_δ. Note however that $\mathbb{R} \setminus \mathbb{Q}$ *is* a G_δ, i.e. the irrationals form a residual set.

The 2nd example above is worth proving in full: the set of continuous functions on $[0,1]$ that are differentiable at *even one point* is meager in $C([0,1])$. Equivalently, the "generic" continuous function is nowhere differentiable. Compare this with the explicit construction of a single such function in [3]; the Baire category argument produces no explicit example, but shows that *almost every* continuous function (in the sense of category) has this pathological behaviour.

Theorem 1.2.5: Genericity of Nowhere Differentiable Functions

Let $X = C([0,1])$ equipped with the uniform norm $\|\cdot\|_u$. Then the set of functions in X that are differentiable at some point of $[0,1]$ is meager. Consequently, the set of continuous nowhere differentiable functions is residual in X.

***Proof*:**
For each $n \in \mathbb{N}$, define:

$$E_n = \{f \in X \mid \exists x_0 \in [0,1],\ |f(x) - f(x_0)| \leq n|x - x_0| \ \forall x \in [0,1]\}$$

We first check that every function differentiable at some point belongs to $\bigcup_n E_n$. If $f'(x_0)$ exists with $|f'(x_0)| = L$, then there exists $\delta > 0$ such that $|f(x) - f(x_0)| \leq (L+1)|x - x_0|$ whenever $|x - x_0| < \delta$. For $|x - x_0| \geq \delta$, the bound $|f(x) - f(x_0)| \leq 2\|f\|_u \leq (2\|f\|_u/\delta)|x - x_0|$ holds trivially. Taking $n > \max(L+1,\ 2\|f\|_u/\delta)$ gives $f \in E_n$. It now suffices to show each E_n is closed with empty interior.

E_n is closed. Let $(f_k) \subseteq E_n$ converge uniformly to $f \in X$. For each k, choose $x_k \in [0,1]$ witnessing $f_k \in E_n$, i.e. $|f_k(x) - f_k(x_k)| \leq n|x - x_k|$ for all x. By compactness of $[0,1]$, pass to a subsequence (still called f_k) with $x_k \to x_0$. Fix any $x \in [0,1]$. Then:

$$\begin{aligned} |f(x) - f(x_0)| &\leq |f(x) - f_k(x)| + |f_k(x) - f_k(x_k)| + |f_k(x_k) - f(x_0)| \\ &\leq \|f - f_k\|_u + n|x - x_k| + |f_k(x_k) - f(x_0)| \end{aligned}$$

As $k \to \infty$, $\|f - f_k\|_u \to 0$ and $|f_k(x_k) - f(x_0)| \leq \|f_k - f\|_u + |f(x_k) - f(x_0)| \to 0$ by uniform convergence and continuity of f. Hence $|f(x) - f(x_0)| \leq n|x - x_0|$ for all x, showing $f \in E_n$.

E_n has empty interior. Let $f \in X$ and $\epsilon > 0$. We construct g with $\|f - g\|_u < \epsilon$ and $g \notin E_n$. Since f is uniformly continuous on $[0, 1]$, there exists $\delta > 0$ such that $|f(x) - f(y)| < \epsilon/8$ whenever $|x - y| < \delta$. Choose an integer $M > 4n$ large enough that $w := \epsilon/(2M) < \delta$. Define the *zigzag function* $\varphi : [0, 1] \to \mathbb{R}$ to be the continuous piecewise linear function with slope alternating between $+M$ and $-M$ on consecutive intervals of width w, starting from $\varphi(0) = 0$. Since φ oscillates between 0 and $Mw = \epsilon/2$, we have $\|\varphi\|_u \leq \epsilon/2$.

Set $g = f + \varphi$, so $\|f - g\|_u = \|\varphi\|_u \leq \epsilon/2 < \epsilon$. Suppose for contradiction that $g \in E_n$, witnessed by some $x_0 \in [0, 1]$. Then x_0 lies in some monotone piece of φ of width w. Pick x to be the endpoint of this piece farther from x_0, so that $|x - x_0| \geq w/2$ and φ has constant slope $\pm M$ on the segment from x_0 to x. Since $|x - x_0| \leq w < \delta$:

$$\begin{aligned} |g(x) - g(x_0)| &\geq |\varphi(x) - \varphi(x_0)| - |f(x) - f(x_0)| \\ &= M|x - x_0| - |f(x) - f(x_0)| \\ &> M \cdot \frac{w}{2} - \frac{\epsilon}{8} = \frac{\epsilon}{4} - \frac{\epsilon}{8} = \frac{\epsilon}{8} \end{aligned}$$

On the other hand, $n|x - x_0| \leq nw = n\epsilon/(2M) < \epsilon/8$ since $M > 4n$. Hence $|g(x) - g(x_0)| > n|x - x_0|$, contradicting $g \in E_n$.

Since each E_n is closed with empty interior, $\bigcup_n E_n$ is meager. The set of nowhere differentiable functions contains the complement of $\bigcup_n E_n$ and is therefore residual, completing the proof.

Hence the vast majority (in the sense of Baire category) of continuous functions have this "wild" behaviour: they oscillate so violently at every scale that no tangent line can be drawn at any point. The zigzag perturbation in the proof is a shadow of this oscillation, forced into existence by the Baire Category Theorem without any explicit fractal construction.

The following lemma is the key technical ingredient for the open mapping theorem. It shows that if the image of the unit ball is *dense* in some ball, then it actually *contains* a slightly smaller ball:

Proposition 1.2.6: Open Mapping Lemma

Let $T : V \to W$ be a bounded linear operator between two Banach spaces, $T \in B(V, W)$. Then if the image of the unit ball $B_1(0)$ is dense in some ball $B_r(0)$ with $r > 0$, i.e. $B_r(0) \subseteq \overline{T(B_1(0))}$, then

$$B_{(1-\epsilon)r}(0) \subseteq T(B_1(0))$$

for every $\epsilon > 0$.

***Proof*:**
Let $U = T(B_1(0))$. Since $T(B_1(0))$ is dense in $B_r(0)$, for any $y \in B_r(0)$ and $\epsilon > 0$, there is a $y_1 \in U$ such that $\|y - y_1\| < \epsilon r$. Consider now ϵU. This set is dense in $B_{\epsilon r}(0)$. Hence, there exists a $y_2 \in \epsilon U$ such that $\|(y - y_1) - y_2\| < \epsilon^2 r$. Continuing by induction, we get a sequence (y_n) where $y_n \in \epsilon^{n-1} U$ and

$$\left\| y - \sum_{k=1}^{n} y_k \right\| < \epsilon^n r$$

Choose now $x_n \in B_1(0)$ such that $T(x_n) = y_n$ and $\|x_n\| \leq \epsilon^{n-1}$. Then $\sum x_n = x \in X$ converges, and by definition $T(x) = y$. Since $\|x\| \leq \sum \epsilon^{n-1} = (1-\epsilon)^{-1}$ [a] we have

$$(1-\epsilon)^{-1}U \supseteq B_r(0)$$

completing the proof.

[a] recall that $1 - b^n = (1-b)(1+b+b^2+\cdots+b^{n-1})$

Theorem 1.2.7: Open Mapping Theorem

Let X and Y be Banach spaces and $T \in B(X,Y)$ be a bounded linear operator with $T(X) = Y$. Then T is an open map (i.e. T maps open sets to open sets)

***Proof*:**
Since T is surjective:

$$Y = T(X) = \bigcup_n T(B_n(0)) \subseteq \bigcup_n \overline{T(B_n(0))} = \bigcup_n n\overline{T(B_1(0))}$$

Then by corollary 1.2.2, not all closed sets $n\overline{T(B_1(0))}$ have an empty interior. Since scaling by n is a homeomorphism, it follows that $\overline{T(B_1(0))}$ itself has nonempty interior. Hence, there exist $y_0 \in Y$ and $\delta > 0$ such that

$$B_\delta(y_0) \subseteq \overline{T(B_1(0))}$$

We now show $B_{\delta/2}(0) \subseteq \overline{T(B_1(0))}$. For any z with $\|z\| < \delta$, both y_0 and $y_0 + z$ lie in $B_\delta(y_0) \subseteq \overline{T(B_1(0))}$. Hence:

$$z = (y_0 + z) - y_0 \in \overline{T(B_1(0))} - \overline{T(B_1(0))} \subseteq \overline{T(B_1(0)) - T(B_1(0))} \subseteq \overline{T(B_2(0))} = 2\overline{T(B_1(0))}$$

where we used linearity of T in the last two steps. It follows that $B_\delta(0) \subseteq 2\overline{T(B_1(0))}$, i.e. $B_{\delta/2}(0) \subseteq \overline{T(B_1(0))}$, so $T(B_1(0))$ is dense in $B_{\delta/2}(0)$.

By the Open Mapping Lemma (proposition 1.2.6), $B_s(0) \subseteq T(B_1(0))$ for every $s < \delta/2$. Since every open set $U \subseteq X$ is a union of open balls, we conclude from linearity of T that $T(U)$ contains a neighborhood around each of its points, thus $T(U)$ is open, completing the proof.

Corollary 1.2.8: Bijective And Continuous

Let $T : X \to Y$ be a bijective bounded operator. Then it has a bounded inverse, that is T is in fact a Banach-space isomorphism

The following also lets us recover a result similar to the one showing us that there is only one norm on a finite dimensional vector space up to equivalence, namely no (non-equivalent) norm on a Banach space can dominate another:

Corollary 1.2.9: Equivalent Banach Spaces

Let X be a vector-space, and a Banach space under two norms $\|\cdot\|_a$ and $\|\cdot\|_b$. Then if $\|\cdot\|_a \leq \alpha\|\cdot\|_b$ for some $\alpha > 0$, then there is a $\beta > 0$ such that $\|\cdot\|_b \leq \beta\|\cdot\|_a$

***Proof*:**
Take $\mathrm{id} : (X, \|\cdot\|_b) \to (X, \|\cdot\|_a)$. This is certainly bijective, continuous by the boundedness assumption, and open by the open mapping theorem. Hence it is a homeomorphism, meaning it has a bounded inverse, giving the desired result.

The next powerful result we get is a quick way of proving a function is bounded:

Theorem 1.2.10: Closed Graph Theorem

Let $T : X \to Y$ be an operator between Banach spaces. Then if the graph

$$\Gamma(T) = \{(x, y) \in X \times Y \mid T(x) = y\}$$

is closed in $X \times Y$, then T is bounded

***Proof*:**
Using the ∞-norm $\|(x, y)\| = \max(\|x\|_X, \|y\|_Y)$ on $X \times Y$, assume that $\Gamma(T)$ is closed in $X \times Y$. Since $X \times Y$ is a Banach space (as both X and Y are), and $\Gamma(T)$ is a closed linear subspace, $\Gamma(T)$ is itself a Banach space. Now consider the projection map $\pi_1 : \Gamma(T) \to X$ given by $\pi_1(x, T(x)) = x$. This is bijective (since T is a function) and norm-decreasing: $\|\pi_1(x, T(x))\| = \|x\| \leq \max(\|x\|, \|T(x)\|)$. Since π_1 is a bounded bijection between Banach spaces, the Banach isomorphism theorem (corollary 1.2.8) gives that π_1^{-1} is bounded. The other projection $\pi_2 : \Gamma(T) \to Y$ given by $\pi_2(x, T(x)) = T(x)$ is also norm-decreasing by the same reasoning. Hence:

$$T = \pi_2 \circ \pi_1^{-1}$$

is a composition of bounded operators, and therefore bounded, completing the proof.

The key take-away is that if we are verifying the continuity of a linear map $T : X \to Y$ (so $x_n \to x$ implies $T(x_n) \to T(x)$), we just need to verify the closedness of the graph. The closedness of the graph implies that if $x_n \to x$ and $T(x_n) \to y$, then $T(x) = y$. Notice how this simplifies proving a function is continuous, since we did not need to show that $\lim_n T(x_n) = T(x)$, simply that it converges to *something*, namely if $T(x_n) \to y$, then $T(x) = y$. Note how completeness is crucial in both these theorems (exercise 29-31 folland for counter-examples).

The converse of the closed graph theorem, that if $T : X \to Y$ is bounded between Banach spaces, then $\Gamma(T)$ is closed, is elementary: let's say T is bounded and that $T(x_n) \to y$. Since $x_n \to x$, $\lim_n T(x_n) = T(x)$. Then:

$$\begin{aligned}
\|T(x) - y\|_Y &\leq \|T(x) - y + T(x_n) - T(x_n)\|_Y \\
&\leq \|T(x - x_n)\|_Y + \|T(x_n) - y\|_Y \\
&\leq C\|x - x_n\|_X + \|T(x_n) - y\| \xrightarrow{n\to\infty} 0
\end{aligned}$$

hence, $\|T(x) - y\| = 0$, implying $T(x) = y$ and showing $\Gamma(T)$ is closed.

Finally, the following result is a way to pass from pointwise boundedness to uniform boundedness, only requiring the completeness of our domain! It can be thought of as one of the "pillars" of functional analysis:

Theorem 1.2.11: Principle Of Uniform Boundedness

Let X, Y be normed vector spaces and $A \subseteq L(X,Y)$. Then if X is a Banach space and $\sup_{T\in A} \|T(x)\| < \infty$ for all $x \in X$, then $\sup_{T\in A} \|T\| < \infty$

This theorem is also called by some the Banach–Steinhaus Theorem.

***Proof*:**
We want to upgrade from a pointwise bound to a uniform bound, showing that $\sup_{T\in A} \|T\| < \infty$. Let

$$E_n = \left\{ x \in X \;\middle|\; \sup_{T\in A} \|T(x)\| \leq n \right\} = \bigcap_{T\in A} \{x \in X \mid \|T(x)\| \leq n\}$$

Each E_n is closed (as an intersection of closed sets, since each T is continuous) and $X = \bigcup_n E_n$ (since $\sup_{T\in A} \|T(x)\| < \infty$ for each x). Then by the Baire Category Theorem, there must exist some $n \in \mathbb{N}$ such that E_n has nonempty interior. Hence, for some $x_0 \in E_n$ and $r > 0$, $\overline{B_r}(x_0) \subseteq E_n$.

Now, consider $u \in X$ such that $\|u\| \leq 1$ and take any $T \in A$. Since $x_0 + ru \in \overline{B_r}(x_0) \subseteq E_n$, we have $\|T(x_0 + ru)\| \leq n$. Also $x_0 \in E_n$, so $\|T(x_0)\| \leq n$. Then:

$$\begin{aligned}
\|T(u)\|_Y = r^{-1}\|T(ru)\|_Y &= r^{-1}\|T(x_0 + ru) - T(x_0)\|_Y \\
&\leq r^{-1}(\|T(x_0 + ru)\|_Y + \|T(x_0)\|_Y) \\
&\leq r^{-1}(n + n)
\end{aligned}$$

Hence, taking the supremum over u in the unit ball of X and over $T \in A$, we get:

$$\sup_{T\in A} \|T\|_{B(X,Y)} \leq 2r^{-1}n < \infty$$

completing the proof.

If X is simply a normed vector-space, we may keep this theorem by simply taking the set supremum of the set of nonmeager points. The key realization is that $\sup_{T\in A} \|T(x)\|$ is a different supremum for each x, but is finite for all x! That is, we are not working with $\sup_{\substack{T\in A \\ x\in X}} \|T(x)\|$ since that would make the result trivial:

$$\sup_{T\in A} \|T\| = \sup_{\substack{T\in A \\ \|x\|=1}} \|T(x)\| \leq \sup_{\substack{T\in A \\ x\in X}} \|T(x)\|$$

This is what makes the Uniform boundedness principle so strong, we got a *uniform* bound by having a *local* bound for all x!

Example 1.2.12: Divergence of Fourier Series
One of the most celebrated applications of the uniform boundedness principle is the existence of continuous functions whose Fourier series diverges at a given point. Consider $X = C(\mathbb{T})$, the Banach space of continuous 2π-periodic functions with the supremum norm. For each n, define the functional $\Lambda_n : C(\mathbb{T}) \to \mathbb{C}$ by

$$\Lambda_n(f) = S_n f(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) D_n(t)\, dt$$

where $S_n f(0)$ is the n-th partial sum of the Fourier series of f evaluated at 0, and $D_n(t) = \sum_{k=-n}^{n} e^{ikt}$ is the Dirichlet kernel. Each Λ_n is a bounded linear functional with $\|\Lambda_n\| = \frac{1}{2\pi}\int_{-\pi}^{\pi} |D_n(t)|\, dt = \|D_n\|_{L^1}$. It is a classical computation that $\|D_n\|_{L^1} \sim \frac{4}{\pi^2}\log n \to \infty$. Hence $\sup_n \|\Lambda_n\| = \infty$. By the *contrapositive* of the uniform boundedness principle, there must exist some $f \in C(\mathbb{T})$ for which $\sup_n |\Lambda_n(f)| = \sup_n |S_n f(0)| = \infty$, i.e. the Fourier series of f diverges at 0. By translation, the same holds at any prescribed point.

Note how the principle is used here: we showed $\sup_n \|\Lambda_n\| = \infty$, and concluded that the pointwise boundedness hypothesis must fail for some f, even though we never constructed such an f explicitly. This is a quintessential non-constructive existence proof via Baire category methods.

Corollary 1.2.13: Pointwise Convergence of Operators

Let X be a Banach space, Y a normed vector space, and $(T_n)_{n\in\mathbb{N}} \subseteq B(X,Y)$ a sequence of bounded operators such that $(T_n(x))_{n\in\mathbb{N}}$ converges in Y for every $x \in X$. Define $T(x) = \lim_n T_n(x)$. Then $T \in B(X,Y)$, i.e. the pointwise limit is itself a bounded linear operator.

***Proof*:**
First, T is linear: for $x, y \in X$ and α, β scalars, $T(\alpha x+\beta y) = \lim_n T_n(\alpha x+\beta y) = \alpha \lim_n T_n(x) + \beta \lim_n T_n(y) = \alpha T(x) + \beta T(y)$.

Since $(T_n(x))$ converges for each x, it is bounded: $\sup_n \|T_n(x)\| < \infty$ for all $x \in X$. Hence, by the principle of uniform boundedness, $\sup_n \|T_n\| = C < \infty$. That is, $\|T_n(x)\| \le C\|x\|$ for all n and all $x \in X$, where C is independent of n. Taking $n \to \infty$ we get:

$$\|T(x)\| \le C\|x\|$$

which shows $T \in B(X,Y)$, completing the proof.

More generally, if $(T_\lambda)_{\lambda\in\Lambda} \subseteq B(X,Y)$ is a net such that each $(T_\lambda(x))$ is bounded and converges for every $x \in X$, the same conclusion holds. For sequences, convergence automatically implies boundedness. It also suffices in the principle of uniform boundedness to know that each sequence $(T_n(x))$ is bounded and that the sequence converges for a dense set of elements in X.

We conclude this section with an application of the Baire Category Theorem that does not involve linear operators. It shows that the set of discontinuities of a pointwise limit of continuous functions is, in a precise sense, small:

Theorem 1.2.14: Continuity of Pointwise Limits

Let (f_n) be a sequence of continuous (complex-valued) functions on a complete metric space X such that $f(x) = \lim_{n\to\infty} f_n(x)$ exists for every $x \in X$. Then the set of points at which f is continuous is a dense G_δ (and hence residual) in X.

***Proof*:**
Recall that f is continuous at x if and only if the *oscillation* of f at x is zero, where

$$\omega_f(x) = \inf_{\delta>0} \sup_{y,z\in B_\delta(x)} |f(y) - f(z)|$$

Thus the set of continuity points of f is $\{x : \omega_f(x) = 0\} = \bigcap_{m=1}^{\infty}\{x : \omega_f(x) < 1/m\}$. Each set $O_m = \{x : \omega_f(x) < 1/m\}$ is open (this follows from the definition of oscillation), so the set of continuity points is a G_δ. It remains to show that each O_m is dense.

Fix $m \geq 1$ and let B be any nonempty open ball in X. We must find a point in $B \cap O_m$. For each $n, k \in \mathbb{N}$, define

$$F_{n,k} = \left\{ x \in \overline{B} \;\middle|\; |f_n(x) - f_k(x)| \leq \frac{1}{3m} \right\}$$

Each $F_{n,k}$ is closed (since $f_n - f_k$ is continuous). Since $f_n(x) \to f(x)$ for every x, for each $x \in \overline{B}$ there exists N such that $|f_n(x) - f_k(x)| \leq 1/(3m)$ for all $n, k \geq N$. Hence:

$$\overline{B} = \bigcup_{N=1}^{\infty} \bigcap_{n,k \geq N} F_{n,k}$$

The sets $G_N = \bigcap_{n,k\geq N} F_{n,k}$ are closed in $\overline{B}$. Since $\overline{B}$ is a complete metric space (closed subset of X), the Baire Category Theorem implies that some G_{N_0} has nonempty interior in $\overline{B}$. Hence there exists an open ball $B' \subseteq B$ such that $|f_n(x) - f_k(x)| \leq 1/(3m)$ for all $n, k \geq N_0$ and all $x \in B'$.

Taking $k \to \infty$, we get $|f_n(x) - f(x)| \leq 1/(3m)$ for all $n \geq N_0$ and $x \in B'$. In particular, f_{N_0} is continuous, so we may shrink B' so that f_{N_0} varies by at most $1/(3m)$ on B'. Then for any $y, z \in B'$:

$$|f(y) - f(z)| \leq |f(y) - f_{N_0}(y)| + |f_{N_0}(y) - f_{N_0}(z)| + |f_{N_0}(z) - f(z)| \leq \frac{1}{3m} + \frac{1}{3m} + \frac{1}{3m} = \frac{1}{m}$$

Hence any point of B' lies in O_m, showing that O_m is dense. Since each O_m is open and dense, $\bigcap_m O_m$ is a dense G_δ by the Baire Category Theorem.

As a consequence, if $f = \lim f_n$ is a pointwise limit of continuous functions (such an f is called a *Baire class 1* function), then f cannot be discontinuous everywhere: its set of continuity points is always residual. This provides another perspective on why "most" functions are badly behaved: a generic continuous function is nowhere differentiable (example 1.2.4), but any pointwise limit of continuous functions must still be continuous on a large set.

1.3 Dual Spaces

The continuous linear functionals on a normed space form its dual, and the first question about them is whether there are enough to be useful. Let X be a Banach space over k, with k either $\mathbb{R}$ or $\mathbb{C}$; its *dual* is $X^* := \operatorname{Hom}(X, k)$, the space of continuous, equivalently bounded, linear functionals on X. In finite dimensions the dual is easily understood: $\dim X^* = \dim X$, so $X^* \cong X$, though not canonically. In infinite dimensions it is far less clear when a functional is continuous at all: d/dx is unbounded, while $\int$ becomes bounded once its domain is restricted. What is wanted is a supply of functionals known in advance to extend to elements of X^*, and the Hahn-Banach extension theorem provides exactly this, showing X^* is always large enough to make its study worthwhile. It is the linear-analytic counterpart of Urysohn's lemma ([5]), which manufactures enough continuous functions to separate points and closed sets in a normal space.

The functionals to be extended are controlled by a one-sided gauge.

Definition 1.3.1: Sublinear Functional

A function $p\colon X \to \mathbb{R}$ is a *sublinear functional* if it is subadditive, $p(x+y) \le p(x) + p(y)$, and positively homogeneous, $p(tx) = t\,p(x)$ for all $t \ge 0$ in $\mathbb{R}$.

A sublinear functional need not be a seminorm: homogeneity is required only for nonnegative scalars, so $p(-x)$ and $p(x)$ may differ. Conversely every seminorm, and hence every norm, is a sublinear functional. For a concrete example that is not a seminorm, $p(x) = \limsup_n x_n$ on the real sequence space ℓ^∞ is subadditive and positively homogeneous, yet $p(-x) = -\liminf_n x_n$ differs from $p(x)$ in general; this p returns below in the construction of Banach limits. The extension theorem is proved first over $\mathbb{R}$; the passage to complex scalars comes afterwards and rests on the following correspondence.

Proposition 1.3.2: Converting Complex and Real Linear Functionals

Let X be a vector space over $\mathbb{C}$. If f is a complex-linear functional on X and $u = \mathrm{Re} f$, then u is a real-linear functional and $f(x) = u(x) - i\,u(ix)$ for all $x \in X$. Conversely, if u is a real-linear functional on X and $f\colon X \to \mathbb{C}$ is defined by $f(x) = u(x) - i\,u(ix)$, then f is complex-linear. When X is normed, $\|u\| = \|f\|$.

***Proof*:**
If f is complex-linear then $u = \mathrm{Re} f$ is real-linear, and $\mathrm{Im} f(x) = -\mathrm{Re}[if(x)] = -u(ix)$, so $f(x) = u(x) - i\,u(ix)$. Conversely, if u is real-linear then $f(x) = u(x) - i\,u(ix)$ is $\mathbb{R}$-linear, and

$$f(ix) = u(ix) - i\,u(-x) = u(ix) + i\,u(x) = if(x)$$

so f is $\mathbb{C}$-linear. Finally, suppose X is normed. Since $|u(x)| = |\mathrm{Re} f(x)| \le |f(x)|$, we have $\|u\| \le \|f\|$. For the reverse inequality, fix x with $f(x) \ne 0$ and set $\alpha = \overline{\mathrm{sgn}(f(x))}$. Then

$$|f(x)| = \alpha f(x) = f(\alpha x) = u(\alpha x)$$

since $f(\alpha x)$ is real, whence $|f(x)| \le \|u\|\,\|\alpha x\| = \|u\|\,\|x\|$ and so $\|f\| \le \|u\|$. Combining the two bounds gives $\|f\| = \|u\|$, as we sought to show.

Theorem 1.3.3: Hahn-Banach Extension Theorem

Let X be a real vector space, p a sublinear functional on X, $M \subseteq X$ a subspace, and f a linear functional on M with $f(x) \le p(x)$ for all $x \in M$. Then there exists a linear functional F on X with $F(x) \le p(x)$ for all $x \in X$ and $F|_M = f$.

***Proof*:**
The argument has two stages: a one-dimensional extension, then a maximality argument by Zorn's lemma (here playing the role of transfinite induction).

Step 1: Extending by one dimension. Let $x \in X \setminus M$; the goal is a linear functional g on $M + \mathbb{R}x$

with $g \leq p$ there and $g|_M = f$. For any $y_1, y_2 \in M$,

$$f(y_1) + f(y_2) = f(y_1 + y_2) \leq p(y_1 + y_2) \leq p(y_1 - x) + p(x + y_2)$$

Rearranging gives $f(y_1) - p(y_1 - x) \leq p(x + y_2) - f(y_2)$, and as this holds for all $y_1, y_2 \in M$,

$$\sup_{y \in M} \{f(y) - p(y - x)\} \leq \inf_{y \in M} \{p(x + y) - f(y)\}$$

so there is a real α with

$$\sup_{y \in M} \{f(y) - p(y - x)\} \leq \alpha \leq \inf_{y \in M} \{p(x + y) - f(y)\}$$

Define $g \colon M + \mathbb{R}x \to \mathbb{R}$ by $g(y + \lambda x) = f(y) + \lambda\alpha$. Then g is linear, $g|_M = f$, and $g(y) \leq p(y)$ on M (the case $\lambda = 0$). For $\lambda > 0$ and $y \in M$,

$$g(y + \lambda x) = \lambda\big[f(y/\lambda) + \alpha\big] \leq \lambda\big[f(y/\lambda) + p(x + y/\lambda) - f(y/\lambda)\big] = p(y + \lambda x)$$

and for $\lambda = -\mu < 0$,

$$g(y + \lambda x) = \mu\big[f(y/\mu) - \alpha\big] \leq \mu\big[f(y/\mu) - f(y/\mu) + p(y/\mu - x)\big] = p(y + \lambda x)$$

so $g(z) \leq p(z)$ for all $z \in M + \mathbb{R}x$.

Step 2: Maximality. The functional g satisfies the same hypotheses as f, now on the larger domain $M + \mathbb{R}x$. Order the pairs (N, h), with $M \subseteq N \subseteq X$ a subspace and h a linear functional on N extending f and dominated by p, by declaring $(N_1, h_1) \leq (N_2, h_2)$ when $N_1 \subseteq N_2$ and $h_2|_{N_1} = h_1$. Every chain has an upper bound, the functional defined on the union of its domains, so Zorn's lemma yields a maximal element (N, F). If $N \neq X$, Step 1 applied to F extends it strictly, contradicting maximality; hence $N = X$, completing the proof.

The extension theorem is most often applied with p a norm, but it does more when p is a one-sided gauge. The construction of *Banach limits* is the standard illustration: a way to assign a limit to every bounded sequence, agreeing with the usual limit when it exists and unchanged by shifting the sequence.

Example 1.3.4: Banach Limits

Let $\ell^\infty = \ell^\infty(\mathbb{N}; \mathbb{R})$ be the bounded real sequences, $c \subseteq \ell^\infty$ the convergent ones, and $S \colon \ell^\infty \to \ell^\infty$ the shift $(Sx)_n = x_{n+1}$. There is a linear functional $\mathrm{LIM} \colon \ell^\infty \to \mathbb{R}$ with:

1. $\mathrm{LIM}\, x = \lim_n x_n$ for every $x \in c$;
2. $\liminf_n x_n \leq \mathrm{LIM}\, x \leq \limsup_n x_n$ for every $x \in \ell^\infty$; in particular LIM is positive and $\|\mathrm{LIM}\| = 1$;
3. $\mathrm{LIM}(Sx) = \mathrm{LIM}\, x$ for every $x \in \ell^\infty$.

To construct it, let $p(x) = \limsup_{N \to \infty} \frac{1}{N} \sum_{n=1}^{N} x_n$, the lim sup of the Cesàro means. Subadditivity of p follows from $\limsup(a_N + b_N) \leq \limsup a_N + \limsup b_N$, and positive homogeneity is clear, so p is a sublinear functional. On c the Cesàro means of a convergent sequence converge to its limit, so $\lim_n x_n = p(x)$ there; viewing the ordinary limit as a linear functional on c, it satisfies $\lim \leq p$ on c. By theorem 1.3.3 it extends to a linear functional LIM on ℓ^∞ with $\mathrm{LIM} \leq p$ everywhere.

Property (1) holds by construction. For (2), the estimate $p(x) \le \limsup_n x_n$ gives $\operatorname{LIM} x \le \limsup_n x_n$, and applying it to $-x$ gives $\operatorname{LIM} x \ge \liminf_n x_n$; taking $x \ge 0$ shows LIM is positive, and since LIM fixes the constant sequence 1 while $|\operatorname{LIM} x| \le \|x\|_\infty$, its norm is exactly 1. For (3), the telescoping average

$$\frac{1}{N}\sum_{n=1}^{N}\big((Sx)_n - x_n\big) = \frac{x_{N+1} - x_1}{N} \xrightarrow[N\to\infty]{} 0$$

forces $p(Sx - x) = 0$ and $p(x - Sx) = 0$, so $\operatorname{LIM}(Sx - x) \le 0$ and $\operatorname{LIM}(x - Sx) \le 0$, giving $\operatorname{LIM}(Sx) = \operatorname{LIM} x$.

To show this extension, take any bounded sequence periodic with period q, so $x_{n+q} = x_n$. Summing q consecutive shifts, $\sum_{j=0}^{q-1} S^j x$ has the value $x_1 + \cdots + x_q$ at every index, so shift invariance and property (1) give

$$q \operatorname{LIM} x = \operatorname{LIM}\Big(\sum_{j=0}^{q-1} S^j x\Big) = x_1 + \cdots + x_q$$

that is, $\operatorname{LIM} x = \frac{1}{q}(x_1 + \cdots + x_q)$, the average over one period. So $(1, 0, 1, 0, \dots)$ has $\operatorname{LIM} x = \frac{1}{2}$ and $(1, 2, 3, 1, 2, 3, \dots)$ has $\operatorname{LIM} x = 2$, though neither sequence converges.

Banach limits also settle a duality question. The functional LIM vanishes on the subspace c_0 of sequences tending to 0, yet is not identically zero, so it is not represented by any $a \in \ell^1$ under the pairing $\langle x, a\rangle = \sum_n x_n a_n$ (an $a \in \ell^1$ annihilating all of c_0 must be 0). Thus ℓ^1 is a proper subspace of $(\ell^\infty)^*$, an early sign that ℓ^∞ is not reflexive.

Before drawing the geometric consequences, the extension theorem must be carried to complex scalars. For the domination bound to behave there, p must be a seminorm rather than a general sublinear functional: when p is a seminorm and $f\colon X \to \mathbb{R}$ is linear, the bound $f \le p$ is equivalent to $|f| \le p$, since $|f(x)| = \pm f(x) = f(\pm x)$ and $p(-x) = p(x)$.

Proposition 1.3.5: Complex Hahn-Banach Theorem

Let X be a vector space over $\mathbb{C}$, p a seminorm on X, $M \subseteq X$ a subspace, and f a complex-linear functional on M with $|f(x)| \le p(x)$ for all $x \in M$. Then there exists a complex-linear functional F on X with $|F(x)| \le p(x)$ for all $x \in X$ and $F|_M = f$.

***Proof*:**
Let $u = \operatorname{Re} f$, a real-linear functional on M with $u \le |f| \le p$. Applying theorem 1.3.3 to u produces a real-linear extension U of u to X with $U \le p$; since $p(-x) = p(x)$, this gives $|U(x)| \le p(x)$ for all x. Set $F(x) = U(x) - i\,U(ix)$. By proposition 1.3.2, F is a complex-linear extension of f, and for x with $F(x) \ne 0$, taking $\alpha = \overline{\operatorname{sgn}(F(x))}$ gives

$$|F(x)| = \alpha F(x) = F(\alpha x) = U(\alpha x) \le p(\alpha x) = p(x)$$

completing the proof.

The geometry now follows. The next result collects the consequences that make the dual space a usable tool.

Proposition 1.3.6: Hahn-Banach Consequences

Let X be a normed vector space over k. Then:

1. If $M \subseteq X$ is a closed subspace and $x \in X \setminus M$, there exists $f \in X^*$ with $f|_M = 0$ and $f(x) \neq 0$; it may be chosen so that, with $\delta = \inf_{y \in M} \|x - y\|$,
$$\|f\| = 1 \qquad f(x) = \delta$$
2. If $0 \neq x \in X$, there exists $f \in X^*$ with $\|f\| = 1$ and $f(x) = \|x\|$.
3. The bounded linear functionals on X separate points.
4. For $x \in X$, the evaluation $\hat{x} \colon X^* \to k$, $\hat{x}(f) = f(x)$, defines a linear isometry $x \mapsto \hat{x}$ from X into the double dual X^{**}.

***Proof*:**

1. Let $\delta = \inf_{y \in M} \|x - y\|$, positive because M is closed and $x \notin M$. On $M + kx$ define $f_0(y + \lambda x) = \lambda\delta$, so $f_0(x) = \delta$ and $f_0|_M = 0$. For $\lambda \neq 0$,
$$|f_0(y + \lambda x)| = |\lambda|\, \delta \leq |\lambda|\, \|\lambda^{-1} y + x\| = \|y + \lambda x\|$$
using $\delta \leq \|x + \lambda^{-1} y\|$, so f_0 has norm at most 1 on $M + kx$, with equality since $\delta = \inf$. Taking $p(z) = \|z\|$, theorem 1.3.3 (or proposition 1.3.5 over $\mathbb{C}$) extends f_0 to $F \in X^*$ with $\|F\| = 1$, $F|_M = 0$, and $F(x) = \delta$.
2. Apply part (1) with $M = \{0\}$, for which $\delta = \|x\|$; the resulting f has $\|f\| = 1$ and $f(x) = \|x\|$.
3. If $x \neq y$, part (2) applied to $x - y \neq 0$ gives $f \in X^*$ with $f(x - y) \neq 0$, so $f(x) \neq f(y)$.
4. Both $\hat{x}$ (in f) and $x \mapsto \hat{x}$ are linear. From $|\hat{x}(f)| = |f(x)| \leq \|f\|\, \|x\|$ we get $\|\hat{x}\| \leq \|x\|$, while part (2) supplies f with $\|f\| = 1$ and $\hat{x}(f) = \|x\|$, so $\|\hat{x}\| = \|x\|$; thus $x \mapsto \hat{x}$ is an isometry.

Parts (1) and (2) sharpen Urysohn's separation into a quantitative statement: the separating functional not only exists but has its norm and value pinned to the geometry of the subspace. This motivates the following definition.

Definition 1.3.7: Annihilator

For a subspace $Y \subseteq X$ of a normed space X, the *annihilator* of Y is
$$Y^{\perp} = \{\varphi \in X^* \mid \varphi(y) = 0 \text{ for all } y \in Y\}$$
Dually, for a subspace $Z \subseteq X^*$, the *pre-annihilator* is
$${}^{\perp}Z = \{x \in X \mid \varphi(x) = 0 \text{ for all } \varphi \in Z\}$$

Part (1) shows that $Y^\perp$ is large enough to detect Y: any point outside the closure of Y is exposed by some functional in $Y^\perp$. Concretely, $Y = {}^\perp(Y^\perp)$ for every norm-closed subspace $Y \subseteq X$. The inclusion $Y \subseteq {}^\perp(Y^\perp)$ is immediate, and conversely, if $x \notin Y$ then part (1) produces $\varphi \in Y^\perp$ with $\varphi(x) \neq 0$, so $x \notin {}^\perp(Y^\perp)$. This identity is the tool behind the closed-range theorem in chapter 5 and its unbounded analogue.

The same functional computes distances, sharpening part (1) into a duality formula.

Corollary 1.3.8: Norm and Distance Duality

Let X be a normed space, $x \in X$, and $M \subseteq X$ a closed subspace. Then

$$\|x\| = \max_{\substack{f \in X^* \\ \|f\| \le 1}} |f(x)| \qquad \text{and} \qquad \inf_{y \in M} \|x - y\| = \max_{\substack{f \in M^\perp \\ \|f\| \le 1}} |f(x)|$$

both maxima attained.

***Proof*:**
Any f with $\|f\| \le 1$ satisfies $|f(x)| \le \|x\|$, and part (2) of proposition 1.3.6 supplies an f of norm 1 with $f(x) = \|x\|$, giving the first identity with the maximum attained. For the second, any $f \in M^\perp$ with $\|f\| \le 1$ has $|f(x)| = |f(x-y)| \le \|x-y\|$ for every $y \in M$, so $|f(x)| \le \inf_y \|x-y\|$, while part (1) supplies an $f \in M^\perp$ of norm 1 attaining $f(x) = \inf_y \|x - y\|$, as we sought to show.

Part (3), that X^* separates points, is the linear face of injectivity: if a construction produces x, y with $f(x) = f(y)$ for every $f \in X^*$, then $x = y$, exactly as coordinates decide equality in finite dimensions. This is the property that makes the weak topology usable. Giving X the coarsest topology in which every $f \in X^*$ is continuous, separation of points is precisely what makes that topology Hausdorff, so weak limits are unique (proposition 1.4.6).

Separation has a second face: bundling all of X^* together represents X as continuous functions on a compact space.

Proposition 1.3.9: Universal Embedding into $C(K)$

Let X be a normed space and $K = (B_{X^*}, w^*)$ the closed unit ball of X^* under the weak* topology. Then K is compact Hausdorff and $x \mapsto \hat{x}|_K$ is a linear isometry $X \to C(K)$. If X is a Banach space, its image is a closed subspace; thus every Banach space is isometrically a closed subspace of some $C(K)$.

***Proof*:**
Compactness of K is Alaoglu's theorem (theorem 1.5.7), proved later in this chapter, and the weak* topology is Hausdorff because X separates the points of X^*. Each $\hat{x}$ is weak* continuous by the definition of that topology, so $\hat{x}|_K \in C(K)$, and

$$\|\hat{x}|_K\|_{C(K)} = \sup_{\|f\| \le 1} |f(x)| = \|x\|$$

by corollary 1.3.8. Hence $x \mapsto \hat{x}|_K$ is a linear isometry. When X is complete, its isometric image is complete and therefore closed in $C(K)$, completing the proof.

1.3.10 Note: Relation to Stone-Weierstrass The embedding places X in $C(K)$ as a *linear* subspace, and $\widehat{X}$ does separate the points of K. It is tempting to invoke the Stone-Weierstrass theorem ([3]) to conclude that X is dense, but that theorem needs a *subalgebra* separating points, and $\widehat{X}$ is not closed under products. What Stone-Weierstrass gives instead is that the closed subalgebra *generated* by $\widehat{X}$ and the constants (with complex conjugates, over $\mathbb{C}$) is dense in $C(K)$; this recovers all of $C(K)$, not X. In general X is a proper closed subspace: linear separation is strictly weaker than the algebraic separation Stone-Weierstrass requires.

The supply of functionals in proposition 1.3.6 depends on the norm being convex in a way the hypotheses quietly use; without local convexity the dual can collapse.

1.3.11 Warning: Enough functionals needs convexity For $0 < p < 1$ the space $L^p([0,1])$, metrized by $d(f,g) = \int_0^1 |f-g|^p$, is a complete topological vector space whose only continuous linear functional is 0. Hahn-Banach produces nothing because its hypotheses fail: d is not a norm and the space is not locally convex. Concretely, let φ be continuous and linear and f arbitrary. Choose c with $\int_0^c |f|^p = \frac{1}{2}\int_0^1 |f|^p$ and split $f = f_1 + f_2$ into the parts on $[0,c]$ and $[c,1]$, so $d(f_i,0) = \frac{1}{2}d(f,0)$. From $\varphi(f_1) + \varphi(f_2) = \varphi(f)$ one part has $|\varphi(f_j)| \geq \frac{1}{2}|\varphi(f)|$; set $g = 2f_j$, so $|\varphi(g)| \geq |\varphi(f)|$ while $d(g,0) = 2^p \cdot \frac{1}{2}d(f,0) = 2^{p-1}d(f,0)$. As $p < 1$ the factor $2^{p-1} < 1$, so iterating from a fixed f yields g_n with $|\varphi(g_n)| \geq |\varphi(f)|$ but $d(g_n,0) \to 0$; continuity forces $\varphi(f) = 0$. The abundance of functionals guaranteed by Hahn-Banach is thus a feature of *locally convex* spaces, the class introduced in section 1.4.

1.3.12 Foreshadowing: The geometric form Parts (1) and (3) are the analytic face of Hahn-Banach: extend a functional, separate a point from a subspace. There is a dual geometric face, in which two disjoint convex sets are separated by a closed hyperplane $\{f = c\}$ and a boundary point of a convex body carries a supporting hyperplane. That form belongs to the theory of locally convex spaces begun in section 1.4, where convex neighbourhoods are plentiful enough to run the same gauge argument on an open convex set.

Part (4) invites a look at the double dual. Writing $\widehat{X} = \{\hat{x} \mid x \in X\}$, the space X^{**} is always complete, so the closure $\overline{\widehat{X}}$ inside X^{**} is a Banach space and $x \mapsto \hat{x}$ embeds X into it isometrically with dense image. This exhibits $\overline{\widehat{X}}$ as the *completion* of X: every normed space sits densely inside a Banach space, obtained here with no new construction. If X is already Banach, then $\widehat{X} = \overline{\widehat{X}}$ is closed.

The image $\widehat{X}$ need not exhaust X^{**}. When it does, so that the canonical isometry $X \to X^{**}$ is onto, X is called *reflexive*. Every finite-dimensional space is reflexive, since $\dim X^{**} = \dim X$. In infinite dimensions reflexivity can fail: c_0 has $c_0^{**} = \ell^\infty \neq c_0$, so it is not reflexive, and the same holds for ℓ^1 and ℓ^∞. By contrast the L^p spaces with $1 < p < \infty$ are reflexive, as the duality $(L^p)^* = L^q$ will show.

Exercise 1.3.1

1. Let $T\colon X \to Y$ be a linear map between Banach spaces such that $f \circ T \in X^*$ for every $f \in Y^*$. Show that T is bounded. (Hint: the closed graph theorem and separation of points.)

2. Show that a subspace $M \subseteq X$ of a normed space is dense if and only if $M^{\perp} = \{0\}$.
3. Deduce from the periodic-sequence value in example 1.3.4 that no Banach limit is multiplicative, i.e. $\mathrm{LIM}(xy) = (\mathrm{LIM}\, x)(\mathrm{LIM}\, y)$ can fail.
4. Let p be a sublinear functional on a real vector space X. Show that $p(x) = \max\{f(x) : f \text{ linear on } X,\ f \leq p\}$ for every x, the maximum attained.
5. For a closed subspace $M \subseteq X$, show that $\psi \mapsto \psi \circ q$, where $q \colon X \to X/M$ is the quotient map, is an isometric isomorphism $(X/M)^* \cong M^{\perp}$.

1.4 Weak Topologies and TVS's

(This post seems like an excellent post to follow! link here)

There are many topologies on which the operations are continuous (i.e. $+$ and scalar multiplication is continuous) that do not arrise from norms. The topology on a vector-space induced by a norm is in fact quite strong. In general, such a space (in infinite dimensions) doesn't have many compact set, and so we loose many of the nice results of working on comapct or pre-compact sets. This is naturally unfavourable in analysis, since many arguments rely on properties of sequences converging. Looking at functional analysis as the study of sequences, norms induce a particular definition of convergence: $\|x - x_n\|_V \to 0$. However, there are many other notions of convergence that are weaker that are quite useful, for example:

1. pointwise convergence
2. convergence in measure
3. smooth convergence
4. convergence on compact sets

None of these types of convergence are induced by a norm. We will instead be induced new types of topologies on our vector-spaces. The two types of topologies we will usually impose on V will be known as the *weak topology* and the *weak* topology*. We will define analogous topologies on $\mathrm{Hom}_k(V, W)$ which somewhat confusingly will be named *strong operator topology* and *weak operator topology*.

Definition 1.4.1: Topological Vector Space (TVS)

Let V be a vector-space over $\mathbb{R}$ or $\mathbb{C}$. Then V is a *topological vector space* (TVS) if the operation of addition and scalar multiplication are continuous. A topological vector-space is said to be *locally convex* if there is a base for the topology consisting of convex sets.

It should be verified that the translation map $x \mapsto x_0 + x$ and the dilation map $x \mapsto \lambda x$ are both homeomorphissm for any scalar λ and vector x_0.

Example 1.4.2: Topological Vector Spaces

1. Let V be a normed vector-space with the topology induced by the norm. Then $\|c \cdot x\| \leq |c|\|x\|$ (in fact $=$) and $\| + (x,y)\| \overset{!}{\leq} 2(\|x\| + \|y\|)$ where $\overset{!}{\leq}$ comes from the triangle inequality and the fact that $\|(x,y)\| = \max(\|x\|, \|y\|)$. Hence, V is a topologicla vector space. If $\| - \|$ was a quasi-norm so that we replace the triangle inequality with $\|x + y\| \leq K(\|x\| + \|y\|)$ then a quasi-normed space V is still a topological vector space.

2. Let V be a semi-normed space so that $\|x\| = 0$ does not necessarily imply $x = 0$. Then V is a topological vector space with the topology induced by the semi-norm. V is Hausdroff if and only if the semi-norm is a norm.

3. Any subbspace $W \subseteq V$ of a topological vector space V is a topological vector space with the subspace topology. Similarly for products and quotients.

4. Let V be a vector-space and $(\mathcal{T}_\alpha)_{\alpha \in A}$ be a collection of topologies on V that makes V a topological vector space. Let $\mathcal{T} = \langle \mathcal{T}_\alpha \rangle_{\alpha \in A}$ be the topology generated by $\bigcup_{\alpha \in A} \mathcal{T}_\alpha$ (this is the weakest topology that contains each $\mathcal{T}_\alpha$). Then $(V, \mathcal{T})$ is also a topological vector space. Importantly, a sequence (resp. net) $x_n \to x$ in $\mathcal{T}$ if and only if $x_n \to x$ in $\mathcal{T}_\alpha$ for all $\alpha \in A$ (recall that $x_n \to x$ in a general topology space if for any neighborhood U of x, there exists an N such that for all $n \geq N$, $x_n \in U$). Commonly, we will have $(\mathcal{T}_\alpha)_{i=1}^\infty$ be generated by countable collection of semi-norms. A complete space generated by a countable collection of semi-norms is called a *Fréchet space*.

5. (Pointwise Convergence) Let X be a set and $\mathbb{C}^X$ the collection of all complex valued functions $f : X \to \mathbb{C}$. Then $\mathbb{C}^X$ is a complex-vector-space. For each $x \in X$, define the seminorm:
$$\|f\|_x := |f(x)|$$
Then the topology generated by all of these seminorms is the *topology of pointwise convergence* on $\mathbb{C}^X$ (it is also the product topology on this space). A sequence $(f_n) \subseteq \mathbb{C}^X$ converges to f in this topology if and only if it converges pointwise. Note that if X has more than one point, then none of the semi-norms individually generate a Hausdroff topology, but taken all together the induced topology is Haudsorff.

6. (Uniform Convergence) Let X be a topological space and $C(X)$ the set of complex-valued continuous function $f : X \to \mathbb{C}$. If X is not compact, then in general we would not expect functions in $C(X)$ to be bounded, so the sup-norm will not necessarily make $C(X)$ into a normed vector-space. Nonetheless, we can still define "balls" $B_r(f) \subseteq C(X)$ by:
$$B_r(f) := \left\{ g \in C(X) \;\middle|\; \sup_{x \in X} |f(x) - g(x)| \leq r \right\}$$
these do form a topological structure on $C(X)$, but not a topological vector space structure since scalar multiplication is no longer continuous. Nonetheless, we have that $(f_n) \subseteq C(X)$ converges in this topology to a limit $f \in C(X)$ if and only if f_n converge uniformaly to f. Thus $\sup_{x \in X} |f_n(x) - f(x)|$ is finite for sufficinetly large n and converges to 0 as $n \to \infty$.

7. (Uniform Convergence on Compact sets) Let X and $C(X)$ be as in the above example. For every compact subset $K \subseteq X$, define a seminorm:
$$\|f\|_{C(K)} = \sup_{x \in K} |f(x)|$$

The topology generated by all these seminorms as k ranges over all compact subsets of X is called the *topology of uniform convergence on compact sets*. It is stronger than the topology of pointwise convergence but weaker than the topology of uniform convergence, and is frequently used in complex analysis. A sequence $(f_n) \subseteq C(X)$ converges to $f \in C(X)$ in this toplogy if and only if f_n converges uniformly to f on each compact set. Is the resulting space a topological vector space?

8. (Smooth Convergence) Let $C^\infty([0,1])$ be the space of smooth functions $f : [0,1] \to \mathbb{C}$. Then we can define the C^k-norm on this space for any non-negative integer k by the formula:

$$\|f\|_{C^k} := \sum_{i=1}^{k} \sup_{x\in[0,1]} |f^{(i)}(x)|$$

where $f^{(i)}$ is the ith derivative of f. The topology generated by all C^k norms $k = 0, 1, 2, ...$, is the *smooth topology*. A sequence f_n converges in this topology to a limit f if $f_n^{(i)}$ converges uniformally to $f^{(i)}$ for each $i \geq 0$. Is this a topological vector space?

9. (Convergence in measure) Let $(X, \mathcal{M}, \mu)$ be a measure space and $L(X)$ the set of measurable functions $f : X \to \mathbb{C}$. Then the sets

$$B_{r,\epsilon}(f) := \{g \in L(X) \mid \mu(\{x \mid |f(x) - g(x)| \geq r\} < \epsilon)\}$$

for $f \in L(X)$, $\epsilon, r > 0$ forms the basis for a topology that makes $L(X)$ into a topological vector space. A sequence $(f_n) \subseteq L(X)$ converges to f in this topology if and only if it converges in measure.

10. Let $[0,1]$ be given the usual Lebesgue measure. Then show that $L^\infty([0,1])$ cannot be given a topological vector space structure in which a sequence $(f_n) \subseteq L^\infty([0,1])$ converges to f in this topology if and only if it converges a.e.

11. Let V be an $\mathbb{R}$-vector space. a subset $U \subseteq V$ is called *algebraically open* if

$$\{t \in \mathbb{R} \mid x + tv \in U\}$$

is open for all $x, v \in V$. Then any open set in a topological vector space is algebraically open. Note that there are algebraically open sets that are not open (take any dense subset of S^1, say D, and $\mathbb{R} \setminus \{D\}$.) This larger set of algebraically open sets forms a topology, but this topology does not give the $\mathbb{R}^2$ structure of a topological vector space.

12. Let X be a LCH space. On $\mathbb{C}^X$, the topology of uniform convergence on compact sets is defined by the seminorms $p_K(f) = \sup_{x\in K} |f(x)|$ as K rangers over compact subsets of X. If X is σ-compact and $\{U_n\}$ are as in proposition ref;HERE (folland p. 168), this topology is defined by the seminorms $p_n(f) = \sup_{x\in\overline{U_n}} |f(x)|$. In this case, $\mathbb{C}^X$ is easly seen to be complete, so it is a Fréchet space. By proposition ref:HERE, so is $C(X)$.

13. The space $L^1_{\text{loc}}(\mathbb{R}^n)$ is a Fréchet space with the topology defined by the seminorms $p_k(f) = \int_{|x|\leq k} |f(x)|dx$ (completeness follows from completness of L^1).

14. Here is another useful place where TVS are defined. Let's say we want to study a space in which d/dx is continuous (and hence bounded) on $C^\infty([0,1])$. However, this is essentially impossible to define the topology through a norm. Indeed, if $f_\lambda(x) = e^{\lambda x}$, then $(d/dx)f_\lambda = \lambda f_\lambda$, and so $\|d/dx\| \geq \lambda$ for all λ, no matter what norm is used on $C^\infty([0,1])$. There are a few ways to rectify this:

(a) One can consider differentiation as an unbounded operator form V to W where W is a suitable banace space and X is a dense subpsac eof W (exercise 30 in Folland)

(b) One can consider differentiation as a bounded linear map from one Banach space V to a different one W such that $V = C^k([0, 10])$ and $W = C^{k-1}([0, 1])$ (execise 9)

(c) One can consider differentiation as a continuous operator on a locally convex space V whose topology is not given by a norm. It is easy to construct families of seminorms on the space of smooth functions st differentiation becomes continuous almost by definition. For example, the seminorms:

$$p_k(f) = \sup_{0 \le x \le 1} |f^{(k)}(x)|$$

where $k \in \{0, 1, 2, ...\}$ makes $C^\infty([0, 1])$ into a Fréchet space (the completeness is proved in exercise 9), and d/dx is continuous by proposition 5.15, since $p_k(f') = p_{k+1}(f)$. Other examples are in exer exercsie 45 in chatper 9 of Folland.

15. The space $\mathbb{R}^\infty = \varinjlim \mathbb{R}^n$ is a Fréchet space; can you see the norms?

Proposition 1.4.3: Topological Vector Space And Hausdorff

Let V be a topological vector space. Then V is Hausdroff if and only if $\{0\}$ is closed

***Proof*:**
exercise

We may ask what properties of Banach spaces extend to topological vector space's. Some nice properties of Banach spaces include the uniform boundedness principle and the open mapping theorem. These can be extended to the slighty larger class of *Fréchet spaces* that we mentioned in the examples above. We will single them out and talk about them in the next section.

If V is a topological vector space, we can take V^* to be the set of all continuous functionals from V to the base field $k \in \{\mathbb{R}, \mathbb{C}\}$. Since V has no norm, V^* has no norm topology. We can however still define interesting topologies on it. This topology is dual to a weaker topology we can put on V:

Definition 1.4.4: Weak And Weak* Topologies

Let V be a topological vector space and V^* its dual. Then

1. the *weak* topology* on V^* is the topology generated by the seminorms $\|\lambda\|_x := |\lambda(x)|$ for all $x \in V$
2. the *weak topology* on V is the topology generated by the seminorms $\|x\|_\lambda = |\lambda(x)|$ for all $\lambda \in V^*$

If V is also a normed vector-space, then the weak topology on V is weaker than the norm topology. If V is a normed space, then so is V^* and $(V^*)^*$. the weak* topology on V^* makes V^* into a topological vector space, and is a topology weaker than the weak topology on V^* (which is defined using the double dual $(V^*)^*$). When V is reflexive ($V \cong V^{**}$), then the weak and weak* topologies on V^* are in fact equivalent.

Definition 1.4.5: Weak Convergence

Let V a vector-space endowed with the weak topology. Then a sequence $(x_n) \subseteq V$ that converges in V with the weak topology is called *weakly convergent* and is said to *weakly converge* to a point x. A sequence weakly converges $x_n \to x$ if and only if

$$\lambda(x_n) \to \lambda(x) \qquad \forall \lambda \in V^*$$

and is usually denoted as $x_n \rightharpoonup x$. Similarly, $\lambda_n \rightharpoonup \lambda$ if $\lambda_n(x) \to \lambda(x)$ for all $x \in V$ (λ_n, as a function of V, converges pointwise to λ)

Proposition 1.4.6: Normed Vector Spaces Weak Topology

Let V be a normed vector-space. Then the weak topology on V and the weak* topology on V^* are both Hausdorff. In particular, we conclude that in the weak and weak* limits, when they exist they are unique

***Proof*:**
Use the Hahn-Banach Theorem

Example 1.4.7: Norm, Weak, Weak* Convergence
Let $V = c_0(\mathbb{N})$, $V^* = \ell^1(\mathbb{N})$, and $V^{**} = \ell^\infty(\mathbb{N})$. Let $e_1, e_2, ...$ be the standard basis of any of these vetor-spaces.

1. The sequence (e_i) converges weakly in V to zero, but not not converge in norm in V
2. (easy to think up some examples)

A natural generalization of the weak* topology on X^* can be imposed $B(X,Y)$ for some TVS Y. Just like for X^*, we can impose a weak topology on it, hewer there is another natural topology we may induce on $B(X,Y)$:

Definition 1.4.8: Strong And Weak Operator Topology

Let $B(X,Y)$ be a all continuous operators between Fréchet spaces. Then

1. the *weak operator topology* is the topology induced by the seminorm $T \mapsto |\lambda(T(x))|$ for all $x \in X$ and $\lambda \in Y^*$
2. If furthermore Y is a Banach space, then the *strong operator topology* (or *operator norm topolgoy*) is the topology induced by the seminorms $T \mapsto \|T(x)\|_Y$ for all $x \in X$

Hence, a sequence $(T_n) \subseteq B(X,Y)$ converges in the strong operator topology to a limit T if and only if $T_n(x) \to T(x)$ in the norm topology on Y, and T_n converges to T in the weak operator topology if $T_n(x) \to T(x)$ in e weak topology on x. Hence, we see that the weaker operator topology is weaker than the strong operator topology, which in turn is (somewaht confusingly) weaker than the operator norm topology.

A final result we shall point out is hat we can reprhase the uniform boundedness principle for convergence now as follows:

Proposition 1.4.9: Uniform Boundedness Principle Reformulation

Let $(T_N) \subseteq B(X,Y)$ be a sequence of bounded linear operators from a Banach space X to a normed space Y, and let $T \in B(X,Y)$ be another bounded lienar operator, and let D be a dense subspace of X. Then the following are equivalent:

1. T_n converges in the strong operator topology fo $B(X,Y)$ to T
2. T_n is bounded in the operator norm (i.e. $\|T_n||_{op}$ is bounded) and the restriction of T_n to D converges in the strong operator topology of $B(D,Y)$ to the restriction of T to D

1.5 Fréchet Spaces

When constructing a space, we often cannot define a space simply through a norm; not all elements may be "finite". However, we can often construct a space using countably many normed spaces, or usually semi-normed space $\|x\| = 0$ does not imply $x = 0$, for example if f is an integrable function). Due to the countable nature of this construction, most types of convergences are accomplished by a family of semi-norms. For example, convergence on compact sets, pointwise convergnce, uniform convergence, and smooth convergence are all convergence in Fréchet spaces. In analysis, there will be many spaces that will Fréchet spaces, notably when you encounter distributions and require to define a topology on the dual space. Hence, a moment to get a better undersatnding of them is worth it.

Theorem 1.5.1: Semi-Norms Generate TVS

Let $\{p_\alpha\}_{\alpha \in A}$ be a family of norms on a vector space V. If $x \in X$, $\alpha \in A$, and $\epsilon > 0$, let

$$U_{x,\alpha,\epsilon} = \{y \in V \mid p_\alpha(y - x) < \epsilon\}$$

and let $\mathcal{T}$ be the topology generated by the sets $U_{x,\alpha,\epsilon}$. Then:

1. For each $x \in X<$ the inite intersections of the sets $U_{x,\alpha,\epsilon}$ $(\alpha \in A, \epsilon > 0)$ form a neigborhood base at x
2. if $\langle x_i \rangle_{i \in I}$ is a net in V ,then $x_i \to x$ if and only if $p_\alpha(x_i - x) \to 0$ for all $\alpha \in A$.
3. $(V, \mathcal{T})$ is a locally convex topological vector-space

***Proof*:**

1. If $x \in \bigcap_1^k U_{x_j,\alpha_j,\epsilon_j}$, let $\delta_j = \epsilon_j - p_\alpha(x - x_j)$. By the triangle inequality

   $$x \in \cap_1^k U_{x,\alpha_j,\delta_j} \subseteq \cap_1^k U_{x-j,\alpha_j,\epsilon_j}$$

   Thus, the assertion now follows from the fact that the topology generated consists of $\emptyset$, X, and all unions of finite intersections.

2. By part (1), it suffices to see that $p_\alpha(x_i - x) \to 0$ if and only if $\langle x_i \rangle$ is eventually in $U_{x,\alpha,\epsilon}$ for all $\epsilon > 0$.

3. Continuity is easy from part (b), for if $x_i \to x$ nd $y_i \to y$, then

$$p_\alpha((x_i + y_i) - (x + y)) \leq p_\alpha(x_i - x) + p_\alpha(y_i - y) \to 0$$

so $x_i + y_i \to x + y$. If furthermore $\lambda_i \to \lambda$, then eventually $\lambda_i \leq C + |\lambda| + 1$, so

$$p + \alpha(\lambda_i x_i - \lambda x) \leq p_\alpha(\lambda_i(x_i - x)) + p_\alpha((\lambda_i - \lambda)x) \leq CP_\alpha(x_i - x) + |\lambda_i - \lambda| p_\alpha(x)$$

Hence, $\lambda_i x_i \to \lambda x$. Furthermore, the sets $U_{x,\alpha,\epsilon}$ are convex, for if $y, z \in U_{x,\alpha,\epsilon}$, then

$$p_\alpha(x - [t + (1-t)z]) \leq p_\alpha(tx - ty) + p_\alpha((1-t)x + (1-t)z) < t\epsilon + (1-t)\epsilon = \epsilon$$

Then local convexity follows from (1)

Definition 1.5.2: Fréchet Space

A complete Hausdorff topological vector-space whose topology is defined by a countable family of seminroms is called a *Fréchet space*.

Proposition 1.5.3: Continuity in Fréchet Space

Let $T : V \to W$ be a linear map between two vector spaces. Let V be a topology induced by a family of semi-norms $(\| - \|_{V_\alpha})_{\alpha \in A}$ and W the topology induced by a family of semi-norm $(\| - \|_{W_\beta})_{\beta \in B}$. Then T is continuous if and only if for each $\beta \in B$, there exists a finite subsets $A_\beta \subseteq A$ and a constant C_β such that

$$\|T(x)\|_{W_\beta} \leq C_\beta \sum_{\alpha \in A_\beta} \|x\|_{V_\alpha} \tag{1.1}$$

for all $x \in V$.

***Proof*:**
Let equation 1.1 hold and consider a convergent net $\langle x_i \rangle \subseteq X$. Then by theorem 1.5.1 $p_\alpha(x_i - x) \to 0$ for all α, thus by the inequality $\|T(x_i) - T(x)\|_{S_\beta} \to 0$ for all β, which implies $T(x_i) \to T(x)$. Since T preserves convergent nets, T is continuous.

Conversely, if T is continuous for every $\beta \in B$ choose a neighbor U_β of 0 in V such that $\|T(x)\|_{W_\beta} < 1$ for all $x \in U_\beta$. By theorem 1.5.1, we may assume that $U_\beta = \cap_{1k} U_{x\alpha_{\epsilon_j}}$. Let $\epsilon = \min(\epsilon_1, \epsilon_2, ..., \epsilon_k)$ so that $\|T(x)\|_{W_\beta} < 1$ whenever $p_{\alpha_j}(x) < \epsilon$ for all j. Now, for $x \in X$, either:

1. $p_{\alpha_j})(x) > 0$ for some j. Then let $y = \epsilon x / (\sum_1^k p_{\alpha_j}(x))$ so tht $p_{\alpha_j}(y) < \epsilon$. Then:

$$\|T(x)\|_{W_\beta} = \sum_1^k \epsilon^{-1} p_{\alpha_j}(x) \|T(y)\|_{W_\beta} \leq \epsilon^{-1} \sum_1^k p_{\alpha_j}(x)$$

2. If $p_{\alpha_j}(x) = 0$ for all j, then $p_{\alpha_j}(rx) = 0$ for all j and all $r > 0$. Thus $r\|T(x)\|_{W_\beta} = \|T(rx)\|_{W_\beta} < 1$ for all $r > 0$, and os $\|T(x)\|_{W_\beta}$. Then certainly

$$\|T(x)\|_{W_\beta} \leq \epsilon^{-1} \sum_{1}^{k} p_{\alpha_j}(x)$$

covering both cases, we are done.

The following proposition is an interesting characteriztion of Fréchtet spaces:

Proposition 1.5.4: Hausdorff and invariant Metric

Let X be Fréhet space given by $\{p_\alpha\}_{\alpha \in A}$. Then

1. X is Haudorff if and only if for each $x \neq 0$, there exists $\alpha \in A$ such that $p_\alpha(x) \neq 0$
2. If X is Hausdorff and A is countable, then X is metrizable with a trnslation invariant metric (i.e. $\rho(x, y) = \rho(x + z, y + z)$ for all $x, y, z \in X$)

***Proof*:**
exercise 43 Folland chatper 5

Note that topologically, all these spaces are the same (as the bellow theorem will sketch). So what really matters is what extra information these semi-norms (or this invariant metric) provides:

Theorem 1.5.5: Anderson-Kadec Theorem

Every infinite-dimensional separable Fréchet space is homeomorphic to the product space $\mathbb{R}^{\mathbb{N}}$ (the countable product of real lines equipped with the product topology).

Note that L^∞ is *inseparable* and hence is not included in this theorem.

***Proof*:**
(Sketch). The proof is historically a combination of results by Kadec (1966) and Anderson (1966), establishing a chain of homeomorphisms. We outline the two main steps:

1. The Banach Case: Kadec proved that every infinite-dimensional separable Banach space is homeomorphic to the Hilbert space ℓ_2. The construction typically involves re-norming the space with a strictly convex equivalent norm (often called a Kadec norm) and defining a homeomorphism via radial projection and coordinate mapping.
2. The Product Case: Anderson proved that ℓ_2 is homeomorphic to $\mathbb{R}^{\mathbb{N}}$. This is often shown using "separable absorption" theorems or the "notback-and-forth" method to manage the coordinate systems.

Since every Fréchet space is a projective limit of Banach spaces (and using the assumption of separability), one can extend the homeomorphism to the general case. Consequently, despite their distinct linear and geometric properties, spaces such as the Schwartz space $\mathscr{S}(\mathbb{R})$, the space of holomorphic functions $H(\mathbb{C})$, and $C^\infty[0, 1]$ are all topologically indistinguishable from $\mathbb{R}^{\mathbb{N}}$.

Proposition 1.5.6: dense convergence, then strong convergence

Let $(T_n)_1^\infty \subseteq L(V,W)$ where $\sup_n \|T_n\| < \infty$ and $T \in L(V,W)$. If $\|T_n(x) - T(x)\| \to 0$ for all x in a dense subset D of X, then $T_n \to T$ strongly.

***Proof*:**
We want to show that $\|T_n(x) - T(x)\| \to 0$ for all x. Let $C = \sup\{\|T\|, \|T_1\|, \|T_2\|, ...\}$. Then given $x \in X$ and $\epsilon > 0$, choose $x' \in D$ such that $\|x - x'\| < \epsilon/3C$. Since $T_n(x') \to T(x)$, choose n large enough so that $\|T_n(x') - T(x')\| < \epsilon/3$. Then

$$\begin{aligned}\|T_n(x) - T(x)\| &\leq \|T_n(x) - T_n(x')\| + \|T_n(x') - T(x')\| + \|T(x') - T(x)\| \\ &\leq 2C\|x - x'\| + \frac{1}{3}\epsilon \\ &< \epsilon\end{aligned}$$

but then $T_n(x) \to T(x)$ strongly, as we sought to show.

One important result is that in the weak operator topology, unit balls are always compact:

Theorem 1.5.7: Alaoglu's Theorem

Let X be a normed space, and B^* the closed unit ball of X^*, $B* = \{f \in X^* \mid \|f\| \leq 1\}$. Then B^* is compact in the weak*-topology.

***Proof*:**
We shall show that B^* is a closed subspace of a compact space. For each $x \in X$, define

$$D_x = \{z \in \mathbb{C} \mid \|z\| \leq \|x\|\}$$

and take $D = \prod_{x \in X} D_x$. Then by Tychnoff's Theorem, D is compact. Now, we may exactly identify the elements of D the complex-valued function $\varphi : X \to \mathbb{C}$ such that $|\varphi(x)| \leq \|x\|$ for all $x \in X$. Then B^* would conssit of the elements of D that are linear. Hence, we must just show that B^* is closed. The induced topology B^* from the product topology on D and the weak* topology on X^* are both the topology of pointwise convergence (for D by the product construction), and hence checking for closedness in X is equivalence to checking closedness of D. Take a net $\langle f_\alpha \rangle$ in B^* that converges to $f \in D$ for any $x, y \in X$ and $a, b \in \mathbb{C}$. Then:

$$f(ax + by) = \lim f_\alpha(ax + by) = \lim[af_\alpha(x) + bf_\alpha(y)] = af(x) + bf(y)$$

showing that f is linear, and certainly $\|f\| \leq 1$, hence $f \in B^*$, making it compact and completing the proof.

1.5.8 Note Since X^* is a vector space, one might hope to conclude that it is locally compact (via the "it suffices to show the result at 0" reduction). This is *not* the case, as Exercise 49(b) of Folland's *Real Analysis* shows.

(The following was copied from nLab, and is very informative!)

Every complete locally convex topological vector space T is the cofiltered projective limit of Banach spaces in the category of locally convex spaces. (Note that Fréchet spaces are additionally required to be metrisable, so this is more general.)

(see this here) Now given that a Fréchet space admits a decreasing sequence of convex balanced and absorbing neighborhoods, it follows immediately that: Every Fréchet space is a sequential projective limit of Banach spaces.

(this is also really cool on homomorphisms of Fréchet space here)

2

L^p space

Look at this blog!! https://terrytao.wordpress.com/2009/01/09/245b-notes-3-lp-spaces/
In this chapter, we will be studying Banach spaces which are defined through the integral: L^p spaces. These spaces are a generalization of L^1 spaces, where if $f \in L^p$, then it must decay "faster" at infinity, and f doesn't blow up too fast at any finite point. In other words, L^p spaces capture a type of "speed of growth". As we have seen in the previous chapters, limits of integrals are central to many constructions. For some constructions, being in L^1 was enough, but for others (like Fourier series) we will require that function be more than finite in integral (i.e. more than $f \in L^1$). L^p spaces will give a nice set of examples of Banach spaces that are relatively simple to define.

My current intuition is that they are treated as the intermediate to being able to bound a function: it is sometimes nontrivial to immediately say a function is bounded, but we can build-up to it by saying it's L^3, which will imply L^5, which will imply L^7, and so on allowing us to conclude it's L^∞, which means a functions is bounded up to a zero-measure set. This is a common phenomena in analysis where we may not be able to bet a function to fit exactly in one space or another (in our case L^1 or L^∞ which are bounded a.e. functions), but we can put the function in intermediary buckets.

2.1 Basics

So far, we have limited our attention to measurable functions f such that $\int |f| < \infty$ and have labelled them L^1. Often, what we are looking are function with more precise "convergent properties", in particular:

$$\int |f|^p < \infty$$

for some $p \geq 1$, including ∞ (which we will define shortly). We start by labelling the colletion of all these functions:

Definition 2.1.1: L^p Space

Let f be measurable. for $p \in (1, \infty)$[a], define the *p-norm* to be:

$$\|f\|_p = \left[\int |f|^p \, d\mu\right]^{1/p}$$

From this norm, define:

$$L^p(X, \mathcal{M}, \mu) := \left\{ f : X \to \mathbb{C} \;\middle|\; f \text{ is measurable and } \|f\|_p < \infty \right\}$$

[a]we will define L^∞ later

As with L^1, we often abbreviate to $L^p(\mu)$, $L^p(X)$, or just L^p. As with L^1, we'll let L^p be the equivalence class of the set of functions such that $|f|^p$ is integrable where $f \sim g$ if and only if $\int |f - g|^p = 0$. Just like L^1, two functions will be equal in L^p (i.e. belong in the same equivalence class) if they are equal almost everywhere. Furthermore, we claimed that $\|\cdot\|_p$ is a norm. That $\|f\|_p = 0$ if and only if $f = 0$ a.e. comes almost immediately, since

$$\left(\int |f|^p\right)^{1/p} = 0 \quad \text{iff} \quad \int |f|^p = 0 \quad \text{iff} \quad \int |f| = 0 \quad \text{iff} \quad f = 0 \text{ a.e.}$$

For $\|cf\|_p = |c| \, \|f\|_p$, we take advantage of our normalisation:

$$\|cf\|_p = \left(\int |cf|^p\right)^{1/p} = \left(\int |c|^p |f|^p\right)^{1/p} = (|c|^p)^{1/p} \left(\int |f|^p\right)^{1/p} = |c| \, \|f\|_p$$

However, the triangle inequality is not so trivial. We will in fact take good part of this section to show that the triangle inequality is satisfied. For now, we take a moment to explore an example of L^p spaces and L^p functions to gain an intuition:

Example 2.1.2: L^p spaces

1. Let $f : [0, \infty) \to \mathbb{R}$ $f(x) = \frac{1}{x^{1/p}}$. Then $f \in L^q$ for $q > p$. This comes down to the fact that:

$$x > x^{1/p} \;\Leftrightarrow\; \frac{1}{x} < \frac{1}{x^{1/p}}$$

   and by the fact that $\int \left|\frac{1}{x^n}\right| < \infty$ if $n > 1$. We have therefore found a class of functions that fits into all possible $p \in [1, \infty)$. In fact, we can find functions that fit into any connected subinterval of $[1, \infty)$! In particular, let $(X, \mathcal{M}, \mu)$ be a measure space and $f : X \to \mathbb{C}$ a complex measurable function on X. Let

$$E := \left\{ p \in [1, \infty)| \;\middle|\; \int_X |f|^p < \infty \right\}$$

   (a) Then E is connected, that is, if $p, q \in E$ such that $p < r < q$ for some r, then $r \in E$

   Proof. Clearly, if $E = \emptyset$ or $|E| = 1$, then E is connected because the only non-trivial clopen set is $\emptyset$ and E, so let $|E| > 1$.

Pick $p, q \in E$ such that $p < q$, and consider a point r such that $p < r < q$. We'll show that $\int_X |f|^r < \infty$ using the following trick:

$$|f| = \chi_{|f|\geq 1}|f| + \chi_{|f|<1}|f|$$

that is, we're splitting $|f|$ to where f is greater than or equal to 1 and strictly less than 1. The equality holds since $A = \{x \mid |f(x)| \geq 1\}$ and $B = \{x \mid |f(x)| < 1\}$ are disjoint. Importantly, notice that:

$$|f|^p = \chi_{|f|\geq 1}|f|^p + \chi_{|f|<1}|f|^p \qquad |f|^q = \chi_{|f|\geq 1}|f|^q + \chi_{|f|<1}|f|^q$$

Thus, by the linearity of the integral, we get that:

$$\int |f|^p = \int \chi_{|f|\geq 1}|f|^p + \int \chi_{|f|<1}|f|^p \qquad \int |f|^q = \int \chi_{|f|\geq 1}|f|^q + \int \chi_{|f|<1}|f|^q$$

It thus suffice show that $\int \chi_{|f|\geq 1}|f|^r < \infty$ and $\int \chi_{|f|<1}|f|^r < \infty$. This is essentially immediate:

$$\chi_{|f|\geq 1}|f|^r \leq \chi_{|f|\geq 1}|f|^q < \infty \qquad \chi_{|f|<1}|f|^r \leq \chi_{|f|<1}|f|^p < \infty$$

Therefore

$$\int |f|^r < \infty$$

showing that $r \in E$, as we sought to show. □

(b) Show that we can pick a measure and an f such that $E = (a, b), [a, b), [a, b]$, or $\{a\}$ for $1 \leq a \leq b \leq \infty$. In other words, all connected subsets of $[1, \infty)$ are possible for E (You may freely use the fact that any Riemann integrable function is also integrable with respect to the Lebesgue measure and its Riemann integral is equal to the Lebesgue integral)
Hint: Think about integrability properties of functions like x^{-a} and $\frac{-1}{\log^2(x)x} = \frac{d}{dx}\frac{1}{\log(x)}$

Proof. Throughout this hole proof we'll be using Riemann integration as a substitute for Lebesgue integration (we showed earlier that this is allowed). We'll be using these function's to construct our intervals:

i. $f(x) = x^{-1/a}$ on $(1, \infty)$: Notice that

$$\int_1^\infty x^{-1/a} dx = \lim_{x\to\infty} \frac{x^{-1/a+1}}{-1/a+1} - \frac{1}{-1/a+1}$$

so $\|f\|_p < \infty$ if and only if $p \in (a, \infty]$.

ii. $f(x) = x^{-1/a}$ on $(0, 1)$: Notice that

$$\int_0^1 x^{-1/a} dx = \frac{1}{-1/a+1} - \lim_{x\to 0} \frac{x^{-1/a+1}}{-1/a+1}$$

so $\|f\|_p < \infty$ if and only if $p \in [1, a)$.

iii. $f(x) = \frac{1}{(\ln^2(x)x)^{1/a}}$ on $(1, \infty)$: This integral doesn't have an elementary solution, however we can still ascertain some basic convergence properties. In particular, if $p = a$, then:

$$\int_1^\infty |f(x)|^a dx = \lim_{x\to\infty}\left[-\frac{1}{\ln(x)}\right] - \lim_{x\to 1}\left[-\frac{1}{\ln(x)}\right]$$

so $\|f\|_p < \infty$ if and only if $p \in [a, \infty]$

iv. $f(x) = \frac{1}{(\ln^2(x)x)^{1/a}}$ on $(0, 1)$: This result is similar to the other 3 cases we've proven. The result for this integral shows that $\|f\|_p < \infty$ if and only if $p \in [1, a]$.

Thus, we have proven all possible "sub-rays" of $[1, \infty]$ can exist. To show all possible sub-intervals (closed, open, or half), we simply add the appropriate function's to produce the result. For example, if we want to have the interval $[a, b)$, simply take:

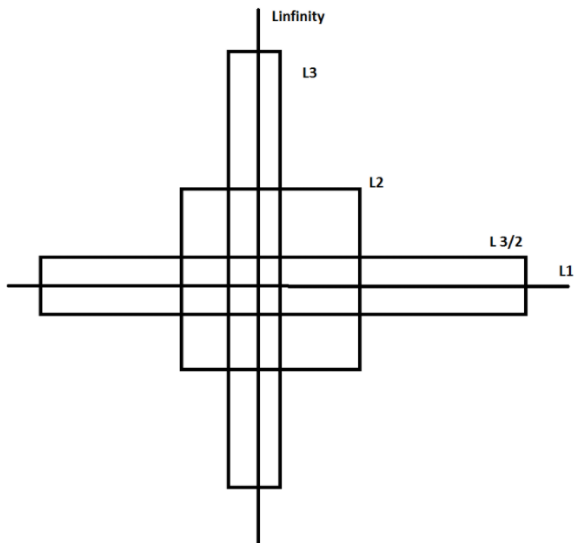
$$f(x) = \frac{1}{(\ln^2(x)x)^{1/a}}\chi_{(1,\infty)} + x^{-1/b}\chi_{(0,1)}$$

which clearly has interval of convergence of $[a, b)$. Similarly, to get a singleton $\{a\}$, take:

$$f(x) = \frac{1}{(\ln^2(x)x)^{1/a}}\chi_{(1,\infty)} + \frac{1}{(\ln^2(x)x)^{1/a}}\chi_{(0,1)}$$

Thus, we have all possible connected subintervals of $[1, \infty]$. $\square$

So if $f \in L^p$, it does not imply that $f \in L^q$ for $p > q$ or $p < q$! Later on, we will show some of the possible relations between L^p spaces. A more accurate intuition is the following [a]: If $f \in L^p$ for any p, then f must decays fast enough at infinity, and f must not blow up too fast at any finite point. As p increases, the first requirement gets less strict, and the second gets more strict. If X is bounded, the first requirement is vacuous, and if X is discrete, the second is vacuous. Sometimes this visual is used to represent these relations:



2. A particular case that will be common for analyzing sequences is when μ is the counting measure on A. In that case, we usually denote $L^p(\mu)$ by $\ell^p(A)$. If $A = \mathbb{N}$, we often simply

abbreviate notation to ℓ^p. If we write $f : \mathbb{N} \to \mathbb{R},\ f(k) = a_k$, then the sequence f (i.e. $\{a_k\}$ is in ℓ^p if:

$$\sum_{k=1}^{\infty} |a_k|^p < \infty$$

Thus, ℓ^1 is the set of convergence series.

[a] credit to this post on stack-exchange: https://mathoverflow.net/questions/314050/intuition-about-lp-spaces

Notice that we are "normalizing" each integral by a factor of $1/p$. This normalization is done to make sure that the constant can be brought out of the norm. It might seem tempted to say that:

$$\left(\int |f|^p\right)^{1/p} \text{``="} \int |f|$$

However, this is most certainly not true (simply take $f(x) = x$). As mentioned before, this makes it non-trivial to show that $\|\cdot\|_p$ is a norm on L^p. However, it is easy to see that L^p is a vector-space. If $f, g \in L^p$, then:

$$|f + g|^p \leq [2\max(|f|, |g|)]^p \leq 2^p(|f|^p + |g|^p)$$

which is finite by assumption, and so $f + g \in L^p$. It is also immediate that $cf \in L^p$. Thus, we may freely add and multiply by L^p functions without worrying about leaving L^p. Because of this, it makes it meaningful to ask whether

$$\|f_1 + f_2\|_p \leq \|f_1\|_p + \|f_2\|_p$$

What we will now do is show that $\|\cdot\|_p$ is does indeed satisfy the triangle inequality for $p \geq 1$. If $p < 1$, then the triangle inequality fails in general:

Example 2.1.3: $p < 1$ then no Triangle Inequality

Take $a, b > 0$ and $0 < p < 1$. Then for any $t \in \mathbb{R}$, we have that:

$$t^{p-1} > (a + t)^{p-1}$$

Integrating both sides from 0 to b, we get:

$$\begin{aligned}
\int_0^b t^{p-1} &> \int_0^b (a + t)^{p-1} \\
\frac{b^p}{p} &> \frac{(a + b)^p}{p} \\
b^p &> (a + b)^p \\
a^p + b^p &> (a + b)^p \\
(a^p + b^p)^{1/p} &> a + b
\end{aligned}$$

Now, if E and F are disjoint union with positive finite measures in X such that $\mu(E)^{1/p} = a$ and $\mu(E)^{1/p} = b$, then $\chi_E, \chi_F \in L^p$. Taking advantage of the fact that $\chi_E\chi_F = 0$ (i.e. the zero function) since $E \cap F = \emptyset$, then:

$$\|\chi_E + \chi_F\|_p = (a^p + b^p)^{1/p} > a + b = \|\chi_E\|_p + \|\chi_F\|_p$$

Thus, the $\|\cdot\|_p$ for $0 < p < 1$ cannot be a norm.

Thus, we stick to showing the case where $p \geq 1$ First, we will state a simple inequality fact similar Jensen's inequality. This is in fact the key idea behind the entire proof:

Lemma 2.1.4: Young's Inequality

If $a, b \geq 0$ and $0 < \lambda < 1$, then:

$$a^\lambda b^{1-\lambda} \leq \lambda a + (1-\lambda)b$$

with equality if and only if $a = b$. In particular, notice that if $\frac{1}{p} + \frac{1}{q} = 1$, then if $\lambda = p^{-1}$, then $(1-\lambda) = q^{-1}$ and so:

$$a^{1/p}b^{1/q} \leq \frac{1}{p}a + \frac{1}{q}b$$

***Proof*:**
This proof is trying to capture what this image is doing in formal language:

https://www.desmos.com/calculator/jdznjzqcju

If $a = b$, equality clearly holds. If $a = 0$ or $b = 0$, inequality clearly holds. If $a, b \neq 0$, then dividing by b and letting $t = a/b$, we get

$$a^\lambda b^{-\lambda} \leq \lambda(a/b) + (1-\lambda) \quad \Leftrightarrow \quad t^\lambda \leq \lambda t + (1-\lambda) \quad \Leftrightarrow \quad t^\lambda - \lambda t \leq (1-\lambda) \tag{2.1}$$

From here, we use some calculus: we see that $t^\lambda - \lambda t$ is strictly increasing for $0 < t < 1$ (since $a, b \geq 0$, $t \geq 0$) and strictly decreasing for $t > 1$, so the maximal value occurs at $t = 1$, and in fact we get that the maximal value is $1 - \lambda$:

$$1^\lambda - \lambda 1 = 1 - \lambda \leq 1 - \lambda$$

and so the inequalities in equation (2.1) hold! Equality happens when $t = a/b = 1$ which implies $a = b$, completing the proof.

Usually, Young's inequality is stated as

$$a^\alpha b^\beta \leq \alpha a + \beta b \qquad 0 \leq \alpha, \beta \leq 1, \quad \alpha + \beta = 1$$

which is exactly what we have, since if we fix some α, then it must be that $\beta = 1 - \alpha$. Another formulation of Young's inequality is:

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$

with $p^{-1} + q^{-1} = 1$. This proof is a "Jensen equality" like proof that more clearly shows the role of concavity and so I'll dedicate a moment to prove this one too:

***Proof*:**
If $a = 0$ or $b = 0$, the result is immediate, so assume $a, b > 0$. Then notice that since log is monotone:

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q} \quad \Leftrightarrow \quad \log(ab) \leq \log\left(\frac{a^p}{p} + \frac{b^q}{q}\right)$$

which is great, since we know log is concave and so Jensen's inequality applies. Since $p^{-1}+q^{-1} = 1$, we get $t = 1/p$ and $(1-t) = 1/q$, and so by Jensen's inequality:

$$\log(ta^p + (1-t)b^q) \geq t\log(a^p) + (1-t)\log(b^q) = \frac{p}{p}\log(a) + \frac{q}{q}\log(b) = \log(ab)$$

completing the proof.

The second interpretation of $p^{-1} + q^{-1} = 1$ in Young's inequality is a nice way to write the straight line (or sub-straight line) equation without the $1-\lambda$ term, but just two numbers. It furthermore let's us relate two numbers that can be much larger. For example, if I choose $p = 100$, then it must be that $q = 100/99$ so that

$$p^{-1} + q^{-1} = 1/100 + 99/100 = 1$$

which, if we consider p and q to be the exponents associated to L^p and L^q, we see that this gives us a way to relate different L^p space!

To see this more succinctly, we use this convexity property given by Young's inequality in the "build-up" lemma towards the triangle inequality. This lemma is important later on in functional analysis since it let's us bound the operation of multiplication of tow functions fg, (namely, $\|fg\|_1$ will be less than a value on the right hand side that seperates f and g). For our current purposess, it is important for the triangle inequality:

Theorem 2.1.5: Hölder's Inequality

Let $1 \leq p, q \leq \infty$ satisfy:

$$\frac{1}{p} + \frac{1}{q} = 1$$

or equivalently $q = p/(p-1)$, where we will interpret $\frac{1}{\infty}$ as 0. Then if f and g are measurable functions on X then:

$$\|fg\|_1 \leq \|f\|_p \, \|g\|_q$$

In particular, if $f \in L^p$ and $g \in L^q$ then $fg \in L^1$. Equality holds if and only if $\alpha|f|^p = \beta|g|^q$ a.e. for some constants α, β with $\alpha\beta \neq 0$

Notice that when $p = q = 2$ (if $p = 2$, it must be that $q = 2$ to satisfy the equation), then this inequality is called the *Cauchy-Schwarz inequality*[1]. Notice too that this is the only time when $p = q$; this will become important when we analyze the $p = 2$ case later.

I also found this intuition on a stack-exchange: the slower f decays, the faster g needs to decay, and the faster f blows up, the more we need g to not blow up. The previous sentence is equally true if we switch f and g, which convinces us that L^p spaces should be reflexive, and by considering $f = g$, we see that the restrictions on f and g should balance at exactly $p = 2$, so L^2 should be self-dual.

Notice too that since we are taking the absolute value of the function on the inside, the integral can be thought of as a sort of "concave function", where the input would be, say in the special case of $\mathbb{R}$ as the domain, some interval. This makes it easier to see why this inequality holds (notice that convex functions must preserve monotonicity, and so does the integral).

[1]Technically we need to define an inner product on L^2 to satisfy this inequality. However, as well see in chapter 5, L^2 is indeed a Hilbert space

***Proof*:**
If $\|f\|_p = 0$ or $\|g\|_p = 0$, then $f = 0$ or $g = 0$ a.e., and so equality holds. Similarly if $\|f\|_p = \infty$ or $\|g\|_p = \infty$, inequality immediately holds. Furthermore, notice that if we find a particular f and g for which the result holds, then the result holds for af and bg since

$$\begin{aligned} \Leftrightarrow\ & \|afbg\|_1 \leq |ab| \|af\|_p \|bg\|_q \\ \Leftrightarrow\ & |ab| \|fg\|_1 \leq |ab| \|f\|_p \|g\|_q \\ \Leftrightarrow\ & \|fg\|_1 \leq \|f\|_p \|g\|_q \end{aligned}$$

So, it suffices to show the result when $\|f\|_p = \|g\|_q = 1$, and equality holds of $|f|^p = |g|^q = 1$. To this end, we use the previous lemma and set $a = |f(x)|^p$ and $b = |g(x)|^q$ and $\lambda = p^{-1}$ (so that $1 - \lambda = 1 - p^{-1} = q^{-1}$). Then by the previous lemma we have:

$$|f(x)|^{p \cdot p^{-1}} |g(x)|^{q \cdot q^{-1}} \leq p^{-1}|f(x)|^p + q^{-1}|g(x)|^q \quad \Leftrightarrow \quad |f(x)g(x)| \leq p^{-1}|f(x)|^p + q^{-1}|g(x)|^q \tag{2.2}$$

Now, integrating both sides, we get:

$$\|fg\|_1 \leq p^{-1} \int |f|^p + q^{-1} \int |f|^q = p^{-1} + q^{-1} = 1 = \|f\|_p \|g\|_q$$

where equality holds when equality holds in equation (2.2) a.e., which holds only when $|f|^p = |g|^q$ a.e., as we sought to show.

For a visual conformation, you can check out:

https://www.desmos.com/calculator/gwqvfv4tie

When p and q satisfy the equality in the equation, we call them *conjugate exponents*. As mentioned before, the conjugate exponents is simply another way or writing the equation of a straight line to apply Young's inequality. From this, we can take advantage of this convexity property yet again to prove the triangle inequality of $\|\cdot\|_p$, which has a different name since for the special case of the p-norm due to the fact it is used on other places and needs to be referenced:

Theorem 2.1.6: Minkowski's Inequality

If $1 \leq p < \infty$ and $f, g \in L^p$. Then:

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p$$

Before proceeding with the proof, a quick word on notation to avoid confusion:

$$\|f\|_p^p = \left(\int |f|^p \right)^{p/p} = \int |f|^p$$

We often work with $\|f\|_p^p = \int |f|^p$ to get rid of the $1/p$ exponent to be able to work with the linearity of the integral, and re-introduce it at the end to get the desried result. We did not have to do this for Hölder's inequality since we reduced to a very nice special case.

***Proof*:**
If $p = 1$, then the result immediately follows by the regular triangle inequality and linearity of the integral. If $f + g = 0$ a.e., then the result also follows, so assume that $f + g \neq 0$ a.e., which using the triangle inequality:

$$|f + g|^p \leq (|f| + |g|)|f + g|^{p-1} = |f||f + g|^{p-1} + |g||f + g|^{p-1}$$

So, we will apply Hölder's inequality using the fact that $q = \frac{p}{p-1}$ and integrating:

$$\begin{aligned}
\|f+g\|_p^p &= \int |f+g|^p d\mu \\
&= \int |f+g| \cdot |f+g|^{p-1} d\mu \\
&\leq \int (|f| + |g|)|f+g|^{p-1} d\mu \\
&= \int |f||f+g|^{p-1} d\mu + \int |g||f+g|^{p-1} d\mu \\
&= \left\| |f|(|f+g|^{p-1}) \right\|_1 + \left\| |g|(|f+g|^{p-1}) \right\|_1 \\
&\leq \||f|\|_p \|f+g\|_q^{p-1} + \|g\|_p \|f+g\|_q^{p-1} && \text{Hölder's Inequality} \\
&= \left(\left(\int |f|^p d\mu \right)^{\frac{1}{p}} + \left(\int |g|^p d\mu \right)^{\frac{1}{p}} \right) \left(\int |f+g|^{(p-1)\left(\frac{p}{p-1}\right)} d\mu \right)^{\frac{p-1}{p}} && \text{recall } q = \frac{p}{p-1} \\
&= \left(\left(\int |f|^p d\mu \right)^{\frac{1}{p}} + \left(\int |g|^p d\mu \right)^{\frac{1}{p}} \right) \left(\int |f+g|^p d\mu \right)^{\frac{p-1}{p}} \\
&= (\|f\|_p + \|g\|_p) \frac{\|f+g\|_p^p}{\|f+g\|_p}
\end{aligned}$$

Thus, multiplying both sides by $\frac{\|f+g\|_p}{\|f+g\|_p^p}$, we get

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p$$

As we sought to show

Thus, we get that $\|\cdot\|_p$ is indeed a norm! Thus L^p is a normed vector space. In fact, just like L^1, it is a complete normed vector space, i.e., a Banach space. We first recall a lemma to simply our task. To understand the lemma, notice that every norm defines a metric:

$$\rho(x, y) = \|x - y\|$$

and so defines a topology, in particular a normal space, that has the notion of sequences well-defined. Next, recall that if $\{x_n\}$ is a sequence in V, it's series is said to converge if there exists an $x \in V$ such that $\sum_{n=1}^{\infty} x_n = x$, and is said to be absolutely convergence if $\sum_{n=1}^{\infty} \|x_n\| < \infty$ (this is lemma 1.1.6). If you did not cover this result yet, it is important to go see it now to understand the following result:

Theorem 2.1.7: L^p is a Banach Space

For $1 \leq p < \infty$, L^p is a complete normed vector-space, i.e., a Banach space. Moreover, if $f_n \to f$ in L^p, then there exists a subsequence that converges pointwise to f a.e.

***Proof*:**
Let $1 \leq p < \infty$. By the previous lemma, it suffices to show that every absolutely convergent series converges:

$$\sum_{k=1}^{\infty} \|f\|_p = B < \infty \quad \Rightarrow \quad \exists f \text{ s.t. } \sum_{k=1}^{\infty} f_k = f \in L^p$$

Let $G_n = \sum_{k=1}^{n} |f_k|$ be the partial sums of the series and $G = \sum_{k=1}^{\infty} |f_k|$ be the series. Then by Minkowski's inequality, we get:

$$\|G_n\|_p \leq \sum_{k=1}^{n} \|f_k\|_p \leq B \quad \forall n \in \mathbb{N}$$

and so $\|G\|_p = \left(\int |G|^p\right)^{1/p} < \infty$ and so $\left(\int |G|^p\right) < \infty$. Since $G_n \leq G_{n+1}$, and $G_n : X \to [0, \infty)$, by the Monotonce Convergence Theorem:

$$\int \lim_{n\to\infty} G_n^p = \lim_{n\to\infty} \int G_n^p \leq B^p < \infty$$

and so $G \in L^p$. By proposition **??**, $G(x) < \infty$ a.e., so the series $\sum_{k=1}^{\infty} f_k$ converges a.e. If we let $F = \sum_{k=1}^{\infty} f_k$, we have that $|F| \leq G$ a.e., and so $F \in L^p$. From this, we can see that:

$$\left\| F - \sum_{k=1}^{\infty} f_k \right\|_p^p = \int \left| F - \sum_{k=1}^{\infty} f_k \right|^p \to 0$$

Thus, the series $\sum_{k=1}^{\infty} f$ converges (to F) in the p-norm, and so by lemma 1.1.6, L^p is complete, as we sought to show.

This is the proof the prof was doing in class, but I got stuck so found a different proof (the above proof) (Commented it out for now)

This also finally showed that L^1 is a complete space under $\|\cdot\|_1$! Note the importance of taking a subsequence: since $f_n \to f$ in the p-norm, i.e. the "sum-averaging" of the function converges, then there can be lots of wiggle room for hwo the functions f_n approach f:

Example 2.1.8: Taking A Subsequence
Construct a sequence of functions such that all the following hold:

1. $f_n : [0, 1] \to \mathbb{R}$ is continuous
2. $0 \leq f_n(x) \leq 1$ for every x
3. $\lim_{n\to\infty} \int_0^1 f_n dm = 0$
4. For every $x \in [0, 1]$ we have that $\lim_{n\to\infty} f_n(x)$ does not exist

(that is, it is definitely important to take a subsequence in order to deduce pointwise convergence from converge in L^1)

Proof. Essentially, we'll be taking advantage that $\sum_n \frac{1}{n}$ does not converge. Let $f_1(x) = 1$, that is, that constant function with constant 1.

For g_2, we split the interval into 2 equal parts. Define

$$g_2(x) = \begin{cases} 1 & x \in [0, 1/2] \\ 0 & x \in (1/2, 1] \end{cases}$$

This function is not continuous. We will make it continuous by simply climbing to it (and for other values, we'll be descending back down too). At the point $1/2$, slope down at a speed of 2^2 until we reach 0:

$$f_2(x) = \begin{cases} 1 & x \in [0, 1/2] \\ -2^2 \left(x - \left(1 - \frac{1}{4}\right)\right) & x \in [1/2, 3/4] \\ 0 & x \in (3/4, 1] \end{cases}$$

From here, we will define our function in a sort of "recursive" sense. Starting form the inteval $[0, 1/2)$, we keep the right end point, and add an interval of length $1/3$ to it: $[1/2, 5/6)$. Now, define f_3 to be 1 on $[1/2, 5/6]$, and zero everywhere else. Next, to define f_4, repeat the process by adding $1/4$ to $5/6$, giving us the interveal $[5/6, 13/12]$. Note that $13/12$ goes beyond 1. We thus wrap it around back to 0: $[0, 1/12] \cup [5/6, 1]$. Now, define a function f_3 to be 1 on this set, and zero everywhere else.

In each of these functions, we extend them to be continuous by attaching them a slope which has 2^n as a factor. In this way, the slope shrinks in size quicker than the size of the rectangles, and so we can morally imagine the problem to be one of indicator functions.

We can continue doing this, defining every f_k by appending $1/k$ and wrapping around when we go past 1 and then making the function continuous. Clearly, $\int f_k \to 0$ as $k \to \infty$ since each inteval will be of total length $1/k + 1/2^k$. However, f_k does not converge anywhere: for any $x \in [0, 1]$, we can choose a subsequence from $f_k(x)$ (say $f_{k_l}(x)$) such that $f_{k_l}(x) = 1$. or $f_{k_l}(x) = 0$

To construct such a sequence, it is easier to imagine our construction without the wrapping around, that is, we keep adding intervals of length $1/k$ as though our co-domain was $[0, \infty)$. Then, our first value of l will be the value for which $x + 1$ is in some interval of length $1/k$ for appropriate k. This interval will exist, since $\sum_k 1/k$ diverges. Similarly, our next value of l for which $x + 2$ will be in some interval of length $1/k$ for appropraite k. We can continue doing this indefinitely, showing that there exists a subsequence of f_k that doesn't converge to 0, meaning f_k cannot converge to 0. On the other hand, it is easy to see that we can choose a subsequence that converges to 0 (since f_k is zero after the trapezoid passes by our point).

Since x was arbitrary, this will work any $x \in [0, 1]$. Thus, f does not converge anywhere, as we sought to show. □

Retruning to analyzing L^p spaces, we can now extend theorem **??** to show that simple function are dense in every L^p!

Theorem 2.1.9: Simple Functions Dense in L^p

For $1 \leq p < \infty$, the set of simple functions $f = \sum_{i=1}^{n} a_i \chi_{E_i}$ where $\mu(E_i) < \infty$ for all i is dense in L^p

***Proof*:**
First of all, the set of simple functions is in L^p for every $1 \leq p < \infty$. Next, choosing any $f \in L^p$, choose some standard representation $\{f_n\}$ where $f_n \to f$ a.e[a]. Then since $|f_n| \leq |f|$, we get that $|f_n| \in L^p$. Furthermore

$$|f - f_n|^p \leq |2f|^p = 2^p|f|^p \in L^1$$

and so by the dominated convergence theorem, we get:

$$\int |f - f_n|^p \to 0 \quad \Rightarrow \quad \left(\int |f - f_n|^p\right)^{1/p} = \|f - f_n\|_p \to 0$$

Moreover, if $f_n = \sum_{k=1}^{\infty} a_k \chi_{E_k}$ was a simple function where all E_i are pairwise disjoint and a_i are nonzero, it must be that $\mu(E_i) < \infty$ since

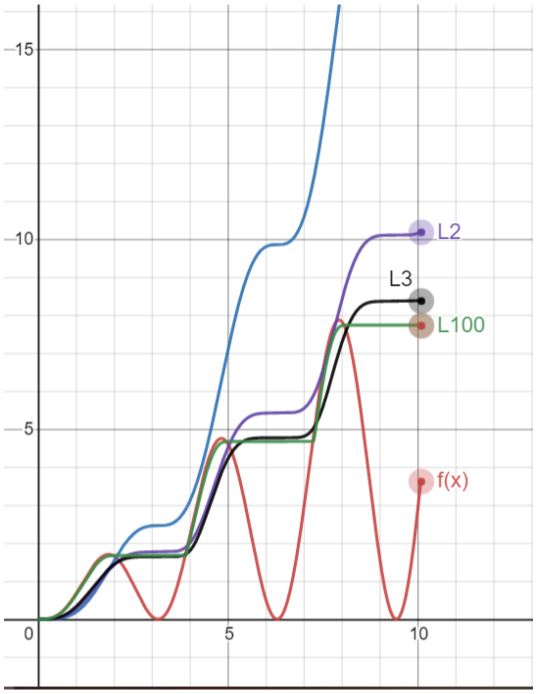
$$\sum_{k=1}^{\infty} |a_k|^p \mu(E_k) = \int |f_n|^p < \infty$$

completing the proof

[a] Recall that the any measurable function has a standard representation, not just integrable functions

Some further remarks about which sets are dense can be enlightening. For $1 \leq p, q < \infty$, L^p is dense in L^q, since each simple set of the form we descriped are subsets of L^p for all $p \geq 1$. If we modify the characteristic function to make them continuous (simply draw a connecting line as we've done in example 2.1.8), then we see that $C_0(\mathbb{R})$ (continuous functions that vanish at infinity) are dense in L^p for every $1 \leq p < \infty$. If we restrict to closed intervals, then $C([a, b])$ is dense in $L^p([a, b])$ since the standard representation becomes uniformaly convergent on bounded domains (see theorem **??**).

So far, we have defined L^p spaces where $0 < p < \infty$ and showed that for $1 \leq p < \infty$, L^p is a Banach space. Interestingly, if we take a look at a graph of $\|f\|_p$ as $p \to \infty$, we'll find that there is a striking pattern that appears:



Notice that as $p \to \infty$, the resulting $\|f\|_p$ value got closer to a max function. In fact, this is almost the case! We might be tempted to then say that $\|f\|_\infty = \sup f$. However, the problem with this is that we might have a measure 0 set that diverges, while the supremum of the rest of the function is attained by a value $k < \infty$. The idea of $\sup f$ on all but a zero measure set will be called the *essential supremum*, and all functions that have a finite essential supremum will be denoted as L^∞. In this way, L^∞ is simply a natural extension of our definition of L^p:

To see the intuition behind this, let us simply consider the p-norm of $\|\vec{v}\|_p = (|3|^p + |10|^p)^{1/p}$

1. Case $p = 1$: $(3^1 + 10^1)^{1/1} = 13$. (The 3 contributes about 23% to the total)
2. Case $p = 2$ (Euclidean): $\sqrt{3^2 + 10^2} = \sqrt{9 + 100} = \sqrt{109} \approx 10.44$. (The 3 matters less now. The 10 is dominating)
3. Case $p = 10$: $(3^{10} + 10^{10})^{1/10} = (59,049 + 10,000,000,000)^{1/10}$. Notice that 3^{10} is negligible dust compared to 10^{10}. $\approx (10^{10})^{1/10} = 10.00000\textit{something}$
4. Case $p = 100$: The 3^{100} is mathematically invisible next to 10^{100}. The sum is basically just 10^{100}. When you take the 1/100 root, you are just returning the base number: 10.

Definition 2.1.10: L^∞ Space

Let $(X, \mathcal{M}, \mu)$ be a measure space and let $f : X \to [0, \infty]$ be a measurable function. Let

$$S = \left\{ r \in \mathbb{R} \;\middle|\; \mu(f^{-1}((r, \infty])) = 0 \right\}$$

Then the *essential supremum* of f, denoted $\|f\|_\infty$ or $\operatorname{ess\,sup} f$ is :

$$\|f\|_\infty := \begin{cases} \infty & \text{if } S = \emptyset \\ \inf S & \text{if } S \neq \emptyset \end{cases}$$

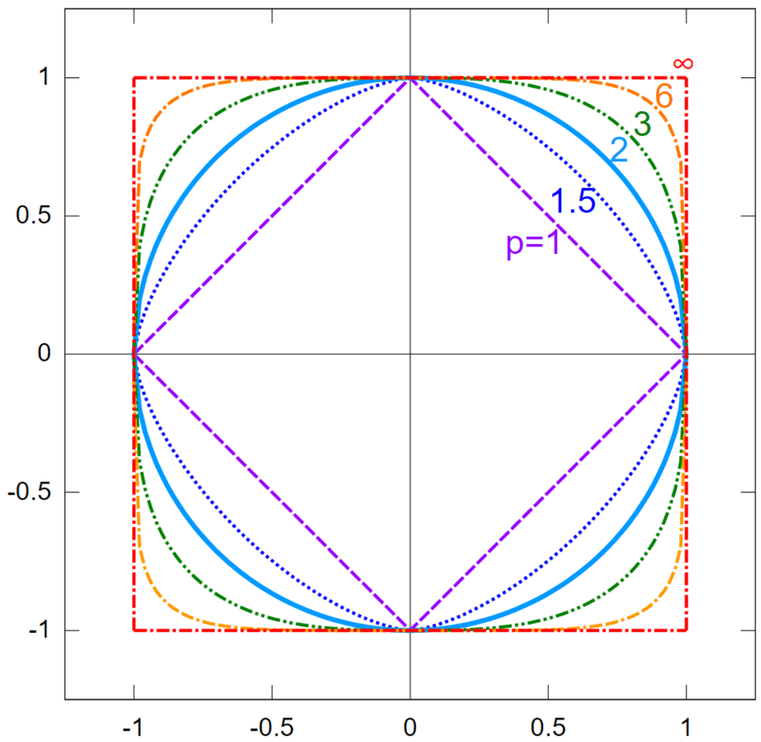
or, equivalently:

$$\|f\|_\infty = \inf_{\substack{E \in \mathcal{M} \\ \mu(X \setminus E) = 0}} \left(\sup_{x \in E} |f(x)| \right)$$

If f is a complex measurable, function, then $\|f\|_\infty = \|\,|f|\,\|_\infty$

Essentially, $\|f\|_\infty$ is the least upper bound of f, ignoring measure zero sets worth of points. Using this definition, we can define L^∞ to be the equivalence classes of functions f such that $\|f\|_\infty < \infty$.

Another quick way of visualizing the relation between all p-norms, $1 \leq p \leq \infty$ is by taking the ℓ^p norm on $\mathbb{R}^2$, taking the function $x \mapsto \|x\|_p = \sqrt[p]{x^p + y^p}$, and considering when $\|x\| = 1$, or in terms of set notation $S_p = \left\{ x \in \mathbb{R}^2 \;\middle|\; \|x\|_p = 1 \right\}$. We'd get:



A couple more quick remarks: $L^\infty(\mu)$ depends on μ only insofar as μ determines which sets have measure zero. If μ and ν are mutually absolutely continuous ($\mu \ll \nu$ and $\nu \ll \mu$[2]) then $L^\infty(\mu) = L^\infty(\nu)$. Secondly, if μ is not semi-finite, it is useful to adpot a slightly useful definition of L^∞ (Folland left this as exercsie 23-25 in chatper 6).

Most properties of L^p for $1 \le p < \infty$ generalises naturally to the L^∞ case:

Proposition 2.1.11: Properties of L^∞

Let $(X, \mathcal{M}, \mu)$ be a measure and L^∞ the set of measurable function satisfying $\|\cdot\|_\infty < \infty$. Then:

1. If f, g are measurable functions on X, then

$$\|fg\|_1 \le \|f\|_1 \|g\|_\infty$$

   If $f \in L^1$ and $g \in L^\infty$, then $\|fg\|_1 = \|f\|_1 \|g\|_\infty$ if and only if $|g(x)| = \|g\|_\infty$ a.e. on the set where $f(x) \neq 0$

2. $\|\cdot\|_\infty$ is a norm on L^∞
3. $\|f_n - f\|_\infty \to 0$ if and only if there exists $E \in \mathcal{M}$ such that $\mu(E^c) = 0$ and $f_n \to f$ uniformally on E
4. L^∞ is a Banach space
5. The simple functions are dense in L^∞

As a consequence of the well-definedness of L^∞, $1^{-1} + \infty^{-1} = 1$ is indeed a well-defined notion (so $\infty^{-1} = 1/\infty = 0$). Thus, 1 and ∞ are conjugate exponents!

***Proof*:**

We will prove (1), from which the rest of the results follow.

1. We'll be using the $\|f\|_\infty = \inf\limits_{\substack{E \in \mathcal{M} \\ \mu(X \setminus E) = 0}} (\sup_{x \in E} |f(x)|)$ definition of the essential supremum to take advantage of the supremum within the definition. Our goal is to show that

$$\int |fg| = \|fg\|_1 \le \|f\|_1 \|g\|_\infty$$

   First, we will limit our attention of $\int |fg|$ to some $E \in \mathcal{M}$ such that $\mu(X \setminus E) = \mu(E^c) = 0$ (i.e. ignoring a zero-measure set). Then:

$$\int |fg| = \int_E |fg| + \int_{E^c} |fg| = \int_E |fg|$$

   Next, we take advantage the fact that we are working with an infimum to see that:

$$\|f\|_\infty \le \sup_{x \in A} |f(x)| < \|f\|_\infty + \epsilon$$

[2] see definition **??**

Therefore, we can take:

$$\int_E |fg| \leq \int_E \left| f \cdot \sup_{x \in E} |g(x)| \right| = \int_E |f| \cdot \left| \sup_{x \in E} |g(x)| \right| = \left| \sup_{x \in E} |g(x)| \right| \int_E |f|$$

where the last equality comes from the fact that the supremum is a constant. Finally, by the property of the infimum, we have:

$$\left| \sup_{x \in E} |g(x)| \right| \int_E |f| \leq (\|g\|_\infty + \epsilon) \int |f|$$

Notice that this inequality does not depend on ϵ, and so:

$$\|fg\|_1 = \int_E |fg| \leq \|f\|_1 \|g\|_\infty$$

as we sought to show

Notice how similar $\| \cdot \|_\infty$ is to the uniform metric $\|f\|_u = \sup_{x \in X} f(x)$. If we are dealing with Borel measures that assign a positive value to all open sets, then if f is continuous we in fact have $\|f\|_\infty = \|f\|_u$ (since $\{x \mid |f(x)| > a\}$ is open). In this way, we can consider the space of bounded continuous functions as a (closed!) subspace of L^∞.

We conclude with some final remarks about the density of simple functions in L^∞. Notice that for L^p where $p < \infty$, simple functions where $\mu(E_i) < \infty$ where dense. This condition is no longer present for density in L^∞; we in fact require simple functions where $\mu(E) = \infty$ now since the function $1(x) = 1$ is in $L^\infty(\mathbb{R})$. This also means that $C_0(\mathbb{R})$ cannot be dense in L^∞. In the case of $L^\infty([a, b])$ $C([a, b])$, not all funcions can be converged to uniformally dense: consider $f : [0, 1] \to \mathbb{R}$, $f(x) = 0$ for $x \in [0, 1/2]$ and $f(x) = 1$ for $x \in [1/2, 1]$. The no continuous function uniformally converges to f in the ∞-norm.

2.1.1 Relating L^p spaces: Interpolation

In this section, we will show how we can relate L^p spaces with each other. Fundamentally, the way we get any order with L^p spaces is through Hölder's inequality. We take full advantage of it in the following proofs. The end-goal is to show that if $p < q < r$ where we allow $r = \infty$, then:

$$L^p \cap L^r \subseteq L^q \subseteq L^p + L^r$$

This should reflect on the example that we have demonstrated in example 2.1.2. In particular, the larger p is, the less f has decay at infinity, but the more f must not explode at any point. In other words p gets bigger, we can let p decay slower, but it must behave better at points where it can explod3. If our space X is bounded, then we don't have to care about decay (this will give us that $L^p \supseteq L^q$ where $p < q$). If X is discrete, we don't have to care about exploding at any point (this will give us that $\ell^p \subseteq \ell^q$ if $p < q$). We thus explore these relation's in this section. These types of relations bewteen space via addition of functions or intersection of spaces is called *interpolation.*

Proposition 2.1.12: L^p **subset**

Let $0 < p < q < r \leq \infty$. Then

$$L^q \subseteq L^p + L^r$$

In particular, each $f \in L^q$ is the sum of a function in L^p and a function in L^q

***Proof*:**
This proof is really similar to that of example 2.1.2[1]. Let $f \in L^q$. Then consider:

$$|f| = \chi_{|f|\geq 1}|f| + \chi_{|f|<1}|f|$$

that is, we're splitting $|f|$ to where f is greater than or equal to 1 and strictly less than 1. The equality holds since $A = \{x \mid |f(x)| \geq 1\}$ and $B = \{x \mid |f(x)| < 1\}$ are disjoint. Importantly, notice that:

$$|f|^p = \chi_{|f|\geq 1}|f|^p + \chi_{|f|<1}|f|^p \qquad |f|^r = \chi_{|f|\geq 1}|f|^r + \chi_{|f|<1}|f|^r$$

Thus, by the linearity of the integral, we get that:

$$\int |f|^p = \int \chi_{|f|\geq 1}|f|^p + \int \chi_{|f|<1}|f|^p \qquad \int |f|^r = \int \chi_{|f|\geq 1}|f|^r + \int \chi_{|f|<1}|f|^r$$

Then we get:

$$\chi_{|f|\geq 1}|f|^p \leq \chi_{|f|\geq 1}|f|^q < \infty \qquad \chi_{|f|<1}|f|^r \leq \chi_{|f|<1}|f|^q < \infty$$

Therefore, $\chi_{|f|\geq 1}f \in L^p$ and $\chi_{|f|<1}f \in L^r$. If $r = \infty$, then clearly $\|\chi_{|f|\leq 1}f\|_\infty \leq 1$, showing the result still holds, as we sought to show.

Proposition 2.1.13: L^p **superset**

Let $0 < p < q < r \leq \infty$. Then

$$L^p \cap L^r \subseteq L^q$$

In particular, if if we find that f is bounded in growth from above and bellow by p and r, $\|f\|_p < \infty$ and $\|f\|_r < \infty$, then f is bounded in growth by q.

***Proof*:**
Let $\|f\|_r < \infty$ and $\|f\|_q < \infty$. We need to show that $\|f\|_p < \infty$. Notice that it is equivalent to show that $\|f\|_r^n < \infty$ and $\|f\|_q^m < \infty$ then $\|f\|_p^\ell < \infty$ for $n, m, \ell \in \mathbb{R}_{\geq 0}$ (since if it's finite, then all powers will be finite). We'll prove it in the case where $r = \infty$ and $r < \infty$. In the infinite case, we simply take advantage of bounding properties of the essential supremum, and in the other case we use Hölder's inequality.

Starting with the $r = \infty$ case, notice that

$$|f|^q = |f|^{q-p+p} = |f|^q|f|^{q-p} \leq \|f\|_\infty^{q-p}|f|^p$$

and so, taking the integral and the qth root, we get:

$$\begin{aligned}
\|f\|_q &= \left(\int |f|^q\right)^{1/q} \\
&\le \left(\int \|f\|_\infty^{q-p}|f|^p\right)^{1/q} \\
&= \|f\|_\infty^{\frac{q-p}{q}}\left(\int |f|^p\right)^{1/q} \\
&= \|f\|_\infty^{\frac{q-p}{q}}\left(\int |f|^p\right)^{p/pq} \\
&= \|f\|_\infty^{1-\frac{p}{q}}\|f\|_p^{p/q} < \infty
\end{aligned}$$

The fact that the last inequality is less than infinity comes from the that $\|f\|_p < \infty$ and $\|f\|_\infty < \infty$, so any power is less than infinity, and so $\|f\|_q < \infty$. Furthermore, notice that the sum of the power's is 1; this fact will come back for another intuition of $\|f\|_\infty$.

For the $r < \infty$ case the key is to find a way to integrate p and r using conjugate exponents. Since

$$p < q < r$$

then

$$\frac{1}{p} > \frac{1}{q} > \frac{1}{r}$$

therefore, define the continuous function $\lambda \mapsto \frac{1-\lambda}{p} + \frac{\lambda}{r}$. Since $(0,1)$ is open, and $1/p > 1/r$, there exists a λ_0 such that

$$\frac{1-\lambda_0}{p} + \frac{\lambda_0}{r} = \frac{1}{q}$$

manipulating the equation around to get conjugate exponents, notice that:

$$\frac{(1-\lambda_0)q}{p} + \frac{\lambda_0 q}{r} = 1 \quad \Leftrightarrow \quad \frac{1}{\frac{p}{(1-\lambda_0)q}} + \frac{1}{\frac{r}{\lambda_0 q}} = 1$$

Thus, we have that $\frac{p}{(1-\lambda_0)q}$ and $\frac{r}{\lambda_0 p}$. With conjugate exponents established, we proceed with trying to bound $\|f\|_q$. As we commented, it is equivalent to bound $\|f\|_q^m$, so let $m = q$ (this will be a common trick to apply Hölder's inequality). Also, since $1 - \lambda_0 + \lambda_0 = 1$, we have that

$q = \lambda_0 q + (1-\lambda_0)q$, giving us the ability to split up our integral:

$$\begin{aligned}
\|f\|_q^q &= \int |f|^q \\
&= \int |f|^{\lambda_0 q + (1-\lambda_0)q} \\
&= \int |f|^{\lambda_0 q}|f|^{(1-\lambda_0)q} \\
&= \left\| |f|^{\lambda_0 q}|f|^{(1-\lambda_0)q} \right\|_1 \\
&\leq \left\| |f|^{\lambda_0 q} \right\|_{\frac{p}{(1-\lambda_0)q}} \left\| |f|^{(1-\lambda_0)q} \right\|_{\frac{r}{\lambda_0 q}} && \text{Hölder's inequality} \\
&= \left(\int |f|^p \right)^{\frac{(1-\lambda_0)q}{p}} \left(\int |f|^r \right)^{\frac{\lambda_0 q}{r}} \\
&= \|f\|_p^{(1-\lambda_0)q} \, \|f\|_r^{\lambda_0 q} < \infty
\end{aligned}$$

Since the last two terms are finite, any power of them are also finite, and so we have bounded $\|f\|_q^q < \infty$, and so $\|f\|_q < \infty$, as we sought to show.

Proposition 2.1.14: ℓ^p Covariant Inclusion

If $0 < p < q \leq \infty$, then $\ell^p(A) \subseteq \ell^q(A)$

***Proof*:**
They key part is to take advantage of the properties of the counting measure. Since the only set of measure 0 in the counting measure is the empty-set, we have that the essential supremum is actually the supremum!

Let's say $f \in \ell^p$, so that $\sum_{a\in A} |f(a)|^p = \|f\|_p < \infty$. We'll start by showing that $\|f\|_\infty < \infty$. Since the $\|f\|_\infty$ is in fact the supremum, we have that:

$$\|f\|_\infty^p = \sup_{a\in A} |f(a)|^p \leq \sum_{a\in A} |f(a)|^p = \|f\|_p < \infty$$

Therefore, $\|f\|_\infty < \infty$.

For the $q < \infty$ case, using proposition 2.1.13 with $\lambda = p/q$, we get:

$$\|f\|_q \leq \|f\|_p^\lambda \|f\|_\infty^{1-\lambda} \leq \|f\|_p$$

therefore, $\|f\|_q < \infty$, completing the proof.

Proposition 2.1.15: L^p Contravariant Inclusion in finite measure

Let $(X, \mathcal{M}, \mu)$ be a measure space where $\mu(X) < \infty$. If $0 < p < q \leq \infty$, then $L^p \supseteq L^q$

***Proof*:**
We need to show that if $f \in L^q$ then $f \in L^p$. Since $f \in L^q$, $\|f\|_q < \infty$, so $\|f\|_q^n < \infty$ for all $n \in \mathbb{N}$. If we can show that $\|f\|_p^m \leq \|f\|_q^n$ for some $m, n \in \mathbb{N}$, then the proof is complete.

We'll apply Hölder's inequality in the caes of $q = \infty$ and $q < \infty$. If $q = \infty$, then:

$$\|f\|_p^p = \int |f|^p = \int |f|^p \cdot 1 \overset{\text{H.}}{\leq} \|f\|_\infty^p \int 1 = \|f\|_\infty^p \mu(X) < \infty$$

since $\int 1 = \mu(X)$. If $q < \infty$, then we do a similar argument. First, since $p < q$, notice that:

$$\frac{1}{\frac{q}{p}} + \frac{1}{\frac{q}{q-p}} = \frac{p}{q} + \frac{q-p}{q} = \frac{q}{q} = 1$$

and so $\frac{q}{p}$ and $\frac{q}{p-q}$ are conjugate exponents. Therefore, by Hölder's inequality:

$$\begin{aligned}
\|f\|_p^p &= \int |f|^p \\
&= \int |f|^p \cdot 1 \\
&= \||f|^p \cdot 1\|_1 \\
&\leq \||f|^p\|_{\frac{q}{p}} \|1\|_{\frac{q}{p-q}} && \text{Hölder's Inequality} \\
&= \left(\int |f|^{p\frac{q}{p}}\right)^{\frac{p}{q}} \left(\int 1\right)^{\frac{q-p}{p}} \\
&= \|f\|_q^p \, \mu(X)^{\frac{q-p}{p}} < \infty
\end{aligned}$$

therefore, $\|f\|_p^p < \infty$, so $\|f\|_p < \infty$, showing that $f \in L^p$, as we sought to show.

Proposition 2.1.16: Relating L^p and L^∞

Let $p \in [1, \infty)$ and assume $\|f\|_p < \infty$. Show that:

$$\lim_{p \to \infty} \|f\|_p = \|f\|_\infty$$

***Proof*:**
For ease of typing, we simplify notation: $\|f\|_p := \|f\|_{L^p(\mu)}$

We'll show that $\lim_{p\to\infty} \|f\|_p \leq \|f\|_\infty$ and $\lim_{p\to\infty} \|f\|_p \geq \|f\|_\infty$. For the $\leq$ direction, note that $|f| \leq \|f\|_\infty$ a.e., and since x^n is a monotone function on $[0, \infty)$, we have $|f|^n \leq \|f\|_\infty^n$. Then for some fixed λ_0:

$$\begin{aligned}\|f\|_p &= \left(\int |f|^p\right)^{1/p} \\ &= \left(\int |f|^{p+\lambda_0-\lambda_0}\right)^{1/p} \\ &= \left(\int |f|^{p-\lambda_0}|f|^{\lambda_0}\right)^{1/p} \\ &\leq \|f\|_\infty^{\frac{p-\lambda_0}{p}} \|f\|_{\lambda_0}^{\frac{\lambda_0}{p}}\end{aligned}$$

where the last inequality is the same trick as part (a) of this question. Since λ_0 is fixed and this inequality holds true for all p, as $p \to \infty$:

$$\lim_{p\to\infty} \|f\|_p \leq \lim_{p\to\infty} \|f\|_\infty^{\frac{p-\lambda_0}{p}} \|f\|_{\lambda_0}^{\frac{\lambda_0}{p}} = \|f\|_\infty$$

For the $\geq$ direction, we have to bound $\|f\|_p$ from bellow by something that includes $\|f\|_\infty$, but ultimately the inequality will be independent of the choices we make. In particular, let $\epsilon > 0$. Take all points which are epsilon close to the essential supremum (which exists by the infimum property). Let M_ϵ be our set, in particular $M_\epsilon = \{x \mid |f(x)| \geq \|f\|_\infty - \epsilon\}$. Since $\|f\|_\infty = k$, then $\mu(M_\epsilon) > 0$. Thus:

$$\|f\|_p = \left(\int |f|^p\right)^{1/p} \geq \left(\int_{M_\epsilon} (\|f\|_\infty - \epsilon)^p\right)^{1/p} \stackrel{!}{=} (\|f\|_\infty - \epsilon)\mu(M_\epsilon)^{1/p}$$

where $\stackrel{!}{=}$ comes from the fact that we are integrating over a constant region. Since $\mu(M_\epsilon) > 0$, we have that $\mu(M_\epsilon)^{1/p} > 0$ for all p, and $\mu(M_\epsilon)^{1/p} \xrightarrow{p\to\infty} 1$. Furthermore, notice our inequalities are independent of choice of ϵ, and so since we can let ϵ be as small as we want:

$$\lim_{p\to\infty} \|f\|_p \geq \|f\|_\infty$$

and since we've shown the $\leq$ direction, we have:

$$\lim_{p\to\infty} \|f\|_p = \|f\|_\infty$$

as we sought to show

Now that we have examined L^p spaces in more detail, we finish off this seciton with some comments on the usefulness these spaces:

1. L^1 is natural for integration; we will essentially work with variation's of L^1 when we integrate. From an algebraic perspective, it can be thought of as a generalizatgion of the notion of "counting" all points in a space, and working in spaces where the count is finite.

2. L^∞ is useful as a broadening of the space of uniform-bound functions. It is a generlization of the idea of working with functions that are bounded.

3. L^p interpolates in between L^1 and L^∞. Many times, L^1 or L^∞ spaces have pathological behaviors (examples of which I yet do not know). L^p spaces behave more nicely, and can sometimes be used to go from L^1 to L^∞ step by step using the inequalities we constructed (ex. we got a function is L^1, we want to show it's L^∞, and we use the proceeding inequalities to build our way up)

4. L^2 is particularly nice since it is a Hilbert space, and so many euclidean intuitions work on L^2 and help us construct many nice properties. A lot of Fourier analysis happens on L^2, though sometimes we do use L^p.

   Another way of thinking of L^2 is that it is a generlization of the notion of a space that *can* have an inner product (this will be made rigorous with a particular universal property, see theorem 5.1.24). In ℓ^2, we would simply get a natural generalization of an inner product on a vector-space, while in L^2, we get the continuous equivalent.

2.2 The dual of Lebesgue Space

By Hölder's inequality, we have that if $f \in L^p$, then for all $g \in L^q$, $fg \in L^1$. We can thus define a function:

$$\varphi_g : L^p \to \mathbb{R} \qquad \varphi_g(f) = \int fg$$

and the operator norm of φ_g is always at most $\|g\|_q$[3]. With the operator norm, the mapping $g \mapsto \varphi_g$ will almost always be an isometry:

Proposition 2.2.1: Operator Norm And Isometry

Let p and q be conjugate exponents, $1 \leq q < \infty$. If $g \in L^q$, then:

$$\|g\|_q = \|\varphi_g\| = \sup\left\{ \left| \int fg \right| \;\middle|\; \|f\|_p = 1 \right\}$$

If μ is semifinite, this result also holds for $q = \infty$

***Proof*:**
The inequality $\|\varphi_g\| \leq \|g\|_q$ is Hölder's inequality: $\left|\int fg\right| \leq \|f\|_p \|g\|_q$ for every f with $\|f\|_p = 1$. For the reverse inequality we may assume $g \neq 0$ (both sides vanish otherwise).

First suppose $1 < q < \infty$. Set

$$f = \frac{|g|^{q-1}\,\overline{\operatorname{sgn}(g)}}{\|g\|_q^{q-1}}$$

Since $p(q-1) = q$, we have $|f|^p = |g|^q / \|g\|_q^q$, so $\|f\|_p = 1$; and since $\overline{\operatorname{sgn}(g)}\, g = |g|$:

$$\int fg = \frac{1}{\|g\|_q^{q-1}} \int |g|^q = \|g\|_q$$

[3] For $p = 2$ the pairing is usually defined as $\int f\overline{g}$ instead; the same convention can be used for any p, but the conjugation matters most at $p = 2$, where it makes the pairing an inner product (chapter 5).

so the supremum is at least (indeed, attains) $\|g\|_q$. If $q = 1$ (so $p = \infty$), take $f = \overline{\mathrm{sgn}(g)}$ instead: then $\|f\|_\infty = 1$ and $\int fg = \int |g| = \|g\|_1$.

Finally, let $q = \infty$ (so $p = 1$) with μ semifinite, and let $\epsilon > 0$. The set $A = \{x \mid |g(x)| > \|g\|_\infty - \epsilon\}$ has positive measure, so by semifiniteness it contains a subset B with $0 < \mu(B) < \infty$. Taking $f = \overline{\mathrm{sgn}(g)}\, \chi_B/\mu(B)$ gives $\|f\|_1 = 1$ and

$$\int fg = \frac{1}{\mu(B)} \int_B |g| \geq \|g\|_\infty - \epsilon$$

As ϵ was arbitrary, $\|\varphi_g\| \geq \|g\|_\infty$.

Conversely, if $f \mapsto \int fg$ is a bounded linear functional on L^p, then $g \in L^q$ in almost all cases:

Theorem 2.2.2: L^p Linear Functional Implies G In L^q

Let p and q be conjugate exponents. Suppose that g is a measurable function on X such that $fg \in L^1$ for all f in the space Σ of simple functions that vanish outside a set of finite measure, and the quantity

$$M_q(g) = \sup\left\{ \left| \int fg \right| \;\middle|\; f \in \Sigma,\ \|f\|_p = 1 \right\}$$

is finite. Also, suppose either that $S_g = \{x \mid g(x) \neq 0\}$ is σ-finite or that μ is semifinite. Then $g \in L^q$ and $M_q(g) = \|g\|_q$.

***Proof*:**
We treat real scalars.[a] If $g = 0$ a.e. the claim is trivial, so assume otherwise. Note first that homogeneity gives $\left|\int fg\right| \leq M_q(g)\|f\|_p$ for every $f \in \Sigma$.

Step 1: the semifinite case reduces to σ-finite S_g. Suppose $q < \infty$ and μ is semifinite, and let $A_n = \{x \mid |g(x)| > 1/n\}$. If $B \subseteq A_n$ has finite measure, then $f = \mu(B)^{-1/p}\mathrm{sgn}(g)\chi_B$ lies in Σ with $\|f\|_p = 1$ (when $\mu(B) > 0$), whence

$$M_q(g) \geq \int fg = \mu(B)^{-1/p} \int_B |g| \geq \frac{1}{n}\mu(B)^{1/q}$$

so $\mu(B) \leq (nM_q(g))^q$. By semifiniteness $\mu(A_n) = \sup\{\mu(B) \mid B \subseteq A_n,\ \mu(B) < \infty\} \leq (nM_q(g))^q < \infty$, and $S_g = \bigcup_n A_n$ is σ-finite. We may therefore assume S_g is σ-finite in Step 2.

Step 2: $q < \infty$. Let E_n be an increasing sequence of finite-measure sets with $\bigcup_n E_n = S_g$. By **??** (applied to g^+ and g^- separately, which have disjoint supports) there are simple $\varphi_n \to g$ pointwise with $|\varphi_n| \leq |g|$ and $\mathrm{sgn}(\varphi_n) = \mathrm{sgn}(g)$ wherever $\varphi_n \neq 0$. Set $g_n = \varphi_n \chi_{E_n}$ and, for n large enough that $\|g_n\|_q > 0$,

$$f_n = \frac{|g_n|^{q-1}\mathrm{sgn}(g_n)}{\|g_n\|_q^{q-1}}$$

with the convention $|g_n|^0 = \chi_{\{x \mid g_n(x) \neq 0\}}$ when $q = 1$. Each f_n is simple, vanishes outside E_n, and has $\|f_n\|_p = 1$ exactly as in the proof of proposition 2.2.1. Since $\mathrm{sgn}(g_n) = \mathrm{sgn}(g)$ where $g_n \neq 0$ and $|g_n| \leq |g|$:

$$f_n g = \frac{|g_n|^{q-1}\,|g|\,\chi_{\{x \mid g_n(x) \neq 0\}}}{\|g_n\|_q^{q-1}} \geq \frac{|g_n|^q}{\|g_n\|_q^{q-1}} \geq 0$$

Hence, by Fatou's Lemma (**??**) applied to $|g_n|^q \to |g|^q$:

$$\|g\|_q \leq \liminf_n \|g_n\|_q = \liminf_n \frac{\int |g_n|^q}{\|g_n\|_q^{q-1}} \leq \liminf_n \int f_n g \leq M_q(g)$$

so $g \in L^q$. Hölder's inequality now gives $M_q(g) \leq \|g\|_q$, so $M_q(g) = \|g\|_q$.

Step 3: $q = \infty$. Let $\epsilon > 0$ and $A = \{x \mid |g(x)| \geq M_\infty(g) + \epsilon\}$. If $\mu(A) > 0$, then in either the σ-finite or the semifinite case A contains a subset B with $0 < \mu(B) < \infty$, and $f = \operatorname{sgn}(g)\chi_B/\mu(B) \in \Sigma$ has $\|f\|_1 = 1$ with

$$\int fg = \frac{1}{\mu(B)} \int_B |g| \geq M_\infty(g) + \epsilon$$

contradicting the definition of $M_\infty(g)$. Hence $\mu(A) = 0$ for every $\epsilon > 0$, i.e. $\|g\|_\infty \leq M_\infty(g)$; the reverse inequality is Hölder's, as before.

[a] For complex scalars, replace $\operatorname{sgn}(g)$ by $\overline{\operatorname{sgn}(g)}$ throughout; the only delicate point is Step 2, where f_n must remain simple, which is arranged by using a simple approximation whose phase is a finite discretization of the phase of g.

Finally, the map $g \mapsto \big(f \mapsto \int fg\big)$ is almost always a surjection onto $(L^p)^*$.

Theorem 2.2.3: Description Of Dual Of Lp

Let p and q be conjugate exponents. If $1 < p < \infty$, for each $\varphi \in (L^p)^*$ there exists $g \in L^q$ such that $\varphi(f) = \int fg$ for all $f \in L^p$, and hence L^q is isometrically isomorphic to $(L^p)^*$. The same conclusion holds for $p = 1$ provided μ is σ-finite

***Proof*:**
Folland p. 190

Corollary 2.2.4: Lp Is Reflexive

If $1 < p < \infty$, L^p is reflexive, that is $(L^p)^{**} \cong L^p$

(Some comment in the case of $p = 1$ and $p = \infty$)

2.3 The Dual of $C_0(X)$

Having identified $(L^p)^* \cong L^q$, we turn to a second concrete dual, that of the space of continuous functions vanishing at infinity. The measure-theoretic input is the Riesz-Markov theorem of **??**, which represents every *positive* linear functional on $C_c(X)$ by a Radon measure; dropping positivity and completing $C_c(X)$ to a Banach space identifies the full dual with a space of signed measures. Throughout, X is a locally compact Hausdorff (LCH) space.

The completion of $C_c(X)$ under the uniform norm is the space of functions vanishing at infinity.

Definition 2.3.1: Functions Vanishing at Infinity

A continuous function $f\colon X \to \mathbb{R}$ *vanishes at infinity* if for every $\varepsilon > 0$ the set $\{x \in X \mid |f(x)| \geq \varepsilon\}$ is compact. The space of such functions is denoted $C_0(X)$; with the uniform norm $\|f\|_u = \sup_{x\in X} |f(x)|$ it is a Banach space, and $C_c(X)$ is dense in it.

To capture the whole dual, and not just its positive part, measures must be allowed to take both signs.

Definition 2.3.2: The Space $M(X)$ of Finite Radon Measures

Let $M(X)$ be the space of all finite signed regular Borel measures on X: a signed measure μ lies in $M(X)$ if its Jordan variations μ^+, μ^- (**??**) are finite Radon measures (**??**). It carries the *total variation norm* $\|\mu\| = |\mu|(X) = \mu^+(X) + \mu^-(X)$.

The identification of the dual is the full Riesz representation theorem, in exact parallel with $(L^p)^* \cong L^q$.

Theorem 2.3.3: Riesz Representation for $C_0(X)^*$

Let X be an LCH space. For every $I \in C_0(X)^*$ there is a unique $\mu \in M(X)$ with

$$I(f) = \int f\, d\mu \qquad \text{for all } f \in C_0(X)$$

and the map $\mu \mapsto I$ is an isometric isomorphism $M(X) \cong C_0(X)^*$, so that $\|I\| = \|\mu\| = |\mu|(X)$.

***Proof*:**
The idea is to split I into positive parts and apply the positive Riesz-Markov theorem to each.

Jordan splitting of the functional. For $f \in C_c(X)$ with $f \geq 0$ set

$$I^+(f) = \sup\{I(g) \mid g \in C_c(X),\ 0 \leq g \leq f\}$$

Boundedness of I gives $I(g) \leq \|I\|\, \|f\|_u$, so the supremum is finite; a routine additivity check extends I^+ to a positive linear functional on $C_c(X)$ (writing $f = f^+ - f^-$). Then $I^- = I^+ - I$ is positive as well, since $I^+(f) \geq I(f)$ for $f \geq 0$.

Representation. By the Riesz-Markov theorem (**??**) there are Radon measures μ^+, μ^- with $I^\pm(f) = \int f\, d\mu^\pm$. Boundedness of I in the uniform norm forces μ^+, μ^- to be finite, so $\mu = \mu^+ - \mu^- \in M(X)$ and $I(f) = \int f\, d\mu$ on $C_c(X)$; density of $C_c(X)$ in $C_0(X)$ and continuity of I extend the representation to all of $C_0(X)$.

Isometry. For $f \in C_0(X)$ with $\|f\|_u \leq 1$,

$$|I(f)| = \left|\int f\, d\mu\right| \leq \int |f|\, d|\mu| \leq |\mu|(X) = \|\mu\|$$

so $\|I\| \leq \|\mu\|$. For the reverse inequality the Hahn decomposition $X = P \cup N$ (**??**) gives, formally, $\int (\chi_P - \chi_N)\, d\mu = |\mu|(X)$; as $\chi_P - \chi_N$ is not continuous, one approximates it in $L^1(|\mu|)$ by $f_n \in C_c(X)$ with $\|f_n\|_u \leq 1$, using the regularity of μ (**??**) and Lusin's theorem, and passes to the limit to obtain $\|I\| \geq \|\mu\|$.

When X is compact, $C_0(X) = C(X)$, so $C(X)^* \cong M(X)$: the finite regular signed measures are the dual of the continuous functions, the measure-theoretic counterpart of the $(L^p)^*$ duality.

2.3.4 Measures as Distributions In chapter 3 it will be natural to replace evaluation of a function with integration against a test function. The present section shows that a continuous functional on $C_0(X)$ is already integration against a measure in $M(X)$, so the measures sit among the distributions as those of order zero.

2.4 Some Useful Inequalities

In this section, we explore more inequalties, since they are at the heart of working with L^p spaces.

We first present a common inequality that shows up often in statistics:

Theorem 2.4.1: Chebychev's Inequality

Let $f \in L^p$ $(0 < p < \infty)$. Then for any $\alpha > 0$

$$\mu(\{x \mid |f(x)| > \alpha\}) \leq \left(\frac{\|f\|_p}{\alpha}\right)^p$$

***Proof*:**
Let $E_\alpha = m(\{x \mid |f(x)| > \alpha\})$. Then:

$$\|f\|_p^p = \int_\Omega |f|^p \geq \int_{E_\alpha} |f|^p \geq \int_{E_\alpha} \alpha^p = \alpha^p m(E_\alpha)$$

re-arranging gives us:

$$\mu(\{x \mid |f(x)| > \alpha\}) \leq \left(\frac{\|f\|_p}{\alpha}\right)^p$$

as we sought to show

The next inequality gives us a general statement on being abel to get a function from $L^p(\mu)$ "transformed" into a function of $L^p(\nu)$:

Theorem 2.4.2: Transitioning L^p Spaces

Let$(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ-finite measures spaces, K be a $M \otimes N$-measurable function on $X \times Y$, and let's say there exists a $C > 0$ such that

$$\int |K(x,y)| d\mu(x) \leq C \qquad \int |K(x,y)| d\nu(y) \leq C$$

for a.e. x and y. Then if $f \in L^p(\nu)$ $(1 \leq p \leq \infty)$, the integral:

$$Tf(x) = \int K(x,y) f(y) d\nu(y)$$

converges absolutely for a.e. $x \in X$, $Tf \in L^p(\mu)$, and $\|Tf\|_p \leq C\|f\|_p$

***Proof*:**
Let $p \in [1, \infty]$, and let q be it's conjugate exponent. Then:

$$|K(x,y)f(y)| = |K(x,y)||f(y)| = |K(x,y)|^{\frac{1}{p}+\frac{1}{q}}|f(y)| = (|K(x,y)|^{1/p})(|K(x,y)|^{1/q}f(y)|)$$

and applying Hölder to the functions in parenthesis, we get:

$$\begin{aligned}
\int |K(x,y)f(y)| d\nu(y) &\leq \left(\int |K(x,y)| d\nu(y) \right)^{1/q} \left(\int |K(x,y)||f(y)|^p d\nu(y) \right)^{1/p} \\
&\leq C^{1/q} \left(\int |K(x,y)|f(y)|^p d\nu(y) \right)^{1/p}
\end{aligned}$$

for a.e. $x \in X$. Hence, taking the integral over $X \times Y$, using Tonelli's theorem, and substituting the result we just found, we get:

$$\begin{aligned}
\int \left[\int |K(x,y)f(y) d\nu(y) \right]^p d\mu(x) &\leq \int \int C^{p/q} |K(x,y)||f(y)|^p d\nu(y) d\mu(x) \\
&= C^{p/q} \int \int |K(x,y)||f(y)|^p d\mu(x) d\nu(y) && \text{Tonelli} \\
&\leq C^{p/q} \int |f(y)|^p \int K(x,u) d\mu(x) d\nu(y) \\
&= C^{p/q+1} \int |f(y)|^p d\nu(y)
\end{aligned}$$

Since $f \in L^p(\nu)$, the last integral is finite. Since this is true for a.e. x, $K(x, \cdot)f \in L^1(\nu)$ for a.e. x. Thus, Tf is well-defined a.e., meaning we can integrate it, and by the inequality we just showed:

$$\int |Tf(x)|^p d\mu(x) = \|Tf\|_p^p \leq C^{p/q+1}\|f\|_p^p$$

taking the pth root of each side get's us the desired inequality

The next result we work towards is a generalization of Minkowski: instead of summing we will be taking integrals. You can also interpret part (a) as you can move the $(|\cdot|^p)^{1/p}$ "down" a layer of integrals, and part (b) as the generalization of monotonicity ($|\int f| \leq \int |f|$).

Theorem 2.4.3: Minkowski's Inequality For Integrals

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ-finite meaure spaces and let f be a [real?] $M \otimes N$-measurable function on $X \times Y$. Then:

1. if $f \geq 0$ and $1 \leq p \leq \infty$, then:

$$\left[\int \left(\int f(x,y)d\nu(y)\right)^p d\nu(x)\right]^{1/p} \leq \int \left[\int f(x,y)^p d\mu(x)\right]^{1/p} d\nu(y)$$

2. If $1 \leq p \leq \infty$, $f(\cdot, y) \in L^p(\mu)$ for a.e. y and the function $y \mapsto \|f(\cdot, y)\|_p$ is in $L^1(\nu)$, then $f(x, \cdot) \in L^1(\nu)$ for a.e. x, the function $x \mapsto \int f(x,y)d\nu(y)$ is in $L^p(\mu)$ and

$$\left\|\int f(\cdot, y)d\nu(y)\right\|_p \leq \int \|f(\cdot, y)\|_p \, d\nu(y)$$

***Proof*:**
If $p = 1$, then this becomes Tonelli's Theorem, so consider $1 < p < \infty$ (the case for $p = \infty$ comes later). Let q be it's conjugate exponent so that $1/p + 1/q = 1$. Suppose $\mu, \nu < \infty$ and $\|f\|_\infty < \infty$ (i.e. it's essential supremum is finite). For notational simplicity, define $H(x) = \int_Y f(x,y)d\nu(y)$. Since the measure is finite and f is esentially bounded, we have that $\|H\|_p < \infty$. Then:

$$\begin{aligned}
\|H\|_p^p &= \int_X \left(\int_Y f(x,y)d\nu(y)\right)^p d\nu(x) \\
&= \int_X \left(\int_Y f(x,y)d\nu(y)\right)(H(x))^{p-1}d\mu(x) && x \text{ indep. of } y \\
&= \int_X \int_Y f(x,y)H(x)^{p-1}d\mu(x)d\nu(y) && \text{Fubini} \\
&\overset{!}{\leq} \int_X \|f(x,\cdot)\|_p \|H\|_p^{p-1} d\nu(y) && \text{Hölder, } q = p/(p-1) \\
&\leq \|H\|_p^{p-1} \int_X \|f(x,\cdot)\|_p d\nu(y)
\end{aligned}$$

where $\overset{!}{\leq}$ I'm not sure how Hölder's inequality was used. Finally, dividng by $\|H(x)\|_p^{p-1}$ get's us:

$$\|H\|_p \leq \int_X \|f(x,\cdot)\|_p d\nu(y)$$

For the general case, Yakov left it as an exercise to do $X = \bigcup_n X_n$, $Y = \bigcup Y_n$, $E_n = \{x \mid f(x) < n\}$, and

$$\tilde{X}_n = X_n \cap E_n \qquad \tilde{Y}_n = Y_n \cap E_n$$

and apply the result to each of these, and the monotone convergence theorem at the end (feels similar to how it was done for Fubini Tonnelli).

For $p = \infty$, This comes down to monotonicity of integrals:

$$\left\| \int f(x,y) d\nu(y) \right\|_\infty \leq \left| \int f(x,y) d\nu(y) \right| \leq \int |f(x,y) d\nu(y) = \int \|f(x,y)\|_\infty d\nu(y)$$

where the last inequality comes from the fact that the integral ignores zero-measure sets.

There are a few more results in the book, but Yakov decided to prove the generalized Hölder's inequality instead for now:

Theorem 2.4.4: Generalized Hölder's Inequality

Let $1 \leq p_i \leq \infty$ for $1 \leq i \leq n$ and $\sum_i^n p_i^{-1} = r^{-1} \leq 1$. If $f_i \in L^{p_i}$, then $\prod_i^n f_i \in L^r$ and

$$\left\| \prod_i^n f_i \right\|_r \leq \prod_i^n \|f_i\|_{p_i}$$

Proof:
If $r = \infty$, then $r^{-1} = 0$, so all the $p_i^{-1} = 0$, so all the $p_i = \infty$, in which case we simply take:

$$\left| \prod_i^n f_i \right| \leq \prod_i^n |f_i|$$

For $r < \infty$, just like for Hölder's inequality, it suffice to assume $r = 1$, since we can do:

$$\sum \frac{1}{\frac{p_i}{r}} = 1$$

In that case, notice that:

$$\begin{aligned} \left\| \prod_i^n f_i \right\|_r^r &= \int |f_1 f_2 \cdots f_n|^r \\ &= \int f_1^r f_2^r \cdots f_n^r \\ &\leq \|f_1\|_{p_1}^r \|f_2\|_{p_2}^r \cdots \|f_n\|_{p_n}^r \end{aligned}$$

Notice that each f_i is raised to the power of p_i/r. (TBD)

Now, we do induction on i. Let the induction hypothesis be:

$$\left(\sum_i^{n-1} p_i^{-1} \right) + p_n^{-1} = r^{-1} + p_n^{-1} = 1$$

Then by Hölder's inequality:

$$\int f_1 \cdots f_n| \leq \|f_1 \cdots f_{n-1}\|_r \|f\|_{p_n} \overset{\text{I.H.}}{\leq} \|f_1\|_{p_1} \cdots \|f_n\|_{p_n}$$

giving our desired result

The final property we'll present a way to represent functions in terms of the integral:

Proposition 2.4.5: Layer Cake Representation

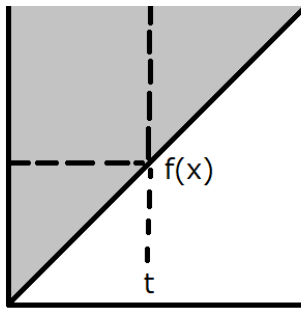
Let $\varphi(t) : [0,\infty] \to [0,\infty]$ be an increasing function which is C^1 on $(0,\infty)$, satisfies $\varphi(0) = 0$ and satisfies $\lim_{t\to\infty} \varphi(t) = \varphi(\infty)$. Let (X, M, μ) be a σ-finite measure space and $f : X \to [0,\infty]$ be a non-negative measurable function. Then:

$$\int_X \varphi(f(x))\, d\mu(x) = \int_0^\infty \mu(\{x \mid f(x) > t\})\varphi'(t)dt$$

One consequence of this equation is that with $\varphi(t) = t^p$ for $1 < p < \infty$ is the following formula for the L^p norm of f:

$$\|f\|_p = \left(p \int_0^\infty \mu(\{x \mid f(x) > t\}\, t^{p-1}\right)^{1/p} dt$$

***Proof*:**
The proof can be captured pictorially as integrating over the following area:



Let $S \subseteq X \times [0,1]$, $S = \{(x,t) \mid f(x) \geq t\}$. With this set, define $F : X \times [0,\infty] \to [0,\infty]$, $F(x,t) = \varphi'(t)\chi_S$. That is, we are going to integrate the "upper triangle". Since φ is an increasing C^1 function and $\varphi'(t) > 0$, we have $F \in L^+(X, [0,\infty])$. Therefore, we can apply Tonneli's Theorem and see that it's equivalent to integrate over the iterated integrals to compute integrals.

To compute it, we need to be clever with the bounds we choose which depend on which iterated integral we choose. The different direction of the iterated integral will ultimately lead to the desired result. If we integrate with respect to $d\mu(x)dt$, then we take slices across t: $(\varphi'(t)\chi_S)_t = \varphi'(t)\chi_{S_t}$, where $(\chi_S)_t = \chi_S(x,t) = \chi_{S_t}$, with $S_t = \{x \mid f(x) \geq t\}$. And so we get:

$$\int_0^\infty \int_X \chi_{S_t}\varphi'(t) = \int_0^\infty \int_{\{x \mid f(x)\geq t\}} \varphi'(t)d\mu(x)dt = \int_0^\infty \mu(\{x \mid f(x) \geq t\})\varphi'(t)dt$$

Now, integrating with respect to $dtd\mu(x)$, we take x slices: $(\varphi'(t)\chi_S)_x = \varphi'(t)\chi_{S_x}$ where

$(\chi_S)_x = \chi_S(x,t) = \chi_{S_x}$, with $S_x = \{x \mid 0 \le f(x)\}$. And so we get:

$$\int_X \int_0^\infty \chi_{S_x} \varphi'(t) dt d\mu = \int_X \int_0^{f(x)} \varphi'(t) dt d\mu(x)$$

and since $[0, f(x)]$ is bounded and $\varphi(0) = 0$, the Lebesgue integral matches the Riemann integral, and so we get that:

$$\int_X \varphi(f(x)) d\mu(x)$$

and by Tonnelli's theorem:

$$\int_X \varphi(f(x)) d\mu(x) = \int_0^\infty \mu(\{x \mid f(x) \ge t\}) \varphi'(t) dt$$

As we sought to show. Using this, if we let $\varphi(t) = t^p$, then we see that φ is C^1 on $(0,\infty)$ and $\varphi(0) = 0$ for $1 \le p < \infty$. Thus, the value of $\|f\|_p$ (where we take f^p instead of $|f|^p$ since $f \ge 0$) is:

$$\begin{aligned} \|f\|_{L^p} = \left(\int |f|^p \right)^{1/p} = \left(\int f^p \right)^{1/p} &= \left(\int_X \varphi(f(x)) d\mu(x) \right)^{1/p} \\ &= \left(p \int_0^\infty \mu(\{x \mid f(x) > t)\}\, t^{p-1} \right)^{1/p} dt \end{aligned}$$

Another use of this formula is by taking $\varphi(t) = t$. Then we will get that:

$$\int f(x) d\mu(x) = \int_0^\infty \mu(\{x \mid f(x) \ge t\}) dt$$

This can be useful if we have information of what is happening to the growth of f as f get's larger, in particular if we know that somehow, the amount of points such that $\mu(\{x \mid f(x) \ge t\})$ is shrinking rapidly enough say that $\mu(\{x \mid f(x) \ge t\}) \le \frac{C}{t^r}$ for some constant C).

2.5 Interpolation: The Riesz-Thorin Theorem

As we saw in the previous section, if a function is in L^p and L^r, it is also in L^q for all $p < q < r$. This is a type of "interpolation" of spaces. A natural question arises: if we have a linear operator T that is bounded on L^p and bounded on L^r, is it also bounded on L^q? The answer is yes, and the Riesz-Thorin theorem provides the precise quantitative relationship between the operator norms.

The proof relies on a powerful complex analysis technique known as the Phragmén-Lindelöf principle, specifically a version called the Three Lines Lemma.

Lemma 2.5.1: Three Lines Lemma

Let φ be a bounded continuous function on the strip $S = \{z \in \mathbb{C} \mid 0 \leq \mathrm{Re}(z) \leq 1\}$ that is holomorphic on the interior of S. If $|\varphi(z)| \leq M_0$ for $\mathrm{Re}(z) = 0$ and $|\varphi(z)| \leq M_1$ for $\mathrm{Re}(z) = 1$, then for all $t \in [0, 1]$ and z with $\mathrm{Re}(z) = t$:

$$|\varphi(z)| \leq M_0^{1-t} M_1^t$$

***Proof*:**

For $\epsilon > 0$, define $\varphi_\epsilon(z) = \varphi(z) M_0^{z-1} M_1^{-z} e^{\epsilon(z^2-1)}$. The term $e^{\epsilon(z^2-1)}$ helps handle the boundedness at infinity within the strip. By the maximum modulus principle applied to rectangles in the strip as the height goes to infinity, one can show $|\varphi_\epsilon(z)| \leq 1$ in the strip. Letting $\epsilon \to 0$ yields the result.

Theorem 2.5.2: Riesz-Thorin Interpolation Theorem

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be measure spaces. Let $p_0, p_1, q_0, q_1 \in [1, \infty]$. Define p_t and q_t for $t \in (0, 1)$ by:

$$\frac{1}{p_t} = \frac{1-t}{p_0} + \frac{t}{p_1}, \qquad \frac{1}{q_t} = \frac{1-t}{q_0} + \frac{t}{q_1}$$

Suppose T is a linear operator defined on the set of simple functions Σ such that for all $f \in \Sigma$:

$$\|Tf\|_{q_0} \leq M_0 \|f\|_{p_0} \quad \text{and} \quad \|Tf\|_{q_1} \leq M_1 \|f\|_{p_1}$$

Then for all $t \in (0, 1)$, T extends to a bounded operator from $L^{p_t}(\mu)$ to $L^{q_t}(\nu)$ satisfying:

$$\|Tf\|_{q_t} \leq M_0^{1-t} M_1^t \|f\|_{p_t}$$

***Proof*:**

Let $t \in (0, 1)$ be fixed. By duality (Proposition 2.2.1), it suffices to estimate $|\int (Tf) g \, d\nu|$ where f and g are simple functions with $\|f\|_{p_t} = 1$ and $\|g\|_{q_t'} = 1$ (where q_t' is the conjugate exponent of q_t).

We construct a family of functions to apply the Three Lines Lemma. Let $f = |f| e^{i\theta}$ and $g = |g| e^{i\psi}$ be the polar decompositions. For any complex number z in the strip $0 \leq \mathrm{Re}(z) \leq 1$, define:

$$\alpha(z) = (1-z)\frac{1}{p_0} + z\frac{1}{p_1}, \qquad \beta(z) = (1-z)\frac{1}{q_0'} + z\frac{1}{q_1'}$$

Note that for $z = t$, $\alpha(t) = 1/p_t$ and $\beta(t) = 1/q_t'$. Now define deformations of f and g:

$$f_z = |f|^{p_t \alpha(z)} e^{i\theta}, \qquad g_z = |g|^{q_t' \beta(z)} e^{i\psi}$$

(If $f = 0$ or $g = 0$, set the function to 0). Note that $f_t = f$ and $g_t = g$.

Define the function $\Phi(z)$ on the strip $0 \leq \mathrm{Re}(z) \leq 1$:

$$\Phi(z) = \int_Y (Tf_z) g_z \, d\nu$$

Since f and g are simple, f_z is a finite linear combination of terms like $c_k\alpha^z$, making $\Phi(z)$ holomorphic and bounded.

We check the boundary conditions. For $\mathrm{Re}(z) = 0$, $f_z \in L^{p_0}$ and $g_z \in L^{q_0'}$. Specifically:

$$\|f_{iy}\|_{p_0}^{p_0} = \int ||f|^{p_t(\frac{1}{p_0}+i\mathrm{Im}(\alpha))}|^{p_0} = \int |f|^{p_t} = \|f\|_{p_t}^{p_t} = 1$$

Similarly $\|g_{iy}\|_{q_0'} = 1$. Thus by Hölder's inequality and the hypothesis on T:

$$|\Phi(iy)| = \left|\int T(f_{iy})g_{iy}\right| \leq \|Tf_{iy}\|_{q_0}\|g_{iy}\|_{q_0'} \leq M_0\|f_{iy}\|_{p_0} \cdot 1 = M_0$$

Similarly, for $\mathrm{Re}(z) = 1$, we find $|\Phi(1 + iy)| \leq M_1$.

By the Three Lines Lemma, $|\Phi(t)| \leq M_0^{1-t}M_1^t$. Since $\Phi(t) = \int(Tf)g$, and this holds for all unit simple functions f, g, we conclude:

$$\|T\|_{L^{p_t}\to L^{q_t}} \leq M_0^{1-t}M_1^t$$

Counter-Example: The Linear Transport Equation

Consider the simple advection equation in one dimension:

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0$$

where $c \neq 0$ is a real constant velocity.

1. **The Operator:** The solution is a translation of the initial state $f(x)$:

$$u(x, t) = f(x - ct)$$

   Therefore, the time-shift operator S_t acts as:

$$(S_t f)(x) = f(x - ct)$$

2. **The Adjoint:** To check if S_t is self-adjoint, we compute its adjoint S_t^* using the standard L^2 inner product $\langle f, g\rangle = \int_{-\infty}^{\infty} f(x)\overline{g(x)}\,dx$.

$$\langle S_t f, g\rangle = \int_{-\infty}^{\infty} f(x - ct)\overline{g(x)}\,dx$$

   Perform a change of variables $y = x - ct$, so $x = y + ct$ and $dx = dy$:

$$\langle S_t f, g\rangle = \int_{-\infty}^{\infty} f(y)\overline{g(y + ct)}\,dy$$

$$\langle S_t f, g\rangle = \langle f, S_{-t}g\rangle$$

   This shows that the adjoint is the inverse shift:

$$S_t^* = S_{-t}$$

3. **Conclusion:** For S_t to be self-adjoint, we require $S_t = S_t^*$. This would imply $S_t = S_{-t}$, or $f(x - ct) = f(x + ct)$ for all functions f. This is clearly false for $c \neq 0$ and $t \neq 0$. Thus, S_t **is not self-adjoint.**

General Context

- **Unitary vs. Self-Adjoint:** In conservative systems (like Quantum Mechanics or the Wave Equation), the evolution operator S_t is typically **unitary**, meaning $S_t^* = S_t^{-1} = S_{-t}$. A unitary operator is rarely self-adjoint (unless $S_t^2 = I$, which is not true for general time evolution).
- **When is it Self-Adjoint?** S_t *is* self-adjoint in specific dissipative systems, such as the **Heat Equation** ($u_t = \Delta u$). In this case, $S_t = e^{t\Delta}$. Since the Laplacian Δ is a self-adjoint operator (with appropriate boundary conditions), the operator exponential $e^{t\Delta}$ is also self-adjoint for real t.

Zak Katz: I see. Does that mean that S_t is not always "diagonalizable"?

Raycast AI: No, that is a common misconception! **"Not self-adjoint" does not mean "not diagonalizable."**

The time-shift operator S_t for the Transport Equation (which I used as a counter-example previously) **is** diagonalizable, even though it is not self-adjoint.

Here is the breakdown of why this works and when it fails.

1. The Key Concept: Normal Operators

In linear algebra and functional analysis, a deeper property governs diagonalizability: **Normality**.

An operator A is **Normal** if it commutes with its adjoint:

$$AA^* = A^*A$$

The **Spectral Theorem** states that **any Normal operator is diagonalizable** (specifically, unitarily diagonalizable).

- **Self-Adjoint Operators** are Normal ($A = A^*$, so $AA = AA$).
- **Unitary Operators** are Normal ($A^* = A^{-1}$, so $AA^{-1} = A^{-1}A = I$).

2. Revisiting the Transport Equation (The "Yes" Case)

Let's look at the Transport Equation again ($u_t + u_x = 0$), where S_t shifts the function: $f(x) \to f(x - t)$.

- We found that $S_t^* = S_{-t}$ (the inverse shift).
- Does it commute?

$$S_t S_t^* = S_t S_{-t} = S_0 = I$$
$$S_t^* S_t = S_{-t} S_t = S_0 = I$$

- Because $S_t S_t^* = S_t^* S_t$, the operator is **Normal**.

How do we "diagonalize" it? We use the **Fourier Transform**. In the frequency domain (Fourier space), the shift operator becomes multiplication:

$$\mathcal{F}(S_t f)(k) = e^{-ikt} \hat{f}(k)$$

Multiplication by a scalar (or function) is the infinite-dimensional equivalent of a diagonal matrix. The Fourier basis functions e^{ikx} are the "eigenvectors" of the time-shift operator.

Conclusion: The Transport operator is not self-adjoint, but it is normal, so it **is** diagonalizable.

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3. When is S_t NOT Diagonalizable? (The "No" Case)

For S_t to fail to be diagonalizable (to be "defective"), the system must be **Non-Normal** in a specific way that creates "Jordan Blocks." This happens when eigenvectors coalesce and you lose a complete basis (this is called an **Exceptional Point** in physics).

Example: A "Defective" System of PDEs Consider a coupled system of two variables $u(x,t)$ and $v(x,t)$:

$$\frac{\partial}{\partial t}\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix}$$

Here, the change in u depends on v, but v is constant in time.

1. **Solve it:**
   - $v(x,t) = v(x,0)$ (constant)
   - $u_t = v(x,0) \implies u(x,t) = u(x,0) + t \cdot v(x,0)$
2. **The Time-Evolution Operator S_t:** We can write S_t as a matrix acting on the vector $\begin{pmatrix} f \\ g \end{pmatrix}$:

$$S_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

3. **Is it Diagonalizable?** Check the matrix $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$.
   - The eigenvalues are $\lambda = 1$ (with algebraic multiplicity 2).
   - However, there is only **one** independent eigenvector: $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.
   - Because we are missing a second eigenvector, this operator is **not diagonalizable** for any $t \neq 0$.

Summary Table

Type of PDE	Operator Property	Self-Adjoint?	Diagonalizable?
Heat Eq. ($u_t = \Delta u$)	Self-Adjoint	**Yes**	**Yes** (Real eigenvalues)
Transport / Wave	Unitary	**No**	**Yes** (Complex eigenvalues)
Defective System	Non-Normal	**No**	**No** (Jordan Block)

Bottom Line: Most standard textbook PDEs (Heat, Wave, Schrödinger) have diagonalizable time-evolution operators because they are typically governed by Normal operators. You usually have to construct specific coupled systems or non-Hermitian boundary conditions to break diagonalizability.

Zak Katz: Let me be clear: S_t shifts *in time* not in space. So to be clear: let a PDE $F(x_1, \ldots, x_n, t) = 0$ PDE w.r.t time t, if $u(x,t)$ is a solution and $f_t(x) = u(x,t)$ is a "state" at time t, and Let S_t be the "time-shift" operator so that $S_t(f_a) = f_{a+t}$ for some state f_a at time a. (NOT the space shift operator $f(x) \to f(x+t)$). Is S_t always diagonalizable?

Raycast AI: Thank you for the clarification. I understand the distinction you are making: you are asking about the **time-evolution operator** (often called the Propagator, $U(t)$ or S_t) that maps the initial state $u(\cdot, 0)$ to the state at time t, $u(\cdot, t)$.

The answer remains: **No, S_t is not always diagonalizable.**

While many standard textbook PDEs (Heat, Wave, Schrödinger) have diagonalizable time-evolution operators because their generators are Normal (Self-Adjoint or Skew-Adjoint), it is very easy to construct physical systems where S_t is **defective** (non-diagonalizable).

This happens specifically when the system exhibits **Linear Instability** or **Critical Damping** (often associated with "Exceptional Points" in modern physics).

The Mathematical Reason: Jordan Blocks

If a PDE can be written in the form $\frac{\partial}{\partial t}\psi = L\psi$, the time-evolution operator is formally $S_t = e^{Lt}$.

For S_t to be diagonalizable, the generator L must possess a complete basis of eigenfunctions. If L has a **Jordan Block** (i.e., it has generalized eigenvectors but not enough true eigenvectors), then the exponential $S_t = e^{Lt}$ will also possess a Jordan block structure for any $t \neq 0$, making it non-diagonalizable.

Here are two concrete counter-examples.

Counter-Example 1: The "Resonant" System (Algebraic Growth)

Consider a system of two fields, $u(x,t)$ and $v(x,t)$, on a domain Ω. The PDE is:

$$\frac{\partial}{\partial t}\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix}$$

where λ is a constant. This is a standard linear PDE system.

1. **Find the Solution:** This is a coupled system.
   - $v_t = \lambda v \implies v(x,t) = e^{\lambda t} v(x,0)$.
   - $u_t = \lambda u + v$. Substituting the solution for v: $u_t = \lambda u + e^{\lambda t} v(x,0)$.
   - The solution for u involves a secular term (a linear t):

$$u(x,t) = e^{\lambda t} u(x,0) + \mathbf{t e^{\lambda t}} v(x,0)$$

2. **The Time-Evolution Operator S_t:** Writing this as an operator matrix acting on the initial vector $\psi_0 = \begin{pmatrix} u_0 \\ v_0 \end{pmatrix}$:

$$S_t = e^{\lambda t} \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

3. **Why it is NOT Diagonalizable:** Look at the matrix part $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$.
   - **Eigenvalues:** It has a single eigenvalue $\mu = 1$ (repeated).
   - **Eigenvectors:** It has only **one** linearly independent eigenvector: $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.
   - Because we need two independent eigenvectors to span the 2D space of fields at every point x, and we only have one, S_t is **not diagonalizable** for any $t \neq 0$.

Physical Interpretation: The term $te^{\lambda t}$ represents "algebraic growth" (or transient growth). Whenever you see a solution of the form $t \times$ (exponential), it is the signature of a non-diagonalizable operator. A diagonalizable operator would only have pure sums of exponentials.

Counter-Example 2: Critical Damping (Classical Mechanics / PDE)

Consider the Damped Harmonic Oscillator equation for a field $u(x,t)$, which is a second-order PDE in time:

$$u_{tt} + 2\gamma u_t + \gamma^2 u = 0$$

(Here $\gamma > 0$ is the damping coefficient).

To analyze S_t, we convert this to a first-order system (state-space form) by defining $v = u_t$:

$$\frac{\partial}{\partial t} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\gamma^2 & -2\gamma \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}$$

The Critical Case: The eigenvalues of the matrix $L = \begin{pmatrix} 0 & 1 \\ -\gamma^2 & -2\gamma \end{pmatrix}$ are found by $\det(L - \omega I) = 0$.

$$\omega^2 + 2\gamma\omega + \gamma^2 = (\omega + \gamma)^2 = 0$$

We have a repeated eigenvalue $\omega = -\gamma$.

Because the matrix L is not the identity matrix but has a repeated eigenvalue, it is a non-trivial Jordan block. The solution is:

$$u(t) = C_1 e^{-\gamma t} + C_2 \mathbf{t}\mathbf{e}^{-\gamma \mathbf{t}}$$

The presence of the $te^{-\gamma t}$ term confirms that the propagator S_t cannot be diagonalized. If it were diagonalizable, we could write the solution purely as $u(t) = Ae^{\omega_1 t} + Be^{\omega_2 t}$, which implies strictly exponential decay without the linear polynomial pre-factor.

Summary

- **Is S_t always diagonalizable? No.**
- **When does it fail?** It fails at **Exceptional Points**, where two or more eigenmodes of the system coalesce (become parallel), making the set of eigenvectors incomplete. This results in solutions that grow or decay with polynomial factors (like t, t^2) multiplying the exponentials.

Zak Katz: Okay. Now, when it *is* diagonalizable, is it *always* the case that we can use Fourier transforms to find an eigenbasis?

Raycast AI: No. Absolutely not.

Just because an operator is diagonalizable does not mean the Fourier Transform provides the basis.

The **Fourier Transform** is a very specific tool that only diagonalizes operators that are **Translation Invariant** (i.e., operators with **Constant Coefficients** defined on the entire space $\mathbb{R}^n$).

If your PDE has **variable coefficients** or is defined on a **bounded domain**, the Fourier basis (e^{ikx}) will generally fail to diagonalize the operator, yet a different eigenbasis might still exist perfectly fine.

Here is the hierarchy of logic:

1. **Diagonalizable:** Means *there exists* a basis $\{\varphi_n(x)\}$ such that the operator acts as multiplication on these functions.
2. **Fourier Diagonalizable:** Means that specific basis is $\{\varphi_k(x) = e^{ikx}\}$.

When does Fourier work?

Fourier analysis works **only** if your operator L commutes with the translation operator T_a (where $T_a f(x) = f(x + a)$).

- **Example:** The Heat Equation $u_t = u_{xx}$.
  - The second derivative behaves the same at $x = 0$ as it does at $x = 100$. Space is homogeneous.
  - Therefore, the eigenfunctions are plane waves e^{ikx}.

When does Fourier fail (but the operator is still diagonalizable)?

Fourier fails whenever **spatial symmetry is broken**. This happens in three common scenarios:

1. Variable Coefficients (Space is not homogeneous) Consider the **Quantum Harmonic Oscillator** (one of the most famous diagonalizable systems in physics).

$$i\frac{\partial\psi}{\partial t} = -\frac{\partial^2\psi}{\partial x^2} + x^2\psi$$

- **Is it Diagonalizable?** Yes. The Hamiltonian is Self-Adjoint. It has a discrete, perfect set of eigenfunctions.
- **Are they Fourier Modes? No.** The term x^2 breaks translation invariance (physics looks different at $x = 0$ vs $x = 100$).
- **The Actual Basis:** The eigenfunctions are **Hermite Functions** (a Gaussian multiplied by a polynomial):
$$\varphi_n(x) = H_n(x)e^{-x^2/2}$$
If you tried to apply a Fourier Transform to this equation, the x^2 multiplication term turns into a Laplacian in k-space ($-\partial_k^2$), so you effectively just swap the PDE for an identical PDE in frequency space without solving it.

2. Bounded Domains (The "Boundary Condition" Problem) Consider the heat equation $u_t = u_{xx}$ on a finite interval $x \in [0, L]$ with Dirichlet boundary conditions $u(0) = u(L) = 0$.

- **Is it Diagonalizable?** Yes.
- **Is it the Fourier Transform?** No. The Continuous Fourier Transform uses e^{ikx} for any real k.
- **The Actual Basis:** The basis is strictly **Sine functions**: $\sin(\frac{n\pi x}{L})$.
  - While related to Fourier *Series*, the Continuous Fourier *Transform* basis (e^{ikx}) is invalid here because e^{ikx} does not satisfy the boundary conditions (it doesn't vanish at 0 and L). You are forced onto a discrete subset of the frequency spectrum.

3. Different Geometries If your PDE is on a sphere (like weather patterns on Earth):

$$\frac{\partial u}{\partial t} = \Delta_{\text{sphere}} u$$

- **Is it Diagonalizable?** Yes.
- **The Actual Basis:** The eigenfunctions are **Spherical Harmonics** $Y_l^m(\theta, \varphi)$.
- Standard planar Fourier modes (e^{ikx}) make no sense here.

Summary Table

PDE Feature	**Is S_t Diag.?**	**Fourier?**	**The Actual Eigenbasis**
Constant Coeffs on $\mathbb{R}$ ($u_t = u_{xx}$)	Yes	**Yes**	Plane Waves (e^{ikx})
Variable Coeffs ($u_t = xu_x$)	Yes	**No**	Power functions (x^λ)
Potential Well ($u_t = i\Delta u + iV(x)u$)	Yes	**No**	Hermite / Legendre / Bessel
Bounded Domain ($x \in [0,1]$)	Yes	**No**	Fourier *Series* (Sines/Cosines)

Conclusion: Diagonalizability guarantees that *some* coordinate system exists where the operator is simple. It does not guarantee that the coordinate system is the one given by Fourier.

Zak Katz: Okay. Is it a necesssary and sufficient condition that S_t is diagonalizable by the Fourier basis iff the diagonalizable operator is translation invariant? Just to make sure I understand what translation invariant means here, we are talking about *space* translation, and we mean that if S_t is our operator and τ_a is our space-translation, then:

$$S_t(\tau_a(f)) = \tau_a(S_t(f))$$

Raycast AI: Yes, precisely.

For a linear operator S_t acting on functions defined on the entire space $\mathbb{R}^n$ (or a periodic domain), the condition is **necessary and sufficient**:

> **Theorem:** A bounded linear operator S_t is diagonalized by the Fourier Transform if and only if it is **Translation Invariant**.

Your definition of translation invariance is exactly correct:

$$[S_t, \tau_a] = S_t\tau_a - \tau_a S_t = 0 \quad \text{for all } a \in \mathbb{R}^n$$

where τ_a is the spatial shift operator.

Here is the logic for why this is an "if and only if" relationship.

1. Why is it Sufficient? (Translation Invariant $\implies$ Fourier Diagonal)

This relies on a fundamental principle of linear algebra: **Commuting operators share a common eigenbasis.**

1. Consider the family of translation operators $\{\tau_a\}_{a\in\mathbb{R}}$.
2. What are the eigenfunctions of a translation? Let's verify that the plane waves e^{ikx} are eigenfunctions of τ_a:
$$\tau_a(e^{ikx}) = e^{ik(x+a)} = e^{ika} \cdot e^{ikx}$$
The eigenvalue is the constant number e^{ika}, and the eigenfunction is e^{ikx}.

3. Since your time-evolution operator S_t commutes with **every** translation operator τ_a (for all shifts a), it must share the same set of eigenfunctions.
4. Therefore, the eigenfunctions of S_t must also be e^{ikx}.
5. If the eigenfunctions are e^{ikx}, then by definition, the operator is diagonalized by the Fourier Transform.

2. Why is it Necessary? (Fourier Diagonal $\implies$ Translation Invariant)

If an operator is diagonal in Fourier space, it acts as a simple multiplication. Let's call the Fourier transform $\mathcal{F}$.

1. Assume S_t is a Fourier Multiplier: $\widehat{S_t u}(k) = m(k)\widehat{u}(k)$ for some multiplier function $m(k)$.
2. The translation operator τ_a in Fourier space is also just a multiplication (by the phase factor e^{ika}).
3. Since regular multiplication of scalars is commutative ($a \cdot b = b \cdot a$), the multiplication by $m(k)$ commutes with the multiplication by e^{ika}.
4. Therefore, S_t and τ_a commute.

Summary

This concept is the bedrock of signal processing and linear systems theory (where these are called **Linear Time-Invariant (LTI)** systems, or in your PDE context, "Linear Space-Invariant" systems).

- **Does S_t commute with space shifts?**
  - **Yes:** You can use Fourier Transforms (e.g., Heat Eq, Wave Eq, Schrödinger with constant potential).
  - **No:** You cannot use Fourier Transforms to find the eigenbasis (e.g., Schrödinger with Harmonic Oscillator potential, variable coefficient Wave Eq).

Zak Katz: Right. Is ther a "theory" for diagonalizable operators S_t?

Raycast AI: Yes, there is a vast and rich "theory" for this, but it doesn't have a single name like "The Theory of Diagonalizable Operators." Instead, it is split into different fields of **Functional Analysis** depending on *why* the operator is diagonalizable.

If you are looking for the generalized version of "Fourier Analysis" for operators that are **not** translation invariant (i.e., not solvable by Fourier Transform), here is the map of the theory.

1. The "Gold Standard": The Spectral Theorem

This is the master theory. It applies if your operator S_t (or its generator L) is **Normal** ($LL^* = L^*L$).

- **The Theory: Spectral Theory of Normal Operators**.
- **What it says:** Any normal operator on a Hilbert space is unitarily equivalent to a multiplication operator.
- **Physical Meaning:** There *always* exists an orthonormal basis of eigenfunctions (or a "continuous basis" / spectral measure) that diagonalizes the operator.
- **How to find the basis:** This is the hard part. It usually requires solving the **Sturm-Liouville Problem** (for 1D) or using **Scattering Theory** (for unbounded domains).

2. The "Variable Coefficient" Theory: Sturm-Liouville Theory

If your PDE has variable coefficients but is in 1D (or separable), this is the standard replacement for Fourier analysis.

- **The Setup:** Your generator looks like $L = \frac{d}{dx}[p(x)\frac{d}{dx}] + q(x)$.
- **The Result:** Just like Fourier series uses $\sin(nx)$ and $\cos(nx)$, Sturm-Liouville theory guarantees a set of orthogonal eigenfunctions $\varphi_n(x)$ weighted by a density function.
- **Example:** Legendre polynomials, Bessel functions, Hermite polynomials (Quantum Harmonic Oscillator). These are the "custom Fourier bases" for specific geometries or potentials.

3. The "Abstract" Theory: Semigroup Theory (C_0-Semigroups)

This is the modern mathematical framework for time-evolution operators S_t.

- **The Setup:** You treat S_t as a "one-parameter semigroup" $T(t)$ acting on a Banach space.
- **The Theorem:** The **Hille-Yosida Theorem**.
- **What it does:** It characterizes exactly when a differential operator L generates a stable time-evolution S_t. While Hille-Yosida doesn't guarantee *diagonalizability* (it only guarantees existence and uniqueness of the flow), it is the starting point for analyzing the spectrum of S_t.
- **Spectral Mapping Theorem:** This theorem relates the eigenvalues of the generator L to the eigenvalues of the propagator S_t (basically $\sigma(S_t) = e^{t\sigma(L)}$). If you know the spectrum of L, you know the spectrum of S_t.

4. The "Non-Normal" Theory: Riesz Bases and Spectral Operators

What if S_t is diagonalizable but the eigenvectors are **not orthogonal**? (This happens in non-conservative systems, like convection-diffusion with non-standard boundary conditions).

- **The Theory: Non-Self-Adjoint Spectral Theory** (or the theory of **Riesz Bases**).

- **What it says:** You can still expand solutions in a basis of eigenfunctions, but because the basis is not orthogonal ($\langle \varphi_n, \varphi_m \rangle \neq \delta_{nm}$), you need a "dual basis" $\{\psi_n\}$ to find the coefficients.
  - Expansion: $f = \sum a_n \varphi_n$
  - Coefficients: $a_n = \langle f, \psi_n \rangle$ (instead of $\langle f, \varphi_n \rangle$).
- **Why it matters:** This is currently a very hot topic in fluid dynamics and laser physics, where systems are often "almost" normal but mathematically non-self-adjoint.

Summary: Which "Theory" do you need?

If your operator S_t is...	**The Theory is called...**	**The Basis is...**
Translation Invariant	Harmonic Analysis	Fourier Transform
Self-Adjoint / Normal	Spectral Theory	Orthonormal Eigenbasis
1D Variable Coeffs	Sturm-Liouville Theory	Special Functions
Not Normal (but diag.)	Non-Self-Adjoint Theory	Riesz Basis (Biorthogonal)
Not Diagonalizable	Jordan Form	Generalized Eigenvectors

2.6 Convolution and Approximate Identities

In this section, we introduce a new way of multiplying two functions that will be very useful and very natural. The naturality comes from extending the addition of $\mathbb{R}$ to functions on $\mathbb{R}$, for simplicity we stick to real functions however nothing changes if we use complex functions instead. Consider the function $\delta_a : \mathbb{R} \to \mathbb{R}$

$$\delta_a(x) = \begin{cases} 1 & x = a \\ 0 & \text{otherwise} \end{cases}$$

The naturality comes from the discrete case, and so we will start with it: recall that if we have two polynomials $p(x)$ and $q(x)$ (which can be thought of as formal sums, bringing in the naturality), then the multiplication formula which groups the terms of the same degree is:

$$\left(\sum_{i=0}^{\deg(p)} p_i x^i\right) \cdot \left(\sum_{i=0}^{\deg(q)} q_j x^j\right) = \sum_{k=0}^{\deg(p)+\deg(q)} \left(\sum_{i+j=k} p_i q_j\right) x^k = \sum_{k=0}^{\deg(p)+\deg(q)} \left(\sum_{l=0}^{k} p_l q_{k-l}\right) x^k$$

If we set p_k or q_{l-k} to zero if k or $l-k$ is negative, then we can re-write the sum as:

$$\left(\sum_{i=0}^{\deg(p)} p_i x^i\right) \cdot \left(\sum_{i=0}^{\deg(q)} q_j x^j\right) = \sum_{k=0}^{\deg(p)+\deg(q)} \left(\sum_{-\infty}^{\infty} p_l q_{k-l}\right) x^k$$

If we let $p[n]$ and $q[n]$ represent the coefficient of the nth degree term, and $h = pq$, then:

$$h[n] = \sum_{k=-\infty}^{\infty} p_k q_{n-k}$$

We can write this as:

$$h[n] = [p * q][n] = \sum_{k=-\infty}^{\infty} p_k q_{n-k}$$

where $p*q$ is a compact notation the right hand side. The summation can be thought of as repersenting the discrete measure. Generalized to the continuous case, this is called *convolution*:

Definition 2.6.1: Convolution

Let f, g be measurable on $\mathbb{R}^n$. Then the *convolution* of f and g is a function $f * g : \mathbb{R}^n \to \mathbb{R}$ defined by:

$$f * g(x) := \int f(x-y)g(y)dy$$

for all x such that the integral exists

To see this in action, Let $f = \chi_{[0,1]}$ and $g = \chi_{[0,1]}$, and you'll find that $f * g$ is a a triangle. You can imagine sliding the graph of f and finding the area of the intersection of f and g, and plotting it. There are many ways to make sure that $f * g$ is a.e. defined. For example, if f was bounded and compactly supported, g can be locally integrable. Note that in many cases, we will need that $K(x, y) = f(x - y)$ be measurable on $\mathbb{R}^n \times \mathbb{R}^n$. If we let $s(x, y) = x - y$, then since s is continuous, if f is Borel measurable, so is K. (Folland makes a comment here saying we can "assume" that this is the case, which I'm guessing is f being Borel measurable in the definition of convolution, because of his theorem 2.12, and he says the same can happen for Lebesgue measurability). Usually, we will use convolution to somehow combine the property of two functions f and g. We will soon see that convolution often preserves the "nice" properties the "nicer" function, so that we would convolution, say, a smooth function f with an integrable function g to get a smooth function $f * g$.

I would also be remiss if I don't mention that convolution is like multiplying two functions in a way the respects operation the *domain*. If f, g are real functions, then $f \cdot g$ respects the multiplication in the *codomain* as it multiplies the outputs. As $f * g$ is the continuous case of polynomial multiplication which multiplies, it respects the multiplication happening in the *domain*. To motivate why this is worth pointing out, consider a finite group G, $f \in \mathrm{Hom}_{\mathbf{Set}}(G, \mathbb{R})$, and interpret it as an element of $\mathbb{C}[G]$ so that $f = \sum_i a_i g_i$. Then we can always make $r \in \mathbb{R}$ act on f via:

$$r \cdot f(g_i) = rf(g_i)$$

However, we can also make $g \in G$ act on f:

$$g \cdot f(g_i) = f(g^{-1}g_i)$$

where we inverse for associativity:

$$\begin{aligned}
(g \cdot (h \cdot f))(x) &= (h \cdot f)(g^{-1}x) \\
&= f(h^{-1}(g^{-1}x)) \\
&= f((h^{-1}g^{-1})x) \\
&= f((gh)^{-1}x) \\
&= ((gh) \cdot f)(x)
\end{aligned}$$

This is the same thing as defining $g \cdot f$ given $f \in \mathbb{C}[G]$ since:

$$g \cdot f = g \cdot \sum_{g_i \in G} a_i g_i = \sum_{g_i \in G} a_i (g g_i) = \sum_{g^{-1} g_i \in G} a_i g_i$$

Thus, we may think of the translation operation as an action from the structure of the *domain*, when the domain has a group structure. It shall turn out that the way of characterizing the orthonormal basis for L^2 that gives rise to the Fourier transform is to choose the one that's invariant under τ_y.

Definition 2.6.2: Translation Operator

If $f : \mathbb{R}^n \to \mathbb{R}$, then

$$y * f(x) = \tau_y f(x) = f(x - y)$$

Notice that:

$$\|\tau_y f\|_p = \|f\|_p \qquad \|\tau_y f\|_u = \|f\|_u$$

Using this, we can give another definition of uniformally continuous: f is uniformally continuous if:

$$\|\tau_y f - f\|_u \to 0$$

Lemma 2.6.3: Compact Support Functions And UC

If $f \in C_c(\mathbb{R}^n)$, then f is uniformally continuous

***Proof*:**
should be easy proof (Folland p.238)

Proposition 2.6.4: Translation Is Continuous In Lp

If $1 \leq p < \infty$, then translation is continuous in the L^p norm, in particular if $f \in L^p$ and $z \in \mathbb{R}^n$, then

$$\lim_{y \to 0} \|\tau_{y+z} f - \tau_z f\|_p = 0$$

Note this proposition is false when $p = \infty$

***Proof*:**
Folland p. 238

The following results all work over $\mathbb{R}^n$ or $\mathbb{T}^n$ (interpreted as the unit cube)

Proposition 2.6.5: Algebraic Properties Of Convolution

Assume all the following integrals are defined and converge absolutely. Then:

1. $f * g = g * f$
2. $(f * g) * h = f * (g * h)$ (this result in particular requires absolute convergence)
3. $f * (g + h) = f * g + f * h$
4. $a(f * g) = (af) * g$ for all $a \in \mathbb{C}$
5. for all $z \in \mathbb{R}^n$, $\tau_z(f * g) = (\tau_z f) * g = f * (\tau_z g)$
6. If A is the closure of $\{x + y \mid x \in \operatorname{supp}(f),\ y \in \operatorname{supp}(g)\}$, then $\operatorname{supp}(f * g) \subseteq A$

***Proof*:**

1. $$\begin{aligned} f * g(x) &= \int f(x-y)g(y)dy \\ &= \int f(z)g(x-z)dz && z = x - y,\ y = x - z \\ &= \int g(x-z)f(z)dz \\ &= g * f(x) \end{aligned}$$
2. Follows from (1) from Fubini's theorem
3. Follows from direct computation
4. Follows from direct computation
5. Follows from direct computation and (1)
6. If $x \notin A$, then for any $y \in \operatorname{supp}(g)$, we have $x - y \notin \operatorname{supp}(f)$, thus $f(x-y)g(y) = 0$, i.e. $f * g(x) = 0$

One of the first properties important to establish is the relation between convolution of a function and p-norms:

Theorem 2.6.6: Young's Convolution Inequality

Let $1 \le p, q, r \le \infty$ be such that $p^{-1} + q^{-1} = r^{-1} + 1$, and let $f \in L^p$ and $g \in L^q$. Then $f * g(x)$ exists for a.e. x, $f * g \in L^r$ and

$$\|f * g\|_r \le \|f\|_p \|g\|_q$$

Notice in the statement that we have something a little different than the usual conjugate exponents. To see why, there is another thing worth pointing out, which is that this condition is stable under scaling. In particular, if $f_\lambda(x) = f(\lambda x)$ and $g_\lambda(x) = g(\lambda x)$. Then:

$$\begin{aligned}
\|f_\lambda\|_p &= \left(\int_{\mathbb{R}^n} f(\lambda x)^p\right)^{1/p} \\
&= \left(\int_{\mathbb{R}^n} \lambda^{-n} f(y)^p\right)^{1/p} && y = \lambda x,\ dy = \lambda^n dx \\
&= \lambda^{-n/p}\|f\|_p
\end{aligned}$$

where λ^n is the determinant of the Jacobian matrix. So:

$$\begin{aligned}
(f_\lambda * g_\lambda)(x) &= \int_{\mathbb{R}^n} f(\lambda x - \lambda y) g(\lambda y) dy \\
&= \lambda^{-n}(f * g)(\lambda x)
\end{aligned}$$

Thus we have:

$$\begin{aligned}
\|f_\lambda * g_\lambda\|_r &= \lambda^{-n}\|(f * g)_\lambda\|_r \\
&= \lambda^{-n/r-n}\|f * g\|_r \\
&\leq \lambda^{-n/p-n/q}\|f\|_p\|g\|_q
\end{aligned}$$

Notice that this inequality must hold for *all* $\lambda > 0$. If the exponents of λ on the left and right sides were different, we could force a contradiction by taking $\lambda \to 0$ or $\lambda \to \infty$ (one side would vanish while the other explodes). Thus, the exponents must be equal:

$$-n\left(1 + \frac{1}{r}\right) = -n\left(\frac{1}{p} + \frac{1}{q}\right)$$

Dividing out the $-n$, we recover the necessary condition:

$$1 + \frac{1}{r} = \frac{1}{p} + \frac{1}{q}$$

Intuitively, the "extra" $+1$ comes from the integration dy in the definition of convolution, which adds n dimensions of volume (unlike pointwise multiplication which preserves the exponent balance of Hölder's inequality).

***Proof*:**
First, consider $r = \infty$, meaning $r^{-1} = 0$, $p^{-1}+q^{-1} = 1$. So, we want to show $\|f*g\|_\infty \leq \|f\|_p\|g\|_q$. So:

$$\begin{aligned}
\|f * g\|_\infty &= \sup_x \left|\int f(x-y)g(y)dy\right| \\
&\leq \|f(x - \cdot)\|_p\|g\|_q \\
&\stackrel{!}{=} \|f\|_p\|g\|_q
\end{aligned}$$

where $\stackrel{!}{=}$ comes from $\int |h(x-y)|dy = \int |h(y)|dy$ where $z = x - y$, $dz = (-1)^n dz$ so the $(-1)^n$ dissapears in the absolute value. (Note: I corrected the L^1 typo to L^∞ here to match the case $r = \infty$).

Next, if p or q is 1 (say $p = 1$), then $1 + q^{-1} = r^{-1} + 1$, so $q^{-1} = r^{-1}$. Then we would have:

$$\|f * g\|_q \leq \|f\|_1 \|g\|_q$$

To see that this happens, we use theorem 2.4.2. In particular, let $K(x, y) = f(x - y)$. Then if $\|g\|_q$ exists, $\|K(x, \cdot)\|_1 = \|f\|_1$ and $\|K(\cdot, y)\| = \|f\|_1$, giving us the result.

Now, assume $1 < p < r$, $1 < q < r$. First, we manipulate it to be able to use Hölder's inequality:

$$|f(y)g(x-y)| = (|f(y)|^p |g(x-y)|^q)^{1/r} \, |f(y)|^{1-p/r} |g(x-y)|^{1-q/r}$$

Next, notice that:

$$\frac{1}{r} + \frac{r-q}{rq} + \frac{r-p}{rp} = \frac{1}{r}\left(1 + \frac{r}{q} - 1 + \frac{r}{p} - 1\right) = 1$$

So, we can apply general Hölder's inequality since $r - p, r - q \neq 0$ which gives us:

$$\begin{aligned} |(f * g)(x)| &\leq \left(\int_{\mathbb{R}^n} |f(y)^p||g(x-y)|^q dy\right)^{1/r} \cdot \|f\|_p^{(r-p)/r} \|g\|_q^{(r-q)/r} \\ \Rightarrow \quad |(f * g)(x)|^r &\leq \left(\int_{\mathbb{R}^n} |f(y)^p||g(x-y)|^q dy\right) \cdot \|f\|_p^{r-p} \|g\|_q^{r-q} \end{aligned}$$

And now, we can integrate the left hand side with respect to x and use Tonnelli's theorem:

$$\Rightarrow \quad \|(f * g)\|_r^r \leq \|f\|_p^p \|f\|_p^{r-p} \|g\|_q^q \|g\|_q^{r-q} = \|f\|_p^r \|g\|_q^r$$

completing the proof.

If we focus on the special case of $p = q = r = 1$, then $f * g \in L^1(\mathbb{R}^n)$. Then proposition 2.6.5 and Young's inequality (theorem 2.6.6) turns $L^1(\mathbb{R}^n)$ into a "commutative associative algebra". Note that this algebra has no multiplicative identity (i.e. We are working over a *rng*). It does have a notion of an identity in a limit, which we will shortly explore. In fact, if we slightly correctly loosen our notion of function to *distribution*, we shall naturally recover an identity (explored in chapter 3). Note that if $f, g \in L^2(\mathbb{R}^n)$ then $f * g \in L^\infty(\mathbb{R}^n)$, namely convoluting forces the functions the result to be bounded.

Before exploring a notion of identity, it must be shown that convoluting two functions where one nicer than the other usually keeps the properties of the nicer function:

Lemma 2.6.7: Convolution Preserves C_c^k

Let $f \in L^1$ and $g \in C_c^k$ (more generally, $\partial^\alpha g$ is bounded for $|\alpha| \leq k$). Then

$$f * g \in C^k$$

and

$$\partial^\alpha (f * g) = (f * \partial^\alpha g)$$

***Proof*:**
In Folland's textbook this follows from differentiation under the integral sign.

This is a matter of computation. We'll do the case of $k = 1$, since the rest is a matter of induction. First, write what you know about $f * g$:

$$\begin{aligned} f * g(x) &= \int_{\mathbb{R}^n} f(x-y)g(y)dy \\ &= \int_{\mathbb{R}^n} f(y)g(x-y)dy \end{aligned}$$

So consider

$$\lim_{h\to 0} \frac{f * g(x+h) - f * g(x)}{h} = \lim_{h\to 0} \int_{\mathbb{R}^n} f(y) \left(\frac{g(x+h-y) - g(x-y)}{h} \right) dy$$

Notice that:

$$\left| \frac{g(x+h-y) - g(x-y)}{h} \right| \leq \frac{|g'(c)| \cdot h}{h}$$

meaning we have a dominating function, and so we can apply dominated convergence theorem to move the limit in!

$$\begin{aligned} \lim_{h\to 0} \frac{f * g(x+h) - f * g(x)}{h} &= \lim_{h\to 0} \int_{\mathbb{R}^n} f(y) \left(\frac{g(x+h-y) - g(x-y)}{h} \right) dy \\ &= \int_{\mathbb{R}^n} f(y) \lim_{h\to 0} \left(\frac{g(x+h-y) - g(x-y)}{h} \right) dy \end{aligned}$$

and since $g \in C_c^1$ (in particular, g is differnetiable), this limit exists, and since g has comapct support, the integral is still well-defined and is C^1 (see the remarks we made at the beginning).

From here we can apply the induction, and the proof is done.

Proposition 2.6.8: Smooth Function Dense in L^p

Compact smooth functions are dense in L^p.

***Proof*:**
hint: simple functions are dense in L^p. Convolute them with a smooth bump function.

With this information, we are going to define a notion of "multiplicative idenity"

Definition 2.6.9: Good Kernels

Let $\{\varphi_t\}_{t>0}$ be a set of functions of the form $\varphi_t : \mathbb{R}^n \to \mathbb{C}$. Then they are called a *good kernel* if $t \in (0, \epsilon_0)$ and:

1. $\sup_{t>0} \|\varphi_t\|_1 < \infty$
2. $\int_{\mathbb{R}^n} \varphi_t(x)dx = 1$
3. $\lim_{t\to 0} \int_{|x|>\delta} |\varphi_t(x)|dx = 0$ for all $\delta > 0$

Theorem 2.6.10: Good Kernels Like Mult. Identity

Let $f \in L^p$ for $1 \leq p < \infty$ (note it is $<$, not $\leq$), then:

$$f * \varphi_t \to f$$

in L^p as $t \to \infty$. Furthermore, if f is continuous at some x_0, then:

$$f * \varphi_t(x_0) \to f(x_0)$$

in L^p

***Proof*:**
It is equivalent to show $f * \varphi_t - f \to 0$, so for all x:

$$\begin{aligned}
f * \varphi_t(x) - f(x) &= \int_{\mathbb{R}^n} f(x-y)\varphi_t(y)dy - f(x) \cdot 1 \\
&= \int_{\mathbb{R}^n} f(x-y)\varphi_t(y)dy - f(x) \cdot \int_{\mathbb{R}^n} \varphi_t(y)dy \\
&= \int_{\mathbb{R}^n} f(x-y)\varphi_t(y)dy - \int_{\mathbb{R}^n} f(x)\varphi_t(y)dy \\
&= \int_{\mathbb{R}^n} f(x-y)\varphi_t(y) - f(x)\varphi_t(y)dy \\
&= \int_{\mathbb{R}^n} [f(x-y) - f(x)]\varphi_t(y)dy \\
&= \int_{|y|\leq\delta} [f(x-y) - f(x)]\varphi_t(y)dy + \int_{|y|>\delta} [f(x-y) - f(x)]\varphi_t(y)dy
\end{aligned}$$

and so, taking the p-norm we get:

$$\|f * \varphi_t - f\|_p \leq \int_{|y|\leq\delta} \|f(\cdot - y) - f(\cdot)\|_p |\varphi_t(y)|dy + \int_{|y|>\delta} 2\|f\|_p |\varphi_t(y)|dy$$

at this point, we are almost done. If we let $c = \sup_{t>0} \int_{\mathbb{R}^n} |\varphi_t(y)|dy$, then we get:

$$\limsup_{t\to 0} \|f * \varphi_t - f\|_p \leq c \limsup_{\delta\to 0,\ |y|\leq\delta} \|f(\cdot - y) - f(\cdot)\|_p = 0$$

where the last inequality comes from the shrinking delta.

It was left as an exercise to do the point-convergence in the continuous case.

Note that if f is uniformally continuous and bouned, then $f * \varphi_t \to f$ uniformally, and if $\int \varphi = a$, then $f * \varphi_t \to af$.

Example 2.6.11: Good Kernels

Let $X(x) \in C_c^\infty(\mathbb{R}^n)$ where $\int X = 1$. Then:

$$\varphi_t(x) = t^{-n} X(x/t)$$

is a good kernel (ex. let $X(x) = \frac{1}{\sqrt{\pi}} e^{-x^2}$). This is just a matter of computation. Note that $\varphi_t = 0$ if $\frac{x}{t} \geq c_0$ for some c_0, and so $t \leq \frac{x}{c_0}$. It [somehow] follows that C_c^∞ is *dense* in L^p $(1 \leq p < \infty)$ (note the $<$ and not $\leq$)

For another Another example of a family of good kernel, take:

$$K_\delta(x) = \delta^{-1/2} e^{\frac{-\pi x^2}{\delta}}$$

These can be visualized here:

https://www.desmos.com/calculator/e8xc37r3ck

Notice that since the function $X \in C_c^\infty(\mathbb{R})$ is smooth on compact supporet then by lemma 2.6.7, $f * X$ is also smooth, and will remain smooth for every φ_n. Since $f * \varphi_n \to f$, we see that a sequence of smooth functions converges to an L^p function, meaning smooth functions are *dense* in L^p!

I found this interesting stack-exchange post on the importance of good Kernels:

https://math.stackexchange.com/questions/3294744/good-kernels-that-exist-in-the-real-world

Using the fact that convolution preserves the "nicer" structure, we strengthen's Urysohn's lemma (or, if you know some differential geometry, generlizes partitions of unity)

Theorem 2.6.12: Smooth Urysohn's Lemma

Let $K \subseteq \mathbb{R}^n$ be compact and $U \subseteq \mathbb{R}^n$ be open and contain K $(U \supseteq K)$. Then there exists a $f \in C_c^\infty$ such that $0 \leq f \leq 1$ and $f \equiv 1$, K and $\text{supp}(K) \subseteq f$.

***Proof*:**

Since K is compact, then $\delta = \rho(K, U^c)$ (the shortest distance from K and U^c) is non-zero. We are going to use a mollification trick: let $V = \left\{x \;\middle|\; \rho(x, K) < \frac{\delta}{3}\right\}$. Choose some φC_c^∞ such that $\int \varphi = 1$ and $\varphi(x) = 0$ for $|x| \geq \delta/3$ (as an example, $\{\psi_t\}$ to be a good kernel and take $\varphi = \left(\int \psi\right)^{-1} \psi_{\delta/3}$).

Now, let $f = \chi_V * \varphi$. Then as we've shown before, $f \in C_c^\infty$, and it is easy to see that $0 \leq f \leq 1$, $f \equiv 1$ on K, and $\text{supp}(f) \subseteq \{x \mid \rho(x, K) \leq 2\delta/3\} \subseteq U$

3

Distribution Theory

Distributions are a generalization of functions that allow us to give more general solutions than classical solutions, similar to how we found a weak solution last week. Locally integrable functions will be distributions (in the appropriate interpretation), but we'll see that there are generalized solutions that don't even fall under the notion of "function".

More broadly, distribution theory is a further exploration of the "dual theory" philosophy that has been a recurring theme in our study of analysis. In functional analysis, we repeatedly found that passing to the dual space reveals structure invisible in the original space: the Riesz representation theorem identified $(L^p)^*$ with L^q, and the Hahn-Banach theorem showed that the dual space is always rich enough to separate points. Distribution theory pushes this idea one step further. Rather than studying functions directly, we study what they do to test functions—that is, we study the linear functionals they induce. By broadening the class of allowable functionals beyond those induced by honest functions, we arrive at a theory in which every distribution is infinitely differentiable, every linear constant-coefficient PDE has a solution, and the Fourier transform becomes an isomorphism. In short, duality buys us completeness of operations that the classical function spaces lack.

Another use we will soon see is the intimate connection between distributions and the Fourier transform. The theory of tempered distributions will allow us to extend the Fourier transform to objects far beyond L^1 or L^2, giving us a powerful and unified framework for studying partial differential equations, signal processing, and harmonic analysis.

3.1 Intuition

For now, let $f \in L^p$. Then as we know, $(L^p)^* \cong L^q$ is an isometric isomorphism where $1 \leq p < \infty$ and q is the conjugate exponent. In particular, f is mapped to the bounded linear functional

$$g \mapsto \int fg.$$

Using the Lebesgue differentiation theorem, by choosing $\varphi_r = m(B_r)^{-1}\chi_{B_r}$, where B_r is a ball of radius r centered at x, we can recover the value of $f(x)$ for a.e. x, as $\lim_{r\to 0}\int f\varphi_r$. In this way, we can think of L^p as being embedded into $(L^q)^*$: the function f is completely determined (up to a.e. equivalence) by the values of the functional it defines.

Now, let us modify the above by letting f be locally integrable (in L^1_{loc}), but restricting our test functions to $\varphi \in C_c^\infty$. Then it is clear that $\varphi \mapsto \int f\varphi$ is still a well-defined linear functional on C_c^∞: the integral converges since φ has compact support and f is integrable on compact sets. Moreover, the pointwise values of f can be recovered a.e. from this functional by the same approximation argument (see Theorem 8.15 in Folland, the Borel Representation Theorem). So locally integrable functions embed faithfully into the space of linear functionals on C_c^∞.

Here is the key observation: unlike in the case for L^p spaces, there are in fact *many* linear functionals on C_c^∞ that are *not* of the form $\varphi \mapsto \int f\varphi$ for any locally integrable f. The most familiar example is the Dirac delta, $\varphi \mapsto \varphi(0)$, which "evaluates at a point"—something no L^1_{loc} function can do. These functionals, subject to some mild continuity conditions we will soon specify, will be our "generalized functions", or *distributions*.

3.2 Definitions

We will first rapid-fire some important definitions to be up to speed with terminology. Recall that $C_c^\infty(E)$ where $E \subseteq \mathbb{R}^n$ is the set of all smooth functions whose support is compact and contained in E. If $U \subseteq \mathbb{R}^n$ is open, $C_c^\infty(U)$ is the union of spaces $C_c^\infty(K)$ where K ranges over all compact subsets of U. Each $C_c^\infty(K)$ is a Fréchet space with the topology defined by the family of seminorms

$$\varphi \mapsto \|\partial^\alpha \varphi\|_u \qquad (\alpha \in \{0, 1, 2, \ldots\}^n).$$

Thus, a sequence $\{\varphi_i\}$ converges in $C_c^\infty(K)$ if and only if $\partial^\alpha\varphi_i \to \partial^\alpha\varphi$ uniformly for all multi-indices α. This is a strong form of convergence: we demand that not only the functions themselves, but all of their derivatives of every order, converge uniformly. The completeness of $C_c^\infty(K)$ under this topology is established through Exercise 9 in Folland, Section 5.1 (the key idea is that a uniform limit of smooth functions, where all derivatives also converge uniformly, is again smooth). With these in hand, we take the following definitions:

1. A sequence $\{\varphi_i\}$ in $C_c^\infty(U)$ *converges in* $C_c^\infty(U)$ to φ if there exists a compact set $K \subseteq U$ such that $\{\varphi_i\} \subseteq C_c^\infty(K)$ and $\varphi_i \to \varphi$ in the topology of $C_c^\infty(K)$, that is, $\partial^\alpha\varphi_i \to \partial^\alpha\varphi$ uniformly for all α. Note the requirement that all the φ_i are supported in a *single* compact set K; the supports are not allowed to "escape to infinity".

2. If X is a locally convex topological vector space and $T : C_c^\infty(U) \to X$ is a linear map, T is *continuous* if $T|_{C_c^\infty(K)}$ is continuous for each compact $K \subseteq U$, that is, if $T(\varphi_i) \to T(\varphi)$

whenever $\varphi_i \to \varphi$ in $C_c^\infty(K)$ for some compact $K \subseteq U$. In other words, continuity is tested "one compact set at a time".

3. A linear map $T : C_c^\infty(U) \to C_c^\infty(U')$ is *continuous* if for each compact $K \subseteq U$, there is a compact $K' \subseteq U'$ such that $T(C_c^\infty(K)) \subseteq C_c^\infty(K')$ and T is continuous from $C_c^\infty(K)$ to $C_c^\infty(K')$. The first condition says that T does not spread out the support in an uncontrolled way; the second says that the map is continuous in the Fréchet topology.

With these, we will define a distribution:

Definition 3.2.1: Distribution

A *distribution* on U is a continuous linear functional on $C_c^\infty(U)$. The space of all distributions on U is denoted by $\mathcal{D}'(U)$, and we set $\mathcal{D}' = \mathcal{D}'(\mathbb{R}^n)$. We impose the weak* topology on $\mathcal{D}'(U)$, that is, the topology of pointwise convergence on $C_c^\infty(U)$.

The *evaluation* of a distribution at a test function φ is often denoted either as:

$$F(\varphi) \qquad \text{or} \qquad \langle F, \varphi \rangle$$

A couple of remarks are in order. First, the $\mathcal{D}'$ notation comes from the fact that Schwartz used the notation $\mathcal{D}$ for C_c^∞, and so $\mathcal{D}'$ denotes the continuous dual of C_c^∞—the space of all continuous linear functionals on it. Next, there is in fact a locally convex topology on $C_c^\infty(U)$ (the so-called *inductive limit topology*) with respect to which sequential convergence is given by the convergence defined above, and the continuity of linear maps $T : C_c^\infty \to X$ and $T : C_c^\infty \to C_c^\infty$ agrees with the definitions in the list above. However, defining this topology rigorously requires the machinery of inductive limits of locally convex spaces, and the construction is somewhat involved without contributing much additional insight to the current exposition of distributions, so it will be omitted. The interested reader may consult Rudin's *Functional Analysis*, Chapter 6, for the full treatment. Furthermore, when we talk about convergence of distributions $F_i \to F$, we will always mean pointwise convergence: $\langle F_i, \varphi \rangle \to \langle F, \varphi \rangle$ for every $\varphi \in C_c^\infty(U)$.

Before giving some examples of distributions, we will remind the reader how to check for continuity of a linear functional in the setting of Fréchet spaces:

Proposition 3.2.2: Distribution Continuity Check

Let $F : C_c^\infty(U) \to \mathbb{R}$ (or $\mathbb{C}$) be linear. Then $u \in \mathcal{D}'(U)$ if and only if for every compact set $K \subseteq U$, there exist constants $C_K > 0$ and $N_K \in \mathbb{N}$ such that

$$|\langle F, \varphi \rangle| \leq C_K \sum_{|\alpha| \leq N_K} \|\partial^\alpha \varphi\|_{L^\infty}$$

for all $\varphi \in C_c^\infty(K)$.

***Proof*:**
Suppose first that the estimate holds. We wish to show that u is continuous on each $C_c^\infty(K)$. Let $\varphi_i \to \varphi$ in $C_c^\infty(K)$. Then for every multi-index α, $\partial^\alpha \varphi_i \to \partial^\alpha \varphi$ uniformly, which means

$\|\partial^\alpha(\varphi_i - \varphi)\|_{L^\infty} \to 0$ for each α. Applying the estimate to $\varphi_i - \varphi$:

$$|\langle u, \varphi_i\rangle - \langle u, \varphi\rangle| = |\langle u, \varphi_i - \varphi\rangle| \leq C_K \sum_{|\alpha|\leq N_K} \|\partial^\alpha(\varphi_i - \varphi)\|_{L^\infty} \to 0,$$

so $\langle u, \varphi_i\rangle \to \langle u, \varphi\rangle$, and u is continuous on $C_c^\infty(K)$. Since K was arbitrary, u is a distribution. Conversely, suppose u is continuous on $C_c^\infty(K)$ but no such estimate holds. Then for each $j \in \mathbb{N}$, there exists $\varphi_j \in C_c^\infty(K)$ with

$$|\langle u, \varphi_j\rangle| > j \sum_{|\alpha|\leq j} \|\partial^\alpha \varphi_j\|_{L^\infty}.$$

Set $\psi_j = \varphi_j / |\langle u, \varphi_j\rangle|$. Then $\langle u, \psi_j\rangle = 1$ for all j (taking absolute values and adjusting the sign if needed, we can arrange $|\langle u, \psi_j\rangle| = 1$). But for $|\alpha| \leq j$:

$$\|\partial^\alpha \psi_j\|_{L^\infty} < \frac{1}{j},$$

so $\psi_j \to 0$ in $C_c^\infty(K)$. By continuity of u, we should have $\langle u, \psi_j\rangle \to \langle u, 0\rangle = 0$, contradicting $|\langle u, \psi_j\rangle| = 1$.

Definition 3.2.3: Order of Distribution

Let $F \in \mathcal{D}'(U)$ be a distribution and compact $K \subseteq U$ so that by proposition 3.2.2 there exists a C_K and N_K such that

$$|\langle F, \varphi\rangle| \leq C_K \sum_{|\alpha|\leq N_K} \|\partial^\alpha \varphi\|_{L^\infty}$$

then N_K is called the *order* of F on K. If there exists an $N \in \mathbb{N}$ such that for all compact $K \subseteq U$:

$$|\langle F, \varphi\rangle| \leq C_K \sum_{|\alpha|\leq N} \|\partial^\alpha \varphi\|_{L^\infty}$$

then N is called the *order* of F. If no such N exists, then F is said to have infinite order.

The way I like to think about the continuity check is as follows: we will soon see that there are many distribution spaces, for example the Schwartz space $\mathcal{S}$ will have a parallel $\mathcal{S}'$ (the *tempered distributions*). What the continuity condition tells us is that a distribution respects the convergence structure of its test function space. If $\varphi_j \to \varphi$ in the test function topology, then $\langle u, \varphi_j\rangle \to \langle u, \varphi\rangle$. Hence, we can translate convergence information from the test function space to information about the distribution.

Example 3.2.4: Distributions

1. *Locally integrable functions.* For every $f \in L^1_{\text{loc}}(U)$, we can define a distribution on U, namely the functional $\varphi \mapsto \int f\varphi$. This is indeed continuous: if $\varphi \in C_c^\infty(K)$, then

$$\left|\int f\varphi\right| \leq \|f\|_{L^1(K)} \|\varphi\|_{L^\infty},$$

so the estimate in Proposition 3.2.2 holds with $N_K = 0$ and $C_K = \|f\|_{L^1(K)}$. Moreover, two functions f and g define the same distribution precisely when they are equal a.e. (this follows from the fact that C_c^∞ is dense in L^q for the appropriate q, or more directly from the Lebesgue differentiation theorem). In this way, $L^1_{\text{loc}}(U)$ embeds into $\mathcal{D}'(U)$, and we often identify a locally integrable function with the distribution it defines.

2. *Radon measures.* Every Radon measure μ on U defines a distribution $\varphi \mapsto \int \varphi \, d\mu$. The continuity check is similar: for $\varphi \in C_c^\infty(K)$, we have $|\int \varphi \, d\mu| \leq |\mu|(K)\|\varphi\|_{L^\infty}$, where $|\mu|(K)$ is the total variation of μ on K. Again, this satisfies the estimate with $N_K = 0$. Note that this generalizes the previous example, since a locally integrable function f defines a Radon measure $d\mu = f \, dm$ via the Radon-Nikodym theorem. In fact, by the Radon-Nikodym theorem, it is the singular part of the measure that create distributions that are not representable by functions.

3. *Derivatives of point masses.* If $x_0 \in U$ and α is a multi-index, the map $\varphi \mapsto \partial^\alpha \varphi(x_0)$ is a distribution. The continuity estimate is immediate: $|\partial^\alpha \varphi(x_0)| \leq \|\partial^\alpha \varphi\|_{L^\infty}$. This distribution does not arise from a locally integrable function: no L^1_{loc} function can "see" the value of a derivative at a single point. It arises from a Radon measure μ precisely when $\alpha = 0$, in which case $\mu = \delta_{x_0}$ is the point mass at x_0. For $|\alpha| \geq 1$, these distributions are genuinely "beyond" measures—they are the first examples of distributions that cannot be represented by any kind of integration.

4. *measure on cantor set* Another example of a singular distribution that is not the point-mass above is the measure on the cantor set. Once differentiation is defined, we shall see that this distribution is the derivative of the devil's staircase function (or cantor functions), see ref:HERE.

As mentioned earlier, we often associate a function f with the distribution it defines, and even label the distribution by f. For this reason, it might be confusing to write $f(\varphi)$ to mean the distribution defined by f evaluated on φ, and so for historical reasons we use the bracket notation $\langle f, \varphi \rangle$ for the distributional pairing. This notation is suggestive of an inner product (and indeed, when $f \in L^2$ and $\varphi \in C_c^\infty$, it literally is the L^2 inner product), though the reader should keep in mind that the pairing is between two different spaces. The use of the inner product notation should bring to mind Riesz-Representation theorem (theorem 5.1.13) where the inner product is the "perfect pairing" between a space and its dual.

Let's next establish some common notation:

1. First, it is often enough that we will reflect a function. Thus, if we put a tilde over a function, we mean:
$$\tilde{\varphi}(x) = \varphi(-x).$$
This notation will appear frequently in the context of convolutions, where the reflection arises naturally from the change of variables $y \mapsto -y$.

2. Second, the point-mass (or Dirac delta) is a very common distribution, which we will denote by δ:
$$\langle \delta, \varphi \rangle = \varphi(0).$$
More generally, δ_{x_0} denotes the point mass at x_0, defined by $\langle \delta_{x_0}, \varphi \rangle = \varphi(x_0)$. When $x_0 = 0$, we simply write δ.

As an example of the use of the delta function, and to illustrate how convergence of distributions works in practice:

Proposition 3.2.5: Convergence To Delta Function

Let $f \in L^1(\mathbb{R}^n)$ with $\int f = a$, and for $t > 0$, let $f_t(x) = t^{-n} f(t^{-1}x)$. Then $f_t \to a\delta$ in $\mathcal{D}'$ as $t \to 0$.

Before the proof, let us unpack what this says. The rescaling $f_t(x) = t^{-n} f(t^{-1}x)$ concentrates the mass of f near the origin as $t \to 0$: the factor t^{-n} ensures that $\int f_t = \int f = a$ for all t, while the argument $t^{-1}x$ compresses the support. So f_t becomes a taller and narrower spike near 0, with constant total mass a. The proposition says that in the distributional sense, this spike converges to a times the delta function—exactly what our intuition would predict.

***Proof*:**
Let $\varphi \in C_c^\infty$. By Theorem 8.15 (Folland), we have:

$$\langle f_t, \varphi \rangle = \int f_t(x)\varphi(x)\,dx = \int t^{-n} f(t^{-1}x)\varphi(x)\,dx.$$

Performing the substitution $y = t^{-1}x$ (so $x = ty$ and $dx = t^n\,dy$):

$$\int t^{-n} f(t^{-1}x)\varphi(x)\,dx = \int f(y)\varphi(ty)\,dy.$$

Now we recognize the right-hand side as $f_t * \tilde{\varphi}(0)$. As $t \to 0$, $\varphi(ty) \to \varphi(0)$ pointwise, and since $|\varphi(ty)| \leq \|\varphi\|_{L^\infty}$ and $f \in L^1$, the dominated convergence theorem gives:

$$\langle f_t, \varphi \rangle = \int f(y)\varphi(ty)\,dy \to \varphi(0) \int f(y)\,dy = a\varphi(0) = a\langle \delta, \varphi \rangle,$$

completing the proof.

For equality of distributions, note that it makes sense to say that $F = G$ on an open set $V \subseteq U$ if and only if $\langle F, \varphi \rangle = \langle G, \varphi \rangle$ for all $\varphi \in C_c^\infty(V)$. If F and G are continuous functions, this implies they are pointwise equal on V, and if F and G are only locally integrable, this means they are equal a.e. on V.

Since a function in $C_c^\infty(V_1 \cup V_2)$ need not be supported in either V_1 or V_2 individually, it is not immediately obvious that if $F = G$ on V_1 and on V_2, then $F = G$ on $V_1 \cup V_2$. However, this is indeed the case, thanks to partitions of unity:

Proposition 3.2.6: Equality On Support

Let $\{V_\alpha\}$ be a collection of open subsets of U and let $V = \bigcup_\alpha V_\alpha$. If $F, G \in \mathcal{D}'(U)$ and $F = G$ on each V_α, then $F = G$ on V.

***Proof*:**
It suffices to show that $F - G = 0$ on V, so we may assume $G = 0$ and show that $F = 0$ on V. Let $\varphi \in C_c^\infty(V)$. Then $\operatorname{supp}(\varphi)$ is a compact subset of $V = \bigcup_\alpha V_\alpha$. By compactness, finitely many of the V_α cover $\operatorname{supp}(\varphi)$, say $V_{\alpha_1}, \ldots, V_{\alpha_m}$. Let $\{\chi_1, \ldots, \chi_m\}$ be a smooth partition of unity subordinate to this finite cover, so that each $\chi_j \in C_c^\infty(V_{\alpha_j})$, $0 \leq \chi_j \leq 1$, and $\sum_{j=1}^m \chi_j = 1$ on a neighborhood of $\operatorname{supp}(\varphi)$. Then $\varphi = \sum_{j=1}^m \chi_j \varphi$, and each $\chi_j \varphi \in C_c^\infty(V_{\alpha_j})$. Since $F = 0$ on each V_{α_j}, we have $\langle F, \chi_j \varphi \rangle = 0$ for each j, and by linearity:

$$\langle F, \varphi \rangle = \sum_{j=1}^m \langle F, \chi_j \varphi \rangle = 0.$$

Since $\varphi \in C_c^\infty(V)$ was arbitrary, $F = 0$ on V.

As a consequence of Proposition 3.2.6, if $F \in \mathcal{D}'(U)$, there is a largest open set on which $F = 0$, namely the union of all open subsets of U on which $F = 0$. The complement of this set in U is called the *support* of F, denoted $\operatorname{supp}(F)$. When F is a locally integrable function, this agrees with the usual notion of [closed] support (up to a set of measure zero).

Definition 3.2.7: Distribution With Compact Support

The set of all distributions on U with compact support is denoted $\mathcal{E}'(U)$

It turns out (though we will not prove it here) that $\mathcal{E}'(U)$ is precisely the dual of $C^\infty(U)$—we no longer need to require that the test functions have compact support, because the distribution itself is compactly supported.

3.3 Operations on Distributions

One of the advantages of generalizing functions to distributions is that even though they are not "functions" in the classical sense, we can perform many of the same operations on them: differentiation, multiplication by a smooth function, convolution, and translation! To achieve this, we use a single unifying principle that we now describe.

Suppose U and V are open subsets of $\mathbb{R}^n$ and T is a linear map from some subspace $\mathcal{V}$ of $L^1_{\text{loc}}(U)$ into $L^1_{\text{loc}}(V)$. Suppose that there is another linear map $T' : C_c^\infty(V) \to C_c^\infty(U)$ (called the *transpose* of T) such that

$$\int (Tf)\varphi = \int f(T'\varphi) \qquad (f \in \mathcal{V},\ \varphi \in C_c^\infty(V)).$$

Suppose further that T' is continuous in the sense defined earlier (i.e., for each compact $K' \subseteq V$, there exists a compact $K \subseteq U$ with $T'(C_c^\infty(K')) \subseteq C_c^\infty(K)$ and T' continuous from $C_c^\infty(K')$ to $C_c^\infty(K)$). Then T can be *extended* to a map from $\mathcal{D}'(U)$ to $\mathcal{D}'(V)$, still denoted by T, by defining:

$$\langle TF, \varphi \rangle = \langle F, T'\varphi \rangle \qquad (F \in \mathcal{D}'(U),\ \varphi \in C_c^\infty(V)).$$

Why does this work? First, TF is a well-defined linear functional on $C_c^\infty(V)$ since $T'\varphi \in C_c^\infty(U)$ and F is a linear functional on $C_c^\infty(U)$. Second, TF is continuous: if $\varphi_i \to \varphi$ in $C_c^\infty(V)$, then $T'\varphi_i \to T'\varphi$ in $C_c^\infty(U)$ by continuity of T', and so $\langle TF, \varphi_i \rangle = \langle F, T'\varphi_i \rangle \to \langle F, T'\varphi \rangle = \langle TF, \varphi \rangle$ since

F is a distribution. Third, this definition is *consistent* with the original map on functions: if F is a locally integrable function in $\mathcal{V}$, then $\langle TF, \varphi\rangle = \langle F, T'\varphi\rangle = \int f(T'\varphi) = \int (Tf)\varphi$, which is exactly the distribution defined by the function Tf.

The continuity of T' also guarantees that the extended map $T : \mathcal{D}'(U) \to \mathcal{D}'(V)$ is continuous with respect to the weak* topology on distributions: if $F_\alpha \to F$ in $\mathcal{D}'(U)$, then for any $\varphi \in C_c^\infty(V)$, $\langle TF_\alpha, \varphi\rangle = \langle F_\alpha, T'\varphi\rangle \to \langle F, T'\varphi\rangle = \langle TF, \varphi\rangle$, so $TF_\alpha \to TF$ in $\mathcal{D}'(V)$.

Using this, we can construct many operations on distributions by transposing the property onto the test functions

1. *(Differentiation)* Let $Tf = \partial^\alpha f$, defined initially on $C^{|\alpha|}(U)$. If $\varphi \in C_c^\infty(U)$, integration by parts gives
$$\int (\partial^\alpha f)\varphi = (-1)^{|\alpha|} \int f(\partial^\alpha \varphi);$$
there are no boundary terms since φ has compact support. Hence $T' = (-1)^{|\alpha|}\partial^\alpha|_{C_c^\infty(U)}$, and we can define the *distributional derivative* $\partial^\alpha F \in \mathcal{D}'(U)$ of any $F \in \mathcal{D}'(U)$ by:
$$\langle \partial^\alpha F, \varphi\rangle := (-1)^{|\alpha|}\langle F, \partial^\alpha \varphi\rangle.$$
By this procedure, every distribution is infinitely differentiable! We can define the derivatives of arbitrary locally integrable functions, even when they are not differentiable in the classical sense. When F happens to be a $C^{|\alpha|}$ function, the distributional derivative agrees with the classical one (by integration by parts), so this truly is a generalization. We will compute the distributional derivatives of some important locally integrable functions shortly.

   A particularly important consequence: distributional derivatives commute. That is, $\partial^\alpha\partial^\beta F = \partial^{\alpha+\beta}F = \partial^\beta\partial^\alpha F$ for all distributions F and all multi-indices α, β. This is immediate from the definition and the fact that classical derivatives of smooth functions commute.

2. *(Multiplication by a smooth function)* Let $\psi \in C^\infty(U)$ and define $Tf = \psi f$. Then $T' = T|_{C_c^\infty(U)}$ (that is, $T'\varphi = \psi\varphi$), since $\int(\psi f)\varphi = \int f(\psi\varphi)$. Evidently T' maps $C_c^\infty(U)$ to $C_c^\infty(U)$ and is continuous (since multiplication by a smooth function preserves both support and uniform convergence of derivatives, by the Leibniz rule). Thus we define $\psi F \in \mathcal{D}'(U)$ for any $F \in \mathcal{D}'(U)$ by:
$$\langle \psi F, \varphi\rangle := \langle F, \psi\varphi\rangle.$$
Note that we require ψ to be C^∞; multiplying a distribution by a continuous or L^∞ function is in general not well-defined. The Leibniz rule extends to distributions: $\partial^\alpha(\psi F) = \sum_{\beta\le\alpha}\binom{\alpha}{\beta}(\partial^{\alpha-\beta}\psi)(\partial^\beta F)$, which is proved by testing against $\varphi \in C_c^\infty$ and using the classical Leibniz rule on the test function side.

3. *(Translation)* Let $y \in \mathbb{R}^n$ and let $V = U + y = \{x + y : x \in U\}$. Take $T = \tau_y$ (recall that $\tau_y f(x) = f(x-y)$). Via the substitution $x \mapsto x + y$, we get
$$\int f(x-y)\varphi(x)\,dx = \int f(x)\varphi(x+y)\,dx,$$
and so $T' = \tau_{-y}|_{C_c^\infty(V)}$. This is evidently continuous, and so for a distribution $F \in \mathcal{D}'(U)$, we define:
$$\langle \tau_y F, \varphi\rangle := \langle F, \tau_{-y}\varphi\rangle.$$
As a quick sanity check: $\langle \tau_y\delta, \varphi\rangle = \langle \delta, \tau_{-y}\varphi\rangle = (\tau_{-y}\varphi)(0) = \varphi(y) = \langle \delta_y, \varphi\rangle$. So translating the delta function by y gives the point mass at y, exactly as expected.

4. *(Composition with linear maps)* Let S be an invertible linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^n$. Let $V = S^{-1}(U)$ and $Tf = f \circ S$. By the change of variables formula (see Chapter 1), for $\varphi \in C_c^\infty(V)$:
$$\int (f \circ S)(x)\varphi(x)\,dx = \int f(y)\varphi(S^{-1}y)|\det S|^{-1}\,dy.$$
So $T'\varphi = |\det(S)|^{-1}\varphi \circ S^{-1}$. For $F \in \mathcal{D}'(U)$, we define:
$$\langle F \circ S, \varphi\rangle := |\det S|^{-1}\langle F, \varphi \circ S^{-1}\rangle.$$
As a special case, taking $S = -I$ (the map $x \mapsto -x$) gives $\langle \tilde{F}, \varphi\rangle = \langle F, \tilde{\varphi}\rangle$, which defines the *reflection* of a distribution.

Convolutions are interesting and deserve more attention as they smooth distributions and recover *functions*. Consider first convoluting with a function. Let $\psi \in C_c^\infty$, and let
$$V = \{x : x - y \in U \text{ for all } y \in \operatorname{supp}(\psi)\}$$
Note that V is open (but may be empty). If $f \in L^1_{\mathrm{loc}}(U)$, then the integral
$$f * \psi(x) = \int f(x-y)\psi(y)\,dy = \int f(y)\psi(x-y)\,dy = \langle f, \tau_x\tilde{\psi}\rangle$$
is well-defined for all $x \in V$. Notice that we are inputting a *point*. Since the last expression involves only the distributional pairing, we can define for $F \in \mathcal{D}'(U)$:
$$F * \psi(x) = \langle F, \tau_x\tilde{\psi}\rangle$$
This is well-defined because $\tau_x\tilde{\psi} \in C_c^\infty(U)$ for $x \in V$.

Proposition 3.3.1: Convoluting With Function Gives Function

Let F be a distribution and ψ a smooth function with compact support on V. Then $F * \psi$ is a smooth functions

***Proof*:**
Since $\tau_x\tilde{\psi} \to \tau_{x_0}\tilde{\psi}$ in C_c^∞ as $x \to x_0$ (and similarly for all derivatives), continuity of F gives $F * \psi(x) \to F * \psi(x_0)$. More precisely, one can show by differentiating under the pairing that $\partial^\alpha(F * \psi) = F * (\partial^\alpha\psi)$ for all multi-indices α, which establishes that $F * \psi \in C^\infty(V)$.

A pertinent example is the following:
$$\delta * \psi(x) = \langle \delta, \tau_x\tilde{\psi}\rangle = \tau_x\tilde{\psi}(0) = \tilde{\psi}(-x) = \psi(x)$$
so δ is the identity element for convolution! This is one of the most important properties of the delta function and lies at the heart of the theory of Green's functions: if L is a differential operator and E is a distribution satisfying $LE = \delta$ (a *fundamental solution*), then $u = E * f$ formally solves $Lu = f$.

We can also define convolution using the same transpose technique from earlier. Let ψ, $\tilde{\psi}$, and V be as above. If $f \in L^1_{\mathrm{loc}}(U)$ and $\varphi \in C_c^\infty(V)$, then by Fubini's theorem:
$$\begin{aligned}\int (f * \psi)(x)\varphi(x)\,dx &= \iint f(y)\psi(x-y)\varphi(x)\,dy\,dx = \int f(y)\left(\int \psi(x-y)\varphi(x)\,dx\right)dy \\ &= \int f(y)(\varphi * \tilde{\psi})(y)\,dy\end{aligned}$$

Thus $Tf = f * \psi$ maps $L^1_{\text{loc}}(U)$ to $L^1_{\text{loc}}(V)$ with transpose $T'\varphi = \varphi * \tilde{\psi}$. For a distribution $F \in \mathcal{D}'(U)$, we therefore get:

$$\langle F * \psi, \varphi \rangle = \langle F, \varphi * \tilde{\psi} \rangle$$

We should verify that the two definitions of convolution are consistent:

Proposition 3.3.2: Consistency of Convolution

Let $F \in \mathcal{D}'(U)$ and $\psi \in C_c^\infty$, and let V be as above. Then the smooth function $x \mapsto \langle F, \tau_x \tilde{\psi} \rangle$ and the distribution defined by $\langle F, \varphi * \tilde{\psi} \rangle$ agree, in the sense that the 2nd convolution definition is the distribution defined by the locally integrable function from the 1st convolution definition.

***Proof*:**
We need to show that for all $\varphi \in C_c^\infty(V)$:

$$\int \langle F, \tau_x \tilde{\psi} \rangle \varphi(x)\, dx = \langle F, \varphi * \tilde{\psi} \rangle.$$

The left-hand side can be written as the limit of Riemann sums. More precisely, for φ supported in a compact set $K' \subseteq V$, the function $x \mapsto \tau_x \tilde{\psi}$ is a continuous map from K' into $C_c^\infty(K)$ for some compact $K \subseteq U$. One checks (using the Riemann sum approximation and continuity of F) that the integral of a continuous $C_c^\infty(K)$-valued function against a scalar function can be "passed inside" the distributional pairing. But the integral $\int \tau_x \tilde{\psi}\, \varphi(x)\, dx$, computed pointwise, is exactly the function $y \mapsto \int \psi(x - y)\varphi(x)\, dx = (\varphi * \tilde{\psi})(y)$. Hence:

$$\int \langle F, \tau_x \tilde{\psi} \rangle \varphi(x)\, dx = \left\langle F, \int \tau_x \tilde{\psi}\, \varphi(x)\, dx \right\rangle = \langle F, \varphi * \tilde{\psi} \rangle.$$

3.4 Density Results

Next, we will show that smooth compactly supported functions $C_c^\infty(U)$ are dense in $\mathcal{D}'(U)$. This is a remarkable fact: it says that every distribution, no matter how singular, can be approximated (in the weak* topology) by smooth functions. We begin with a preparatory lemma.

Lemma 3.4.1: Build-Up For Density

Suppose that $\varphi \in C_c^\infty$, $\psi \in C_c^\infty$ with $\int \psi = 1$, and let $\psi_t(x) = t^{-n}\psi(t^{-1}x)$ for $t > 0$. Then:

1. Given any open neighborhood U of $\operatorname{supp}(\varphi)$, we have $\operatorname{supp}(\varphi * \psi_t) \subseteq U$ for sufficiently small $t > 0$.

2. $\varphi * \psi_t \to \varphi$ in C_c^∞ as $t \to 0$.

***Proof*:**

1. Note that $\operatorname{supp}(\varphi * \psi_t) \subseteq \operatorname{supp}(\varphi) + \operatorname{supp}(\psi_t) = \operatorname{supp}(\varphi) + t \cdot \operatorname{supp}(\psi)$. Since $\operatorname{supp}(\varphi)$ is compact, the set $\operatorname{supp}(\varphi) + t \cdot \operatorname{supp}(\psi)$ is the Minkowski sum of $\operatorname{supp}(\varphi)$ with a set of diameter shrinking to 0 as $t \to 0$. Since U is an open neighborhood of the compact set $\operatorname{supp}(\varphi)$, there exists $\varepsilon > 0$ such that the ε-neighborhood of $\operatorname{supp}(\varphi)$ is contained in U. For t small enough that $t \cdot \operatorname{supp}(\psi)$ lies within a ball of radius ε, we get $\operatorname{supp}(\varphi * \psi_t) \subseteq U$.

2. We need to show that $\partial^\alpha(\varphi * \psi_t) \to \partial^\alpha \varphi$ uniformly for every multi-index α. Since differentiation commutes with convolution, $\partial^\alpha(\varphi * \psi_t) = (\partial^\alpha \varphi) * \psi_t$. So it suffices to show that $g * \psi_t \to g$ uniformly whenever $g \in C_c^\infty$. We have:

$$(g * \psi_t)(x) - g(x) = \int [g(x-y) - g(x)]\psi_t(y)\, dy = \int [g(x-tz) - g(x)]\psi(z)\, dz,$$

using the substitution $y = tz$. Since g is smooth and compactly supported, it is uniformly continuous, so for any $\varepsilon > 0$ there exists $\delta > 0$ such that $|g(x-tz) - g(x)| < \varepsilon$ whenever $t|z| < \delta$. For $z \in \operatorname{supp}(\psi)$ and t small enough, this is satisfied, giving $|g * \psi_t(x) - g(x)| \leq \varepsilon \int |\psi|$ for all x. By part (a), all the $\varphi * \psi_t$ are eventually supported in a single compact set, so the convergence is in C_c^∞.

Proposition 3.4.2: Density Of Smooth Compactly Supported Functions In Distributions

Let $U \subseteq \mathbb{R}^n$ be open. Then $C_c^\infty(U)$ is dense in $\mathcal{D}'(U)$ in the weak* topology.

***Proof*:**
Let $F \in \mathcal{D}'(U)$; we must find a net (or sequence) $\{\varphi_i\}$ in $C_c^\infty(U)$ such that $\langle \varphi_i, \eta\rangle \to \langle F, \eta\rangle$ for all $\eta \in C_c^\infty(U)$, where $\langle \varphi_i, \eta\rangle = \int \varphi_i \eta$.

Fix $\psi \in C_c^\infty(\mathbb{R}^n)$ with $\int \psi = 1$ and let $\psi_t(x) = t^{-n}\psi(t^{-1}x)$. We also need a family of cutoff functions: let $\{\chi_j\}_{j=1}^\infty$ be a sequence in $C_c^\infty(U)$ with $0 \leq \chi_j \leq 1$ and $\chi_j \to 1$ pointwise on U (e.g., take $\chi_j = 1$ on $\{x \in U : |x| \leq j \text{ and } \operatorname{dist}(x, U^c) \geq 1/j\}$ and smooth it out). Consider the distributions $F_j = \chi_j F$ (defined by $\langle F_j, \eta\rangle = \langle F, \chi_j \eta\rangle$). Each F_j has compact support in U, and $F_j \to F$ in $\mathcal{D}'(U)$ since for fixed $\eta \in C_c^\infty(U)$, $\chi_j \eta \to \eta$ in $C_c^\infty(U)$ for large j (eventually $\chi_j = 1$ on $\operatorname{supp}(\eta)$).

Now consider $F_j * \psi_t$. By our earlier discussion, this is a C^∞ function. For t sufficiently small, it has compact support in U (since F_j has compact support in U and the convolution only slightly enlarges the support). By Proposition 3.2.5 and the properties of convolution, $F_j * \psi_t \to F_j$ in $\mathcal{D}'$ as $t \to 0$: for any $\eta \in C_c^\infty(U)$,

$$\langle F_j * \psi_t, \eta\rangle = \langle F_j, \eta * \tilde{\psi}_t\rangle \to \langle F_j, \eta\rangle$$

by lemma 3.4.1, since $\eta * \tilde{\psi}_t \to \eta$ in C_c^∞. By a diagonal argument (choosing $t_j \to 0$ slowly enough), we obtain a sequence $g_j = F_j * \psi_{t_j} \in C_c^\infty(U)$ with $g_j \to F$ in $\mathcal{D}'(U)$.

3.5 Distributional Derivatives of Classical Functions

We promised earlier that the distributional derivative lets us differentiate functions that are not classically differentiable, and now we deliver on that promise. The computations in this section are among the most important examples in all of distribution theory, and the reader should internalize them thoroughly.

Recall that for $F \in \mathcal{D}'(\mathbb{R}^n)$ and a multi-index α, the distributional derivative is defined by

$$\langle \partial^\alpha F, \varphi\rangle = (-1)^{|\alpha|}\langle F, \partial^\alpha \varphi\rangle.$$

Example 3.5.1: Derivative Of The Heaviside Function

Let $H : \mathbb{R} \to \mathbb{R}$ be the Heaviside step function, $H(x) = \mathbf{1}_{[0,\infty)}(x)$. Then $H \in L^1_{\text{loc}}(\mathbb{R})$ and hence defines a distribution. Classically, H is differentiable everywhere except at $x = 0$, where it has a jump discontinuity of height 1. Let us compute H' in the distributional sense. For $\varphi \in C_c^\infty(\mathbb{R})$:

$$\langle H', \varphi\rangle = -\langle H, \varphi'\rangle = -\int_0^\infty \varphi'(x)\, dx = -[\varphi(x)]_{x=0}^{x=\infty} = -\big(\underbrace{\varphi(\infty)}_{=0} - \varphi(0)\big) = \varphi(0) = \langle \delta, \varphi\rangle.$$

Therefore $H' = \delta$ in $\mathcal{D}'(\mathbb{R})$. This is the prototypical example: the distributional derivative detects the jump discontinuity and places a point mass there. More generally, if f is piecewise C^1 with jump discontinuities of height c_j at points x_j, then $f' = f'_{\text{classical}} + \sum_j c_j \delta_{x_j}$ in $\mathcal{D}'$.

Example 3.5.2: Derivative Of $|x|$ And The Sign Function

Consider $f(x) = |x|$ on $\mathbb{R}$. This function is continuous but not differentiable at $x = 0$. For $\varphi \in C_c^\infty(\mathbb{R})$:

$$\begin{aligned}
\langle f', \varphi\rangle = -\langle f, \varphi'\rangle &= -\int_{-\infty}^\infty |x|\varphi'(x)\, dx \\
&= -\int_{-\infty}^0 (-x)\varphi'(x)\, dx - \int_0^\infty x\varphi'(x)\, dx \\
&= \int_{-\infty}^0 x\varphi'(x)\, dx - \int_0^\infty x\varphi'(x)\, dx.
\end{aligned}$$

Integrating by parts on each piece (the boundary terms at $\pm\infty$ vanish since φ has compact support, and the boundary terms at 0 cancel):

$$= \Big[x\varphi(x)\Big]_{-\infty}^0 - \int_{-\infty}^0 \varphi(x)\, dx - \Big[x\varphi(x)\Big]_0^\infty + \int_0^\infty \varphi(x)\, dx = -\left(\int_{-\infty}^0 \varphi\right) + \left(\int_0^\infty \varphi\right) = \int_{-\infty}^\infty \operatorname{sgn}(x)\varphi(x)\, dx,$$

where $\operatorname{sgn}(x) = \mathbf{1}_{(0,\infty)}(x) - \mathbf{1}_{(-\infty,0)}(x)$ is the sign function. Hence $|x|' = \operatorname{sgn}(x)$ in $\mathcal{D}'$. Note that $\operatorname{sgn}(x) = 2H(x) - 1$, which is consistent: differentiating again gives $|x|'' = \operatorname{sgn}'(x) = 2H'(x) = 2\delta$.

The next computation requires introducing an important distribution that is not given by a locally integrable function in the standard sense.

Definition 3.5.3: Principal Value Distribution

The *principal value of* $1/x$, denoted $\mathrm{p.v.}(1/x)$, is the distribution on $\mathbb{R}$ defined by

$$\left\langle \mathrm{p.v.}\frac{1}{x}, \varphi \right\rangle = \lim_{\varepsilon\to 0^+} \int_{|x|>\varepsilon} \frac{\varphi(x)}{x}\, dx.$$

We should verify that the limit exists and that this defines a distribution. Since φ is smooth, we can write $\varphi(x) = \varphi(0) + x\psi(x)$ where $\psi(x) = \int_0^1 \varphi'(tx)\, dt$ is also smooth. Then

$$\int_{|x|>\varepsilon} \frac{\varphi(x)}{x}\, dx = \underbrace{\varphi(0)\int_{|x|>\varepsilon} \frac{dx}{x}}_{=0 \text{ by symmetry}} + \int_{|x|>\varepsilon} \psi(x)\, dx \to \int_{-\infty}^{\infty} \psi(x)\, dx$$

as $\varepsilon \to 0$. Since ψ is smooth and φ has compact support, the integral converges. The continuity estimate $|\langle \mathrm{p.v.}(1/x), \varphi\rangle| \le C(\|\varphi\|_\infty + \|\varphi'\|_\infty)$ on any fixed compact set follows from this decomposition, confirming that $\mathrm{p.v.}(1/x) \in \mathcal{D}'(\mathbb{R})$. Notice that the order of this distribution is 1 (it requires control of one derivative), unlike the distributions arising from locally integrable functions, which have order 0.

Example 3.5.4: Derivative Of $\log|x|$

The function $\log|x|$ is locally integrable on $\mathbb{R}$ (since $\int_{-1}^1 |\log|x||\, dx < \infty$) and hence defines a distribution. We claim that its distributional derivative is $\mathrm{p.v.}(1/x)$. For $\varphi \in C_c^\infty(\mathbb{R})$:

$$\begin{aligned}\langle (\log|x|)', \varphi\rangle = -\langle \log|x|, \varphi'\rangle &= -\int_{-\infty}^{\infty} \log|x|\, \varphi'(x)\, dx \\ &= -\lim_{\varepsilon\to 0^+} \int_{|x|>\varepsilon} \log|x|\, \varphi'(x)\, dx.\end{aligned}$$

Integrating by parts (with the boundary terms at $\pm\infty$ vanishing):

$$\begin{aligned}&= -\lim_{\varepsilon\to 0^+} \left(\Big[\log|x|\, \varphi(x)\Big]_{|x|=\varepsilon}^{|x|=\infty} - \int_{|x|>\varepsilon} \frac{\varphi(x)}{x}\, dx \right) \\ &= -\lim_{\varepsilon\to 0^+} \big(-\log\varepsilon\, [\varphi(\varepsilon) + \varphi(-\varepsilon)]\big) + \lim_{\varepsilon\to 0^+} \int_{|x|>\varepsilon} \frac{\varphi(x)}{x}\, dx.\end{aligned}$$

Since $\varphi(\varepsilon) + \varphi(-\varepsilon) \to 2\varphi(0)$ and $\varepsilon\log\varepsilon \to 0$, the first term requires more care. We have $\log\varepsilon\, [\varphi(\varepsilon) + \varphi(-\varepsilon)] = 2\varphi(0)\log\varepsilon + O(\varepsilon\log\varepsilon)$. But this term appears with a minus sign outside, so:

$$= \lim_{\varepsilon\to 0^+} \log\varepsilon\, [\varphi(\varepsilon) + \varphi(-\varepsilon)] + \lim_{\varepsilon\to 0^+} \int_{|x|>\varepsilon} \frac{\varphi(x)}{x}\, dx.$$

For the boundary term, note that $[\log|x|\, \varphi(x)]_{-\infty}^{-\varepsilon} + [\log|x|\, \varphi(x)]_{\varepsilon}^{\infty} = -\log\varepsilon\, \varphi(-\varepsilon) - \log\varepsilon\, \varphi(\varepsilon)$. Thus:

$$= \lim_{\varepsilon\to 0^+} \left[\log\varepsilon \big(\varphi(\varepsilon) + \varphi(-\varepsilon)\big) + \int_{|x|>\varepsilon} \frac{\varphi(x)}{x}\, dx \right].$$

Since $\varphi(\varepsilon) + \varphi(-\varepsilon) = 2\varphi(0) + O(\varepsilon^2)$ (the odd part cancels) and $\int_\varepsilon^1 dx/x = -\log\varepsilon$ (with the symmetric contribution from $[-1, -\varepsilon]$ cancelling), the $\log\varepsilon$ terms cancel exactly, and we are left with

$$\langle (\log|x|)', \varphi\rangle = \lim_{\varepsilon\to 0^+} \int_{|x|>\varepsilon} \frac{\varphi(x)}{x}\,dx = \left\langle \text{p.v.}\frac{1}{x}, \varphi \right\rangle.$$

Therefore $(\log|x|)' = \text{p.v.}(1/x)$ in $\mathcal{D}'(\mathbb{R})$.

It is also instructive to see how multiplication interacts with the principal value:

Proposition 3.5.5: Multiplication By x

As distributions on $\mathbb{R}$, we have $x \cdot \text{p.v.}(1/x) = 1$, that is, the distribution $x \cdot \text{p.v.}(1/x)$ agrees with the constant function 1.

***Proof*:**
For $\varphi \in C_c^\infty(\mathbb{R})$:

$$\langle x \cdot \text{p.v.}(1/x),\, \varphi\rangle = \left\langle \text{p.v.}\frac{1}{x},\, x\varphi \right\rangle = \lim_{\varepsilon\to 0^+} \int_{|x|>\varepsilon} \frac{x\varphi(x)}{x}\,dx = \int_{-\infty}^{\infty} \varphi(x)\,dx = \langle 1, \varphi\rangle,$$

since φ is integrable and the integrand $\varphi(x)$ has no singularity.

This is a distributional identity with no classical analogue: the "function" $1/x$ is not locally integrable, so the product $x \cdot (1/x)$ is not defined pointwise in L^1_{loc}. But the distributional version makes perfect sense and gives the expected answer. On the other hand, note that $x \cdot \delta = 0$ (since $\langle x\delta, \varphi\rangle = \langle \delta, x\varphi\rangle = 0$), so x is a zero-divisor in $\mathcal{D}'$—a phenomenon that does not occur for ordinary functions.

4

Sobolev Spaces

The spaces of the previous chapters measure the size of a function; the Sobolev spaces measure a function *together with its derivatives*, and they are where the differential equations of analysis and geometry live. The construction completes the smooth functions in a norm that controls not only $\|f\|_{L^p}$ but $\|\partial^\alpha f\|_{L^p}$ up to a fixed order, so that a limit of smooth functions keeps a definite derivative "in the L^p sense" even where it is nowhere classically differentiable. Two ingredients already in hand make this work: the Lebesgue spaces of chapter 1, which supply the norm, and the distributional derivative of chapter 3, which supplies a notion of derivative robust enough to survive the completion.

This chapter develops what a reader needs before the elliptic and parabolic equations of the partial-differential-equations volumes: the definition and completeness of $W^{k,p}$; the density of smooth functions, proved with the mollifiers of section 2.6; the embedding theorems that trade Sobolev regularity for classical regularity or higher integrability; the trace that assigns boundary values; and the Rellich-Kondrachov compactness behind existence proofs. Everything here is Fourier-free, resting only on L^p and the distributional derivative; the complementary scale H^s, defined through the Fourier transform, is built in the harmonic-analysis part.

4.1 Weak Derivatives and the Spaces $W^{k,p}$

Recall from chapter 3 that every distribution $F \in \mathcal{D}'(U)$ has distributional derivatives $\partial^\alpha F$, defined by $\langle \partial^\alpha F, \varphi\rangle = (-1)^{|\alpha|}\langle F, \partial^\alpha \varphi\rangle$ for $\varphi \in C_c^\infty(U)$. A locally integrable function is a distribution, so its derivatives of every order exist as distributions; Sobolev theory is organized by one question: for which α is $\partial^\alpha f$ *again a function*, and how integrable is it?

Definition 4.1.1: Weak Derivative

Let $U \subseteq \mathbb{R}^n$ be open, let $f \in L^1_{\mathrm{loc}}(U)$, and let α be a multi-index. A function $g \in L^1_{\mathrm{loc}}(U)$ is the *weak α-derivative* of f if

$$\int_U f\, \partial^\alpha \varphi \, dx = (-1)^{|\alpha|} \int_U g\, \varphi \, dx \qquad \text{for every } \varphi \in C_c^\infty(U).$$

Equivalently, the distributional derivative $\partial^\alpha f$ is represented by the function g. When such a g exists we write $\partial^\alpha f := g$, and for first-order derivatives $\partial_i f$ and $\nabla f = (\partial_1 f, \dots, \partial_n f)$.

A classical C^k function is weakly differentiable, its weak and classical derivatives agreeing: integration by parts against $\varphi \in C_c^\infty(U)$ produces no boundary terms, which is precisely the identity above. The weak derivative is thus an extension of the classical one, and like the classical derivative it is determined almost everywhere.

Lemma 4.1.2: Uniqueness of the Weak Derivative

If $g_1, g_2 \in L^1_{\mathrm{loc}}(U)$ are both weak α-derivatives of f, then $g_1 = g_2$ almost everywhere.

***Proof*:**
Put $h = g_1 - g_2 \in L^1_{\mathrm{loc}}(U)$. Subtracting the two defining identities gives $\int_U h\, \varphi \, dx = 0$ for every $\varphi \in C_c^\infty(U)$. Fix a compact $K \subset U$ and a mollifier $\{\psi_\varepsilon\}$, a good kernel (definition 2.6.9) supported in $B(0, \varepsilon)$. For $x \in K$ and $\varepsilon < \mathrm{dist}(K, \partial U)$ the function $y \mapsto \psi_\varepsilon(x - y)$ belongs to $C_c^\infty(U)$, so

$$(h * \psi_\varepsilon)(x) = \int_U h(y)\, \psi_\varepsilon(x - y)\, dy = 0 \qquad (x \in K).$$

As $\varepsilon \to 0$, $h * \psi_\varepsilon \to h$ in $L^1(K)$ by theorem 2.6.10, so $h = 0$ almost everywhere on K. Exhausting U by compact sets gives $h = 0$ almost everywhere on U.

Example 4.1.3: Weak Differentiability of a Corner and a Jump

On $U = (-1, 1)$ the corner $f(x) = |x|$ has weak derivative $\mathrm{sgn}(x)$: for $\varphi \in C_c^\infty(-1, 1)$, integrating by parts over $(-1, 0)$ and $(0, 1)$ separately (the boundary terms at ± 1 vanish by compact support, and those at 0 cancel),

$$\int_{-1}^{1} |x|\, \varphi'(x)\, dx = \int_{-1}^{0} \varphi \, dx - \int_0^1 \varphi \, dx = -\int_{-1}^{1} \mathrm{sgn}(x)\, \varphi(x)\, dx,$$

so $|x| \in W^{1,p}(-1, 1)$ for every p. The jump sgn, by contrast, has *no* weak derivative: $\int_{-1}^1 \mathrm{sgn}(x)\, \varphi'(x)\, dx = -2\varphi(0)$, and no $g \in L^1_{\mathrm{loc}}$ satisfies $-\int g\varphi = -2\varphi(0)$ for all φ (the distributional derivative of sgn is the Dirac mass $2\delta_0$, which is not a function). Hence $|x| \in W^{1,p} \setminus W^{2,p}$: weak differentiability is a real but limited amount of regularity, exactly the feature that makes these spaces useful.

With derivatives in hand we form the spaces. Since the weak derivative is linear in f (immediate from definition 4.1.1), what follows is a linear subspace of L^p.

Definition 4.1.4: Sobolev Space $W^{k,p}$

Let $U \subseteq \mathbb{R}^n$ be open, let k be a nonnegative integer, and let $1 \leq p \leq \infty$. The *Sobolev space* is

$$W^{k,p}(U) = \{f \in L^p(U) \mid \partial^\alpha f \in L^p(U) \text{ for all } |\alpha| \leq k\},$$

the derivatives being weak (definition 4.1.1), normed by

$$\|f\|_{W^{k,p}(U)} = \Big(\sum_{|\alpha|\leq k} \|\partial^\alpha f\|^p_{L^p(U)}\Big)^{1/p} \quad (p<\infty), \qquad \|f\|_{W^{k,\infty}(U)} = \max_{|\alpha|\leq k} \|\partial^\alpha f\|_{L^\infty(U)}.$$

When $p = 2$ we write $H^k(U) := W^{k,2}(U)$, an inner-product space under $\langle f, g\rangle_{H^k} = \sum_{|\alpha|\leq k} \int_U \partial^\alpha f\, \overline{\partial^\alpha g}\, dx$.

The norm records a function and all of its weak derivatives up to order k as a single L^p quantity; $W^{0,p} = L^p$, and raising k demands more regularity. That the smooth members of $W^{k,p}$ are dense in it, so that $W^{k,p}$ is the completion of smooth functions promised in the introduction, is the subject of the next section; first we record that the space is already complete as defined.

Theorem 4.1.5: $W^{k,p}$ is a Banach Space

For every open $U \subseteq \mathbb{R}^n$, every nonnegative integer k, and every $1 \leq p \leq \infty$, $W^{k,p}(U)$ is a Banach space, and $H^k(U)$ is a Hilbert space.

***Proof*:**
The map $f \mapsto \|f\|_{W^{k,p}}$ is a norm: absolute homogeneity is clear, the triangle inequality follows from that of the L^p norms (theorem 2.1.6) combined with the triangle inequality of the ℓ^p sum, and $\|f\|_{W^{k,p}} = 0$ forces $\|f\|_{L^p} = 0$, hence $f = 0$. Only completeness remains.

Let (f_m) be Cauchy in $W^{k,p}(U)$. For each $|\alpha| \leq k$,

$$\|\partial^\alpha f_m - \partial^\alpha f_l\|_{L^p} \leq \|f_m - f_l\|_{W^{k,p}},$$

so $(\partial^\alpha f_m)$ is Cauchy in $L^p(U)$. As $L^p(U)$ is complete (theorem 2.1.7), there are $g_\alpha \in L^p(U)$ with $\partial^\alpha f_m \to g_\alpha$ in L^p; write $f := g_0$.

It remains to identify g_α as $\partial^\alpha f$. Fix $\varphi \in C_c^\infty(U)$. For each m, the weak-derivative identity reads

$$\int_U f_m\, \partial^\alpha \varphi\, dx = (-1)^{|\alpha|} \int_U (\partial^\alpha f_m)\, \varphi\, dx.$$

Both φ and $\partial^\alpha\varphi$ are bounded with compact support, so they lie in $L^{p'}(U)$; Hölder's inequality (theorem 2.1.5) then passes the limit under each integral, giving $\int_U f_m\, \partial^\alpha \varphi \to \int_U f\, \partial^\alpha\varphi$ and $\int_U (\partial^\alpha f_m)\varphi \to \int_U g_\alpha\, \varphi$. Hence

$$\int_U f\, \partial^\alpha \varphi\, dx = (-1)^{|\alpha|} \int_U g_\alpha\, \varphi\, dx \qquad \text{for all } \varphi \in C_c^\infty(U),$$

which says exactly that $\partial^\alpha f = g_\alpha$ weakly. Thus $f \in W^{k,p}(U)$, and

$$\|f_m - f\|_{W^{k,p}}^p = \sum_{|\alpha| \le k} \|\partial^\alpha f_m - g_\alpha\|_{L^p}^p \to 0,$$

so $f_m \to f$ in $W^{k,p}$ and the space is complete. For $p = 2$ the norm is induced by the inner product of definition 4.1.4, so $H^k(U)$ is a Hilbert space.

The completeness proof used nothing but the completeness of L^p and the stability of the weak-derivative identity under L^p limits, the reward for defining derivatives by duality against test functions rather than pointwise. The next section shows that the members of $W^{k,p}$ are, on reasonable domains, limits of smooth functions, so that theorems can be proved by checking them on C^∞ and passing to the limit.

4.2 Approximation by Smooth Functions

This section proves that smooth functions are dense in $W^{k,p}(U)$, so that a theorem about Sobolev functions can be reduced to the smooth case and recovered by passing to the limit. The mechanism throughout is that mollification commutes with the weak derivative.

Theorem 4.2.1: Local Approximation by Mollification

Let $U \subseteq \mathbb{R}^n$ be open, let $f \in W^{k,p}(U)$ with $1 \le p < \infty$, and fix a mollifier $\{\psi_\varepsilon\}$, a good kernel (definition 2.6.9) with $\psi_\varepsilon \in C_c^\infty(\mathbb{R}^n)$ supported in $B(0, \varepsilon)$. On any open $V \Subset U$, for $\varepsilon < \operatorname{dist}(V, \partial U)$ the mollification $f * \psi_\varepsilon$ is defined and smooth on V, and for every multi-index $|\alpha| \le k$,

$$\partial^\alpha (f * \psi_\varepsilon) = (\partial^\alpha f) * \psi_\varepsilon \qquad \text{on } V$$

Consequently $f * \psi_\varepsilon \to f$ in $W^{k,p}(V)$ as $\varepsilon \to 0$.

***Proof*:**
Fix $V \Subset U$ and write $d = \operatorname{dist}(V, \partial U) > 0$; throughout, $\varepsilon < d$. For $x \in V$ the ball $B(x, \varepsilon)$ lies in U, so

$$(f * \psi_\varepsilon)(x) = \int_U f(y)\, \psi_\varepsilon(x - y)\, dy = \int_{B(x,\varepsilon)} f(y)\, \psi_\varepsilon(x - y)\, dy$$

is a genuine integral against a function supported in U, and it is finite because $f \in L^1_{\text{loc}}(U)$ and ψ_ε is bounded.

Step 1: The commutation identity. Fix $x \in V$ and $|\alpha| \le k$. Since $y \mapsto \psi_\varepsilon(x - y)$ is smooth and, for x in a small neighbourhood of any point of V, supported in a fixed compact subset of U, differentiation under the integral sign applies to the parameter x. Each x-derivative ∂_{x_i} falls on $\psi_\varepsilon(x - y)$, so

$$\partial_x^\alpha (f * \psi_\varepsilon)(x) = \int_U f(y)\, \partial_x^\alpha \psi_\varepsilon(x - y)\, dy$$

Now $\partial_x^\alpha \psi_\varepsilon(x - y) = (-1)^{|\alpha|}\, \partial_y^\alpha \psi_\varepsilon(x - y)$, since each factor of $x - y$ contributes a sign under the switch from an x-derivative to a y-derivative. For $x \in V$ the function $\varphi(y) := \psi_\varepsilon(x - y)$

belongs to $C_c^\infty(U)$: it is smooth and supported in $B(x,\varepsilon) \subseteq U$. The weak derivative identity of definition 4.1.1 applied to this φ therefore reads

$$\int_U f(y)\,\partial_y^\alpha \psi_\varepsilon(x-y)\,dy = (-1)^{|\alpha|}\int_U (\partial^\alpha f)(y)\,\psi_\varepsilon(x-y)\,dy$$

Combining the last three displays, the two factors of $(-1)^{|\alpha|}$ cancel and

$$\partial_x^\alpha(f*\psi_\varepsilon)(x) = \int_U (\partial^\alpha f)(y)\,\psi_\varepsilon(x-y)\,dy = \big((\partial^\alpha f)*\psi_\varepsilon\big)(x)$$

which is the claimed identity. Smoothness of $f*\psi_\varepsilon$ on V is now immediate: since every x-derivative falls on ψ_ε, the identity $\partial_x^\alpha(f*\psi_\varepsilon) = f*\partial_x^\alpha\psi_\varepsilon$ holds for *every* multi-index α (not only $|\alpha|\le k$), so all partial derivatives of $f*\psi_\varepsilon$ exist and, being mollifications of the locally integrable f, are continuous.

Step 2: Convergence in $W^{k,p}(V)$. Since $f\in W^{k,p}(U)$, each $\partial^\alpha f$ with $|\alpha|\le k$ lies in $L^p(U)$, hence in $L^p(W)$ for any W with $V \Subset W \Subset U$; fix such a W. Restricting to $\varepsilon < \operatorname{dist}(V,\partial W)$, the mollification $(\partial^\alpha f)*\psi_\varepsilon$ is computed on V from the values of $\partial^\alpha f$ on W only. By the approximate-identity convergence of theorem 2.6.10, $(\partial^\alpha f)*\psi_\varepsilon \to \partial^\alpha f$ in $L^p(V)$ as $\varepsilon\to 0$. Using the Step 1 identity on the left,

$$\|f*\psi_\varepsilon - f\|_{W^{k,p}(V)}^p = \sum_{|\alpha|\le k}\|\partial^\alpha(f*\psi_\varepsilon) - \partial^\alpha f\|_{L^p(V)}^p = \sum_{|\alpha|\le k}\|(\partial^\alpha f)*\psi_\varepsilon - \partial^\alpha f\|_{L^p(V)}^p$$

and each of the finitely many summands tends to 0. Hence $f*\psi_\varepsilon \to f$ in $W^{k,p}(V)$, completing the proof.

The identity $\partial^\alpha(f*\psi_\varepsilon) = (\partial^\alpha f)*\psi_\varepsilon$ is the whole engine of Sobolev approximation. Mollification is a smoothing operation, so its output is a bona fide C^∞ function to which the classical calculus applies; the identity says that this smooth function carries, as its classical derivatives, exactly the mollified weak derivatives of f. Convergence of each weak derivative in L^p then upgrades automatically to convergence in the Sobolev norm, because that norm is nothing but the L^p norms of the derivatives bundled together. The one limitation is visible in the hypothesis $V\Subset U$: the mollification $(f*\psi_\varepsilon)(x)$ reaches out to the ball $B(x,\varepsilon)$, so near ∂U it would sample points outside U where f is not defined. Local approximation is therefore free; the work in the rest of the section is to recover global statements from it. The restriction $p<\infty$ is inherited from theorem 2.6.10, where the translation $y\mapsto f(\cdot - y)$ fails to be L^∞-continuous.

Example 4.2.2: Mollifying a Corner

Return to $f(x)=|x|$ on $U=(-1,1)$, which lies in $W^{1,p}(U)$ with weak derivative $\partial f = \operatorname{sgn}(x)$ (example 4.1.3). Fix the standard mollifier $\psi_\varepsilon(x)=\varepsilon^{-1}\psi(x/\varepsilon)$ with $\psi\in C_c^\infty(-1,1)$ even, $\psi\ge 0$, $\int\psi = 1$. On any $V=(-a,a)$ with $a<1$ and $\varepsilon<1-a$, the mollification is

$$(f*\psi_\varepsilon)(x) = \int_{-\varepsilon}^{\varepsilon}|x-y|\,\psi_\varepsilon(y)\,dy$$

a smooth function that agrees with $|x|$ for $|x|>\varepsilon$ (there $|x-y|$ is affine on the support of ψ_ε and the first moment of the even kernel vanishes) and rounds off the corner on $(-\varepsilon,\varepsilon)$. Its derivative is $(f*\psi_\varepsilon)'(x) = (\operatorname{sgn}*\psi_\varepsilon)(x)$, exactly as theorem 4.2.1 predicts: a smooth function increasing from -1 to 1 across the corner, equal to $\operatorname{sgn}(x)$ outside $(-\varepsilon,\varepsilon)$. As $\varepsilon\to 0$ the smoothed corner and its derivative converge to $|x|$ and $\operatorname{sgn}(x)$ in $L^p(V)$, hence $f * \psi_\varepsilon \to f$ in $W^{1,p}(V)$.

Local approximation leaves two gaps. Mollification only produces smooth functions on the interior, so it does not by itself yield a smooth approximant on all of U; and even where it does, the approximant need not extend smoothly to the boundary. The first gap is closed by patching local mollifications together with a partition of unity, which is the content of the Meyers-Serrin theorem. Historically, two definitions of a Sobolev space competed: the space $W^{k,p}(U)$ of L^p functions with weak derivatives in L^p, and the space $H^{k,p}(U)$ obtained as the abstract completion of $\{f \in C^\infty(U) \mid \|f\|_{W^{k,p}} < \infty\}$ in the Sobolev norm. That these two coincide for every open U, with no regularity assumption on the boundary, is the theorem's content, and it is why one no longer distinguishes "H" from "W".

Theorem 4.2.3: Meyers-Serrin ($H = W$)

Let $U \subseteq \mathbb{R}^n$ be open, let $k \geq 0$, and let $1 \leq p < \infty$. Then $C^\infty(U) \cap W^{k,p}(U)$ is dense in $W^{k,p}(U)$: for every $f \in W^{k,p}(U)$ and every $\varepsilon > 0$ there is a $g \in C^\infty(U) \cap W^{k,p}(U)$ with $\|f - g\|_{W^{k,p}(U)} < \varepsilon$.

***Proof*:**
The idea is to mollify f separately on each shell of an exhaustion of U, where the previous theorem applies, and to glue the local smooth pieces with a partition of unity, arranging the individual errors to sum to less than ε.

Step 1: An open exhaustion and a partition of unity. Set, for integers $m \geq 1$,

$$U_m = \{x \in U \mid \operatorname{dist}(x, \partial U) > 1/m \text{ and } |x| < m\}$$

with the conventions $U_0 = U_{-1} = \varnothing$; then $U_m \Subset U_{m+1}$ and $\bigcup_m U_m = U$. The open sets

$$V_m = U_{m+1} \setminus \overline{U_{m-1}} \qquad (m \geq 1)$$

form a locally finite open cover of U: each $V_m \Subset U$, and any point of U lies in V_m for at most three consecutive indices. Let $\{\zeta_m\}_{m \geq 1}$ be a smooth partition of unity subordinate to $\{V_m\}$, so $\zeta_m \in C_c^\infty(V_m)$, $0 \leq \zeta_m \leq 1$, and $\sum_m \zeta_m \equiv 1$ on U (the sum being locally finite). Each $\zeta_m f$ then lies in $W^{k,p}(U)$ with support in V_m, because multiplication by the smooth compactly supported ζ_m preserves weak differentiability: for $|\alpha| \leq k$ the Leibniz rule

$$\partial^\alpha(\zeta_m f) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} \partial^\beta \zeta_m \, \partial^{\alpha - \beta} f$$

holds weakly (the distributional Leibniz rule for multiplication by a smooth function, chapter 3), its right-hand side a finite sum of L^p functions.

Step 2: Mollify each piece with a controlled error. Fix $\varepsilon > 0$. The function $\zeta_m f$ is supported in the compact-in-U set $\overline{V_m}$, so theorem 4.2.1 gives $\zeta_m f * \psi_\delta \to \zeta_m f$ in $W^{k,p}(U)$ as $\delta \to 0$ (the mollification is computed on a neighbourhood of $\overline{V_m}$ still compact in U). Choose $\delta_m > 0$ small enough that

$$g_m := (\zeta_m f) * \psi_{\delta_m} \in C_c^\infty(U), \qquad \|g_m - \zeta_m f\|_{W^{k,p}(U)} < \frac{\varepsilon}{2^m}$$

and, in addition, δ_m small enough that $\operatorname{supp} g_m \subseteq W_m := U_{m+2} \setminus \overline{U_{m-2}}$ (possible because $\operatorname{supp}(\zeta_m f) \subseteq V_m$ sits well inside W_m). The sets W_m are again locally finite: only finitely many W_m meet any given U_l.

Step 3: Sum and estimate. Define

$$g = \sum_{m \geq 1} g_m$$

On any U_l the sum is finite (only the g_m with $W_m \cap U_l \neq \varnothing$ contribute), so g is well-defined and $g \in C^\infty(U)$. Since $\sum_m \zeta_m f = f$ with the same local finiteness, for every $x \in U$

$$g(x) - f(x) = \sum_{m \geq 1} \big(g_m(x) - (\zeta_m f)(x)\big)$$

is a finite sum. Fix l. On U_l only the finitely many indices m with $W_m \cap U_l \neq \varnothing$ or $V_m \cap U_l \neq \varnothing$ enter, and the triangle inequality in $W^{k,p}(U_l)$ gives

$$\|g - f\|_{W^{k,p}(U_l)} \leq \sum_{m \geq 1} \|g_m - \zeta_m f\|_{W^{k,p}(U)} < \sum_{m \geq 1} \frac{\varepsilon}{2^m} = \varepsilon$$

the bound being uniform in l. Letting $l \to \infty$ and applying the monotone convergence theorem to $\sum_{|\alpha| \leq k} |\partial^\alpha (g - f)|^p$ over the increasing exhaustion $U_l \uparrow U$,

$$\|g - f\|_{W^{k,p}(U)} = \lim_{l \to \infty} \|g - f\|_{W^{k,p}(U_l)} \leq \varepsilon$$

In particular $g - f \in W^{k,p}(U)$, so $g \in W^{k,p}(U)$; and $\|g - f\|_{W^{k,p}(U)} \leq \varepsilon$. As $\varepsilon > 0$ was arbitrary, $C^\infty(U) \cap W^{k,p}(U)$ is dense, completing the proof.

The strength of Meyers-Serrin is that it asks nothing of the domain: for any open set whatsoever, however irregular its boundary, the smooth Sobolev functions are dense. This is what dissolves the historical distinction between the completion-of-smooth-functions space $H^{k,p}$ and the weak-derivative space $W^{k,p}$. The price is that the approximating g is smooth only in the *interior* of U: nothing controls its behaviour as x approaches ∂U, and indeed g and its derivatives may blow up there. For many purposes this is enough, since the norm only sees the interior. But to integrate by parts up to the boundary, or to make sense of boundary values, one needs approximants smooth on the closure $\overline{U}$, and that does require the boundary to be regular.

Example 4.2.4: Interior Smoothness Is Not Boundary Smoothness

Take $n = 1$, $U = (0, 1)$, and $f(x) = x^{1/2}$. Then $f \in W^{1,p}(0,1)$ for $1 \leq p < 2$, since $f' = \frac{1}{2}x^{-1/2} \in L^p(0,1)$ exactly when $p/2 < 1$. Mollifying f over the exhaustion of $(0,1)$ as in theorem 4.2.3 produces smooth approximants on the open interval, but they inherit the growth of f' near 0: the mollified derivative $(f' * \psi_\delta)(x)$ is of order $\delta^{-1/2}$ for x within δ of the endpoint, so no interior mollification is bounded up to $x = 0$, and none extends to a function in $C^\infty([0,1])$. The Meyers-Serrin conclusion is exactly as advertised and no better: density by functions smooth on the *interior*.

The remedy is the inward shift. For small $t > 0$ the translate $f_t(x) = f(x + t) = (x + t)^{1/2}$ is smooth on all of $[0, 1]$, because its only singularity, at $x = -t$, now sits outside the closed interval; and $\|f_t - f\|_{W^{1,p}(0,1)} \to 0$ as $t \to 0$ by L^p-continuity of translation. Thus f_t (already smooth here, no mollification needed) is a $C^\infty([0,1])$ function within any prescribed $W^{1,p}$ tolerance of f, which is precisely what theorem 4.2.5 delivers in general. The shift trades a function singular *on* the boundary for one singular just outside it, at the cost of a small L^p error, converting interior smoothness into smoothness up to $\overline{U}$.

For a bounded domain with regular boundary, one can do better than interior smoothness. The strategy near a boundary point is to flatten the boundary with a C^1 change of coordinates, translate the function a small distance *into* the domain so that its support pulls away from the boundary, and then mollify at a scale finer than the translation. The translated-and-mollified function is smooth on a neighbourhood of the piece of $\overline{U}$ under consideration, and the translation costs little in $W^{k,p}$ because translation is continuous on L^p.

Theorem 4.2.5: Global Approximation up to the Boundary

Let $U \subseteq \mathbb{R}^n$ be bounded with C^1 boundary ∂U, let $k \geq 0$, and let $1 \leq p < \infty$. Then $C^\infty(\overline{U})$ is dense in $W^{k,p}(U)$: for every $f \in W^{k,p}(U)$ and every $\varepsilon > 0$ there is a $g \in C^\infty(\overline{U})$ with $\|f - g\|_{W^{k,p}(U)} < \varepsilon$.

***Proof*:**
Fix $f \in W^{k,p}(U)$ and $\varepsilon > 0$.

Step 1: Localize to a boundary chart and translate inward. Fix a point $x_0 \in \partial U$. Because ∂U is C^1, after relabelling coordinates there is $r > 0$ and a C^1 function $\gamma\colon \mathbb{R}^{n-1} \to \mathbb{R}$ with

$$U \cap B(x_0, r) = \{x \in B(x_0, r) \mid x_n > \gamma(x_1, \dots, x_{n-1})\}$$

so near x_0 the domain lies above the graph of γ. Set $V = U \cap B(x_0, r/2)$. For a point $x \in V$ and $t > 0$, define the inward-shifted point

$$x^t = x + t\, e_n$$

and the shifted function $f_t(x) = f(x^t)$ on V. Choose r small enough (using continuity of $\nabla\gamma$, i.e. the C^1 property: in coordinates where $\nabla\gamma(x_0') = 0$ the graph has small slope over a small ball) that there is a cone opening in the $+e_n$ direction: there is $\lambda > 0$ so that for all $x \in V$ and all $0 < t \leq \lambda$, the ball $B(x^t, t/2)$ lies inside $U \cap B(x_0, r)$. Then $f_t \in W^{k,p}(V)$ with $\partial^\alpha f_t = (\partial^\alpha f)_t$ (translation commutes with weak differentiation, by the change of variables $y \mapsto y - te_n$ in definition 4.1.1), and by the L^p-continuity of translation (a standard consequence of the density of C_c^∞ in L^p, proposition 2.6.8: approximate each $\partial^\alpha f$ by a uniformly continuous compactly supported function, for which the shift converges uniformly),

$$\|f_t - f\|_{W^{k,p}(V)} = \Big(\sum_{|\alpha| \leq k} \|(\partial^\alpha f)_t - \partial^\alpha f\|^p_{L^p(V)} \Big)^{1/p} \to 0 \qquad (t \to 0)$$

Fix $t \leq \lambda$ with $\|f_t - f\|_{W^{k,p}(V)} < \varepsilon/2$.

Step 2: Mollify the shifted function. By construction $B(x^t, t/2) \subseteq U$ for every $x \in V$, so for $\delta < t/2$ the mollification $g^\delta := f_t * \psi_\delta$ samples f only at points of U: for $x \in \overline{V}$,

$$g^\delta(x) = \int_{B(0,\delta)} f(x + te_n - y)\, \psi_\delta(y)\, dy$$

is well-defined because $x + te_n - y \in B(x^t, t/2) \subseteq U$. Thus g^δ is defined and smooth on a neighbourhood of $\overline{V}$ (the mollification of a locally integrable function is smooth, by lemma 2.6.7), including the boundary piece $\partial U \cap B(x_0, r/2)$. By theorem 4.2.1 applied to f_t on V (whose mollification is interior to the slightly larger set where f_t is defined), $g^\delta \to f_t$ in $W^{k,p}(V)$ as

$\delta \to 0$; fix $\delta < t/2$ with $\|g^\delta - f_t\|_{W^{k,p}(V)} < \varepsilon/2$. Then $g^\delta \in C^\infty(\overline{V})$ and, by the triangle inequality,

$$\|g^\delta - f\|_{W^{k,p}(V)} \leq \|g^\delta - f_t\|_{W^{k,p}(V)} + \|f_t - f\|_{W^{k,p}(V)} < \varepsilon$$

Step 3: Patch the boundary charts with the interior. The compact boundary ∂U is covered by finitely many such half-ball neighbourhoods $V_1, \ldots, V_N$, on each of which Steps 1 and 2 produce a $g_j \in C^\infty(\overline{V_j})$ with $\|g_j - f\|_{W^{k,p}(U \cap V_j)} < \varepsilon$. Adjoin one more open set $V_0 \Subset U$ with $\overline{U} \subseteq V_0 \cup V_1 \cup \cdots \cup V_N$ and, on it, the interior mollification $g_0 \in C^\infty(V_0)$ of theorem 4.2.1 with $\|g_0 - f\|_{W^{k,p}(V_0)} < \varepsilon$. Let $\{\eta_j\}_{j=0}^N$ be a smooth partition of unity subordinate to $\{V_j\}_{j=0}^N$ over a neighbourhood of $\overline{U}$, and set

$$g = \sum_{j=0}^{N} \eta_j \, g_j$$

Each $\eta_j g_j$ is smooth up to $\overline{U}$ (either interior, or smooth on $\overline{V_j}$), so $g \in C^\infty(\overline{U})$. Using $\sum_j \eta_j = 1$ and the Leibniz rule as in the proof of theorem 4.2.3, on U

$$\partial^\alpha (g - f) = \sum_{j=0}^{N} \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} \partial^\beta \eta_j \, \partial^{\alpha-\beta} (g_j - f)$$

Since the derivatives of the η_j are bounded (there are finitely many, each in C_c^∞) by a constant M depending only on k and the cover, and each g_j satisfies $\|g_j - f\|_{W^{k,p}(U \cap V_j)} < \varepsilon$,

$$\|g - f\|_{W^{k,p}(U)} \leq C(k, N, M)\, \varepsilon$$

for a constant independent of ε. As $\varepsilon > 0$ was arbitrary, $C^\infty(\overline{U})$ is dense in $W^{k,p}(U)$, completing the proof.

The mechanism to remember is the inward shift. Mollification alone fails at the boundary only because it reaches outside U; translating f a distance t into the domain first pulls the relevant values away from ∂U, buying room to mollify at the finer scale $\delta < t/2$ without ever leaving U. The C^1 hypothesis enters exactly to guarantee a consistent inward direction near each boundary point, so that a single shift $+te_n$ (after flattening) works throughout a boundary chart. Domains with cusps or inward spikes, where no such uniform inward cone exists, can fail this theorem; the C^1 regularity is what rules them out. This global approximation is the workhorse behind the extension and trace theorems of the next section, and behind every argument in this chapter that proves an estimate first for smooth functions and then extends it by density.

The density theorems distinguish two natural classes of Sobolev functions: those approximable by smooth functions with no constraint at the boundary, which is all of $W^{k,p}(U)$, and those approximable by smooth functions that *vanish near the boundary*. The latter class is where boundary conditions live, and it deserves a name.

Definition 4.2.6: The Space $W_0^{k,p}$

Let $U \subseteq \mathbb{R}^n$ be open, let $k \geq 0$, and let $1 \leq p < \infty$. The space $W_0^{k,p}(U)$ is the closure of $C_c^\infty(U)$ in the norm of $W^{k,p}(U)$:

$$W_0^{k,p}(U) := \overline{C_c^\infty(U)}^{\|\cdot\|_{W^{k,p}(U)}}$$

When $p = 2$ one writes $H_0^k(U) := W_0^{k,2}(U)$.

By construction $W_0^{k,p}(U)$ is a closed subspace of the Banach space $W^{k,p}(U)$, hence a Banach space in its own right, and $H_0^k(U)$ a Hilbert space. Its elements are the Sobolev functions that are limits, in the Sobolev norm, of functions compactly supported inside U, and one reads this as the functions "with zero boundary values" together with, for $k \geq 2$, vanishing normal derivatives up to order $k-1$. The trace theorem of the next section will make this interpretation precise for $k = 1$ by identifying $W_0^{1,p}(U)$ with the kernel of the boundary-value map (theorem 4.3.9); for now it is a definition, and the one case where it collapses onto the full space is worth recording at once.

Proposition 4.2.7: $W_0^{k,p}(\mathbb{R}^n) = W^{k,p}(\mathbb{R}^n)$

For every $k \geq 0$ and $1 \leq p < \infty$, $W_0^{k,p}(\mathbb{R}^n) = W^{k,p}(\mathbb{R}^n)$.

***Proof*:**

The inclusion $W_0^{k,p}(\mathbb{R}^n) \subseteq W^{k,p}(\mathbb{R}^n)$ is immediate, as $W_0^{k,p}$ is a subspace. For the reverse, it suffices to approximate an arbitrary $f \in W^{k,p}(\mathbb{R}^n)$ by elements of $C_c^\infty(\mathbb{R}^n)$ in the $W^{k,p}$ norm; the smooth approximation runs directly on the whole space, and the only extra work is to make the approximants compactly supported by a cutoff.

Fix $\varepsilon > 0$. First mollify. On $\mathbb{R}^n$ there is no boundary to obstruct the convolution, so the commutation identity holds globally: for $U = \mathbb{R}^n$ every ball $B(x, \delta)$ lies in $\mathbb{R}^n$, so the differentiation-under-the-integral and weak-derivative steps of theorem 4.2.1 Step 1 go through for all x, giving $\partial^\alpha(f * \psi_\delta) = (\partial^\alpha f) * \psi_\delta$ on all of $\mathbb{R}^n$ for $|\alpha| \leq k$. Each $\partial^\alpha f \in L^p(\mathbb{R}^n)$, so the approximate-identity convergence of theorem 2.6.10 on $L^p(\mathbb{R}^n)$ gives $(\partial^\alpha f) * \psi_\delta \to \partial^\alpha f$ in $L^p(\mathbb{R}^n)$ as $\delta \to 0$; summing over $|\alpha| \leq k$, $f * \psi_\delta \to f$ in $W^{k,p}(\mathbb{R}^n)$. Choose $h := f * \psi_\delta \in C^\infty(\mathbb{R}^n) \cap W^{k,p}(\mathbb{R}^n)$ with $\|h - f\|_{W^{k,p}(\mathbb{R}^n)} < \varepsilon/2$. Now cut off. Fix a function $\chi \in C_c^\infty(\mathbb{R}^n)$ with $\chi \equiv 1$ on $B(0,1)$, $0 \leq \chi \leq 1$, supported in $B(0,2)$, and set $\chi_R(x) = \chi(x/R)$, so $\partial^\alpha \chi_R$ is bounded by $C_\alpha R^{-|\alpha|} \leq C_\alpha$ for $R \geq 1$ and $\chi_R \equiv 1$ on $B(0,R)$. Then $\chi_R h \in C_c^\infty(\mathbb{R}^n)$, and by the Leibniz rule

$$\partial^\alpha(\chi_R h - h) = \partial^\alpha\big((\chi_R - 1)h\big) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} \partial^\beta(\chi_R - 1)\, \partial^{\alpha-\beta} h$$

where the $\beta = 0$ term is $(\chi_R - 1)\,\partial^\alpha h$ and the $\beta \neq 0$ terms carry a factor $\partial^\beta \chi_R$ supported in the annulus $R \leq |x| \leq 2R$. Since $\chi_R - 1$ vanishes on $B(0,R)$ and is bounded by 1, and each $\partial^\alpha h \in L^p(\mathbb{R}^n)$, dominated convergence gives $\|(\chi_R - 1)\,\partial^\alpha h\|_{L^p(\mathbb{R}^n)} \to 0$ as $R \to \infty$; the annular terms are bounded by $C \sum_{|\gamma|<|\alpha|} \|\partial^\gamma h\|_{L^p(R \leq |x| \leq 2R)} \to 0$ likewise. Summing over $|\alpha| \leq k$, choose R with $\|\chi_R h - h\|_{W^{k,p}(\mathbb{R}^n)} < \varepsilon/2$. Then $\chi_R h \in C_c^\infty(\mathbb{R}^n)$ and

$$\|\chi_R h - f\|_{W^{k,p}(\mathbb{R}^n)} \leq \|\chi_R h - h\|_{W^{k,p}(\mathbb{R}^n)} + \|h - f\|_{W^{k,p}(\mathbb{R}^n)} < \varepsilon$$

so f lies in the closure of $C_c^\infty(\mathbb{R}^n)$, as we sought to show.

The proposition says that on the whole space there is no boundary to see, so the constraint defining $W_0^{k,p}$ is empty and the two spaces agree. On a bounded domain the situation is the opposite extreme: $W_0^{k,p}(U)$ is a genuinely proper closed subspace of $W^{k,p}(U)$, precisely because a nonzero constant function belongs to $W^{k,p}(U)$ but cannot be approximated by compactly supported functions without paying for the boundary layer where it must drop to zero. The gap between the two spaces is what the trace measures, and the Poincaré inequality of the last section of this chapter will quantify how the vanishing boundary condition of $W_0^{1,p}$ controls a function by its gradient alone.

4.3 Extensions and Traces

A Sobolev function on a domain U is defined only inside U, and two operations that the elliptic theory needs are not visible from the definition alone: extending such a function to all of $\mathbb{R}^n$ without losing a derivative, and restricting it to the boundary ∂U where a function without pointwise values must still be assigned boundary data. This section builds both, resting on the same device, flattening the C^1 boundary in local charts, and on the density of functions smooth up to the boundary proved in the previous section.

Throughout, $U \subseteq \mathbb{R}^n$ is bounded with C^1 boundary ∂U: near each boundary point there is a ball in which, after relabelling coordinates, ∂U is the graph $x_n = \gamma(x')$ of a C^1 function γ of the first $n-1$ coordinates $x' = (x_1, \dots, x_{n-1})$, with U lying on the side $x_n > \gamma(x')$. The boundary carries its surface measure, and ∂U is a compact $(n-1)$-dimensional C^1 manifold.

The extension is built one boundary chart at a time. In a chart where the boundary has been straightened to a piece of the hyperplane $x_n = 0$, a $W^{1,p}$ function on the upper half is extended to the lower half by reflection. A naive even reflection $f(x', -x_n)$ keeps f continuous across the interface but jumps in $\partial_n f$; the higher-order reflection below instead matches the first weak derivatives across $\{x_n = 0\}$, the condition for the reflected function to lie in $W^{1,p}$ of the full ball. The two lemmas that follow isolate this half-space reflection and the passage across a flattened C^1 graph; the theorem then patches the charts together with a partition of unity.

Write $\mathbb{R}^n_+ = \{x \in \mathbb{R}^n \mid x_n > 0\}$ and $\mathbb{R}^n_- = \{x \in \mathbb{R}^n \mid x_n < 0\}$, and for a ball $B = B(0, r)$ put $B_+ = B \cap \mathbb{R}^n_+$ and $B_- = B \cap \mathbb{R}^n_-$.

Lemma 4.3.1: Higher-Order Reflection Across a Hyperplane

Let $B = B(0, r)$ and let $f \in W^{1,p}(B_+)$ with $1 \leq p < \infty$, and suppose f is supported away from the upper cap: $f = 0$ almost everywhere on $B_+ \setminus B(0, \rho)$ for some $\rho < r/2$. Define Ef on B by

$$(Ef)(x) = \begin{cases} f(x', x_n) & x_n > 0 \\ 3f(x', -x_n) - 2f(x', -2x_n) & x_n < 0 \end{cases}$$

where the reflected values are read off as 0 wherever their argument leaves B_+. Then $Ef \in W^{1,p}(B)$, it agrees with f on B_+, is supported in $B(0, \rho)$, and

$$\|Ef\|_{W^{1,p}(B)} \leq C\|f\|_{W^{1,p}(B_+)}$$

with C depending only on n.

***Proof*:**

The plan is to prove the estimate and the extension property for $f \in C^1(\overline{B_+})$, where Ef can be differentiated classically, and then remove the smoothness by the density of such functions.

Step 1: The reflection matches value and normal derivative for smooth f. Suppose first $f \in C^1(\overline{B_+})$ vanishing on $B_+ \setminus B(0, \rho)$ with $\rho < r/2$. On B_- set $g(x) = 3f(x', -x_n) - 2f(x', -2x_n)$, so $Ef = f$ on B_+ and $Ef = g$ on B_-. The support hypothesis makes both reflected pieces well-defined on B_-: $f(x', -x_n)$ is nonzero only when $|(x', -x_n)| = |x| < \rho$, and $f(x', -2x_n)$ only when $|(x', -2x_n)|^2 = |x'|^2 + 4x_n^2 < \rho^2$, which forces $|x| < \rho$ as well, so in both cases the argument lies in $B(0, \rho)_+ \subseteq B_+$ (elsewhere the reflected value is 0). In particular g is supported in $B(0, \rho) \cap B_-$, and Ef in $B(0, \rho)$. As $x_n \to 0^-$ both reflected arguments tend to $(x', 0)$, and

$$\lim_{x_n \to 0^-} g(x', x_n) = 3f(x', 0) - 2f(x', 0) = f(x', 0)$$

so Ef is continuous across $\{x_n = 0\}$. Differentiating g classically on B_-,

$$\partial_n g(x) = -3\,\partial_n f(x', -x_n) + 4\,\partial_n f(x', -2x_n)$$

whose boundary limit is $(-3 + 4)\,\partial_n f(x', 0) = \partial_n f(x', 0)$, matching the upper value $\partial_n f(x', 0)$; the tangential derivatives $\partial_i g$ for $i < n$ carry the factors 3 and -2 through the chain rule with total $3 - 2 = 1$, so they too match $\partial_i f(x', 0)$. Thus $Ef \in C^1(\overline{B})$ and its classical gradient is bounded on B by a multiple of $\sup_{B_+}(|f| + |\nabla f|)$.

Step 2: The classical gradient is the weak gradient. Because Ef is continuous across $\{x_n = 0\}$ and C^1 on each of B_+, B_- separately, its weak gradient on all of B is the function that equals the classical gradient off the interface. To see this, fix $\varphi \in C_c^\infty(B)$ and integrate by parts on B_+ and B_- separately:

$$\int_B (Ef)\,\partial_i\varphi\,dx = \int_{B_+} f\,\partial_i\varphi\,dx + \int_{B_-} g\,\partial_i\varphi\,dx$$

Integration by parts on B_+ produces $-\int_{B_+}(\partial_i f)\varphi$ plus a boundary integral over $\{x_n = 0\}$ with integrand $f(x', 0)\varphi\,\nu_i$, where ν is the outward normal of B_+; the same on B_- produces $-\int_{B_-}(\partial_i g)\varphi$ plus a boundary integral with integrand $g(x', 0)\varphi\,(-\nu_i)$. The two boundary integrals cancel because $f(x', 0) = g(x', 0)$ by Step 1. Hence $\int_B (Ef)\,\partial_i\varphi = -\int_B (\partial_i^{\mathrm{cl}} Ef)\,\varphi$, so the weak derivative $\partial_i(Ef)$ is the classical one, as claimed.

Step 3: The norm estimate for smooth f. With the weak and classical derivatives identified, change variables in the reflected pieces. Under $x_n \mapsto -x_n$ the map $(x', x_n) \mapsto (x', -x_n)$ has unit Jacobian, and under $x_n \mapsto -2x_n$ the map $(x', x_n) \mapsto (x', -2x_n)$ has Jacobian $\frac{1}{2}$ in the x_n variable. By Step 1 both substitutions carry the region where the corresponding reflected value is nonzero into $B(0,\rho)_+ \subseteq B_+$, so for the value term

$$\int_{B_-} |g|^p\,dx \le 2^{p-1} \int_{B_-} \left(3^p|f(x', -x_n)|^p + 2^p|f(x', -2x_n)|^p\right) dx \le C \int_{B_+} |f|^p\,dx$$

using $(a+b)^p \le 2^{p-1}(a^p + b^p)$ and the substitutions (the second contributing a factor $\frac{1}{2}$ from its Jacobian, and integrating over a region contained in B_+ because f vanishes outside $B(0,\rho)_+$). The identical estimate applied to the derivative formulas of Step 1, whose coefficients are again bounded by constants depending only on n, gives $\int_{B_-} |\nabla g|^p \le C \int_{B_+} |\nabla f|^p$. Summing, $\|Ef\|^p_{W^{1,p}(B)} \le C\|f\|^p_{W^{1,p}(B_+)}$ with $C = C(n)$.

Step 4: Removal of smoothness by density. For general $f \in W^{1,p}(B_+)$ vanishing on $B_+ \setminus B(0,\rho)$, the domain B_+ has C^1 (indeed Lipschitz) boundary, so by theorem 4.2.5 there are $\tilde{f}_m \in C^\infty(\overline{B_+})$ with $\tilde{f}_m \to f$ in $W^{1,p}(B_+)$. Fix a cutoff $\chi \in C_c^\infty(B(0,\rho'))$ with $\rho < \rho' < r/2$, $0 \le \chi \le 1$, and $\chi \equiv 1$ on $\overline{B(0,\rho)}$, and set $f_m := \chi\tilde{f}_m$. Since $f = \chi f$, the product rule gives $f_m \to f$ in $W^{1,p}(B_+)$, and each $f_m \in C^\infty(\overline{B_+})$ vanishes on $B_+ \setminus B(0,\rho')$ with $\rho' < r/2$, so the reflection of Steps 1 to 3 applies to each f_m. The map E is linear, and Step 3 gives $\|Ef_m - Ef_l\|_{W^{1,p}(B)} \le C\|f_m - f_l\|_{W^{1,p}(B_+)}$, so (Ef_m) is Cauchy in $W^{1,p}(B)$; let F be its limit. Restricting to B_+, where $Ef_m = f_m$, gives $F = f$ on B_+. The estimate passes to the limit as $\|F\|_{W^{1,p}(B)} \le C\|f\|_{W^{1,p}(B_+)}$, and the explicit reflection formula for F on B_- follows because $f_m(x', -x_n) \to f(x', -x_n)$ and $f_m(x', -2x_n) \to f(x', -2x_n)$ in $L^p(B_-)$ under the same changes of variable. Thus $F = Ef$ as defined, and $Ef \in W^{1,p}(B)$ with the stated bound, completing the proof.

The reflection is engineered so that both the value and the full gradient are continuous across the interface, which Step 2 shows is exactly the condition making the two half-space derivatives assemble into a single weak derivative on the whole ball; the even reflection fails precisely at the normal derivative, and the coefficients $3, -2$ are the unique pair (a, b) correcting it while preserving continuity of the value: matching the value forces $a + b = 1$ and matching the normal derivative forces $-a - 2b = 1$ (differentiating the two reflected pieces at $x_n = 0$), two linear equations solved only by $a = 3$, $b = -2$. The support hypothesis, confining the reflected arguments to the half-ball where f lives, costs nothing in the application: the localized pieces that the extension operator reflects are cut off by a partition of unity and already vanish near the top of their chart. This is the entire content of the extension near a straight boundary. The passage to a curved C^1 boundary is a change of variables that straightens it, recorded in the next lemma.

Lemma 4.3.2: Straightening a C^1 Boundary

Let $w \in \partial U$. There is an open ball V about w and a C^1 diffeomorphism $\Phi\colon V \to B(0,r)$ with C^1 inverse $\Psi = \Phi^{-1}$, such that $\Phi(V \cap U) = B(0,r)_+$ and $\Phi(V \cap \partial U) = B(0,r) \cap \{x_n = 0\}$. Composition with Φ and Ψ carries $W^{1,p}(V \cap U)$ boundedly to $W^{1,p}(B(0,r)_+)$ and back, with constants depending on Φ.

***Proof*:**
By the C^1 boundary hypothesis, after relabelling coordinates there is a ball V about w and a C^1 function γ of $x' = (x_1, \dots, x_{n-1})$ with $V \cap \partial U = \{x \in V \mid x_n = \gamma(x')\}$ and $V \cap U = \{x \in V \mid x_n > \gamma(x')\}$. Define

$$\Phi(x) = (x', x_n - \gamma(x')), \qquad \Psi(y) = (y', y_n + \gamma(y'))$$

Both are C^1 (as γ is C^1) and mutually inverse, with Jacobian determinant identically 1. The shear image $\Phi(V)$ is an open set containing $\Phi(w) = 0$ but need not be a ball, so fix a radius r small enough that $B(0,r) \subseteq \Phi(V)$ and *re-restrict*, replacing V by $\Phi^{-1}(B(0,r))$; on this smaller neighborhood Φ is a C^1 diffeomorphism onto exactly $B(0,r)$, and Φ sends $x_n > \gamma(x')$ to $y_n > 0$ and $x_n = \gamma(x')$ to $y_n = 0$, giving the stated images. For a function u on $V \cap U$, the composition $u \circ \Psi$ is defined on $B(0,r)_+$; by the chain rule for weak derivatives under C^1 changes of variable, $u \in W^{1,p}(V \cap U)$ if and only if $u \circ \Psi \in W^{1,p}(B(0,r)_+)$, with

$$\nabla(u \circ \Psi)(y) = (D\Psi(y))^{\top} (\nabla u)(\Psi(y))$$

and since $D\Phi, D\Psi$ are continuous hence bounded on the relevant compact sets and the Jacobians are 1, the L^p norms of u and its gradient are comparable to those of $u \circ \Psi$ up to a constant depending on $\sup |D\Phi|, \sup |D\Psi|$. The inverse map $v \mapsto v \circ \Phi$ is bounded likewise, as we sought to show.

With the local reflection and the local straightening in hand, the global extension is assembled by a finite partition of unity subordinate to boundary charts, extending each localized piece by reflection in its straightened chart and extending the interior piece by zero.

Theorem 4.3.3: Extension Operator

Let $U \subseteq \mathbb{R}^n$ be bounded with C^1 boundary and let $1 \le p < \infty$. Fix a bounded open set W with $U \Subset W$. There is a bounded linear operator

$$E \colon W^{1,p}(U) \to W^{1,p}(\mathbb{R}^n)$$

such that for every $f \in W^{1,p}(U)$:

1. $Ef = f$ almost everywhere on U;
2. Ef is supported in W;
3. $\|Ef\|_{W^{1,p}(\mathbb{R}^n)} \le C \|f\|_{W^{1,p}(U)}$, with $C = C(n, p, U, W)$.

***Proof*:**
Step 1: A finite cover by boundary charts. The boundary ∂U is compact, and by lemma 4.3.2 each $w \in \partial U$ has a ball V_w carrying a straightening diffeomorphism $\Phi_w \colon V_w \to B(0, r_w)$. Shrinking each V_w so that $V_w \Subset W$, the balls $\{V_w\}_{w \in \partial U}$ cover ∂U; extract a finite subcover $V_1, \dots, V_N$ with straightenings $\Phi_1, \dots, \Phi_N$. The compact set $K = \overline{U} \setminus \bigcup_{j=1}^{N} V_j$ lies in the open set U, so choose an open V_0 with $K \Subset V_0 \Subset U$. Then $V_0, V_1, \dots, V_N$ is a finite open cover of $\overline{U}$, all contained in W. Choose a C^∞ partition of unity $\zeta_0, \zeta_1, \dots, \zeta_N$ subordinate to it: each $\zeta_j \in C_c^\infty(V_j)$, $0 \le \zeta_j \le 1$,

and $\sum_{j=0}^{N} \zeta_j = 1$ on a neighborhood of $\overline{U}$. For each boundary chart $j \geq 1$, the support of ζ_j is a compact subset of V_j, so its image $\Phi_j(\mathrm{supp}\zeta_j \cap \overline{U})$ is a compact subset of $B(0, r_j)$, contained in some $\overline{B(0, s_j)}$ with $s_j < r_j$. Post-composing Φ_j with the dilation $x \mapsto (r_j/3s_j)\, x$ (a straightening onto $B(0, r_j/3)$, then relabelled to radius r_j) moves this image inside $B(0, r_j/3)$, so that any function supported in it vanishes on $B(0, r_j)_+ \setminus B(0, \rho_j)$ for some $\rho_j < r_j/2$.

Step 2: Extend each localized piece. Given $f \in W^{1,p}(U)$, decompose $f = \sum_{j=0}^{N} \zeta_j f$ on U. The interior piece $\zeta_0 f$ has compact support in $V_0 \Subset U$; extending it by zero outside U gives a function $E_0 f \in W^{1,p}(\mathbb{R}^n)$ with $\|E_0 f\|_{W^{1,p}(\mathbb{R}^n)} \leq C\|\zeta_0 f\|_{W^{1,p}(U)} \leq C\|f\|_{W^{1,p}(U)}$, the product rule $\nabla(\zeta_0 f) = \zeta_0 \nabla f + f \nabla \zeta_0$ bounding the gradient since $\nabla \zeta_0$ is bounded. For a boundary piece $\zeta_j f$ with $j \geq 1$, transport to the straightened chart: $u_j := (\zeta_j f) \circ \Psi_j$ lies in $W^{1,p}(B(0, r_j)_+)$ by lemma 4.3.2 and, by the radius choice of Step 1, is supported in $B(0, \rho_j)$ with $\rho_j < r_j/2$, so it satisfies the support hypothesis of lemma 4.3.1. That lemma gives a reflection $Eu_j \in W^{1,p}(B(0, r_j))$ satisfying $\|Eu_j\|_{W^{1,p}(B(0,r_j))} \leq C\|u_j\|_{W^{1,p}(B(0,r_j)_+)}$. Transport back: $E_j f := (Eu_j) \circ \Phi_j$, extended by zero outside V_j, is a function on $\mathbb{R}^n$ that agrees with $\zeta_j f$ on $V_j \cap U$, is supported in $V_j \Subset W$, and by the two comparisons of lemma 4.3.2 satisfies

$$\|E_j f\|_{W^{1,p}(\mathbb{R}^n)} \leq C\|Eu_j\|_{W^{1,p}(B(0,r_j))} \leq C\|u_j\|_{W^{1,p}(B(0,r_j)_+)} \leq C\|\zeta_j f\|_{W^{1,p}(V_j \cap U)} \leq C\|f\|_{W^{1,p}(U)}$$

Extension by zero across ∂V_j preserves the $W^{1,p}$ class because Eu_j, hence $E_j f$, already vanishes in a neighborhood of ∂V_j.

Step 3: Assemble and estimate. Set $Ef := \sum_{j=0}^{N} E_j f$. Each E_j is linear in f (reflection, chart transport, and multiplication by ζ_j are all linear), so E is linear. On U every $E_j f$ agrees with $\zeta_j f$, so $Ef = \sum_j \zeta_j f = f$ almost everywhere on U, giving (1). Each $E_j f$ is supported in $V_j \subseteq W$, so Ef is supported in W, giving (2). Finally, by the triangle inequality and the $(N+1)$ bounds of Step 2,

$$\|Ef\|_{W^{1,p}(\mathbb{R}^n)} \leq \sum_{j=0}^{N} \|E_j f\|_{W^{1,p}(\mathbb{R}^n)} \leq C\,(N+1)\,\|f\|_{W^{1,p}(U)}$$

with C absorbing the chart constants, giving (3) with $C = C(n, p, U, W)$. Thus E is a bounded linear extension operator, completing the proof.

The extension operator makes the domain U transparent to any theorem already known on $\mathbb{R}^n$: prove an inequality for $W^{1,p}(\mathbb{R}^n)$, apply it to Ef, and restrict back to U using $Ef = f$ there, at the cost of the operator norm C. This is exactly how the Rellich-Kondrachov compactness of section 4.5 will descend from a statement on all of $\mathbb{R}^n$ to bounded C^1 domains. The dependence of C on U is real and not an artifact: the constant blows up as the boundary degenerates (a domain with an outward cusp has no bounded extension operator), which is why the C^1 hypothesis, controlling $\sup |D\Phi_j|$, cannot be dropped.

Example 4.3.4: Extension of a Radial Bump on the Ball

Let $U = B(0, 1) \subseteq \mathbb{R}^n$ and $f(x) = 1 - |x|$, which lies in $W^{1,p}(U)$ for every p: it is Lipschitz, with $\nabla f(x) = -x/|x|$ of modulus 1 almost everywhere, so $\|f\|^p_{W^{1,p}(U)} = \int_U (1 - |x|)^p + \int_U 1 < \infty$. The construction straightens the sphere ∂U in each boundary chart; because f is already Lipschitz on $\overline{U}$ with $f = 0$ on ∂U, the simplest admissible extension is $Ef(x) = \max(1 - |x|, 0)$, which is Lipschitz on all of $\mathbb{R}^n$, equals f on U, is supported in $\overline{B(0, 1)} \Subset W = B(0, 2)$, and satisfies $\|Ef\|_{W^{1,p}(\mathbb{R}^n)} = \|f\|_{W^{1,p}(U)}$ since Ef adds no mass outside U. The reflection machinery is needed only when f does not vanish on the boundary, where an even continuation would break the normal derivative; here $f|_{\partial U} = 0$ lets extension by zero do the job, and the general operator agrees with it up to the boundary chart constants.

The trace is the second boundary operation. A function in $W^{1,p}(U)$ is an equivalence class modulo sets of Lebesgue measure zero, and ∂U has n-dimensional Lebesgue measure zero, so "restrict f to ∂U" has no meaning on the nose. The trace theorem gives it meaning: on functions smooth up to the boundary, restriction is a well-defined map into $L^p(\partial U)$; the estimate below shows this map is bounded for the $W^{1,p}$ norm, so it extends by density to all of $W^{1,p}(U)$. Before the theorem, one preliminary space records where the trace actually lands, which is finer than $L^p(\partial U)$.

The trace of a $W^{1,p}$ function lands in more than $L^p(\partial U)$: it carries a fractional order of differentiability, one derivative shy of the full jump from U (dimension n) to ∂U (dimension $n-1$) discounted by the integrability exponent. That fractional regularity is measured by a seminorm with no reference to integer derivatives, integrating a difference quotient of order s against the surface measure of ∂U. Here $d = n - 1$ is the dimension of ∂U.

Definition 4.3.5: Fractional Sobolev Space $W^{s,p}(\partial U)$

Let $U \subseteq \mathbb{R}^n$ be bounded with C^1 boundary, write $d = n-1$ for the dimension of ∂U, and let $0 < s < 1$ and $1 \le p < \infty$. The *Gagliardo (Slobodeckij) seminorm* of a measurable $f \colon \partial U \to \mathbb{R}$ is

$$[f]_{W^{s,p}(\partial U)} = \left(\int_{\partial U} \int_{\partial U} \frac{|f(x) - f(y)|^p}{|x-y|^{d+sp}} \, dS(x) \, dS(y) \right)^{1/p}$$

where dS is the surface measure and $|x-y|$ the Euclidean distance in $\mathbb{R}^n$. The *fractional Sobolev space* is

$$W^{s,p}(\partial U) = \left\{ f \in L^p(\partial U) \;\middle|\; [f]_{W^{s,p}(\partial U)} < \infty \right\}$$

normed by $\|f\|_{W^{s,p}(\partial U)} = \left(\|f\|^p_{L^p(\partial U)} + [f]^p_{W^{s,p}(\partial U)} \right)^{1/p}$.

The exponent $d+sp$ in the denominator is calibrated so that the seminorm sees exactly s derivatives: replacing f by $f(\lambda\,\cdot)$ scales $[f]_{W^{s,p}}$ by $\lambda^{s-d/p}$ (the change of variables $u = \lambda x$ turns the p-th power of the seminorm into λ^{sp-d} times itself, matching the scaling of $\|\nabla^s f\|_{L^p}$ for the would-be integer analog), which is what makes s the order of differentiability. Finiteness of the double integral demands that $f(x) - f(y)$ decay faster than $|x-y|^{s+d/p}$ near the diagonal on average, a genuine but fractional smoothness: $W^{s,p}$ sits strictly between L^p and the integer space $W^{1,p}$ as s runs through $(0,1)$.

Example 4.3.6: The Threshold Exponent of a Power Singularity

Take the unit circle $\partial U = \partial B(0,1) \subseteq \mathbb{R}^2$, so $d = 1$, parametrized by arc length $\theta \in [0, 2\pi)$, and the function with a mild corner at $\theta = 0$, $f(\theta) = |\theta|^a$ near 0 for a fixed $0 < a < 1$ (extended smoothly elsewhere). Near the diagonal the seminorm integrand behaves like $|f(\theta) - f(\eta)|^p / |\theta - \eta|^{1+sp}$, and a direct estimate of $\int\int$ over a neighborhood of $\theta = \eta = 0$ shows the double integral converges precisely when $a > s$: the difference $|\theta^a - \eta^a|$ is comparable to $|\theta - \eta|^a$ for a corner of this shape, because $t \mapsto |t|^a$ with $0 < a < 1$ is a-Hölder continuous ($||\theta|^a - |\eta|^a| \le |\theta - \eta|^a$ by concavity of t^a) and the bound is saturated when one of the two points sits at the corner (e.g. $\eta \approx 0$, where $||\theta|^a - |\eta|^a| = |\theta|^a \sim |\theta - \eta|^a$), so the integrand is comparable to $|\theta - \eta|^{ap-1-sp}$, integrable across the diagonal exactly when $ap - 1 - sp > -1$, i.e. $a > s$. Thus $f \in W^{s,p}(\partial U)$ if and only if $s < a$: the corner of exponent a has fractional smoothness of every order below a and none above, pinning the seminorm down as a sharp gauge of the singularity.

The trace theorem now states what the boundary restriction is and where it lands. Its heart is a single integration by parts on a boundary chart, bounding the L^p norm of the boundary values by the $W^{1,p}$ norm on the interior; density (theorem 4.2.5) then produces the operator on all of $W^{1,p}(U)$, and a refinement of the same estimate reads off the sharper fractional target.

Theorem 4.3.7: Trace Theorem

Let $U \subseteq \mathbb{R}^n$ be bounded with C^1 boundary and let $1 \le p < \infty$. There is a bounded linear operator

$$T\colon W^{1,p}(U) \to L^p(\partial U)$$

the *trace*, such that $Tf = f|_{\partial U}$ for every $f \in C^1(\overline{U}) \cap W^{1,p}(U)$, and

$$\|Tf\|_{L^p(\partial U)} \le C\|f\|_{W^{1,p}(U)}$$

with $C = C(n, p, U)$. Moreover T maps $W^{1,p}(U)$ into $W^{1-1/p,p}(\partial U)$ for $1 < p < \infty$, boundedly for the fractional norm of definition 4.3.5.

***Proof*:**

Step 1: The boundary estimate on a flat chart. Suppose first $f \in C^1(\overline{U})$, and work in a boundary chart straightened by lemma 4.3.2, so it suffices to bound $\int_{\{x_n=0\}} |f|^p\,dx'$ over a piece of the flattened boundary in terms of the $W^{1,p}$ norm on the half-ball B_+ above it. Fix a cutoff $\xi \in C_c^\infty(B)$ with $\xi \equiv 1$ on the smaller half-ball where the boundary piece sits and $0 \le \xi \le 1$. For x' in that piece, the fundamental theorem of calculus in the x_n-direction gives, since ξf vanishes for large x_n,

$$|f(x', 0)|^p = -\int_0^\infty \partial_n\big(\xi(x', x_n)\,|f(x', x_n)|^p\big)\,dx_n$$

Expanding the derivative, $\partial_n(\xi|f|^p) = (\partial_n\xi)|f|^p + \xi\,p|f|^{p-1}\operatorname{sgn}(f)\,\partial_n f$, and integrating in x' over the boundary piece,

$$\int |f(x', 0)|^p\,dx' \le \int_{B_+} \Big(|\partial_n\xi|\,|f|^p + p\,\xi\,|f|^{p-1}|\partial_n f|\Big)\,dx$$

The first term is bounded by $C\int_{B_+}|f|^p$. For the second, when $p > 1$ Young's inequality $ab \le \frac{a^{p'}}{p'} + \frac{b^p}{p}$ with $a = |f|^{p-1}$, $b = |\partial_n f|$ (so $a^{p'} = |f|^p$) gives $|f|^{p-1}|\partial_n f| \le \frac{p-1}{p}|f|^p + \frac{1}{p}|\partial_n f|^p$; the degenerate case $p = 1$ is immediate, the integrand $p\,\xi\,|f|^{p-1}|\partial_n f|$ reducing to $\xi\,|\partial_n f| \le |\partial_n f|$ with no exponent to split. Either way,

$$\int |f(x', 0)|^p\,dx' \le C\int_{B_+} \big(|f|^p + |\nabla f|^p\big)\,dx = C\|f\|^p_{W^{1,p}(B_+)}$$

Summing over a finite cover of ∂U by such charts, and using that the surface measure dS is comparable on each chart to dx' up to the bounded Jacobian factor of the graph parametrization (lemma 4.3.2), yields

$$\int_{\partial U} |f|^p\,dS \le C\|f\|^p_{W^{1,p}(U)} \qquad (f \in C^1(\overline{U}))$$

Step 2: Extension to all of $W^{1,p}(U)$. Define $Tf = f|_{\partial U}$ for $f \in C^1(\overline{U})$, a linear map into $L^p(\partial U)$ bounded by Step 1. By theorem 4.2.5, $C^\infty(\overline{U})$, hence $C^1(\overline{U})$, is dense in $W^{1,p}(U)$: given $f \in W^{1,p}(U)$ take $f_m \in C^\infty(\overline{U})$ with $f_m \to f$ in $W^{1,p}(U)$. Then (f_m) is Cauchy in $W^{1,p}(U)$, and Step 1 applied to $f_m - f_l$ gives $\|Tf_m - Tf_l\|_{L^p(\partial U)} \le C\|f_m - f_l\|_{W^{1,p}(U)}$, so (Tf_m) is Cauchy in the complete space $L^p(\partial U)$; define $Tf = \lim_m Tf_m$. The limit is independent of the approximating sequence: if $g_m \to f$ as well, interleaving the two sequences is again $W^{1,p}$ convergent to f, so the traces converge to a common limit. The bound $\|Tf\|_{L^p(\partial U)} \le C\|f\|_{W^{1,p}(U)}$ passes to the limit, and T is linear as a limit of linear maps. For $f \in C^1(\overline{U}) \cap W^{1,p}(U)$ the constant sequence $f_m = f$ shows $Tf = f|_{\partial U}$, so T extends pointwise restriction.

Step 3: The sharp fractional target. Fix $1 < p < \infty$; the claim is $[Tf]_{W^{1-1/p,p}(\partial U)} \le C\|f\|_{W^{1,p}(U)}$, so that $s = 1 - 1/p$ is the order of the trace. By Step 2 and density it suffices to prove this for $f \in C^1(\overline{U})$, the seminorm passing to the limit as in Step 2. Work again in a straightened chart, where the boundary is $\{x_n = 0\}$ and $g := f|_{x_n=0}$; the seminorm is comparable to the flat-chart integral $\int\int |g(x') - g(y')|^p/|x' - y'|^{d+sp}\, dx'\, dy'$ with $d = n - 1$; the comparability uses that the C^1 graph is a bi-Lipschitz parametrization, so $|x - y|$ on ∂U and $|x' - y'|$ in the chart agree up to a constant (lemma 4.3.2). For $x' \ne y'$ set $h = |x' - y'|$ and pass from $g(x')$ to $g(y')$ through the value of f at height h above each point, splitting the difference into a vertical defect and a horizontal difference that must be handled separately:

$$g(x') - g(y') = \big(g(x') - f(x', h)\big) + \big(f(x', h) - f(y', h)\big) - \big(g(y') - f(y', h)\big)$$

Estimate the two types of piece after raising to the p-th power and dividing by $|x' - y'|^{d+sp} = h^{d+p-1}$ (using $sp = p-1$). *Vertical defect.* Since $g(x') - f(x', h) = -\int_0^h \partial_n f(x', t)\, dt$, integrating y' over the sphere $|y' - x'| = h$ in polar coordinates $dy' = h^{d-1}\, dh\, d\omega$ cancels the h^{d-1} against $h^{-(d+p-1)}$ to leave the weight $h^{-p}\, dh$, so the vertical contribution at fixed x' is

$$\int \frac{|g(x') - f(x', h)|^p}{h^{d+p-1}}\, dy' \le C \int_0^\infty \frac{1}{h^p} \Big| \int_0^h \partial_n f(x', t)\, dt \Big|^p dh \le C \Big(\frac{p}{p-1}\Big)^p \int_0^\infty |\partial_n f(x', t)|^p\, dt$$

the last inequality being the one-dimensional Hardy inequality $\int_0^\infty h^{-p} \big(\int_0^h \psi\big)^p\, dh \le (p/(p-1))^p \int_0^\infty \psi^p$ applied to $\psi = |\partial_n f(x', \cdot)|$, whose constant is finite precisely because $p > 1$; this is where the logarithm that a direct column bound $\int_0^h |\partial_n f|^p\, dt$ against dh/h would produce is absorbed. Integrating in x' over the boundary piece bounds the vertical contribution by $C \int_{B_+} |\partial_n f|^p$. *Horizontal difference.* The middle term $f(x', h) - f(y', h) = \int_{[x',y']} \nabla' f(w', h) \cdot dw'$ runs along the length-h tangential segment at fixed height h, so Hölder gives $|f(x', h) - f(y', h)|^p \le h^{p-1} \int_{[x',y']} |\nabla' f(w', h)|^p\, dw'$. Dividing by h^{d+p-1} and integrating y' over $|y' - x'| = h$ leaves the weight $h^{-1}\, dh$ against the segment integral; because the integrand is evaluated at the same height $h = |x' - y'|$ that sets the scale, each height is visited once and the h-integral reconstructs the slab integral of $|\nabla' f|^p$ by Fubini, giving

$$\int\int \frac{|f(x', h) - f(y', h)|^p}{h^{d+p-1}}\, dx'\, dy' \le C \int_{B_+} |\nabla' f|^p$$

with no logarithm, since the horizontal integrand carries no whole-column tail. Combining the vertical and horizontal contributions bounds the flat-chart seminorm by $C \int_{B_+} |\nabla f|^p$. Summing the finitely many charts and using the comparability of dS with dx',

$$[Tf]^p_{W^{1-1/p,p}(\partial U)} \le C \int_U |\nabla f|^p\, dx \le C\|f\|^p_{W^{1,p}(U)}$$

Combined with the L^p bound of Step 1, $\|Tf\|_{W^{1-1/p,p}(\partial U)} \le C\|f\|_{W^{1,p}(U)}$, so T maps $W^{1,p}(U)$ boundedly into $W^{1-1/p,p}(\partial U)$, completing the proof.

The trace theorem is what makes boundary value problems well posed for Sobolev functions: prescribing $Tf = g$ on ∂U is a closed linear constraint (the trace is bounded), so the Dirichlet condition "$f = g$ on ∂U" has meaning for $f \in W^{1,p}(U)$ even though f has no pointwise values on a measure-zero set. The two targets record two levels of precision: $L^p(\partial U)$ is enough to state boundary conditions and prove the kernel characterization below, while the sharp target $W^{1-1/p,p}(\partial U)$ is the natural home for the boundary data of the inverse problem of extending g inward. Theorem 4.3.7 proves only one half of what makes it sharp, that T maps $W^{1,p}(U)$ boundedly *into* $W^{1-1/p,p}(\partial U)$; the converse, that T is also onto this space, so that every fractional datum $g \in W^{1-1/p,p}(\partial U)$ is the trace of some $W^{1,p}(U)$ function, is a separate and deeper theorem, established by constructing a bounded right inverse (a trace-extension operator) and not proved here.[1] Granting it, $W^{1-1/p,p}(\partial U)$ is exactly the image of T. The loss of $1/p$ of a derivative is the price of the codimension-one restriction, and it is sharp.

Example 4.3.8: Trace of a Radial Bump on the Ball

Return to $U = B(0,1) \subseteq \mathbb{R}^n$ and $f(x) = 1 - |x|$ of example 4.3.4, which lies in $W^{1,p}(U)$ for every p and is continuous up to the boundary, so $f \in C(\overline{U}) \cap W^{1,p}(U)$. The trace is then the pointwise restriction: on $\partial U = \{|x| = 1\}$ every point has $|x| = 1$, so

$$Tf(x) = f|_{\partial U}(x) = 1 - |x| = 0 \qquad (|x| = 1)$$

Hence $Tf = 0$, and f lies in the kernel of the trace. This is consistent with the extension example 4.3.4 continued f by zero across ∂U without breaking the $W^{1,p}$ class: a boundary value of 0 is exactly what lets extension by zero preserve one derivative. By the characterization below, $f \in W_0^{1,p}(U)$, so f is a $W^{1,p}$-limit of functions compactly supported inside the open ball, even though f itself does not vanish in the interior. A function whose boundary trace were a nonzero constant c, by contrast, would have $\|Tf\|^p_{L^p(\partial U)} = |c|^p\,|\partial U| > 0$, placing it outside the kernel.

The kernel of the trace has a clean description: it is exactly the closure of $C_c^\infty(U)$, the space $W_0^{1,p}(U)$ of definition 4.2.6. This says the two competing notions of "vanishing on the boundary" coincide, the analytic one (zero trace) and the approximation-theoretic one (approximable by functions compactly supported inside U).

[1] The extension operator of theorem 4.3.3 does not supply this right inverse: it extends a function already given on U, whereas a right inverse must build a $W^{1,p}(U)$ function from boundary data alone. The standard construction lifts g off ∂U by a convolution extension calibrated to the fractional norm, the boundary trace theorem of Gagliardo; a full proof is beyond the scope of this section.

Theorem 4.3.9: Zero-Trace Characterization

Let $U \subseteq \mathbb{R}^n$ be bounded with C^1 boundary and let $1 < p < \infty$. Then

$$W_0^{1,p}(U) = \ker T = \{f \in W^{1,p}(U) \mid Tf = 0\}$$

where T is the trace of theorem 4.3.7 and $W_0^{1,p}(U)$ is the closure of $C_c^\infty(U)$ in $W^{1,p}(U)$ from definition 4.2.6.[a]

[a] The restriction to $p > 1$ enters at the boundary Hardy inequality of Step 3, whose constant $(p/(p-1))^p$ blows up as $p \to 1$; Hardy's inequality genuinely fails at $p = 1$, so the cutoff argument below does not cover that case. The characterization does still hold for $p = 1$, but by a different route (a truncation-and-mollification argument avoiding Hardy), which is not pursued here.

***Proof*:**

Step 1: $W_0^{1,p}(U) \subseteq \ker T$. For $\varphi \in C_c^\infty(U)$ the trace is the pointwise restriction (Step 2 of theorem 4.3.7), and $\varphi \equiv 0$ near ∂U, so $T\varphi = \varphi|_{\partial U} = 0$. Since T is bounded (theorem 4.3.7) and $C_c^\infty(U)$ is dense in $W_0^{1,p}(U)$ by definition (definition 4.2.6), continuity gives $Tf = 0$ for every $f \in W_0^{1,p}(U)$: if $\varphi_m \to f$ in $W^{1,p}(U)$ with $\varphi_m \in C_c^\infty(U)$, then $Tf = \lim_m T\varphi_m = 0$. Hence $W_0^{1,p}(U) \subseteq \ker T$.

Step 2: $\ker T \subseteq W_0^{1,p}(U)$*, reduction to compactly supported functions.* Let $f \in W^{1,p}(U)$ with $Tf = 0$; the goal is to approximate f in $W^{1,p}(U)$ by functions in $C_c^\infty(U)$. By theorem 4.2.5 choose $f_m \in C^\infty(\overline{U})$ with $f_m \to f$ in $W^{1,p}(U)$; then $Tf_m = f_m|_{\partial U} \to Tf = 0$ in $L^p(\partial U)$ by continuity of T. It suffices to produce, for each ε, a function $g \in W^{1,p}(U)$ with compact support in U and $\|f - g\|_{W^{1,p}(U)} < \varepsilon$, since mollifying g (theorem 4.2.1 applied on a neighborhood of supp$g$) then yields a $C_c^\infty(U)$ approximant, placing f in the closure $W_0^{1,p}(U)$.

Step 3: Cutting off near the boundary using the vanishing trace. Work in a boundary chart straightened by lemma 4.3.2, so near a boundary point the domain is the half-ball B_+, the boundary is $\{x_n = 0\}$, and $f \in W^{1,p}(B_+)$ has trace $g = Tf = 0$ on $\{x_n = 0\}$. The vanishing trace gives the boundary Hardy estimate: since the smooth approximants satisfy $f_m(x', 0) = Tf_m \to 0$, the fundamental theorem of calculus $f_m(x', t) = f_m(x', 0) + \int_0^t \partial_n f_m(x', \tau)\, d\tau$ passes to the limit to give, for f with $Tf = 0$,

$$\int_{B_+ \cap \{0 < x_n < h\}} \frac{|f(x)|^p}{x_n^p}\, dx \leq C \int_{B_+ \cap \{0 < x_n < 2h\}} |\partial_n f|^p\, dx$$

the one-dimensional Hardy inequality applied in the x_n-variable (its constant $(p/(p-1))^p$ depends only on p and is finite precisely because $p > 1$), using $f(x', 0) = 0$. In particular $|f(x)|/x_n$ is p-integrable near the boundary. Now let $\eta_h(x_n)$ be a smooth cutoff, $\eta_h = 0$ for $x_n \leq h$, $\eta_h = 1$ for $x_n \geq 2h$, with $|\eta_h'| \leq C/h$, and set $g_h = \eta_h f$. Then g_h vanishes in the strip $0 < x_n < h$, so it has positive distance from the flattened boundary, and

$$\|f - g_h\|_{W^{1,p}(B_+)}^p \leq \int_{\{x_n < 2h\}} |f|^p + \int_{\{x_n < 2h\}} |1 - \eta_h|^p |\nabla f|^p + \int_{\{h < x_n < 2h\}} |\eta_h'|^p |f|^p$$

The first two integrals tend to 0 as $h \to 0$ by dominated convergence (the domains shrink to a null set). The third is controlled by the Hardy estimate: $|\eta_h'|^p \leq C/h^p \leq C/x_n^p$ on $\{h < x_n < 2h\}$, so $\int_{\{h<x_n<2h\}} |\eta_h'|^p |f|^p \leq C \int_{\{h<x_n<2h\}} |f|^p / x_n^p \to 0$ as $h \to 0$, again because the domain shrinks to a null set while the integrand is dominated by the integrable $|f|^p/x_n^p$. Hence $g_h \to f$ in $W^{1,p}(B_+)$, and each g_h vanishes near the flattened boundary.

Step 4: Globalization. Patch the chart cutoffs with the finite boundary partition of unity $\zeta_1, \dots, \zeta_N$ of theorem 4.3.3 together with the interior cutoff $\zeta_0 \in C_c^\infty(U)$: transporting each g_h back through the charts and combining as $\sum_{j \geq 1} \zeta_j g_h^{(j)} + \zeta_0 f$, the resulting function g lies in $W^{1,p}(U)$, has support at positive distance from ∂U hence compact in U, and satisfies $\|f - g\|_{W^{1,p}(U)} < \varepsilon$ for h small, by Step 3 in each boundary chart and the boundedness of the transport maps. This furnishes the compactly supported approximant required in Step 2; mollifying it produces a $C_c^\infty(U)$ function within 2ε of f. Since ε was arbitrary, $f \in \overline{C_c^\infty(U)} = W_0^{1,p}(U)$, and $\ker T \subseteq W_0^{1,p}(U)$.

The two inclusions give $W_0^{1,p}(U) = \ker T$, completing the proof.

The characterization anchors $W_0^{1,p}(U)$ as the natural space for the Dirichlet problem: a solution with homogeneous boundary data $f|_{\partial U} = 0$ is sought exactly in $W_0^{1,p}(U)$, and the theorem certifies that this closure of $C_c^\infty(U)$ is the same as the analytically defined set of zero-trace functions. It also explains the caveat in definition 4.2.6 that $W_0^{1,p}(\mathbb{R}^n) = W^{1,p}(\mathbb{R}^n)$: on all of $\mathbb{R}^n$ there is no boundary, the trace is the zero map, and its kernel is everything, so every $W^{1,p}(\mathbb{R}^n)$ function is a limit of compactly supported smooth ones. On a bounded domain the boundary is genuine, the trace is nonzero, and $W_0^{1,p}(U)$ is a proper closed subspace on which the Poincaré inequality of section 4.5 will hold.

4.4 Sobolev Embeddings

The completion of section 4.1 records that a Sobolev function has L^p derivatives; it says nothing about the function itself beyond membership in L^p. This section extracts the extra regularity that possession of a weak gradient forces: control of ∇f in L^p buys either higher integrability of f or, past a threshold, outright continuity. Which of the two occurs is decided by comparing the order of differentiation against the dimension n, and the two regimes are governed by the Gagliardo-Nirenberg-Sobolev inequality when the derivative is too weak to reach continuity and by Morrey's inequality when it is strong enough.

The mechanism behind both is a single scaling count. If f is rescaled to $f_\lambda(x) = f(\lambda x)$, then $\|f_\lambda\|_{L^q(\mathbb{R}^n)} = \lambda^{-n/q}\|f\|_{L^q(\mathbb{R}^n)}$ while $\|\nabla f_\lambda\|_{L^p(\mathbb{R}^n)} = \lambda^{1-n/p}\|\nabla f\|_{L^p(\mathbb{R}^n)}$. An inequality of the form $\|f\|_{L^q} \leq C\|\nabla f\|_{L^p}$ can hold for all f and all $\lambda > 0$ only if the two powers of λ agree, forcing $-n/q = 1 - n/p$, that is $q = \frac{np}{n-p}$. This exponent is the only candidate; the content of the theorem is that the inequality holds for it.

Definition 4.4.1: Sobolev Conjugate

Let $1 \leq p < n$. The *Sobolev conjugate* of p is

$$p^* = \frac{np}{n-p}$$

It satisfies $p^* > p$ and $\frac{1}{p^*} = \frac{1}{p} - \frac{1}{n}$.

The relation $\frac{1}{p^*} = \frac{1}{p} - \frac{1}{n}$ is the form to remember: one weak derivative improves the integrability exponent by the fixed amount $\frac{1}{n}$, independent of p. As $p \to n^-$ the conjugate $p^* \to \infty$, the signal that at $p = n$ integrability alone breaks down and a different phenomenon (borderline embedding into every finite exponent, then continuity for $p > n$) takes over.

Example 4.4.2: The Sobolev Conjugate in Low Dimensions

In dimension $n = 3$ the Hilbert exponent $p = 2$ has $p^* = \frac{3 \cdot 2}{3-2} = 6$, so a function on $\mathbb{R}^3$ with gradient in L^2 automatically lies in L^6. In dimension $n = 2$ the same $p = 2$ gives $n - p = 0$, so p^* is undefined and $p = n$ is the borderline case excluded from definition 4.4.1. For $p = 1$ in dimension n one finds $p^* = \frac{n}{n-1}$, the exponent that will appear at the base of the induction below.

The strategy is to prove the gradient estimate first on smooth compactly supported functions, where the fundamental theorem of calculus applies directly, and then to remove the smoothness by the density theory of section 4.2. The smooth estimate rests on a one-dimensional bound integrated against itself in every coordinate direction, tied together by an iterated Hölder inequality of Loomis-Whitney type.

Theorem 4.4.3: Gagliardo-Nirenberg-Sobolev Inequality

Let $1 \leq p < n$. There is a constant $C = C(n, p)$ such that

$$\|f\|_{L^{p^*}(\mathbb{R}^n)} \leq C \, \|\nabla f\|_{L^p(\mathbb{R}^n)} \qquad \text{for every } f \in W^{1,p}(\mathbb{R}^n)$$

where $p^* = \frac{np}{n-p}$. In particular the inclusion $W^{1,p}(\mathbb{R}^n) \hookrightarrow L^{p^*}(\mathbb{R}^n)$ is a bounded linear map.

***Proof*:**

Step 1: The case $p = 1$ for $f \in C_c^\infty(\mathbb{R}^n)$. Here $p^* = \frac{n}{n-1}$. Fix $f \in C_c^\infty(\mathbb{R}^n)$ and $x = (x_1, \ldots, x_n)$. For each index i, integrating the derivative along the i-th coordinate line from $-\infty$ and using compact support,

$$f(x) = \int_{-\infty}^{x_i} \partial_i f(x_1, \ldots, x_{i-1}, t, x_{i+1}, \ldots, x_n) \, dt$$

so, bounding the integrand by its absolute value and extending the range to all of $\mathbb{R}$,

$$|f(x)| \leq \int_{-\infty}^{\infty} |\partial_i f(x_1, \ldots, t, \ldots, x_n)| \, dt =: g_i(\hat{x}_i)$$

where g_i depends on all coordinates except x_i (denoted $\hat{x}_i$). Taking the product over the n directions and raising to the power $\frac{1}{n-1}$,

$$|f(x)|^{\frac{n}{n-1}} \leq \prod_{i=1}^{n} g_i(\hat{x}_i)^{\frac{1}{n-1}}$$

The task is to integrate this inequality over $\mathbb{R}^n$. Notice that each factor g_i is independent of the variable x_i, so integrating in x_1 first leaves the factor $g_1^{1/(n-1)}$ constant and applies the generalized Hölder inequality (with $n-1$ exponents all equal to $n-1$) to the remaining $n-1$ factors, each of which depends on x_1:

$$\int_{\mathbb{R}} |f|^{\frac{n}{n-1}} \, dx_1 \leq g_1(\hat{x}_1)^{\frac{1}{n-1}} \prod_{i=2}^{n} \Big(\int_{\mathbb{R}} g_i \, dx_1 \Big)^{\frac{1}{n-1}}$$

Integrating successively in $x_2, \dots, x_n$ and applying the same generalized Hölder inequality at each stage, every factor is eventually integrated in the one variable on which it does not already depend, and one arrives at

$$\int_{\mathbb{R}^n} |f|^{\frac{n}{n-1}}\,dx \le \prod_{i=1}^{n} \Big(\int_{\mathbb{R}^n} |\partial_i f|\,dx \Big)^{\frac{1}{n-1}}$$

This is the Loomis-Whitney arrangement: an $L^{n/(n-1)}$ bound on f by the geometric mean of the n one-directional gradient integrals. Bounding each $\int_{\mathbb{R}^n} |\partial_i f| \le \int_{\mathbb{R}^n} |\nabla f|$ and taking the resulting product of n equal factors,

$$\Big(\int_{\mathbb{R}^n} |f|^{\frac{n}{n-1}}\,dx \Big)^{\frac{n-1}{n}} \le \int_{\mathbb{R}^n} |\nabla f|\,dx$$

that is $\|f\|_{L^{1^*}(\mathbb{R}^n)} \le \|\nabla f\|_{L^1(\mathbb{R}^n)}$, with constant $C(n,1) = 1$.

Step 2: General $1 < p < n$ for $f \in C_c^\infty(\mathbb{R}^n)$ by applying Step 1 to a power of $|f|$. Fix $t > 1$ to be chosen and apply the $p = 1$ estimate to the function $|f|^t$, which is C^1 with compact support and gradient $\nabla(|f|^t) = t\,|f|^{t-1}\,\mathrm{sgn}(f)\,\nabla f$. Step 1 gives

$$\Big(\int_{\mathbb{R}^n} |f|^{\frac{tn}{n-1}}\,dx \Big)^{\frac{n-1}{n}} \le t \int_{\mathbb{R}^n} |f|^{t-1}\,|\nabla f|\,dx$$

Apply Hölder's inequality (theorem 2.1.5) to the right side with exponents p and $p' = \frac{p}{p-1}$:

$$\int_{\mathbb{R}^n} |f|^{t-1} |\nabla f|\,dx \le \Big(\int_{\mathbb{R}^n} |f|^{(t-1)p'}\,dx \Big)^{1/p'} \Big(\int_{\mathbb{R}^n} |\nabla f|^p\,dx \Big)^{1/p}$$

Choose t so that the two powers of $|f|$ match, namely $\frac{tn}{n-1} = (t-1)p'$. Solving gives $t = \frac{p(n-1)}{n-p} > 1$, and with this choice

$$\frac{tn}{n-1} = (t-1)p' = \frac{np}{n-p} = p^*$$

Writing $I = \big(\int_{\mathbb{R}^n} |f|^{p^*} \big)^{1/p^*} = \|f\|_{L^{p^*}}$, the left side is $I^{p^* \cdot \frac{n-1}{n}}$ and the first factor on the right is $I^{p^*/p'}$, so the inequality reads

$$I^{\,p^* \frac{n-1}{n}} \le t\, I^{\,p^*/p'}\, \|\nabla f\|_{L^p}$$

A direct computation of exponents gives $p^* \frac{n-1}{n} - \frac{p^*}{p'} = 1$, so dividing by $I^{p^*/p'}$ (which is finite, since $f \in C_c^\infty$ makes I finite) leaves

$$\|f\|_{L^{p^*}(\mathbb{R}^n)} \le t\, \|\nabla f\|_{L^p(\mathbb{R}^n)} = \frac{p(n-1)}{n-p}\, \|\nabla f\|_{L^p(\mathbb{R}^n)}$$

giving the inequality on $C_c^\infty(\mathbb{R}^n)$ with constant $C(n,p) = \frac{p(n-1)}{n-p}$.

Step 3: Extension to $W^{1,p}(\mathbb{R}^n)$ by density. Let $f \in W^{1,p}(\mathbb{R}^n)$. Since $W_0^{1,p}(\mathbb{R}^n) = W^{1,p}(\mathbb{R}^n)$ (definition 4.2.6), there are $f_m \in C_c^\infty(\mathbb{R}^n)$ with $f_m \to f$ in $W^{1,p}(\mathbb{R}^n)$; concretely one mollifies and truncates via theorem 4.2.1. Applying Step 2 to the difference $f_m - f_l$,

$$\|f_m - f_l\|_{L^{p^*}(\mathbb{R}^n)} \le C(n,p)\, \|\nabla f_m - \nabla f_l\|_{L^p(\mathbb{R}^n)} \le C(n,p)\, \|f_m - f_l\|_{W^{1,p}(\mathbb{R}^n)}$$

so (f_m) is Cauchy in $L^{p^*}(\mathbb{R}^n)$ and converges there to some h; passing to a subsequence, $f_m \to h$ almost everywhere, while $f_m \to f$ in L^p forces (along a further subsequence) $f_m \to f$ almost everywhere, so $h = f$ almost everywhere and $f \in L^{p^*}(\mathbb{R}^n)$. Applying Step 2 to f_m and letting $m \to \infty$, with $\|f_m\|_{L^{p^*}} \to \|f\|_{L^{p^*}}$ and $\|\nabla f_m\|_{L^p} \to \|\nabla f\|_{L^p}$,

$$\|f\|_{L^{p^*}(\mathbb{R}^n)} \leq C(n,p)\,\|\nabla f\|_{L^p(\mathbb{R}^n)}$$

which is the claimed inequality on all of $W^{1,p}(\mathbb{R}^n)$. Boundedness of the inclusion is then $\|f\|_{L^{p^*}} \leq C\|\nabla f\|_{L^p} \leq C\|f\|_{W^{1,p}}$, completing the proof.

The inequality controls the full L^{p^*} norm of f by the L^p norm of its gradient alone, with no term for f itself on the right. This is possible only on $\mathbb{R}^n$, where there is no room for constants: a nonzero constant has vanishing gradient but infinite L^{p^*} norm, so the estimate would fail were $\mathbb{R}^n$ replaced by a set of finite measure without also restricting to functions that decay or vanish at the boundary. On a bounded domain the corresponding statement carries a lower-order term, or restricts to $W_0^{1,p}$; the Poincaré inequality of section 4.5 supplies exactly the missing control. The gain in integrability, from p up to p^*, is the quantitative content: one weak derivative is worth a definite jump in the integrability exponent, and iterating this jump is what produces the full hierarchy at the end of the section.

Example 4.4.4: Sharpness of the Exponent

No exponent larger than p^* can appear in theorem 4.4.3. The scaling argument of the section opening shows why: if $\|f\|_{L^q(\mathbb{R}^n)} \leq C\|\nabla f\|_{L^p(\mathbb{R}^n)}$ held for a fixed C and some $q \neq p^*$, testing on $f_\lambda(x) = f(\lambda x)$ gives $\lambda^{-n/q}\|f\|_{L^q} \leq C\lambda^{1-n/p}\|\nabla f\|_{L^p}$, that is

$$\|f\|_{L^q} \leq C\lambda^{n/q+1-n/p}\|\nabla f\|_{L^p}$$

If $q > p^*$ then $n/q + 1 - n/p < 0$ and letting $\lambda \to \infty$ forces $\|f\|_{L^q} = 0$; if $q < p^*$ then the exponent is positive and $\lambda \to 0$ does the same. Only $q = p^*$, where the exponent of λ vanishes, survives.

At $p = n$ the conjugate p^* diverges and the argument above degenerates. Past this threshold, when $p > n$, a weak gradient in L^p is strong enough to force not just integrability but pointwise continuity, indeed Hölder continuity with a definite exponent. The precise target is a Hölder space, which the following definition records.

Definition 4.4.5: Holder Space $C^{0,\gamma}$

Let $0 < \gamma \leq 1$. A function $f\colon \mathbb{R}^n \to \mathbb{R}$ is *Hölder continuous* with exponent γ if the seminorm

$$[f]_{C^{0,\gamma}(\mathbb{R}^n)} = \sup_{x \neq y} \frac{|f(x) - f(y)|}{|x - y|^\gamma}$$

is finite. The *Hölder space* $C^{0,\gamma}(\mathbb{R}^n)$ is the space of bounded continuous f with $[f]_{C^{0,\gamma}(\mathbb{R}^n)} < \infty$, normed by $\|f\|_{C^{0,\gamma}(\mathbb{R}^n)} = \sup_x |f(x)| + [f]_{C^{0,\gamma}(\mathbb{R}^n)}$. More generally, for a nonnegative integer m, $C^{m,\gamma}(\mathbb{R}^n)$ consists of the $f \in C^m$ whose derivatives up to order m are bounded with $\partial^\alpha f \in C^{0,\gamma}$ for every $|\alpha| = m$, normed by $\|f\|_{C^{m,\gamma}} = \sum_{|\alpha|\leq m} \sup_x |\partial^\alpha f| + \sum_{|\alpha|=m} [\partial^\alpha f]_{C^{0,\gamma}}$.

The exponent γ measures how rapidly f may oscillate: $\gamma = 1$ is Lipschitz continuity, and smaller γ permits sharper local variation. A member of $W^{1,p}(\mathbb{R}^n)$ is only an equivalence class of functions agreeing almost everywhere, so the assertion that it embeds into $C^{0,\gamma}$ means each class has a (necessarily unique) continuous representative lying in that space, and it is this representative the oscillation estimate controls.

Example 4.4.6: The Model Holder Function

On $\mathbb{R}$ the function $f(x) = |x|^\gamma$ (say cut off smoothly away from the origin) has $[f]_{C^{0,\gamma}} < \infty$ but is not Lipschitz for $\gamma < 1$: near 0 the difference quotient $\frac{||x|^\gamma - 0|}{|x|^1} = |x|^{\gamma-1}$ blows up, whereas $\frac{||x|^\gamma - 0|}{|x|^\gamma} = 1$ stays bounded. This is the extremal behavior Morrey's inequality permits, and the exponent $\gamma = 1 - \frac{n}{p}$ produced below is exactly the regularity of such a cusp at the borderline.

The route to continuity runs through a comparison between the value of f at a point and its average over a small ball. Write

$$f_{B(x,r)} = \frac{1}{|B(x,r)|} \int_{B(x,r)} f \, dy$$

for the average of f over the ball $B(x,r)$ of radius r centered at x, where $|B(x,r)|$ is its Lebesgue measure. The key estimate bounds the deviation of f from this average by an integral of the gradient, a Sobolev analogue of the mean value theorem.

Lemma 4.4.7: Averaged Gradient Estimate

Let $p > n$ and $f \in C^1(\mathbb{R}^n)$. For every $x \in \mathbb{R}^n$ and $r > 0$,

$$\frac{1}{|B(x,r)|} \int_{B(x,r)} |f(y) - f(x)| \, dy \leq C(n) \int_{B(x,r)} \frac{|\nabla f(y)|}{|x-y|^{n-1}} \, dy$$

***Proof*:**
Fix x and a point $y = x + s\omega$ in $B(x,r)$, where $|\omega| = 1$ and $0 < s \leq r$. The fundamental theorem of calculus along the segment from x to y gives

$$f(x+s\omega) - f(x) = \int_0^s \frac{d}{d\tau} f(x+\tau\omega) \, d\tau = \int_0^s \nabla f(x+\tau\omega) \cdot \omega \, d\tau$$

so $|f(x+s\omega) - f(x)| \leq \int_0^s |\nabla f(x+\tau\omega)| \, d\tau$. Integrate this over the unit sphere $\omega \in S^{n-1}$, whose surface measure is written $d\omega$:

$$\int_{S^{n-1}} |f(x+s\omega) - f(x)| \, d\omega \leq \int_{S^{n-1}} \int_0^s |\nabla f(x+\tau\omega)| \, d\tau \, d\omega \leq \int_{S^{n-1}} \int_0^r |\nabla f(x+\tau\omega)| \, d\tau \, d\omega$$

Insert the factor $\tau^{n-1}/\tau^{n-1} = 1$ into the last integrand and pass to the polar-coordinate variable $z = x + \tau\omega$, for which $dz = \tau^{n-1} \, d\tau \, d\omega$ and $|z - x| = \tau$:

$$\int_{S^{n-1}} \int_0^r |\nabla f(x+\tau\omega)| \, d\tau \, d\omega = \int_{S^{n-1}} \int_0^r \frac{|\nabla f(x+\tau\omega)|}{\tau^{n-1}} \tau^{n-1} \, d\tau \, d\omega = \int_{B(x,r)} \frac{|\nabla f(z)|}{|z-x|^{n-1}} \, dz$$

Now multiply the resulting inequality

$$\int_{S^{n-1}} |f(x+s\omega) - f(x)| \, d\omega \leq \int_{B(x,r)} \frac{|\nabla f(z)|}{|z-x|^{n-1}} \, dz$$

by s^{n-1} and integrate in s from 0 to r. The right side is independent of s, so it picks up the factor $\int_0^r s^{n-1}\,ds = r^n/n$, while the left side becomes $\int_{B(x,r)} |f(y)-f(x)|\,dy$ in polar coordinates. Thus

$$\int_{B(x,r)} |f(y)-f(x)|\,dy \le \frac{r^n}{n}\int_{B(x,r)} \frac{|\nabla f(z)|}{|z-x|^{n-1}}\,dz$$

Dividing by $|B(x,r)| = \omega_n r^n$, where ω_n is the volume of the unit ball, gives the claim with $C(n) = \frac{1}{n\,\omega_n}$, as we sought to show.

With the deviation from an average controlled, the oscillation of f between two points follows by placing both points inside a common ball and comparing each to the average over it.

Theorem 4.4.8: Morrey's Inequality

Let $n < p < \infty$ and set $\gamma = 1 - \frac{n}{p} \in (0,1)$. There is a constant $C = C(n,p)$ such that every $f \in W^{1,p}(\mathbb{R}^n)$ has a representative in $C^{0,\gamma}(\mathbb{R}^n)$, and this representative satisfies

$$|f(x) - f(y)| \le C\,|x-y|^{\gamma}\,\|\nabla f\|_{L^p(\mathbb{R}^n)} \qquad \text{for all } x, y \in \mathbb{R}^n$$

together with $\sup_x |f(x)| \le C\|f\|_{W^{1,p}(\mathbb{R}^n)}$. Consequently the inclusion $W^{1,p}(\mathbb{R}^n) \hookrightarrow C^{0,\gamma}(\mathbb{R}^n)$ is a bounded linear map.

***Proof*:**

Step 1: The oscillation estimate for $f \in C^1(\mathbb{R}^n)$. Fix $x, y \in \mathbb{R}^n$, set $r = |x-y|$, and let $W = B(x,r) \cap B(y,r)$, a set of measure $|W| \ge c_n r^n$ for a dimensional constant $c_n > 0$ (both balls have radius r and centers at distance r, so their intersection contains a fixed fraction of a ball). The triangle inequality through the average $f_W = \frac{1}{|W|}\int_W f$ gives

$$|f(x)-f(y)| \le |f(x)-f_W| + |f_W - f(y)| = \frac{1}{|W|}\left|\int_W (f(x)-f(z))\,dz\right| + \frac{1}{|W|}\left|\int_W (f(y)-f(z))\,dz\right|$$

Estimate the first term. Since $W \subseteq B(x,r)$ and $|W| \ge c_n r^n \ge \frac{c_n}{\omega_n}|B(x,r)|$,

$$\frac{1}{|W|}\int_W |f(x)-f(z)|\,dz \le \frac{\omega_n}{c_n}\cdot\frac{1}{|B(x,r)|}\int_{B(x,r)} |f(x)-f(z)|\,dz \le C(n)\int_{B(x,r)} \frac{|\nabla f(z)|}{|x-z|^{n-1}}\,dz$$

the last step by lemma 4.4.7. Apply Hölder's inequality (theorem 2.1.5) with exponents p and p' to this integral over $B(x,r)$:

$$\int_{B(x,r)} \frac{|\nabla f(z)|}{|x-z|^{n-1}}\,dz \le \|\nabla f\|_{L^p(\mathbb{R}^n)}\Big(\int_{B(x,r)} |x-z|^{-(n-1)p'}\,dz\Big)^{1/p'}$$

The remaining integral is finite precisely because $p > n$: in polar coordinates centered at x,

$$\int_{B(x,r)} |x-z|^{-(n-1)p'}\,dz = \left|S^{n-1}\right|\int_0^r \tau^{-(n-1)p'}\tau^{n-1}\,d\tau = \left|S^{n-1}\right|\int_0^r \tau^{(n-1)(1-p')}\,d\tau$$

and the exponent satisfies $(n-1)(1-p') > -1$ exactly when $p > n$, so the integral converges to a constant times $r^{(n-1)(1-p')+1}$. Raising to the power $1/p'$ and simplifying the exponent,

$\frac{(n-1)(1-p')+1}{p'} = 1 - \frac{n}{p} = \gamma$, so

$$\Big(\int_{B(x,r)} |x-z|^{-(n-1)p'}\, dz \Big)^{1/p'} = C(n,p)\, r^\gamma$$

Combining, the first term is bounded by $C(n,p)\, r^\gamma \|\nabla f\|_{L^p}$. The second term is identical with y in place of x and $B(y,r)$ in place of $B(x,r)$, giving the same bound. Adding the two and recalling $r = |x-y|$,

$$|f(x) - f(y)| \le C(n,p)\, |x-y|^\gamma\, \|\nabla f\|_{L^p(\mathbb{R}^n)}$$

Step 2: The sup bound for $f \in C^1(\mathbb{R}^n) \cap W^{1,p}$. Fix x and take $r = 1$. Then

$$|f(x)| \le |f(x) - f_{B(x,1)}| + |f_{B(x,1)}|$$

The first term is bounded via lemma 4.4.7 and the Hölder step of Step 1 (with $r = 1$) by $C(n,p)\|\nabla f\|_{L^p}$; the second is $\big|\frac{1}{|B(x,1)|}\int_{B(x,1)} f\big| \le \frac{1}{|B(x,1)|}\|f\|_{L^1(B(x,1))} \le C(n)\|f\|_{L^p(\mathbb{R}^n)}$ by Hölder on the unit ball. Taking the supremum over x,

$$\sup_x |f(x)| \le C(n,p)\big(\|\nabla f\|_{L^p} + \|f\|_{L^p}\big) \le C(n,p)\|f\|_{W^{1,p}(\mathbb{R}^n)}$$

Step 3: Passage to $W^{1,p}(\mathbb{R}^n)$ by density. Let $f \in W^{1,p}(\mathbb{R}^n)$ and choose $f_m \in C_c^\infty(\mathbb{R}^n)$ with $f_m \to f$ in $W^{1,p}(\mathbb{R}^n)$, which is possible since $W_0^{1,p}(\mathbb{R}^n) = W^{1,p}(\mathbb{R}^n)$ (definition 4.2.6) and by theorem 4.2.1. Steps 1 and 2 apply to each f_m; applied to $f_m - f_l$ they give

$$\sup_x |f_m(x) - f_l(x)| \le C\|f_m - f_l\|_{W^{1,p}}, \qquad [f_m - f_l]_{C^{0,\gamma}} \le C\|\nabla f_m - \nabla f_l\|_{L^p}$$

so (f_m) is Cauchy in the norm of $C^{0,\gamma}(\mathbb{R}^n)$. This space is complete, so $f_m \to \tilde f$ in $C^{0,\gamma}(\mathbb{R}^n)$, hence uniformly, and $\tilde f$ is continuous with $[\tilde f]_{C^{0,\gamma}} < \infty$. Uniform convergence gives $f_m \to \tilde f$ pointwise, while $f_m \to f$ in L^p gives $f_m \to f$ almost everywhere along a subsequence; thus $\tilde f = f$ almost everywhere and $\tilde f$ is the sought continuous representative. Letting $m \to \infty$ in the estimates of Steps 1 and 2, now for $\tilde f$, yields the oscillation bound and the sup bound, and boundedness of the inclusion follows, completing the proof.

Morrey's inequality is the mechanism by which analysis recovers classical smoothness from weak hypotheses: a function known only to have a p-th power integrable gradient, $p > n$, is after modification on a null set genuinely continuous, with a quantitative modulus of continuity. The exponent $\gamma = 1 - \frac{n}{p}$ is sharp and degenerates to 0 as $p \to n^+$, matching the divergence of p^* from the other side. The two theorems together partition the range of p at the dimension: below n one gains integrability, above n one gains continuity, and the borderline $p = n$ sits between, embedding into every finite exponent but into no Hölder space (the model cusp $|x|^\gamma$ with $\gamma \to 0$ shows why continuity fails there, cf. example 4.4.6).

The two first-order theorems combine to describe the entire scale $W^{k,p}$. The organizing quantity is kp, the total order of differentiation weighted against the dimension, and the outcome is decided by comparing kp to n. When $kp < n$ each derivative can be traded, via theorem 4.4.3, for a fixed gain in the integrability exponent, and iterating k times lands in an L^q space with a computable q; when $kp > n$ enough derivatives can be spent to cross the Morrey threshold and land in a classical Hölder space; and when $kp = n$ the borderline holds, with membership in every finite L^q but not, in general, continuity.

Theorem 4.4.9: General Sobolev Embedding

Let $k \geq 1$ be an integer and $1 \leq p < \infty$, and let $f \in W^{k,p}(\mathbb{R}^n)$.

1. If $kp < n$, then $f \in L^q(\mathbb{R}^n)$ for $q = \frac{np}{n-kp}$, and there is a constant $C = C(n,p,k)$ with $\|f\|_{L^q(\mathbb{R}^n)} \leq C\|f\|_{W^{k,p}(\mathbb{R}^n)}$. Consequently $W^{k,p}(\mathbb{R}^n) \hookrightarrow L^q(\mathbb{R}^n)$ for every $p \leq q \leq \frac{np}{n-kp}$.
2. If $kp = n$, then $f \in L^q(\mathbb{R}^n)$ for every $p \leq q < \infty$, with $\|f\|_{L^q(\mathbb{R}^n)} \leq C(n,p,k,q)\|f\|_{W^{k,p}(\mathbb{R}^n)}$.
3. If $kp > n$, write $m = k - \lfloor n/p \rfloor - 1$ and let

$$\gamma = \begin{cases} \lfloor n/p \rfloor + 1 - \dfrac{n}{p} & \text{if } n/p \notin \mathbb{Z} \\ \text{any } \gamma \in (0,1) & \text{if } n/p \in \mathbb{Z} \end{cases}$$

Then f has a representative in $C^{m,\gamma}(\mathbb{R}^n)$, with $\|f\|_{C^{m,\gamma}(\mathbb{R}^n)} \leq C(n,p,k,\gamma)\|f\|_{W^{k,p}(\mathbb{R}^n)}$. Hence $W^{k,p}(\mathbb{R}^n) \hookrightarrow C^{m,\gamma}(\mathbb{R}^n)$.

***Proof*:**
The argument iterates the first-order theorems, tracking the exponent one derivative at a time. Throughout, if $f \in W^{j,p}$ then each $\partial^\alpha f$ with $|\alpha| \leq j-1$ lies in $W^{1,p}$, so the first-order estimates apply to f and to all its lower-order derivatives at once.

Step 1: The subcritical case $kp < n$. Apply theorem 4.4.3 to f and to each first derivative. Since $f \in W^{k,p}$ has $f, \partial^\alpha f \in W^{1,p}$ for $|\alpha| \leq k-1$, the theorem gives $f \in W^{k-1,p_1}$ with $\frac{1}{p_1} = \frac{1}{p} - \frac{1}{n}$: each derivative of order at most $k-1$ gains one unit of integrability, so $\|f\|_{W^{k-1,p_1}} \leq C\|f\|_{W^{k,p}}$. Repeating, after j steps $f \in W^{k-j,p_j}$ with

$$\frac{1}{p_j} = \frac{1}{p} - \frac{j}{n}$$

valid as long as $p_j < n$ at each stage, that is $\frac{1}{p} - \frac{j}{n} > \frac{1}{n}$, which holds for $j \leq k-1$ because $kp < n$ means $\frac{1}{p} - \frac{k-1}{n} > \frac{1}{n}$. After k steps $f \in W^{0,p_k} = L^{p_k}$ with $\frac{1}{p_k} = \frac{1}{p} - \frac{k}{n} = \frac{n-kp}{np}$, so $p_k = \frac{np}{n-kp} = q$ and $\|f\|_{L^q} \leq C\|f\|_{W^{k,p}}$. The embedding into intermediate L^q for $p \leq q \leq \frac{np}{n-kp}$ follows by interpolation: $f \in L^p \cap L^q$ gives $f \in L^{q'}$ for every q' between with $\|f\|_{L^{q'}} \leq \|f\|_{L^p}^{1-\theta}\|f\|_{L^q}^{\theta} \leq C\|f\|_{W^{k,p}}$.

Step 2: The critical case $kp = n$. Iterate as in Step 1 while $p_j < n$. The obstruction is the final step: after $k-1$ iterations $f \in W^{1,p_{k-1}}$ with $\frac{1}{p_{k-1}} = \frac{1}{p} - \frac{k-1}{n} = \frac{1}{n}$, so $p_{k-1} = n$ and theorem 4.4.3 no longer applies with a finite conjugate. The following borderline fact finishes the argument: if $g \in W^{1,n}(\mathbb{R}^n)$ then $g \in L^q(\mathbb{R}^n)$ for every $n \leq q < \infty$, with $\|g\|_{L^q} \leq C(n,q)\|g\|_{W^{1,n}}$. To prove it, run the power substitution behind theorem 4.4.3 at $p = n$. For $g \in C_c^\infty(\mathbb{R}^n)$ and $t > 1$, the $p = 1$ Gagliardo estimate applied to $|g|^t$ gives

$$\left(\int_{\mathbb{R}^n} |g|^{\frac{tn}{n-1}}\right)^{\frac{n-1}{n}} \leq t\int_{\mathbb{R}^n} |g|^{t-1}|\nabla g| \leq t\left(\int_{\mathbb{R}^n} |g|^{(t-1)n'}\right)^{1/n'} \|\nabla g\|_{L^n}$$

by Hölder (theorem 2.1.5) with exponents n and $n' = \frac{n}{n-1}$. Since $(t-1)n' = (t-1)\frac{n}{n-1}$ and $\frac{tn}{n-1}$ differ, this bounds $\|g\|_{L^{tn/(n-1)}}$ by a power of $\|g\|_{L^{(t-1)n/(n-1)}}$ times $\|\nabla g\|_{L^n}$. Starting from $t = n$, where the input exponent $(t-1)n/(n-1) = n$ is controlled by $\|g\|_{W^{1,n}}$, the output exponent is $\frac{n^2}{n-1} > n$; feeding the output exponent back in as the new input and raising t at each stage, the output exponent $\frac{tn}{n-1}$ increases without bound, so after finitely many steps it exceeds any prescribed $q < \infty$, and interpolation between that exponent and n (as in Step 1) yields $\|g\|_{L^q} \le C(n,q)\|g\|_{W^{1,n}}$. Density of C_c^∞ in $W^{1,n}$ (definition 4.2.6, theorem 4.2.1) extends the bound to all $g \in W^{1,n}(\mathbb{R}^n)$, exactly as in Step 3 of theorem 4.4.3. Applying this to $g = \partial^\alpha f$ for $|\alpha| = k-1$, which lie in $W^{1,n}$, gives $\partial^\alpha f \in L^q$ for every finite q and every $|\alpha| \le k-1$; combined with $f \in L^p$ this gives $f \in L^q$ for every finite q, with the stated bound.

Step 3: The supercritical case $kp > n$, with $n/p \notin \mathbb{Z}$. Let $\ell = \lfloor n/p \rfloor$, so $\ell < n/p < \ell + 1$ and $\ell < k$ (because $kp > n$ gives $k > n/p > \ell$). Iterate theorem 4.4.3 exactly ℓ times: at each stage the current exponent r_j satisfies $\frac{1}{r_j} = \frac{1}{p} - \frac{j}{n}$, which stays positive and below $\frac{1}{n}$ only while $j < n/p$, so all ℓ steps are legitimate and land $f \in W^{k-\ell,r}$ with $\frac{1}{r} = \frac{1}{p} - \frac{\ell}{n}$. Here $r > n$ precisely because $\frac{1}{r} = \frac{1}{p} - \frac{\ell}{n} < \frac{1}{n}$, using $\ell < n/p$. Now $k - \ell = k - \lfloor n/p \rfloor = m + 1$, so $f \in W^{m+1,r}$ with $r > n$. Each $\partial^\alpha f$ with $|\alpha| \le m$ then lies in $W^{1,r}$, $r > n$, so theorem 4.4.8 gives $\partial^\alpha f \in C^{0,\gamma}(\mathbb{R}^n)$ with $\gamma = 1 - \frac{n}{r} = \ell + 1 - \frac{n}{p}$ and the estimate $[\partial^\alpha f]_{C^{0,\gamma}} \le C\|\nabla \partial^\alpha f\|_{L^r} \le C\|f\|_{W^{k,p}}$, together with $\sup|\partial^\alpha f| \le C\|f\|_{W^{k,p}}$. Assembling over all $|\alpha| \le m$ gives $f \in C^{m,\gamma}(\mathbb{R}^n)$ with $\|f\|_{C^{m,\gamma}} \le C\|f\|_{W^{k,p}}$.

Step 4: The supercritical case $kp > n$, with $n/p \in \mathbb{Z}$. Now $\lfloor n/p \rfloor = n/p$ and $m = k - \frac{n}{p} - 1$. Iterate theorem 4.4.3 exactly $\frac{n}{p} - 1$ times, which is legitimate since each intermediate exponent stays below n; this lands $f \in W^{k-(n/p-1),n}$, that is, $f \in W^{m+2,n}$. Each $\partial^\alpha f$ with $|\alpha| \le m+1$ then lies in $W^{1,n}$, so by the borderline fact of Step 2 it lies in $L^s(\mathbb{R}^n)$ for every finite s, with $\|\partial^\alpha f\|_{L^s} \le C(n,s)\|f\|_{W^{k,p}}$. Fix any $s > n$; then each $\partial^\beta f$ with $|\beta| \le m$ lies in $W^{1,s}$ (its derivative $\partial^\alpha f$, $|\alpha| = |\beta| + 1 \le m+1$, is in L^s, and $\partial^\beta f$ itself is in L^s), so theorem 4.4.8 gives $\partial^\beta f \in C^{0,\gamma}(\mathbb{R}^n)$ with $\gamma = 1 - \frac{n}{s}$, any value in $(0,1)$ being attainable by the choice of $s > n$, and $\|\partial^\beta f\|_{C^{0,\gamma}} \le C\|\partial^\beta f\|_{W^{1,s}} \le C\|f\|_{W^{k,p}}$. Assembling over $|\beta| \le m$ gives $f \in C^{m,\gamma}(\mathbb{R}^n)$ for the chosen $\gamma \in (0,1)$, with the stated estimate, completing the proof.

The theorem converts the abstract membership $f \in W^{k,p}$ into concrete information a differential equation can use, either an integrability exponent or a number of classical derivatives with a Hölder modulus. The count kp against n is the single dial. In the elliptic regularity theory of the partial-differential-equations volumes, a solution is first shown to lie in some $W^{k,p}$ by energy estimates; the embedding then reads off how many classical derivatives it actually has, and the passage from $kp < n$ to $kp > n$ as k increases is precisely the bootstrap that turns a weak solution into a smooth one. The Rellich-Kondrachov theorem of section 4.5 sharpens the subcritical case further, upgrading the bounded inclusion into L^q to a compact one for $q < p^*$, which is what existence proofs by compactness require.

Example 4.4.10: Reading Off Regularity in Dimension Three

Take $n = 3$ and $p = 2$, the setting of most of mathematical physics. For $k = 1$ one has $kp = 2 < 3$, so theorem 4.4.9(1) gives $H^1(\mathbb{R}^3) = W^{1,2} \hookrightarrow L^q$ for $2 \le q \le \frac{3\cdot 2}{3-2} = 6$, recovering the L^6 bound of example 4.4.2. For $k = 2$ one has $kp = 4 > 3$, so case (3) applies with $\lfloor n/p \rfloor = \lfloor 3/2 \rfloor = 1$, hence $m = 2 - 1 - 1 = 0$ and $\gamma = 1 + 1 - \frac{3}{2} = \frac{1}{2}$; thus $H^2(\mathbb{R}^3) \hookrightarrow C^{0,1/2}(\mathbb{R}^3)$, and a function with two weak derivatives in L^2 is Hölder-$\frac{1}{2}$ continuous. This is why H^2 is the natural space in which to seek continuous solutions of second-order equations in three dimensions.

4.5 Compactness and the Poincaré Inequality

The embeddings of section 4.4 are bounded but not compact on $\mathbb{R}^n$: a fixed bump translated to infinity keeps its Sobolev norm while wandering off, so no subsequence converges. On a bounded domain there is no room to escape, and the same one-derivative gain that produced the embedding now produces *compactness*. This section proves that on a bounded domain the embedding of $W^{1,p}$ into L^q is compact below the critical exponent, and extracts the two inequalities that turn this compactness into existence theorems: the Poincaré inequality, which controls a function by its gradient once boundary values are killed, and its mean-zero variant, the Poincaré-Wirtinger inequality.

The compactness statement underlies existence proofs in the calculus of variations and the elliptic theory: a bounded sequence of approximate solutions, bounded in $W^{1,p}$ by an energy estimate, has a subsequence converging *strongly* in L^q, and strong convergence is what lets one pass to the limit in a nonlinear term. What makes it work is that a $W^{1,p}$ bound controls not just the size of a function but the size of its translates: a gradient bound forces $\|\tau_h f - f\|_{L^p}$ to be small uniformly, and uniform smallness of translates is exactly the hypothesis of the Fréchet-Kolmogorov compactness criterion in L^p.

Theorem 4.5.1: Rellich-Kondrachov Compactness

Let $U \subseteq \mathbb{R}^n$ be bounded with C^1 boundary, and let $1 \leq p < n$. Then for every q with $1 \leq q < p^*$ the embedding

$$W^{1,p}(U) \hookrightarrow L^q(U)$$

is compact: every bounded sequence in $W^{1,p}(U)$ has a subsequence converging in $L^q(U)$.

***Proof*:**
Let (f_m) be a sequence with $\|f_m\|_{W^{1,p}(U)} \leq M$ for all m; the goal is a subsequence Cauchy in $L^q(U)$. Throughout, C denotes a constant that may change from line to line but never depends on m.

Step 1: Extend to compactly supported functions on $\mathbb{R}^n$*.* By the extension theorem (theorem 4.3.3) there is a bounded linear $E\colon W^{1,p}(U) \to W^{1,p}(\mathbb{R}^n)$ with $Ef = f$ almost everywhere on U. A cutoff makes E land in functions supported in a fixed bounded open set V with $\overline{U} \subseteq V$: multiply by a fixed cutoff $\zeta \in C_c^\infty(V)$ equal to 1 on a neighborhood of $\overline{U}$, which changes E only by a bounded operator and does not disturb $Ef = f$ on U. Write $g_m := Ef_m$. Then each g_m is supported in V and

$$\|g_m\|_{W^{1,p}(\mathbb{R}^n)} \leq \|E\| \, \|f_m\|_{W^{1,p}(U)} \leq CM$$

so (g_m) is bounded in $W^{1,p}(\mathbb{R}^n)$, with all g_m supported in the one bounded set V. A subsequence of (f_m) converges in $L^q(U)$ as soon as the corresponding subsequence of (g_m) converges in $L^q(V)$, since $f_m = g_m$ on U; it therefore suffices to extract a convergent subsequence of (g_m) in $L^q(V)$.

Step 2: A uniform bound at the critical exponent. Each $g_m \in W^{1,p}(\mathbb{R}^n)$, so by the Gagliardo-Nirenberg-Sobolev inequality (theorem 4.4.3),

$$\|g_m\|_{L^{p^*}(\mathbb{R}^n)} \leq C \, \|\nabla g_m\|_{L^p(\mathbb{R}^n)} \leq C \, \|g_m\|_{W^{1,p}(\mathbb{R}^n)} \leq CM$$

Thus (g_m) is bounded in $L^{p^*}(\mathbb{R}^n)$, uniformly in m. This is the reservoir of integrability that the interpolation in Step 5 will spend.

Step 3: Control of translates. Fix $h \in \mathbb{R}^n$ and write $\tau_h g(x) = g(x-h)$. For a smooth compactly supported φ the fundamental theorem of calculus along the segment from x to $x-h$ gives

$$\varphi(x) - \varphi(x-h) = \int_0^1 h \cdot \nabla\varphi(x - th)\, dt$$

so, taking absolute values and applying Minkowski's integral inequality (theorem 2.1.6),

$$\|\tau_h\varphi - \varphi\|_{L^p(\mathbb{R}^n)} \le |h| \int_0^1 \|\nabla\varphi(\cdot - th)\|_{L^p(\mathbb{R}^n)}\, dt = |h|\, \|\nabla\varphi\|_{L^p(\mathbb{R}^n)}$$

the last equality because translation preserves the L^p norm. Since $W_0^{1,p}(\mathbb{R}^n) = W^{1,p}(\mathbb{R}^n)$ (definition 4.2.6), every element of $W^{1,p}(\mathbb{R}^n)$ is a limit in the $W^{1,p}$ norm of such $\varphi \in C_c^\infty(\mathbb{R}^n)$; both sides of the estimate are continuous in that norm, so it passes to the limit and

$$\|\tau_h g_m - g_m\|_{L^p(\mathbb{R}^n)} \le |h|\, \|\nabla g_m\|_{L^p(\mathbb{R}^n)} \le |h|\, CM$$

holds for every m and every h. The translates of the g_m are thus small uniformly in m: given $\varepsilon > 0$, taking $|h| \le \varepsilon/(CM)$ makes $\|\tau_h g_m - g_m\|_{L^p(\mathbb{R}^n)} \le \varepsilon$ for all m at once.

Step 4: Precompactness in $L^1(W_0)$. The estimate of Step 3, together with the common support in V and the uniform L^p bound (which gives a uniform $L^1(V)$ bound since V is bounded), is precisely the hypothesis of the Fréchet-Kolmogorov criterion for precompactness in L^1. Rather than invoke it, mollify directly. Let W_0 be the closed 1-neighborhood of V, a fixed compact set, and let $\{\psi_\delta\}_{0<\delta\le 1}$ be the good kernel (definition 2.6.9) with $\psi_\delta(x) = \delta^{-n}\psi(x/\delta)$ for a fixed $\psi \in C_c^\infty(\mathbb{R}^n)$, $\psi \ge 0$, $\int \psi = 1$, supported in the unit ball; then g_m and $g_m * \psi_\delta$ are all supported in W_0. For each m,

$$(g_m * \psi_\delta)(x) - g_m(x) = \int_{\mathbb{R}^n} \big(g_m(x-y) - g_m(x)\big)\, \psi_\delta(y)\, dy$$

so integrating and using $\int \psi_\delta = 1$, Minkowski's integral inequality (theorem 2.1.6), and Step 3,

$$\begin{aligned} \|g_m * \psi_\delta - g_m\|_{L^1(\mathbb{R}^n)} &\le \int_{|y|\le\delta} \|\tau_y g_m - g_m\|_{L^1(\mathbb{R}^n)}\, \psi_\delta(y)\, dy \\ &\le \int_{|y|\le\delta} |W_0|^{1-1/p}\, \|\tau_y g_m - g_m\|_{L^p(\mathbb{R}^n)}\, \psi_\delta(y)\, dy \le C\,\delta \end{aligned}$$

where Hölder on the common bounded support W_0 turned the L^1 norm into an L^p norm at the cost of $|W_0|^{1-1/p}$, and the bound is uniform in m. Fix $\varepsilon > 0$ and choose δ so that $C\delta \le \varepsilon/3$. For this frozen δ the mollified family $(g_m * \psi_\delta)_m$ is uniformly bounded and equicontinuous on W_0: uniformly bounded because

$$|(g_m * \psi_\delta)(x)| \le \|g_m\|_{L^1(\mathbb{R}^n)}\, \|\psi_\delta\|_{L^\infty(\mathbb{R}^n)} \le C\delta^{-n}$$

and equicontinuous because

$$|(g_m * \psi_\delta)(x) - (g_m * \psi_\delta)(x')| \le \|g_m\|_{L^1(\mathbb{R}^n)}\, \|\tau_{x'-x}\psi_\delta - \psi_\delta\|_{L^\infty(\mathbb{R}^n)}$$

and ψ_δ is uniformly continuous. By the Arzelà-Ascoli theorem a subsequence of $(g_m * \psi_\delta)_m$ converges uniformly on W_0, hence in $L^1(W_0)$. Combining with the uniform bound $\|g_m * \psi_\delta - g_m\|_{L^1} \le \varepsilon/3$, along this subsequence

$$\limsup_{k,l} \|g_{m_k} - g_{m_l}\|_{L^1(W_0)} \le \frac{2\varepsilon}{3} + \limsup_{k,l} \|g_{m_k} * \psi_\delta - g_{m_l} * \psi_\delta\|_{L^1(W_0)} = \frac{2\varepsilon}{3} < \varepsilon$$

A diagonal argument over $\varepsilon = 1, \frac{1}{2}, \frac{1}{3}, \dots$ then produces a single subsequence, still written (g_m), that is Cauchy in $L^1(W_0)$, hence in $L^1(V)$ since $V \subseteq W_0$.

Step 5: Interpolate up to $L^q(V)$. Fix q with $1 \le q < p^*$. Interpolation between L^1 and L^{p^*} writes $\frac{1}{q} = \frac{\theta}{1} + \frac{1-\theta}{p^*}$ for a unique $\theta \in (0,1]$, and Hölder's inequality (theorem 2.1.5) gives, for any g,

$$\|g\|_{L^q(V)} \le \|g\|_{L^1(V)}^{\theta} \, \|g\|_{L^{p^*}(V)}^{1-\theta}$$

Applying this to differences of the extracted subsequence, and using the uniform L^{p^*} bound of Step 2 (which controls $\|g_{m_k} - g_{m_l}\|_{L^{p^*}(V)} \le 2CM$),

$$\|g_{m_k} - g_{m_l}\|_{L^q(V)} \le \|g_{m_k} - g_{m_l}\|_{L^1(V)}^{\theta} \, (2CM)^{1-\theta}$$

The right side tends to 0 as $k, l \to \infty$ because the sequence is Cauchy in $L^1(V)$ and $\theta > 0$. Hence (g_{m_k}) is Cauchy, so convergent, in $L^q(V)$, and its restrictions to U give a subsequence of (f_m) convergent in $L^q(U)$, as we sought to show.

The mechanism is worth isolating. A $W^{1,p}$ bound buys two things: an integrability gain past p^*, and control of translates through the gradient. The first is a fixed reservoir of smallness at high exponents; the second, via mollification, upgrades the reservoir to genuine L^1 compactness on a bounded set. Interpolation then converts L^1 compactness plus an L^{p^*} bound into L^q compactness for every q strictly below p^*. The exclusion of the endpoint $q = p^*$ is not an artifact of the proof: at the critical exponent, concentration can defeat compactness. Fixing a bump $u \in C_c^\infty(\mathbb{R}^n)$ and rescaling by $u_\lambda(x) = \lambda^{(n-p)/p} u(\lambda x)$ keeps both $\|\nabla u_\lambda\|_{L^p}$ and $\|u_\lambda\|_{L^{p^*}}$ constant in λ while concentrating the mass to a point as $\lambda \to \infty$, so no subsequence converges in L^{p^*}; the compact embedding is a subcritical phenomenon.

Example 4.5.2: Compactness on the Interval

Take $U = (0,1) \subset \mathbb{R}$, so $n = 1$. Here the subcritical range $1 \le p < n$ is empty, but the compactness is still visible directly through Morrey's inequality (theorem 4.4.8), which for $n = 1$ and $p > 1 = n$ turns a $W^{1,p}(0,1)$ bound into a uniform Hölder bound. Concretely, a sequence (f_m) bounded in $W^{1,2}(0,1)$ satisfies, for $x < y$,

$$|f_m(y) - f_m(x)| = \left| \int_x^y f_m'(t)\, dt \right| \le \|f_m'\|_{L^2(0,1)} \, |y - x|^{1/2} \le M \, |y - x|^{1/2}$$

by Cauchy-Schwarz, so (f_m) is equicontinuous with a uniform Hölder-1/2 modulus; it is uniformly bounded because the Morrey embedding turns the $W^{1,2}(0,1)$ bound into a uniform sup-bound $\|f_m\|_{L^\infty(0,1)} \le CM$. Arzelà-Ascoli then extracts a uniformly convergent subsequence, in particular convergent in $L^q(0,1)$ for every $q < \infty$: in one dimension a gradient bound is a Hölder bound, and Hölder bounds are precompact. The proof of theorem 4.5.1 replaces this pointwise Hölder estimate, unavailable when $p \le n$, by the L^p control of translates in Step 3.

The first payoff of compactness is a coercivity estimate. For a function that vanishes at the boundary, the gradient alone controls the full L^p norm: there is nothing left to compensate a large function with a small gradient once constants are forbidden. This is the inequality that makes the Dirichlet energy $\int_U |\nabla f|^2$ an equivalent norm on $W_0^{1,2}(U)$, and with it the existence theory for the Dirichlet problem. Its proof needs no compactness at all, only the one-dimensional fundamental theorem of calculus, so it precedes the Wirtinger variant.

Theorem 4.5.3: Poincaré Inequality

Let $U \subseteq \mathbb{R}^n$ be bounded and let $1 \leq p < \infty$. There is a constant $C = C(n,p,U)$ such that

$$\|f\|_{L^p(U)} \leq C \, \|\nabla f\|_{L^p(U)}$$

for every $f \in W_0^{1,p}(U)$.

***Proof*:**
Since U is bounded, choose $a > 0$ with $U \subseteq \{x \in \mathbb{R}^n \mid |x_1| < a\}$, the slab of width $2a$ in the first coordinate. By the definition of the zero-trace space (definition 4.2.6), $C_c^\infty(U)$ is dense in $W_0^{1,p}(U)$, so it suffices to prove the inequality for $f \in C_c^\infty(U)$ and then pass to the limit; extend such an f by zero to all of $\mathbb{R}^n$, keeping it smooth and compactly supported inside the slab.

Fix $x = (x_1, x') \in U$ with $x' = (x_2, \ldots, x_n)$. Because $f(-a, x') = 0$, the fundamental theorem of calculus in the first variable gives

$$f(x_1, x') = \int_{-a}^{x_1} \partial_1 f(t, x') \, dt$$

Take absolute values and apply Hölder's inequality (theorem 2.1.5) in t over the interval $(-a, a)$, of length $2a$:

$$|f(x_1, x')| \leq \int_{-a}^{a} |\partial_1 f(t, x')| \, dt \leq (2a)^{1/p'} \left(\int_{-a}^{a} |\partial_1 f(t, x')|^p \, dt \right)^{1/p}$$

Raise to the pth power and integrate in x_1 over $(-a, a)$. The right side does not depend on x_1, so integrating contributes a factor $2a$:

$$\int_{-a}^{a} |f(x_1, x')|^p \, dx_1 \leq (2a)^{p/p'} \, (2a) \int_{-a}^{a} |\partial_1 f(t, x')|^p \, dt = (2a)^p \int_{-a}^{a} |\partial_1 f(t, x')|^p \, dt$$

using $p/p' + 1 = p$. Now integrate in the remaining variables x' and recall f vanishes outside the slab:

$$\|f\|_{L^p(U)}^p = \int_{\mathbb{R}^{n-1}} \int_{-a}^{a} |f(x_1, x')|^p \, dx_1 \, dx' \leq (2a)^p \int_{\mathbb{R}^{n-1}} \int_{-a}^{a} |\partial_1 f|^p \, dt \, dx' = (2a)^p \, \|\partial_1 f\|_{L^p(U)}^p$$

Since $\|\partial_1 f\|_{L^p(U)} \leq \|\nabla f\|_{L^p(U)}$, taking pth roots gives $\|f\|_{L^p(U)} \leq 2a \, \|\nabla f\|_{L^p(U)}$ with $C = 2a$. For general $f \in W_0^{1,p}(U)$, choose $f_m \in C_c^\infty(U)$ with $f_m \to f$ in $W^{1,p}(U)$; the inequality holds for each f_m, and both $\|f_m\|_{L^p(U)} \to \|f\|_{L^p(U)}$ and $\|\nabla f_m\|_{L^p(U)} \to \|\nabla f\|_{L^p(U)}$, so it passes to the limit, completing the proof.

The constant depends only on the width of the smallest slab containing U, not on its shape: Poincaré's inequality holds on any bounded domain, and its constant degrades only as the domain grows in one direction. On $W_0^{1,p}(U)$ the estimate says $\|\nabla f\|_{L^p(U)}$ is a norm equivalent to the full $\|f\|_{W^{1,p}(U)}$ norm, since $\|f\|_{W^{1,p}}^p = \|f\|_{L^p}^p + \|\nabla f\|_{L^p}^p \leq (1 + C^p) \|\nabla f\|_{L^p}^p$. The gradient alone therefore measures everything, which is why the Dirichlet energy is the natural object on the zero-trace space. The hypothesis that boundary values vanish is essential: on all of $W^{1,p}(U)$ the inequality is false, since a nonzero constant function has zero gradient but positive L^p norm. The constant functions are the only obstruction, and quotienting them out restores the estimate, which is the content of the next theorem.

Example 4.5.4: The Optimal Constant on an Interval

On $U = (0, L) \subset \mathbb{R}$ with $p = 2$, the Poincaré inequality reads $\|f\|_{L^2(0,L)} \leq C \|f'\|_{L^2(0,L)}$ for $f \in H^1_0(0, L)$. The proof above gives $C = 2a = L$ after centering the interval, but the sharp constant is smaller. Expand $f \in H^1_0(0, L)$ in the sine basis $e_k(x) = \sqrt{2/L}\, \sin(k\pi x/L)$, which vanishes at both endpoints and diagonalizes $-d^2/dx^2$ with eigenvalues $(k\pi/L)^2$. Writing $f = \sum_k c_k e_k$, Parseval gives $\|f\|^2_{L^2} = \sum_k c_k^2$ and $\|f'\|^2_{L^2} = \sum_k (k\pi/L)^2 c_k^2 \geq (\pi/L)^2 \sum_k c_k^2$, so

$$\|f\|_{L^2(0,L)} \leq \frac{L}{\pi} \|f'\|_{L^2(0,L)}$$

with equality for $f = e_1$. The sharp constant L/π is the reciprocal square root of the first Dirichlet eigenvalue $(\pi/L)^2$; the crude slab estimate $C = L$ overshoots it by the factor π. The general principle, that the best Poincaré constant is the reciprocal square root of the first eigenvalue of the Dirichlet Laplacian, connects these inequalities to spectral theory: the compactness of theorem 4.5.1 is exactly what makes $-\Delta$ with zero boundary values have compact resolvent, so its spectrum is a discrete sequence of Poincaré-type constants.

When boundary values are not prescribed the constant functions must be removed by hand, subtracting the average. The resulting inequality controls the deviation of a function from its mean by the gradient, and it holds on any bounded connected domain with reasonable boundary. Unlike the Poincaré inequality, no explicit constant is available from a one-dimensional estimate: the argument is by compactness and contradiction, and it is here that Rellich-Kondrachov is used.

Theorem 4.5.5: Poincaré-Wirtinger Inequality

Let $U \subseteq \mathbb{R}^n$ be bounded, connected, and with C^1 boundary, and let $1 \leq p < \infty$. Write $f_U = \frac{1}{|U|} \int_U f\, dx$ for the average of f over U. There is a constant $C = C(n, p, U)$ such that

$$\|f - f_U\|_{L^p(U)} \leq C \|\nabla f\|_{L^p(U)}$$

for every $f \in W^{1,p}(U)$.

***Proof*:**

Suppose the inequality fails for every constant. Then for each m there is $f_m \in W^{1,p}(U)$ with

$$\|f_m - (f_m)_U\|_{L^p(U)} > m \|\nabla f_m\|_{L^p(U)}$$

Normalize by setting

$$g_m := \frac{f_m - (f_m)_U}{\|f_m - (f_m)_U\|_{L^p(U)}}$$

which is legitimate because the left side of the failed inequality is positive. Then $g_m \in W^{1,p}(U)$ has three properties: its average $(g_m)_U = 0$ (subtracting a constant sets the mean to zero, and normalizing preserves it), its L^p norm $\|g_m\|_{L^p(U)} = 1$, and dividing the failed inequality by its own left side,

$$\|\nabla g_m\|_{L^p(U)} = \frac{\|\nabla f_m\|_{L^p(U)}}{\|f_m - (f_m)_U\|_{L^p(U)}} < \frac{1}{m}$$

so $\nabla g_m \to 0$ in $L^p(U)$. In particular

$$\|g_m\|^p_{W^{1,p}(U)} = \|g_m\|^p_{L^p(U)} + \|\nabla g_m\|^p_{L^p(U)} \le 1 + \frac{1}{m^p} \le 2$$

so (g_m) is bounded in $W^{1,p}(U)$.

The compactness of the embedding $W^{1,p}(U) \hookrightarrow L^p(U)$ furnishes a convergent subsequence in every case. If $p < n$, then $q = p < p^*$ lies in the compact range of the Rellich-Kondrachov theorem (theorem 4.5.1), so a subsequence of (g_m) converges in $L^p(U)$. If $p > n$, the $kp > n$ case of theorem 4.4.9 embeds $W^{1,p}(U)$ into a Hölder space, and a bounded sequence there is precompact in $C(\overline{U})$ by Arzelà-Ascoli, hence in $L^p(U)$. The two boundary cases $p = n$ remain. For $p = n \ge 2$, pick $p' < n$ with $(p')^* > p$ (possible since $(p')^* \to \infty$ as $p' \uparrow n$); then $W^{1,p}(U) \hookrightarrow W^{1,p'}(U)$ on the bounded set U, and $p < (p')^*$ puts $q = p$ in the compact range for exponent p', so theorem 4.5.1 again applies. The last case is $n = 1$, $p = 1$: here neither route is available, but the Rellich mechanism applies directly. Extend the g_m to a common bounded support in $\mathbb{R}$ as in Step 1 of the proof of theorem 4.5.1; the translate estimate of its Step 3 gives $\|\tau_h g_m - g_m\|_{L^1(\mathbb{R})} \le |h|\,\|\nabla g_m\|_{L^1}$, uniformly bounded in m, which together with the common support is the hypothesis of the Fréchet-Kolmogorov criterion for precompactness in L^1, yielding a subsequence convergent in $L^1(U) = L^p(U)$. In every case, pass to a subsequence, still called (g_m), with $g_m \to g$ in $L^p(U)$ for some $g \in L^p(U)$.

The limit g inherits both normalizations by L^p convergence: $\|g\|_{L^p(U)} = \lim \|g_m\|_{L^p(U)} = 1$, and averaging is L^p-continuous on the bounded set U (by Hölder against the constant $1 \in L^{p'}(U)$), so $g_U = \lim (g_m)_U = 0$. It remains to identify the weak gradient of g. For any $\varphi \in C_c^\infty(U)$ and any index i, the weak-derivative identity (definition 4.1.1) for g_m reads $\int_U g_m\, \partial_i \varphi = -\int_U \partial_i g_m\, \varphi$. Let $m \to \infty$: the left side tends to $\int_U g\, \partial_i \varphi$ since $g_m \to g$ in L^p and $\partial_i \varphi \in L^{p'}$, while the right side tends to 0 since $\partial_i g_m \to 0$ in L^p and $\varphi \in L^{p'}$. Hence $\int_U g\, \partial_i \varphi = 0$ for all φ, which says $\partial_i g = 0$ weakly for every i, so $\nabla g = 0$. Thus $g \in W^{1,p}(U)$ with vanishing weak gradient on the connected set U.

A $W^{1,p}$ function with zero weak gradient on a connected open set is constant almost everywhere. Indeed, mollifying, $g * \psi_\delta$ is smooth with classical gradient $\nabla(g * \psi_\delta) = (\nabla g) * \psi_\delta = 0$ on any $V \Subset U$ (by theorem 4.2.1, which commutes the derivative past the mollifier), hence constant on each component of V; letting $\delta \to 0$ and using connectedness of U shows g equals a single constant almost everywhere. But then $g = g_U = 0$ almost everywhere, contradicting $\|g\|_{L^p(U)} = 1$. The assumed failure is therefore impossible, and the inequality holds with some finite constant C, completing the proof.

The proof is nonconstructive: it produces a finite constant but no formula for it, in contrast to the explicit $2a$ of the Poincaré inequality. This is the price of the compactness argument, and it is characteristic of the method: an inequality is proved by ruling out its failure, and the witness to failure is destroyed by extracting a convergent subsequence whose limit cannot exist. Connectedness is used exactly once, in the final step, to force a gradient-free function to be a single constant rather than a locally constant one; on a disconnected domain the average over the whole set is the wrong reference value, and the inequality fails with a function equal to different constants on different components. The mean f_U is the natural replacement for the boundary value of the Poincaré inequality: both the zero-trace condition and the zero-mean condition eliminate the constants, which are the sole functions the gradient cannot see, and the estimate holds on the quotient by constants in either case.

Example 4.5.6: Mean-Zero Functions on a Ball

On the unit ball $U = B(0,1) \subset \mathbb{R}^n$ with $p = 2$, the Poincaré-Wirtinger inequality $\|f - f_U\|_{L^2(U)} \leq C\,\|\nabla f\|_{L^2(U)}$ again has a spectral reading. Its sharp constant is the reciprocal square root of the first *nonzero* Neumann eigenvalue μ_1 of $-\Delta$ on the ball: the mean-zero constraint $f_U = 0$ is exactly orthogonality to the constant eigenfunction, whose Neumann eigenvalue is $\mu_0 = 0$, so on the orthogonal complement the Rayleigh quotient $\|\nabla f\|_{L^2}^2 / \|f\|_{L^2}^2$ is bounded below by μ_1. A mean-zero f therefore satisfies $\|f\|_{L^2}^2 \leq \mu_1^{-1} \|\nabla f\|_{L^2}^2$, giving $C = \mu_1^{-1/2}$. The contradiction argument of theorem 4.5.5 recovers the finiteness of this constant without computing μ_1: it shows only that the infimum of the Rayleigh quotient over mean-zero functions is positive, which is exactly $\mu_1 > 0$, the statement that 0 is a simple Neumann eigenvalue on a connected domain.

5

Hilbert Spaces

There are many reasons why Hilbert spaces are interesting. We shall start with a motivation stemming from looking for the ideal space where the product of two functions is still integrable, and then take a moment to elaborate on some more advanced intuitions of Hilbert spaces for readers with more background.

We are naturally interested in integrating products of functions:

$$\int_X fg$$

The problem we may encounter when working over L^1 is that $\int_X fg$ need not be finite. We may be tempted to say "let us work with the functions for which this integral *is* finite". That is, let $L'(X)$ be the space of all measurable functions where for any choice $f, g \in L'(X)$:

$$\int_X |f\overline{g}| < \infty \qquad \text{i.e.} \qquad f\overline{g} \in L^1$$

Then we have just made a circular-reasoning mistake, for we are quantifying existence of elements in $L'(X)$ using elements of $L'(X)$. Instead, we may ask: given a set $S \subseteq \mathcal{M}(X)$, what is the subset of measurable functions $T \subseteq \mathcal{M}(X)$ such that:

$$T = \left\{ f \in \mathcal{M}(X) \;\middle|\; \int_X |f\overline{g}| < \infty,\ \forall g \in S \right\}$$

We can label this space $L^1_S(X)$. The two extremes of this are if $S = \{0\}$, then $L^1_{\{0\}}(X) = \mathcal{M}(X)$ is all measurable functions, and if S is all measurable functions, then $L^1_{\mathcal{M}(X)}(X) = \{0\}$ only contains the zero function. The ideal is a middle ground choice of S where $L^1_S(X) = S$, so that we don't need to keep track of the two spaces we are choosing our functions from. Fortunately, this is *exactly* the space L^2:[1] taking $S = L^2(X)$,

$$L^1_{L^2(X)}(X) = \left\{ f \in \mathcal{M}(X) \;\middle|\; \int_X |f\overline{g}| < \infty \;\forall g \in L^2 \right\} = L^2$$

Half of this is elementary: by the Cauchy-Schwarz inequality[2]

$$\int_X |fg| \leq \left(\int_X |f|^2 \right)^{1/2} \left(\int_X |g|^2 \right)^{1/2} = \|f\|_2 \|g\|_2$$

every $f \in L^2$ pairs integrably with all of L^2, showing $L^2(X) \subseteq L^1_{L^2(X)}(X)$. The substantive half is the reverse inclusion: if $\int_X |f\overline{g}| < \infty$ for *every* $g \in L^2$, then f itself must lie in L^2. This is the *converse* of Hölder's inequality, theorem 2.2.2.[3] Hence $L^1_{L^2(X)}(X) = L^2$: the space L^2 is a fixed point of $S \mapsto L^1_S(X)$.

It is, moreover, the *only* fixed point. Suppose $L^1_S(X) = S$. Every $f \in S$ then pairs with itself, so $\int_X |f|^2 = \int_X |f\overline{f}| < \infty$; that is, $S \subseteq L^2$. But $S \mapsto L^1_S(X)$ is inclusion-reversing, so $S = L^1_S(X) \supseteq L^1_{L^2(X)}(X) = L^2$. Hence $S = L^2$.

From this perspective, L^2 is the ideal space in which to multiply functions and they remain integrable (i.e. $f, g \in L^2$, $fg \in L^1$)! We shall call such multiplication the *inner product*:

$$\langle f, g \rangle = \int_X f\overline{g}$$

The conjugation in the definition is used so that $f\overline{f} = |f|^2$. Note that we are not looking for the product of two integrable functions ($f, g \in L^1$) to remain within L^1. As we saw in section 2.6, *convolution* has that property: if $f, g \in L^1(G)$ then $f * g \in L^1(G)$ for G a locally compact group, making $L^1(G)$ a *group algebra*.

5.0.1 Post-Hoc Motivation A lot of the following motivation came *a posteriori*, that is, I felt much more comfortable with Hilbert spaces after learning about L^p spaces, and generalizing some results from Riemannian Geometry to Banach spaces to notice that L^2 is the only space with flat geometry, hence really is the "correct" generalization of Euclidean space/geometry in infinite dimensions.

Banach spaces provide a robust framework for measuring the magnitude of elements, allowing us to quantify convergence and bound operations via the triangle and Minkowski inequalities. However, a fundamental deficiency in general Banach spaces is the lack of a linear measurement theory to quantify *similarity*. While we can measure the total size of a sum $\|f + g\|$, we often lack the granular information required to isolate the contribution of f from g. For instance, determining how much a vector projects onto a specific basis element is straightforward in Euclidean space but algebraically obstructed in general norms.

[1] We assume in this motivation that the measure space is σ-finite. On a measure space consisting of a single atom of infinite measure, say, $L^2 = \{0\}$ while $L^1_{\{0\}}(X) = \mathcal{M}(X)$, and no fixed point exists at all.

[2] Proven as theorem 5.1.2 in section 5.1; the reader has most likely already encountered it, so we use it here for intuition.

[3] Strictly, theorem 2.2.2 also assumes the finiteness of the supremum M_q; that the integrability hypothesis alone forces this bound is a closed-graph argument, which we omit here.

To make this deficiency precise, consider how the inner product arises in Euclidean geometry. Define the *energy* of a vector as $E(x) = \frac{1}{2}\|x\|_2^2$. Then the directional derivative of this energy at x in the direction of a perturbation y is:

$$\frac{d}{dt}E(x+ty)\Big|_{t=0} = \frac{d}{dt}\frac{1}{2}\langle x+ty, x+ty\rangle\Big|_{t=0} = \langle x, y\rangle$$

Thus, the inner product $\langle x, y\rangle$ physically represents the "sensitivity" of the energy of x to a perturbation by y. In particular, the energy satisfies the clean linear decomposition:

$$E(x+y) = E(x) + E(y) + \langle x, y\rangle$$

Note the interaction term $\langle x, y\rangle$ is *bilinear*. In general Banach spaces, such a linear decomposition of the interaction is impossible due to the non-flat geometry of the unit ball.

The natural question, then, is: *what replaces the inner product in L^p?* The correct generalization is to replace the quadratic energy with the *p-energy*:

$$F(f) = \frac{1}{p}\|f\|_p^p = \frac{1}{p}\int |f|^p$$

and ask: how does $F(f)$ change if we (linearly) perturb f by a function g? To answer this, we must compute the gradient ∇F, which requires first understanding the natural duality of L^p spaces.

What does it mean to go in the "same direction" in a function space? (For simplicity we work throughout with real L^p spaces for $1 < p < \infty$.[4]) Rigorously, given $f \neq 0$, we are asking for the continuous linear functional $\varphi_f \in (L^p)^*$ of unit norm that *maximizes* the evaluation at f. We can interpret this φ_f as the *optimal filter* or sensor for f, i.e. it captures the full "energy" of f without loss ($\varphi_f(f) = \|f\|_p$ and $\|\varphi_f\|_{(L^p)^*} = 1$). Such a functional exists in any Banach space by the Hahn-Banach Theorem (theorem 1.3.3); that it is *unique* is special to $1 < p < \infty$, as the next computation shows (at $p = 1$ or $p = \infty$ the maximizer need not be unique).

By theorem 2.2.3 (and the equality condition of Hölder's inequality), this functional is represented by a positive multiple of:

$$g(x) = |f(x)|^{p-1}\mathrm{sgn}(f(x))$$

Note that $g \in L^q$ since:

$$|g|^q = |f|^{q(p-1)} = |f|^{qp-q} \stackrel{!}{=} |f|^p$$

where $\stackrel{!}{=}$ holds since $1/p + 1/q = 1$ iff $p + q = pq$ iff $p = q(p-1)$. Hence as $f \in L^p$:

$$\int |g|^q = \int |f|^p < \infty$$

The same computation fixes the multiple: $\int fg = \int |f|^p = \|f\|_p^p$ while $\|g\|_q = \|f\|_p^{p/q} = \|f\|_p^{p-1}$, so the unit-norm sensor is represented by $\|f\|_p^{1-p}\, g$, and indeed $\varphi_f(f) = \|f\|_p^{1-p}\int fg = \|f\|_p$.

For the geometry to come, however, it is the unnormalized representer that matters. We define the *duality map*:

$$J_p \colon L^p \to L^q, \qquad J_p(f) = |f|^{p-1}\mathrm{sgn}(f)$$

[4]The reasoning can extend to nuclear Fréchet spaces, which admit descriptions as projective limits of Hilbert spaces. For general metric spaces, a different approach is needed via CAT(k) geometry; see [10] for details. For complex scalars, replace $\mathrm{sgn}(f)$ below by $\overline{\mathrm{sgn}(f)}$; at $p = 2$ the duality map then becomes conjugation $f \mapsto \overline{f}$, which is why the Hermitian inner product conjugates one of its slots.

so that $J_p(f)$ represents the functional $\|f\|_p^{p-1}\varphi_f$ for $f \neq 0$, and $J_p(0) = 0$. As $(L^p)^* \cong L^q$, we can equivalently view this as a map $J_p \colon L^p \to (L^p)^*$, $f \mapsto \big(h \mapsto \int J_p(f)\, h\big)$.

The key is to realize that the duality map J_p is the *geometric normal* to the level sets of the p-energy. To see this, consider the function:

$$F(x) = \frac{1}{p}\|x\|_p^p = \frac{1}{p}\sum |x_i|^p$$

(where the ℓ^p norm is used for visualization, but the logic holds for integrals). From multivariable calculus, the unit sphere is the level set $F(x) = \frac{1}{p}$, and the gradient $\nabla F(x)$ is the vector strictly perpendicular (normal) to the level set at point x. Computing the gradient (the Fréchet derivative):

$$\frac{\partial}{\partial x_i}\left(\frac{1}{p}|x_i|^p\right) = |x_i|^{p-1}\mathrm{sgn}(x_i)$$

So, the gradient vector is:

$$\nabla F(x) = \big(|x_1|^{p-1}\mathrm{sgn}(x_1), \dots, |x_n|^{p-1}\mathrm{sgn}(x_n)\big)$$

This is exactly our duality map:

$$\boxed{\nabla F(x) = J_p(x)}$$

This identification is the key to the entire theory. It tells us that the optimal functional for f (the element of $(L^p)^*$ that best "measures" f) points precisely along the gradient of the energy at f: the direction of steepest increase in p-energy. The duality map is a *geometric* object, not merely an algebraic one.

The p-Pairing and Its Interpretation. With the gradient in hand, we can now answer our original question: how does the p-energy $F(f)$ change under a perturbation by g? The first-order variation is[5]:

$$dF(f; g) = \langle \nabla F(f), g\rangle_{\mathrm{dual}} = \int \nabla F(f) \cdot g$$

Substituting $\nabla F = J_p(f)$, this defines a natural pairing:

$$\langle f, g\rangle_p := \int J_p(f) \cdot g$$

Note that it is important to pass through $J_p(f)$ since $\int fg$ need not be finite in general. However, by Hölder's inequality:

$$\int |J_p(f) \cdot g| < \infty$$

This pairing has a precise geometric interpretation. The gradient $J_p(f)$ is a cotangent vector (a 1-form $dF|_f \in T_f^* L^p$) and the perturbation g is a tangent vector $g \in T_f L^p$. Thus the pairing is the evaluation of a 1-form on a tangent vector, under the canonical evaluation $T_f^* L^p \times T_f L^p \to \mathbb{R}$:

$$\langle f, g\rangle_p = dF|_f(g)$$

[5]The dF here can be thought of as the Gâteaux derivative, commonly defined for functions between locally convex topological vector spaces.

This canonical evaluation pairing between the tangent and cotangent spaces exists in *every* geometric setting without any choice of metric. What J_p adds is the ability to *produce* a specific cotangent vector from a tangent vector, turning the pairing from a bilinear form on $T^*_f L^p \times T_f L^p$ into a (generally nonlinear) form on $T_f L^p \times T_f L^p$: that is, internalizing the measurement within the space itself.

The Infinitesimal Meaning of V^*. We are now in a position to make a precise statement about the geometric content of the dual space. In a bare vector space without additional structure, the dual V^* is merely an algebraic object: a collection of linear maps with no geometric significance beyond probing linearly. However, in L^p equipped with its norm, the identity $J_p = \nabla F$ reveals that *every element of* $(L^p)^*$ *arises as the gradient of the energy functional*, a differential $dF|_f$. This endows $(L^p)^*$ with genuine infinitesimal meaning: its elements are not just linear maps, but *cotangent vectors* that encode the first-order sensitivity of the geometry.

Crucially, cotangent vectors are a different species of infinitesimal object than tangent vectors. A tangent vector $g \in T_f L^p \cong L^p$ represents an *infinitesimal direction of motion*: it answers "which way?". A cotangent vector $\varphi \in T^*_f L^p \cong L^q$ represents an *infinitesimal measurement*: it answers "how fast is this quantity changing?". These are related by the evaluation pairing $\varphi(g) = dF|_f(g)$, but they transform in opposite ways: tangent vectors push forward under maps, while cotangent vectors pull back. The duality map $J_p : T_f L^p \to T^*_f L^p$ provides a (nonlinear) identification between these two types of infinitesimal data, allowing us to transfer the directional interpretation onto the dual space. However, this transfer *depends essentially on the norm geometry*: it is canonical only for $p = 2$.

Geometrically, if we view the identity map $f \mapsto f$ as the radial vector field (a *section* of the tangent bundle TL^p), then the duality map J_p transforms this into a covector field (a *section* of the cotangent bundle T^*L^p). In the language of Riemannian geometry, this transformation is the *musical isomorphism* $\flat$ (flat), which maps a vector field X to a 1-form $X^\flat$ by lowering indices via the metric tensor. The four corollaries that follow are all consequences of this single geometric fact.

Corollary 1: Nonlinearity of the Musical Isomorphism. The duality map is *not* linear:

$$J_p(2f) = |2f|^{p-1}\mathrm{sgn}(2f) = 2^{p-1}|f|^{p-1}\mathrm{sgn}(f) = 2^{p-1}J_p(f)$$

This is only linear in the special case of $p = 2$, since:

$$J_2(nf) = |nf|^{2-1}\mathrm{sgn}(nf) = nJ_2(f)$$

In a general Banach space ($p \neq 2$), the musical isomorphism is *nonlinear*: the operation of lowering an index depends on the magnitude of the vector itself, meaning the "metric" changes as we move through the space. However, when $p = 2$, this map is linear. This implies a linear bundle isomorphism $TL^2 \cong T^*L^2$, effectively trivializing the distinction between vectors and covectors. Only in L^2 is the musical isomorphism a linear operator, meaning the "force" required to correct an error is linearly proportional to the error itself.

Corollary 2: Non-Symmetry of the Pairing. The pairing $\langle f, g\rangle_p$ is *not* symmetric for $p \neq 2$. This is an immediate consequence of the nonlinearity of J_p: the 1-form $dF|_f$ evaluated on g has no reason to equal the 1-form $dF|_g$ evaluated on f, since the gradient depends nonlinearly on its base point. If however $p = 2$, then $J_p(x)$ maps identically to itself and we recover the symmetric form $\int fg$. This asymmetry is the first signal that the geometry of L^p is *not flat* for $p \neq 2$.

Corollary 3: Failure of Rotational Invariance. This lack of symmetry in the pairing $\langle f, g\rangle_p$ is directly linked to the fact that L^2 is the unique L^p space that is *rotationally invariant.* Geometrically, rotational invariance means that the properties of a vector depend only on its length, not on its alignment with the coordinate axes. Mathematically, this requires that the duality map commutes with rotations. If R is a Euclidean rotation, we expect the "normal vector" of the rotated point to be the rotated "normal vector" of the original point:

$$J_p(Rf) \stackrel{?}{=} R(J_p(f))$$

In L^2, since J_2 is linear (essentially the identity), this holds trivially: $J_2(Rf) = Rf = R(J_2(f))$. However, for $p \neq 2$, the map $f \mapsto |f|^{p-1}\text{sgn}(f)$ applies a non-linear distortion component-wise. Rotating the vector mixes the components, and applying the power law *after* mixing is not the same as applying it *before*.

To see this concretely, consider ℓ^3 ($p = 3$) in $\mathbb{R}^2$ and let $f = (1, 0)$. If we first compute the functional and then rotate by $\pi/4$ (45°), we get:

$$R(J_3(f)) = R(1^2, 0^2) = R(1, 0) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

However, if we rotate the vector first, the components become $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$. Applying the duality map J_3 (squaring components) yields:

$$J_3(Rf) = J_3\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = \left(\frac{1}{2}, \frac{1}{2}\right)$$

Since $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}) \neq (\frac{1}{2}, \frac{1}{2})$, we have $J_p(Rf) \neq R(J_p(f))$. This confirms that for $p \neq 2$, the geometry is *anisotropic*: the shape of the unit ball looks different depending on the angle from which you view it. Equivalently, for $p \neq 2$ a Euclidean rotation is not an isometry of ℓ^p; the genuine isometries of ℓ^p (signed permutations of the coordinates) do commute with J_p.

Corollary 4: Birkhoff-James Orthogonality. With this understanding of the position-dependent metric, we can clarify the notion of orthogonality. In Euclidean geometry, orthogonality is symmetric ($x \perp y \iff y \perp x$) and additive. In general Banach spaces, without a global bilinear form, we must rely on *Birkhoff-James orthogonality*: we say $x \perp y$ if moving in the direction of y does not decrease the length of x ($\|x + \lambda y\| \geq \|x\|$ for all scalars λ). Geometrically, this means the line passing through x in the direction of y is tangent to the unit sphere at x.

Analytically, utilizing the duality map J_p, this condition is equivalent to requiring that the tangent vector y be annihilated by the cotangent vector generated by x:

$$\langle J_p(x), y \rangle = 0$$

This is precisely the statement that y lies in the kernel of the 1-form $dF|_x$, a natural geometric condition. However, it reveals the fundamental asymmetry in non-Hilbert geometry. In L^2, because J_2 is linear, the condition is symmetric. However, in L^p ($p \neq 2$), the non-linearity of J_p breaks reciprocity. The set of vectors orthogonal to x (kernel of $J_p(x)$) does not align symmetrically with the set of vectors orthogonal to y. Consequently, it is common to have $x \perp y$ but $y \not\perp x$: in ℓ^3 on $\mathbb{R}^2$, take $x = (2, 1)$ and $y = (1, -4)$; then $\langle J_3(x), y \rangle = 4 \cdot 1 + 1 \cdot (-4) = 0$ while $\langle J_3(y), x \rangle = 1 \cdot 2 + (-16) \cdot 1 = -14 \neq 0$. This lack of symmetry illustrates why Hilbert spaces are the unique setting where Euclidean geometric intuition remains valid.

The Riemannian Perspective. The corollaries above all concern the first derivative of the energy (the gradient J_p). The full geometry of the space, however, is determined by how this gradient *changes*: the second derivative, which acts as the metric tensor. In this framework, the metric tensor $g_{ij}(x)$ at a point x dictates how to measure the length of tangent vectors. In our context, this metric is generated by the Hessian of the energy function $F(x) = \frac{1}{p}\|x\|_p^p$. For L^2, this Hessian is proportional to the identity operator everywhere; the metric is *constant* (flat). However, for L^p ($p \neq 2$), the second derivative depends on position x, specifically scaling as $|x|^{p-2}$. This implies that the "metric" changes as we move around the space.

It is worth noting why the theory breaks down at the boundary cases. For L^1, the unit ball is a diamond with corners; at these corners, the gradient is not unique (it is a set-valued subgradient). For L^∞, the unit ball has flat faces; on these faces, the gradient is non-unique or vanishes in certain directions. This lack of smoothness (L^1) or rotundity (L^∞) prevents a well-defined duality map, leaving L^p ($1 < p < \infty$) as the sweet spot for smooth geometry.

Topological vs. Geometric Structure. To visualize the practical consequences of this non-flat geometry, it is helpful to consider the "shape" of the unit ball in infinite dimensions. In finite dimensions, all norms are equivalent, meaning the unit ball of one norm can always be fit inside a scaled unit ball of another. In infinite dimensions, this equivalence breaks. The curvature of the norm dictates how "energy" is allowed to concentrate without bound. The L^∞ unit ball acts like a strict box, while the L^1 unit ball allows for infinitely long thin spikes.

Ideally, we would like to know if these distortions are topological or "merely" geometric. This is answered by the *Mazur map*:

$$M_{p,q} \colon L^p \to L^q, \qquad M_{p,q}(f) = \operatorname{sgn}(f) \cdot |f|^{p/q}$$

which restricts to a *uniform* homeomorphism between the unit spheres of L^p and L^q: we can continuously deform the "spikes" of the L^1 sphere into the round sphere of L^2, with uniform control in both directions. The distinction between spheres and whole spaces here is essential: as topological spaces all separable infinite-dimensional L^p are homeomorphic (theorem 1.5.5), but the whole spaces are *not* uniformly homeomorphic to one another for $p \neq q$, since uniform homeomorphisms preserve enough local linear structure to distinguish the L^p scale (a rigidity phenomenon initiated by Ribe; see Benyamini and Lindenstrauss [1] for both the Mazur map and this rigidity theory). However, while the Mazur map preserves the uniform structure of the spheres, it destroys the linear structure (additivity): it is homogeneous of degree p/q rather than degree one. The special role of L^2 in this family is thus not topological but geometric: it is the unique L^p whose duality map is linear, as the preceding analysis showed.

This geometric disparity explains why convergence depends on the norm. Convergence $f_n \to 0$ is not about movement, but about entering arbitrarily small ϵ-balls. Consider the sequence $f_n(x) = x^n$ on $[0, 1]$. In the L^1 geometry, this sequence travels down one of the "spikes" of the unit ball: the functions get thinner and thinner, their area shrinking to zero despite their height remaining constant. In the L^∞ geometry, however, these functions are trapped on the surface of the unit sphere; they cannot enter the ϵ-ball because the L^∞ "box" does not extend down the "spikes" of concentration.

Consequences for Hilbert Spaces. The unique geometry of Hilbert spaces (self-duality, isotropy, and the inner product) unlocks powerful results unavailable in general Banach spaces. The preceding analysis explains *why*: in L^2, the duality map J_2 is the identity, making the musical isomorphism linear, the pairing symmetric, and the geometry flat. The following are key consequences we will explore:

1. *The Projection Theorem (Best Approximation):* In general Banach spaces, finding the element in a subspace M closest to a point x is difficult (the closest point might not be unique, e.g.,

the flat edge of L^1). In Hilbert spaces, the strict convexity and "roundness" of the unit ball guarantee that for any closed subspace M, there exists a *unique* element $P_M x$ minimizing the distance $\|x - m\|$. Furthermore, the error vector $x - P_M x$ is orthogonal to the subspace.

2. *The Riesz Representation Theorem (Self-Duality):* The duality map $J\colon H \to H^*$, $y \mapsto \langle -, y\rangle$, is bijective and conjugate-linear (linear, over the reals). This means we can completely identify linear functionals with vectors. Every continuous linear functional φ acts exactly like an inner product: $\varphi(x) = \langle x, y\rangle$ for a unique y. This collapses the study of H^* to the study of H.

3. *Orthonormal Bases and Fourier Series:* In L^p, we may have a Schauder basis[6], but determining the coefficients is non-trivial. In Hilbert spaces, orthogonality allows us to decouple coefficients. This is key for Fourier series, where any element is reconstructed by projection: $x = \sum \langle x, e_n\rangle e_n$. Additionally, the *Parseval Identity* allows us to measure size by summing squared coefficients: $\|x\|^2 = \sum |\langle x, e_n\rangle|^2$.

4. *Existence of Adjoints:* In Banach spaces, the adjoint of $T : X \to Y$ maps $Y^* \to X^*$. Because Hilbert spaces are self-dual, we can pull this operator back to the original space. Thus, for every bounded operator T on a Hilbert space, there is a unique adjoint T^* on the *same space* satisfying $\langle Tx, y\rangle = \langle x, T^* y\rangle$. This is foundational for spectral theory.

In summary, while Banach spaces provide the necessary topological framework (completeness), Hilbert spaces add the rigid geometric structure required to generalize linear algebra to infinite dimensions. The duality map J_p, understood as the gradient of the p-energy, reveals the precise mechanism: it is the *cotangent structure* of the space (the infinitesimal measurement theory) that determines the geometry. The canonical evaluation pairing between tangent and cotangent vectors is the universal source of "derivative-like" phenomena; J_p is the geometric data that internalizes this pairing within the space itself. In L^2 alone, this internalization is linear, making vectors and covectors (directions and measurements) two aspects of a single object.

5.1 Basic definition and properties

Definition 5.1.1: Inner Product Space

Let H be a complex vector space with an inner product $\langle \cdot, \cdot \rangle : H \times H \to \mathbb{C}$ where

1. $\langle x, y\rangle = \overline{\langle y, x\rangle}$
2. $\langle x + y, z\rangle = \langle x, z\rangle + \langle x, z\rangle$
3. $\langle ax, y\rangle = a\langle x, y\rangle$, $a \in \mathbb{C}$
4. $\langle x, x\rangle \in (0, \infty)$ for all nonzero $x \in H$
5. $\langle x, x\rangle = 0$ if and only if $x = 0$

We'll define $\|x\|_2 = \sqrt{\langle, x, x\rangle}$ where the 2 subscript hints that this will become the L^2-norm and is added to not hold ambiguity with the L_1 norm. This is naturally a norm since:

[6] A sequence $\{b_n\}$ where every $v \in V$ has a unique expansion $v = \sum \alpha_n b_n$.

1. $\|x\|_2 = \sqrt{\langle x, x\rangle} = 0$ if and only if $x = 0$
2. $\|ax\|_2 = \sqrt{\langle ax, ax\rangle} = \sqrt{a\bar{a}\langle x, x\rangle} = \sqrt{a^2\langle x, x\rangle} = |a|\sqrt{\langle x, x\rangle} = |a|\|x\|_2$

Proving the triangle inequality is a bit more difficult: IF we square both sides, we it is equivalent to show:

$$\|x + y\|_2^2 \leq \|x\|_2^2 + 2\|x\|_2\|y\|_2 + \|y\|_2^2$$

expanding out the left hand side, we get:

$$\|x + y\|_2^2 = \|x\|_2^2 + 2\mathrm{Re}\langle x, y\rangle + \|y\|_2^2$$

so we have to show that $\mathrm{Re}\langle x, y\rangle \leq \|x\|_2\|y\|_2$. In fact, a bit more is true: $|\langle x, y\rangle| \leq \|x\|_2\|y\|_2$. Though we are using this inequality to prove the triangle inequality, notice that this in fact gives us a general bound on the value of the inner product! This inequality is one of the central tools used in the study of Hilbert spaces:

Theorem 5.1.2: Cauchy-Schwarz Inequality

$$|\langle x, y\rangle| \leq \|x\|_2\|y\|_2$$

***Proof*:**
If $\|y\|_2 = 0$ then $y = 0$, and then $|\langle x, 0\rangle| = 0$, so assume $\|y\|_2 \neq 0$. Then for all $t \in \mathbb{R}$

$$\begin{aligned} 0 &\leq \langle x - ty, x - ty\rangle \\ &= \|x\|_2^2 + 2t\mathrm{Re}\langle x, y\rangle + t^2\|y\|_2^2 \\ &= \|x\|_2^2 + \frac{2\mathrm{Re}\langle x, y\rangle^2}{\|y\|_2^2} + \frac{\mathrm{Re}\langle x, y\rangle^2}{\|y\|_2^2} && t = \frac{\mathrm{Re}\langle x, y\rangle}{\|y\|_2^2} \\ \Rightarrow |\mathrm{Re}\langle x, y\rangle| &\leq \|x\|_2\|y\|_2 \end{aligned}$$

To finish the proof, pick $\alpha \in \mathbb{C}$ with $|\alpha| = 1$ such that

$$|\mathrm{Re}(\alpha\langle x, y\rangle)| = |\langle x, y\rangle|$$

Notice this does not break the inequality, and so we can repeat the entire process but with α taken into account with t, and so:

$$|\langle x, y\rangle \leq \|x\|_2\|y\|_2$$

as we sought to show

With this, the rest of the proof that $\|-\|_2$ induced by $\langle -, -\rangle$ is a norm follows

Theorem 5.1.3: Triangle Inequality For Induced Norm

$$\|x + y\|_2 \leq \|x\|_2 + \|y\|_2$$

***Proof*:**

$$\begin{aligned}\|x+y\|_2^2 &= \langle x+y, x+y\rangle \\ &\leq \|x\|_2^2 + 2\text{Re}\langle x,y\rangle + \|y\|_2^2 \\ &\leq \|x\|_2^2 + 2\|x\|_2\|y\|_2 + \|y\|_2^2 \\ &= (\|x\|_2 + \|y\|_2)^2\end{aligned}$$

The Cauchy-Shwartz inequality is the tool we use to bring in geometry as we know it into Hilbert space. For example, in real inner product spaces, we will define the angle between two vectors to be θ where θ satisfies:

$$\langle u, v\rangle = \cos(\theta)\|u\|_2\|v\|_2$$

Many other cool properties of the Cauchy-Shwartz inequality can be found here: (commented)

Now, notice we can define a metric by doing $\rho(x,y) = \|x-y\|_2$:

1. $\rho(x,x) = \|x-x\|_2 = 0$ if and only if $x - x = 0$ so $x = x$
2. $\rho(x,y) = \|x-y\|_2 \geq 0$
3. $\rho(x,y) = \|x-y\|_2 = \|x-y+y-z\|_2 \leq \|x-y\|_2 + \|y-z\|_2 = \rho(x,y) = \rho(y,z)$

Thus, we can define a natural topology on any pre-Hilbert space. If that topology is complete, we give it a name:

Definition 5.1.4: Hilbert Space

Let H be an inner product space. Then H is a Hilbert space if it is *complete* with respect to the induced metric.

Example 5.1.5: Hilbert Spaces

1. $\mathbb{R}^n$ is a (real) Hilbert space with the inner product being:

   $$\langle x, y\rangle = \sum_i x_i y_i$$

   verifying this is a Hilbert space is left as an exercise.

2. $L^2_\mu(X)$ for a complex measure μ is a Hilbert space with the inner product being:

   $$\langle f, g\rangle_{L^2} = \int_X f(x)\overline{g(x)}d\mu(x)$$

   To show it's well-defined, recall that if $a, b \geq 0$:

   $$2|ab| = 2|a||b| \leq |a|^2 + |b|^2$$

   Thus:

   $$|f(x)\overline{g(x)}| = |f(x)||g(x)| \leq |f(x)|^2 + |g(x)|^2$$

As integrating keeps inequality when integrating positive functions:

$$\|f\overline{g}\|_1 \leq \int_X |f(x)|^2 + \int_X |g(x)|^2 < \infty$$

showing it is well-defined. Alternatively, using Cauchy-Schwarz:

$$\|f\overline{g}\|_1 \leq \|f\|_2\|g\|_2 < \infty$$

3. As an exercise, show that $\ell^2(\mathbb{Z})$ is a Hilbert Space.

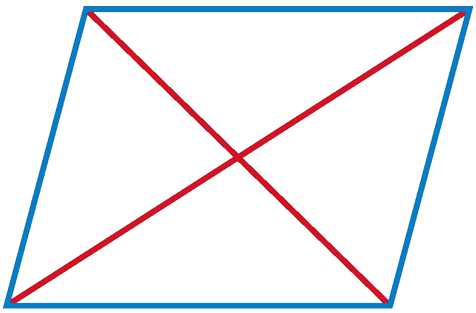
Hilbert spaces are very euclidean in nature, for example:

Theorem 5.1.6: Parallelogram Law

Let $\|-\|$ be a norm. Then $\|-\|$ is induced by an inner product if and only if:

$$\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2)$$

Visually, this means that the “area” of a parallelogram is can be measured in two ways:



***Proof*:**

$$\begin{aligned}
\|x+y\|^2 + \|x-y\|^2 &= \langle x+y, x+y\rangle + \langle x-y, x-y\rangle \\
&= 2\langle x,x\rangle + \langle x,y\rangle - \langle x,y\rangle + \langle y,x\rangle - \langle y,x\rangle + 2\langle y,y\rangle \\
&= 2\langle x,x\rangle + 2\langle y,y\rangle \\
&= 2\|x\|^2 + 2\|y\|^2
\end{aligned}$$

It turns out that a norm satisfying the parallelogram law will in fact define an inner product. This shows how norms define geometries that are more general than euclidean:

Example 5.1.7: Exploring Parallelogram Law

1. (The polarization Identity)$\mathcal{H}$ be a Hilbert space. For any $x, y \in \mathcal{H}$,

$$\langle x, y\rangle = \frac{1}{4}(\|x+y\|^2 - \|x-y\|^2 + i\|x+iy\|^2 - i\|x-iy\|^2)$$

Proof. This is simply a matter of computation:

$$\begin{aligned}
&\frac{1}{4}(\|x+y\|^2 - \|x-y\|^2 + i\|x+iy\|^2 - i\|x-iy\|^2)\\
&= \frac{1}{4}(\langle x+y, x+y\rangle - \langle x-y, x-y\rangle + i\langle x+iy\rangle - i\langle x-iy, x-iy\rangle)\\
&= \frac{1}{4}(\langle x,x\rangle + \langle x,y\rangle + \langle y,x\rangle + \langle y,y\rangle - (\langle x,x\rangle - \langle x,y\rangle - \langle y,x\rangle + \langle y,y\rangle)\\
&\quad + i(\langle x,x\rangle + \langle x,iy\rangle + \langle iy,x\rangle + \langle iy,iy\rangle - (\langle x,x\rangle - \langle x,iy\rangle - \langle iy,x\rangle + \langle iy,iy\rangle))\\
&= \frac{1}{4}(2\langle x,y\rangle + 2\langle y,x\rangle + i(2\langle x,iy\rangle + 2\langle iy,x\rangle))\\
&= \frac{1}{4}(2\langle x,y\rangle + 2\langle y,x\rangle + i(-i)2\langle x,y\rangle + 2ii\langle y,x\rangle)\\
&= \frac{1}{4}(2\langle x,y\rangle + 2\langle y,x\rangle + 2\langle x,y\rangle - 2\langle y,x\rangle)\\
&= \frac{1}{4}4\langle x,y\rangle\\
&= \langle x,y\rangle
\end{aligned}$$

□

2. If $\mathcal{H}'$ is another Hilbert space, a linear map from $\mathcal{H}$ to $\mathcal{H}'$ is unitary if and only if it is isometric and surjective

Proof. Let U be unitary so that $\langle Ux, Uy\rangle_2 = \langle x, y\rangle_1$ for any $x, y \in \mathcal{H}$. Letting $x = y$ gives us $\|Ux\|_2 = \|x\|_1$, giving us that it's an isometry. Since U is invertible, it is surjective, and so U is a surjective isometry.

Conversely, let U be a surjective isometry. We need to show its invertible (it suffices to show it's injective), an isometry, and that U and U^{-1} are bounded. Clearly, U is bounded since $\|U(x)\| \leq 1\|x\|$. For injectivity, let $Ux = Uy$, so $Ux - Uy = 0$. Taking the norm on both sides, we get:

$$0 = \|0\|_2 = \|Ux - Uy\|_2 = \|U(x-y)\|_2 = \|x-y\|_1$$

where the last equality is due to U being an isometry. Since $\|z\| = 0$ if and only if $z = 0$, we have $x - y = 0$, i.e. $x = y$, proving that U is injective. Along with surjectivity, that makes U invertible. With this, we can also show that U^{-1} is bounded: for any $y \in \mathcal{H}'$, there exists an $x \in \mathcal{H}$ such that $U(x) = y$. Thus:

$$\|U^{-1}(y)\| = \|x\| \overset{!}{=} \|U(x)\| = \|y\|$$

where $\overset{!}{=}$ comes from U being an isometry, showing that $\|U^{-1}(y)\| \leq 1 \cdot \|y\|$, making it also bounded.

Finally, to show that $\langle Ux, Uy\rangle_2 = \langle x, y\rangle_1$, we will use part (a):

$$\begin{aligned}\langle x, y\rangle &= \frac{1}{4}(\|x+y\|^2 - \|x-y\|^2 + i\|x+iy\|^2 - i\|x-iy\|^2) \\ &\overset{!}{=} \frac{1}{4}(\|U(x+y)\|^2 - \|U(x-y)\|^2 + i\|U(x+iy)\|^2 - i\|U(x-iy)\|^2) \\ &= \frac{1}{4}(\|U(x)+U(y)\|^2 - \|U(x)-U(y)\|^2 + i\|U(x)+iU(y))\|^2 - i\|U(x)-iU(y)\|^2) \\ &= \langle U(x), U(y)\rangle\end{aligned}$$

where $\overset{!}{=}$ comes from U being an isometry. Thus:

$$\langle Ux, Uy\rangle = \langle x, y\rangle$$

As we sought to show. □

Proposition 5.1.8: Inner Product is Continuous

Let V be a Hilbert space. Then $\langle -, -\rangle : V \times V \to \mathbb{R}$ (or $\mathbb{C}$) is continuous

***Proof*:**
As $V \times V$ has the product topology, it is enough to show that if $x_n \to x$ and $y_n \to y$ in V then

$$\langle x_n, y_n\rangle \to \langle x, y\rangle$$

Take:

$$\begin{aligned}|\langle x, y\rangle - \langle x_n, y_n\rangle| &= |\langle x, y\rangle - \langle x_n, y\rangle + \langle x_n, y\rangle - \langle x_n, y_n\rangle| \\ &= |\langle x - x_n, y\rangle + \langle x_n, y - y_n\rangle| \\ &\leq |\langle x - x_n, y\rangle| + |\langle x_n, y - y_n\rangle| \qquad \leq \|x - x_n\|\|y\| + \|x_n\|\|y - y_n\| \\ &\to 0\end{aligned}$$

where the last line comes since $x_n \to x$ and $y_n \to y$, completing the proof.

The next concept we consider is a way of defining two vectors being somehow "unrelated" or going in some notion of "direction" that doesn't give information for the other vector:

Definition 5.1.9: Orthogonal

We say that x is *orthogonal* to y if $\langle x, y\rangle = 0$. We will write $x \perp y$

If two vectors are orthogonal, the triangle inequality becomes instant:

Theorem 5.1.10: Pythagoras

$$x \perp y \Rightarrow \|x+y\|^2 = \|x\|^2 + \|y\|^2$$

***Proof*:**

$$\|x+y\|^2 = \langle x+y, x+y\rangle = \|x\|^2 + \|y\|^2$$

We might wonder if we take $E \subseteq H$, can we find all vectors in H that are somehow "dual" to E? Define it's *orthogonal compliment* to be:

$$E^\perp = \{x \in H \mid \langle x, y\rangle = 0 \ \forall y \in E\}$$

Notice that E is *always* a closed subspace:

Proof. The fact it's a subspace follows from linearity of the second component.t

for closed, suppose there is a sequecne $\{x_n\} \subseteq E^\perp$ where $x_n \to x \in H$. Then:

$$|\langle x, y\rangle| \leq |\langle x - x_n, y\rangle| \leq \|x - x_n\|\|y\| \to 0 \quad \text{as } n \to \infty$$

□

We can also find closed (and not closed) subspace of hilbert spaces:

Example 5.1.11: Subspaces Examples

1. If $\mathcal{H}$ is a finite-dimensional Hilbert space, then all subspace are closed
2. If $\mathcal{H}$ is infinite-dimensional, consider $\mathcal{H} = L^1([0,1])$ with the L^2 inner product. and take $P([0,1]) \subseteq L^1([0,1])$ to be the subspace of all polynomial functions. This is is certainly not equal to $L^1([0,1])$, however polynomial functions are dense in $L^1([0,1])$ (they can arbitraily approximate continuous bump characteristic functions, and hence simple functions). However, their closure is $L^1([0,1])$, showing that $P([0,1])$ cannot be closed.
3. Assume $E \subseteq \mathcal{H}$ is an open subspace. Then since $0 \in E$, there exists a $r > 0$ such that

$$B_r(0) \subseteq E$$

However, for any nonzero element $x \in \mathcal{H}$,

$$\frac{r}{2\|x\|}x \in B_r(0) \subseteq E$$

Since E is closed under scalar multiplication, this shows that $x \in E$, so $\mathcal{H} \subseteq E$, so $\mathcal{H} = E$, showing that the only open subspace of $\mathcal{H}$ is $\mathcal{H}$ itself.

Thus, we are interested in closed subspace since are closed under limits, meaning we can use our tools from analysis to get the following decomposition result:

Theorem 5.1.12: Hilbert Space Decomposition

Let $E \subseteq \mathcal{H}$ be a closed subspace. Then:

$$\mathcal{H} = E \oplus E^\perp$$

In particular, for any $x \in H$, there is a unique $y \in E$ and $z \in E^\perp$ such that $x = y + z$.

***Proof*:**
Let $x \in \mathcal{H}$, and let $\delta = \inf\{\|x-y\| = \rho(x,y) \mid y \in E\}$ (which exists by completeness of $\mathbb{R}^n$ and $\mathcal{H}$). let $\{y_n\} \subseteq E$ be a sequence such that $\|x-y_n\| \to \delta$. We will show that $\{y_n\}$ is a cauchy sequence, and then find a limit point for it. By the parralelogram law:

$$2(\|x-y_n\| + \|x-y_m\|) = \|y_n - y_m\|^2 + \|(y_n+y_n) - 2x\|^2$$

Since $y_n + y_m \in E$, for sufficinetly large N so that $n, m > N$:

$$\begin{aligned}\|y_n - y_m\|^2 &= 2\|y_n - x\|^2 + 2\|y_m - x\|^2 - 4\|\frac{1}{2}(y_n + y_n) - x\|^2 \\ &\leq 2\|y_n - x\|^2 + 2\|y_m - x\|^2 - 4\delta^2\end{aligned}$$

and as $n, m \to \infty$, this sequence goes to $4\delta^2 - 4\delta^2 = 0$, and so $\|y_n - y_m\|\rho(y_n, y_m) \to 0$, meaning $\{y_i\}_i$ is a cauchy sequence. Take $y \lim y_n$. Then since E is closed, $y \in E$. By construction $\|x-y\| = \delta$. Now let $z = x - y$.

I claim that $z \in E^\perp$, that is $\langle z, u\rangle = 0$ for all $u \in E$. So, consider $\langle u, z\rangle$ for any $u \in E$. If $\langle u, z\rangle$ is complex, multiply it by a [complex] unit-length scalar to make it real ($\langle u, z\rangle = 0$ if and only if $a\langle u, z\rangle = 0$). With the fact that it's real, we will do a sneaky trick where we take advantage of optimization of differentiable functions. In particular, let

$$f : \mathbb{R} \to \mathbb{R} \quad f(t) = \|z + tu\|^2 = \|z\|^2 + 2t\langle z, u\rangle + \|u\|^2$$

f is certainly a differentiable function, being a polynomial with a square-root on $\mathbb{R}$. Furthermore, $z = x - y$ is the minimal value of $\|x-y\|$, and so the minimal value of $\|x-y\|^2$. It follows that $z + ut$ is not minimal, and so this equation is has a minimum when $t = 0$. Thus, we get $0 = f'(0) = 2\langle u, z\rangle$, implying that $\langle u, z\rangle = 0$. Since u was arbitrary, we get that $z \in E^\perp$.

To show that z is the minimal distance from x, let's say there was another $z' \in E^\perp$ that satisfied the equation. Then by the Pythagorean theorem:

$$\begin{aligned}\|x - z'\| &= \|(x-z) + (z-z')\|^2 \\ &= \|x-z\|^2 + \|z-z'\|^2 && x - y \in E, z - z' \in E^\perp \\ &\geq \|x-z\|^2\end{aligned}$$

So equality holds if and only if $z = z'$, meaning z is of minimal distance to x. A similar reasoning shows the same for y.

To show uniqueness, let's say $y + z = x = y' + z'$. Then $y - y' = z - z' \in E \cap E^\perp = \{0\}$, so $y - y' = z - z' = 0$, which can be used to show that $y = y'$ and $z = z'$, completing the proof.

We said that orthogonal vector are somehow dual to each other. However, there is the more classical notion of "dual" vector spaces. Let $\mathcal{H}^*$ be set of all linear functions from $\mathcal{H}$ to the underlying field (usually $\mathbb{C}$, but sometimes $\mathbb{R}$); such linear functions are called functionals. It turns out that all linear functionals can be written in the form $f_y(x) = \langle x, y\rangle$, and we can define a norm on $\mathbb{H}^*$ such that $\|f_y\| = \|y\|$. Thus, $y \mapsto f_y$ will define a conjugate-linear isometry from $\mathcal{H}$ to $\mathcal{H}^*$. The following theorem shows tha this map is surjective:

Theorem 5.1.13: Natural Isomorphism To $\mathcal{H}^*$ (Riesz Representation)

Let $f \in \mathcal{H}^*$. Then there exists a $y \in \mathcal{H}$ such that $f(x) = \langle x, y \rangle$ for all $x \in X$.

***Proof*:**
It easy easy to check uniqueness: if $\langle x, y \rangle = \langle x, y' \rangle$ for all x, then let $x = y - y'$. Manipulating the equation, we get that $\|y - y'\| = 0$, so $y = y'$.

If f is the zero functional, then take $y = 0$. If otherwise, then let $\mathcal{M} = \{x \mid f(x) = 0\}$ be the kernel of f. It is clearly a subspace, and any squence will also necesarily converge within $\mathcal{M}$, and so it is also closed. Furthermore, $\mathcal{M}$ is a proper subspace, since $f \neq 0$. Thus, $\mathcal{M}^\perp \neq \{0\}$ by the previous proof.

Now, pick any $z \in \mathcal{M}^\perp$ with $\|z\| = 1$. Then if we set $u = f(x)z - f(z)x$, we have that $u \in \mathcal{M}$. Hence:

$$0 = \langle u, z \rangle = f(x)\|z\|^2 - f(z)\langle x, z \rangle = f(x) - \langle x, \overline{f(z)}z \rangle$$

re-arrangine and setting $y = \overline{f(z)}z$, we get

$$f(x) = \langle y, z \rangle$$

as we sought to show.

The uniqueness makes this map to be injective, and by linearity of f and the first component of the inner product makes it is an isomorphism. Furthermore, since $\|f_y\| = \|y\|$, limits are also preserved, and so this in fact defines an isomorphism between $\mathcal{H} \cong \mathcal{H}^*$. Notice how this differs from vector-spaces where there is no natural isomorphism between V and V^* (there is one between V and V^{**}).

Since we are working over a vector-space, we should be able to define a basis. Since we are studying infinite dimensional vector-spaces, we often care more about "countable linear combination" rather than "finite linear combination" as we have for vector-spaces. To distinguish the two, we will call *Hamel basis* a set of linearly independent elements for which all *finite linear combinations* can represent every element of the vector space. We will call a *Schauder basis* a set of linearly independent elements for which all *countable inear combinations* can represent every element of the vector space. If every element in a Schauder basis has norm 1 and is orthogonal, we call it an *orthonormal basis*. Since we have an inner-product that is complete, it would be nicest if this vector-space has an orthonormal basis. We strive to this end in the following build-up:

Theorem 5.1.14: Bessel's Inequality

If $\{u_\alpha\}_{\alpha \in A}$ is an orthonormal set in $\mathcal{H}$, then for any $x \in \mathcal{H}$:

$$\sum_{\alpha \in A} |\langle x, u_\alpha \rangle|^2 \leq \|x\|^2$$

As a consequence, $\{\alpha \mid \langle x, u_\alpha \rangle \neq 0\}$ is countable.

***Proof*:**
It suffices to show that $\sum_{\alpha\in F}|\langle x,u_\alpha\rangle|^2 \leq \|x\|^2$ for any finite subset $F\subseteq A$, since if it fails for some countable amount, it will fail for some finite amount. Then we see that:

$$\begin{aligned}
0 &\leq \left\| x - \sum_{\alpha\in F}\langle x,u_\alpha\rangle u_\alpha \right\| \\
&= \|x\|^2 - 2\mathrm{Re}\left\langle x, \sum_{\alpha\in F}\langle x,u_\alpha\rangle u_\alpha \right\rangle + \left\| \sum_{\alpha\in F}\langle x,u_\alpha\rangle u_\alpha \right\| \\
&= \|x\|^2 - 2\sum_{\alpha\in F}|\langle x,u_\alpha\rangle|^2 + \sum_{\alpha\in F}|\langle x,u_\alpha\rangle|^2 && \text{Pythagorean Theorem} \\
&= \|x\|^2 - \sum_{\alpha\in F}|\langle x,u_\alpha\rangle|^2
\end{aligned}$$

and manipulating the final equation we get:

$$\sum_{\alpha\in F}|\langle x,u_\alpha\rangle|^2 \leq \|x\|^2$$

as we sought to show

This in particular is important for uncountable sets. In the following we establish when equality holds, and thus when an orthonormal set is an orthonormal basis

Theorem 5.1.15: Orthonormal Basis Condition

Let $\{u_\alpha\}_{\alpha\in A}$ be an orthonormal set in $\mathcal{H}$. Then the following are equivalent:

1. Finite linear combinations of $\{u_\alpha\}$ are dense in $\mathcal{H}$.
2. (Completeness) If $\langle x,u_\alpha\rangle = 0$ for all α, then $x=0$
3. For each $x\in\mathcal{H}$, $x=\sum_{\alpha\in A}\langle x,u_\alpha\rangle u_\alpha$ where the right hand side sum has only countably many nonzero terms and converges in the norm topology no matter the order of the terms
4. (Parseval's Identity) $\|x\|^2 = \sum_{\alpha\in A}|\langle x,u_\alpha\rangle|^2$ for all $x\in\mathcal{H}$

***Proof*:**

(1) $\Rightarrow$ (2) Assume $\langle f,u_\alpha\rangle = 0$. As finite linear combinations of $\{u_\alpha\}$ are dense in $\mathcal{H}$, for any $f\in\mathcal{H}$ there is a sequence g_n such that $\|f-g_n\| \xrightarrow{n\to\infty} 0$. As $\langle f,u_\alpha\rangle = 0$, then $\langle f,-g_n\rangle = 0$. By bi-linearity and Cauchy-Schwarz:

$$\|f\|^2 = \langle f,f\rangle = \langle f, f-g_n\rangle \leq \|f\|\|f-g_n\| \xrightarrow{x\to\infty} 0 \tag{5.1}$$

hence $f=0$

(2) $\Rightarrow$ (3) As the nonzero portion of the sum is countable, we can reduce to the countable

case. For any $f \in \mathcal{H}$, let:

$$S_n(f) = \sum_{k=1}^{n} a_k u_k \qquad a_k = \langle f, u_k \rangle$$

Let's first show $S_n(f) \to g$ for some $g \in \mathcal{H}$, then show $g = f$. First, we show an alternative proof of Bessel's identity with a satisfying trick. Notice that certainly $S_n(f) \perp (f - S_n(f))$. Thus:

$$\|f\|^2 = \|S_n(f)\|^2 + \|f - S_n(f)\|^2 = \sum_{k=1}^{n} a_k u_k + \|f\|^2$$

Taking $n \to \infty$ we get:

$$\|f\| \geq \sum_{k=1}^{\infty} a_k u_k$$

Importantly, $\sum_{k=1}^{\infty}$ *converges*, and hence $S_n(f)$ forms a Cauchy sequence in $\mathcal{H}$, so by completeness $S_n(f) \to g$. To show $f = g$, notice that for any k, for sufficiently large n:

$$\langle f - S_n(f), u_k \rangle = a_k - a_k = 0$$

Thus:

$$\langle f - g, u_k \rangle = 0 \qquad \forall k$$

so by (2), $f - g = 0$, so $f = g$.

(3) $\Rightarrow$ (4) Assuming (3) so that $= \sum_{k}^{\infty} \langle f, u_k \rangle$, then $\|f - S_n(f)\| \to 0$ so plugging this into equation (5.1):

$$\|f\|^2 = \sum_{k}^{\infty} |a_k|^2$$

(4) $\Rightarrow$ (1) Notice that $S_n(f)$ is a finite linear combinations elements u_k, hence using equation (5.1) again we get the desired implication.

Any orthonormal set satisfying these conditions is given a name:

Definition 5.1.16: Orthonormal Basis

Let $\{u_\alpha\}_{\alpha \in A}$ be an orthonormal set. Then if the set satisfies any of the 3 conditions in theorem 5.1.15, it is called an *orthonormal basis*.

An argument similar to the one for vector-spaces gives that every Hilbert spaces has an orthonormal basis, essentially an orthonormal basis is an orthonormal set that is maximal:

Theorem 5.1.17: Hilbert Space Basis

Every Hilbert space has an orthonormal basis.

***Proof*:**
Application of Zorn's lemma: The collection of ordered orthonormal sets ordered by inclusion has a maximal element, and maximality is equivalent to the first property of theorem 5.1.15.

Example 5.1.18: Orthonormal Basis
Given a Hilbert space $L^2(G)$ where $G = \mathbb{R}^n$ or $\mathbb{T}^n$, we would want to find an orthonormal basis to get decomposition of functions. go over all of these!

1. Legendre basis (can be thought of as "global" Taylor approximation) (must add integrate image. Also a good example of orthonormal basis that is not the Fourier one)

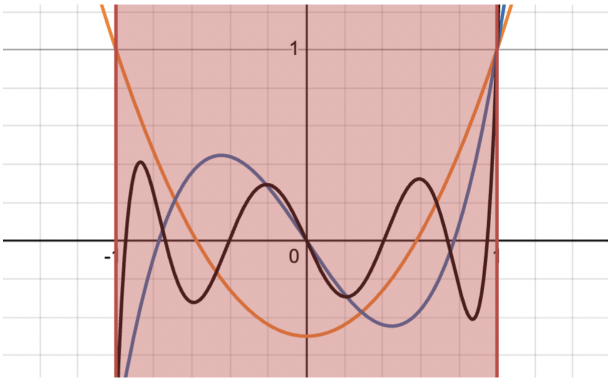


Figure 5.1: Legendre polynomials integrate to 0

2. Hermite basis
3. Jacobi Basis
4. Chebyshev

5.1.19 Basis On Banach Space Without an inner product, we cannot define an orthonormal basis on L^p. We can ask if we can at least define a *Schauder basis*, i..e any $f \in L^p$ is a (potentially) linear of a *single* sequence $\{b_n\}$ with different weights:

$$f = \sum a_n b_n$$

This was actually a very difficult problem and is related to the following question: is every compact operator on a Banach space the limit of finite operators. It isn't too hard to check this is the case in Hilbert spaces (exercise ref:HERE). However, this was proven in the negative by Per Enflo [6] (for a refinement of the proof see [8]). A consequence of this is that not every Banach space has a Schauder basis [2]. Another consequence of this paper for Banach spaces is that not all Banach space embed into $\ell^p(\mathbb{Z})$, $L^p(\mathbb{R}^d)$, or $L^p(K)$ for a compact set K; the family of Banach spaces is in fact larger (ex. $C(K)$ Is a Banach space not belonging to any of these spaces) .

Now that we have established a basis for Hilbert spaces, we will separate them into 3 categories: Those with finite, countable, and uncountable basis. In most cases, we will be working with countable basis due to the following property:

Proposition 5.1.20: Separable Hilbert Spaces

A Hilbert space $\mathcal{H}$ is separable if and only if it has a countable orthonormal basis, in which case every orthonormal basis for $\mathcal{H}$ is countable.

***Proof*:**
p. 176

Though all separable Hilbert spaces are "equivalent" (isometric), it is often the specific manifestation of the Hilbert space that matters as it frames how operators on the space transform functions (ex. the choice of basis, the choice of inner product).

Theorem 5.1.21: Enflo's Rigidity Theorem

A Banach space is uniformly homeomorphic to a Hilbert space if and only if it is linearly isomorphic to a Hilbert space.

***Proof*:**
This was a seminal result by Per Enflo (1969) [7]. It settled the question of whether the "uniform" structure of a space determines its linear structure in the context of Hilbert spaces. Since L^p (for $p \neq 2$) is not isomorphic to L^2, this theorem implies they are not uniformly homeomorphic either.

5.1.22 Hierarchy of Sameness It is useful to summarize how "same" two infinite-dimensional separable spaces can be. Let X and Y be infinite-dimensional separable Banach spaces (e.g., L^p and L^2).

1. Topological Homeomorphism : They are homeomorphic by the *Anderson-Kadec Theorem*, i.e. all such spaces are homeomorphic to $\mathbb{R}^{\mathbb{N}}$. They are topologically indistinguishable.
2. Uniform Homeomorphism: Generally no. By *Enflo's Rigidity Theorem*, if X is a Hilbert space and Y is not linearly isomorphic to it (like $L^p, p \neq 2$), they are *not* uniformly homeomorphic. *Note:* Their unit balls *are* uniformly homeomorphic via the **Mazur Map**, but this does not extend to the whole space.
3. Linear Isomorphism (The "Algebraic" level): Is there a linear bijection? Depends. L^p is not isomorphic to L^q for $p \neq q$.
4. Isometric Isomorphism (The "Geometry" level): They don't they preserve distance. The geometry of the unit ball (round vs. diamond vs. box) is strictly preserved here.

We next talk about the isomorphisms in the category of Hilbert spaces:

Definition 5.1.23: Unitary Map

Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be two Hilbert spaces with inner products $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$, and let $U : \mathcal{H}_1 \to \mathcal{H}_2$. Then U is called an *isometry* if if it is a linear map that preserves inner products:

$$\langle Ux, Uy \rangle_2 = \langle x, y \rangle_1$$

If it is also bijective, it is a *unitary map*.

Note that by taking $x = y$, we get that $\|x\|_1 = \|Ux\|_2$, i.e. it is an isometry. In fact, the converse is true too: every surjective isometry is unitary. With the definition of an isomorphism, we can see that all Hilbert spaces are isomorphic to $\ell^2(A)$ given basis A:

Theorem 5.1.24: Classifying Hilbert Spaces

Let $\{e_\alpha\}_{\alpha \in A}$ be an orthonormal basis for $\mathcal{H}$. Then the correspondence

$$x \mapsto \hat{x} \qquad \hat{x}(\alpha) = \langle x, u_\alpha \rangle$$

is a unitary map from $\mathcal{H}$ to $\ell^2(A)$.

***Proof*:**
We'll show it's a surjective isometry, which is equivalent to it being unitary.

First off, the map $x \to \hat{x}$ is linear since the inner product is linear in each component, and it is an isometry by Parseval identity: $\|x\|^2 = \sum |\hat{x}(x)|^2$.

For surjectivity, let $f \in \ell^2(A)$, so $\sum_{\alpha \in A} |f(\alpha)^2| < \infty$, so the Pythagorean shows that all partial sumns of the series $\sum f(\alpha) u_\alpha$ are Cauchy [a], so $x = \sum_\alpha f(\alpha) u_\alpha$ exists in $\mathcal{H}$ and $\hat{x} = f$.

[a] Note that only countably many terms are nonzero

As an application that ties together several threads from previous chapters, we close off with the proof a theorem of Grothendieck showing that the Hilbert space structure of L^2 imposes rigid constraints on subspaces of L^p. The result says, roughly, that a closed subspace of L^p cannot consist entirely of bounded functions unless it is finite-dimensional. The proof combines the closed graph theorem from the Baire category section (theorem 1.2.10), the L^p inclusion theory (proposition 2.1.13), and the Bessel inequality developed above.

Theorem 5.1.25: Grothendieck's Theorem on Closed Subspaces of L^p

Let $(X, \mathcal{F}, \mu)$ be a finite measure space, i.e. $\mu(X) < \infty$. Suppose that E is a closed subspace of $L^p(X, \mu)$ for some $1 \leq p < \infty$, and that $E \subseteq L^\infty(X, \mu)$. Then E is finite-dimensional.

***Proof*:**
Step 1: The L^∞ norm is controlled by the L^p norm on E. Consider the inclusion $\iota : (E, \|\cdot\|_p) \to L^\infty(X, \mu)$ defined by $\iota(f) = f$. We verify that its graph is closed. Suppose $f_n \to f$ in L^p with $f_n \in E$, and simultaneously $f_n \to g$ in L^∞. Since E is closed in L^p, $f \in E$. Convergence in L^p gives $f_{n_k} \to f$ a.e. along a subsequence, while convergence in L^∞ gives $f_n \to g$ a.e. Hence $f = g$

a.e., showing the graph is closed. By the closed graph theorem (theorem 1.2.10), ι is bounded: there exists $C > 0$ such that:

$$\|f\|_\infty \leq C\|f\|_p \qquad \text{for all } f \in E$$

Step 2: E carries a Hilbert space structure. Since $\mu(X) < \infty$ and $E \subseteq L^\infty$, the L^2 norm is finite on E. We show the L^p and L^2 norms are equivalent on E.

For one direction: $\|f\|_2^2 = \int |f|^2 \leq \|f\|_\infty^2 \mu(X) \leq C^2\|f\|_p^2\mu(X)$, giving $\|f\|_2 \leq C\mu(X)^{1/2}\|f\|_p$.

For the reverse: if $p \leq 2$, then Hölder directly gives

$$\|f\|_p \leq \mu(X)^{1/p-1/2}\|f\|_2$$

If $p > 2$, then interpolate:

$$\|f\|_p^p \leq \|f\|_\infty^{p-2}\|f\|_2^2 \leq (C\|f\|_p)^{p-2}\|f\|_2^2$$

so that $\|f\|_p^2 \leq C^{p-2}\|f\|_2^2$, i.e. $\|f\|_p \leq C^{(p-2)/2}\|f\|_2$.

In either case, the L^p and L^2 norms are equivalent on E. Since E is complete in $\|\cdot\|_p$, it is complete in $\|\cdot\|_2$, so $(E, \langle\cdot,\cdot\rangle_{L^2})$ is a Hilbert space. Write C' for a constant satisfying $\|f\|_\infty \leq C'\|f\|_2$ for all $f \in E$ (obtained by combining Step 1 with the norm equivalence).

Step 3: Bounding the dimension via Bessel's inequality. Let $\{e_1, \ldots, e_N\}$ be any orthonormal set in $(E, \|\cdot\|_2)$. Note that N is independent of the choice of C'; this will be the key fact to bound $\dim(E)$. For almost every $x \in X$, the evaluation functional $\varphi_x : E \to \mathbb{C}$ defined by $\varphi_x(f) = f(x)$ satisfies:

$$|\varphi_x(f)| \leq \|f\|_\infty \leq C'\|f\|_2$$

so φ_x is a bounded linear functional on the Hilbert space E with $\|\varphi_x\| \leq C'$. By the Riesz representation theorem (theorem 5.1.13), there exists $K_x \in E$ with $\varphi_x(f) = \langle f, K_x\rangle$ and $\|K_x\|_2 = \|\varphi_x\| \leq C'$. In particular,

$$\varphi_x(e_n) = e_n(x) = \langle e_n, K_x\rangle$$

for each n. Applying Bessel's inequality (theorem 5.1.14) to K_x:

$$\sum_{n=1}^N |e_n(x)|^2 = \sum_{n=1}^N |\langle K_x, e_n\rangle|^2 \leq \|K_x\|_2^2 \leq C'^2$$

Integrating over X and using $\|e_n\|_2 = 1$:

$$N = \sum_{n=1}^N \|e_n\|_2^2 = \int_X \sum_{n=1}^N |e_n(x)|^2 \, d\mu \leq C'^2\mu(X)$$

Since N was the size of an arbitrary orthonormal set, $\dim(E) \leq C'^2\mu(X) < \infty$.

The element K_x appearing in the proof is a *reproducing kernel*: it allows one to recover pointwise values of functions in E via the inner product. The argument shows that the existence of such a kernel, together with the finiteness of the measure, forces the space to be finite-dimensional. The bound $\dim(E) \leq C'^2\mu(X)$ is quantitative: the larger the ratio $\|f\|_\infty/\|f\|_2$ can be on E, the fewer orthonormal directions the space can accommodate.

It is worth pausing to appreciate how three seemingly unrelated tools conspire here. The closed graph theorem (theorem 1.2.10) upgrades the set-theoretic inclusion $E \subseteq L^\infty$ into a norm inequality. The L^p interpolation inequalities (proposition 2.1.13) transfer this to equivalence of all L^q norms on E, revealing a hidden Hilbert space. And Bessel's inequality (theorem 5.1.14) converts the uniform pointwise bound into a dimension bound via integration.

5.2 Operators on Hilbert Spaces

In this section, we try to generalize many notion's we've seen before, including:

1. adjoints, self-adjoint, and normal operators
2. sign and absolute value and when can we decompose a function into this form
3. diagonalizable
4. polar decomposition (think of how for every $z \in \mathbb{C}$, $z = \arg(z)|z|$)

Adjoints

Hilbert spaces have very simple dual structure, since $\mathbb{H} \cong \mathbb{H}^*$ in a very natural way with the inner product of $\mathbb{H}$. Thus, the problem of finding the structure of dual spaces is fully finished for hilbert spaces. Due to this very simple dual structure, we will be able to properly define an "opposite" operator given any operator $T : \mathbb{H}_1 \to \mathbb{H}_2$. In essense, there is an interesting symmetry in the structure of any map: Given a map $T : \mathbb{H}_1 \to \mathbb{H}_2$ between two hilbert spaces, there is a natural dual map $T^* : \mathbb{H}_2^* \to \mathbb{H}_1^*$. Hence, we have

$$(-)^* : \mathrm{Hom}(\mathbb{H}_1, \mathbb{H}_2) \to \mathrm{Hom}(\mathbb{H}_2^*, \mathbb{H}_1^*)$$

and since in Hilbert spaces $\mathbb{H}_i \cong \mathbb{H}_i^*$, we may envision this map as "flipping" the direction of arrows in a natural way:

$$(-)^* : \mathrm{Hom}(\mathbb{H}_1, \mathbb{H}_2) \to \mathrm{Hom}(\mathbb{H}_2, \mathbb{H}_1)$$

Not only does this map exist, but it is an isometric isomrophism (in **Ban**)! We will start with the special case of $\mathrm{Hom}(\mathbb{H}, k) = \mathbb{H}^*$. We will find that $(-)^*$ is a map

$$(-)^* : \mathrm{Hom}(\mathbb{H}, \mathbb{H}) \to \mathrm{Hom}(\mathbb{H}, \mathbb{H})$$

The following shows how we can define such a dual and some consequences:

Lemma 5.2.1: Endomorphism and And Inner Products

Let $\mathbb{H}$ be a complex hilbert space. Then there is a bijective isometric correspondence between $\mathrm{End}_k(\mathbb{H})$ and bounded sesquilinear forms on $\mathbb{H}$ given by tne map $T \to B_T$ where

$$B_T(x, y) = \langle x, T(y) \rangle$$

***Proof*:**
Let $T \in \mathrm{End}_k(\mathbb{H})$. Then certainly B_T is a sequilinear form on $\mathbb{H}$ bounded by $\|T\|$ since

$$\begin{aligned} \|B_T\| &= \sup\{|B_T(x,y)| \mid \|x\| \leq 1,\ \|y\| \leq 1\} \\ &= \sup\{|\langle x, T(y)\rangle| \mid \|x\| \leq 1, \|y\| \leq 1\} \\ &\leq \|T\| \end{aligned}$$

On the other hand

$$\begin{aligned} \|T(x)\|^2 &= \langle T(x), T(x)\rangle \\ &= B_T(T(x), x) \\ &\leq \|B_T\|\|T(x)\|\|x\| \\ &\leq \|B_T\|\|T\|\|x\|^2 \end{aligned}$$

showing that $\|T\| \leq \|B_T\|$, and so $\|T\| = \|B_T\|$.

Theorem 5.2.2: Existence Of Adjoint

Let $T \in \mathrm{End}_k(\mathbb{H})$. Then there exists a unique T^* in $\mathrm{End}_k(\mathbb{H})$ such that

$$\langle T(x), y\rangle = \langle x, T^*(y)\rangle$$

for all $x, y \in \mathbb{H}$. The map $T \mapsto T^*$ is conjugate linear, contravariant with respect to multiplication, an isometry of $\mathrm{End}_k(\mathbb{H})$ onto itself, an involution, and satisfies

$$\|T^*T\| = \|T\|^2$$

***Proof*:**
Fix $T \in \mathrm{End}_k(\mathbb{H})$, and let $B_T(x,y)$ be a continuous sesquilinear form defined as $(x,y) \mapsto \langle T(x), y\rangle$. Then by lemma 5.2.1, there exists a $T^* \in \mathrm{End}_k(\mathbb{H})$ such that $T^* \to B_T$. In particular

$$B_T(x,y) = \langle x, T^*(y)\rangle$$

Thus,

$$\langle T(x), y\rangle = \langle x, T^*(y)\rangle$$

giving us existence of such an operator. We now establish all the properties named in the theorem. By conjugate symmetry

$$\begin{aligned} \langle x, T^{**}(y)\rangle &= \langle T^*(x), y\rangle \\ &= \overline{\langle y, T^*(x)\rangle} \\ &= \overline{\langle T(y), x\rangle} \\ &= \langle x, T(y)\rangle \end{aligned}$$

Thus, T^{**} and T correspond to the same sesquilienar form, thus $T^{**} = T$. This also shows that $T \mapsto T^*$ is an involution. Furthermore:

$$\langle x(ST)^*y\rangle = \langle S(T(x)), y\rangle = \langle T(x), S^*(y)\rangle = \langle x, T^*(S^*(y))\rangle$$

thus $(S \circ T)^* = T^* \circ S^*$ showing $(-)^*$ is a contravariant endofunctor. Since the operator norm is submiltiplicative, we have $\|T^*T\| \leq \|T\|\|T^*\|$. On the other hand, applying $\angle T(x), y\rangle = \langle x, T^*(y)\rangle$ to T^* we get

$$\|T(x)\|^2 = \langle T(x), T(x)\rangle = \langle T^*(T(x)), x\rangle = \|T^*T\|\|x\|^2$$

showing that $\|T\|^2 \leq \|T^*T$
. Combining these we get $\|T\| \leq \|T^*\|$, hence $\|T\| = \|T^*\|$ (since $T^{**} = T$). This finishes the proof.

Note that if $\mathbb{H}$ is a finite dimensional complex Hilbert space, then each operator T has a matrix representation, say $(\alpha_{ij)})$. Then $T^* = (\overline{\alpha_{ji}})$, i.e. $T^* = \overline{T^t}$ where T^t is the transpose given the matrix representation.

Definition 5.2.3: Adjoint And Self-Adjoint

Let $T \in B(\mathbb{H})$ be a continuous linear operator between Hilbert spaces. Then T^* is called the *adjoint* of T. If $T = T^*$, then T is said to be *self adjoint* (or *Hermitian*).

Proposition 5.2.4: Adjoint Properties I

Let $T \in \mathrm{End}_k(\mathbb{H})$. Then
$$\ker(T^*) = (T(\mathbb{H}))^\perp$$

***Proof*:**
Since $\langle T(x), y\rangle = \langle x, T^*(y)\rangle$ we have $y \in \ker(T^*)$ when $y \in (T(\mathbb{H}))^\perp$. Conversely, if $y \in (T(\mathbb{H}))^\perp$, then $T^*(y) \in \mathbb{H}^\perp = \{0\}$.

For ease of reference, we shall list some of the results we have proven in the theorem:

Proposition 5.2.5: Adjoint Properties II

Let $T, S \in B(\mathbb{H})$. Then:

1. $(T + S)^* = T^* + S^*$
2. $(cT)^* = \overline{c}T^*$
3. $(TS)^* = S^*T^*$
4. $(T^*)^* = T$
5. $I^* = I$

***Proof*:**
If you're not convinced of any of these, double check the theorem or try proving it again.

Proposition 5.2.6: Adjoint Properties III

Let $T \in \mathrm{End}_k(\mathbb{H})$. Then the following are equivalent:

1. T is invertible (so $T^{-1} \in \mathrm{End}_k(\mathbb{H})$)
2. T^* is invertible
3. T and T^* are bounded away from zero.
4. T and T^* are injective, and $T(\mathbb{H})$ is closed
5. T is injective and $T(\mathbb{H}) = \mathbb{H}$

***Proof*:**
Since T is invertible, $T^{-1}T = TT^{-1} = I$, and so $(T^{-1})^*$ is the inverse of T^*, giving (1) $\Leftrightarrow$ (2).

For (1) $\Rightarrow$ (3), we say that an operator F is bounded away from zero if

$$\|x\| \leq C\|F(x)\|$$

for some constant C. In our case:

$$\|x\| = \|T^{-1} \circ T(x)\| \leq \|T^{-1}\|\|T(x)\|$$

Since (1) $\Leftrightarrow$ (2), the same holds for T^*.

For (3) $\Rightarrow$ (4), We have $\|T(x)\| \geq \epsilon\|x\|$ and $\|T^*(x)\| \geq \epsilon\|x\|$ for some $\epsilon > 0$ and all $x \in \mathbb{H}$. Thus certainly both T and T^* are injective: if $T(x) = T(y)$ then

$$0 = \|T(x) - T(y)\| = \|T(x-y)\| \geq \epsilon\|x-y\|$$

hence $\|x-y\| = 0$, so $x = y$. Furthermore, $\|T(x) - T(y)\| \geq \|x-y\|$ imples that $T(\mathbb{H})$ is complete, hence a closed subspace of $\mathbb{H}$.

For (4) $\Rightarrow$ (5), by corollary **??** ($\overline{Y} = (Y^\perp)^\perp$) and proposition 5.2.4 we get

$$\overline{T(\mathbb{H})} = (T(\mathbb{H})^\perp)^\perp = (\ker(T^*))^\perp = \{0\}^\perp = \mathbb{H}$$

where the last equality come from T^* bein injective.

Finally, for (5) $\rightarrow$ (1), we use the open mapping theorem.

This allows us to generlize this result to the followin

Proposition 5.2.7: Inversing Maps

Let $\mathbb{H}_1$ and $\mathbb{H}_2$ be hilbert spaces and consider an operator $T \in \mathrm{Hom}_k(\mathbb{H}_1, \mathbb{H}_2)$. Then there exists a unique operator $T^* \in B(\mathbb{H}_2, \mathbb{H}_1)$ such that

$$\langle T(x), y \rangle = \langle x, T^*(y) \rangle$$

show that the map $T \to T^*$ is conjugate linear isometry of $\mathrm{Hom}_k(\mathbb{H}_1, \mathbb{H}_2)$ to $\mathrm{Hom}_k(\mathbb{H}_2, \mathbb{H}_1)$ and satisfies

$$\|T^*T\| = \|T^2 = \|TT^*\|$$

***Proof*:**
Let

$$\hat{T} = \begin{pmatrix} 0 & 0 \\ T & 0 \end{pmatrix} \in \mathrm{End}_k(\mathbb{H}_1 \oplus \mathbb{H}_2)$$

By theorem 5.2.2, there is a unique function $\hat{T}^*$ such that

$$\langle \widehat{T}(x_1, x_2), (y_1, y_2) \rangle = \langle (x_1, x_2), \widehat{T}^*(y_1, y_2) \rangle$$

By definition of $\widehat{T}$, we see

$$\widehat{T}^* = \begin{pmatrix} 0 & T^* \\ 0 & 0 \end{pmatrix}$$

Hence we have that T^* is a function with domain and codomain $T^* : \mathbb{H}_2 \to \mathbb{H}_1$. It follows that

$$\langle T(x), y \rangle = \langle x, T^*(y) \rangle$$

and by theorem 5.2.2, T and T^* respect all the desired properties, completing the proof.

For the rest of this seciton, we will focus on $T \in B(\mathbb{H})$ insead of $T \in B(\mathbb{H}_1, \mathbb{H}_2)$. The reason for this is to find suitable decompositino theorems of T generlizing the notion of eigenvalues.

Recall that over finite dimensional vector spaces, $TT^* = T^*T$ implied T was diagonalizable, with the usual special case of $T = T^*$ (self-adjoint) being looked into. We would like to know if this notion of diagonalizability for normal operators remains true for infinite dimensional Hilbert spaces. We shall see that this is not the case, some more conditions is required (for ex. a special case of digaonalizable operators will be comapct normal operators). Before we get into the technicalities, I want to put a word on how I think about $TT^* = T^*T$. In my mind, when two functions commute, it means that one function is somehow "invariant" with respect to the other. Think of PDE's with equations that commute with the differential operator that makes it invariant to the Lorenzian metric. In this case, I like to think of T being somehow "symmetric". Since $TT^*(x) = T^*T(x)$ for all $x \in \mathbb{H}$, and T^* is in some ways the dual (think of ref:HERE, the generalization), we see that T is "invariant" to the effect of the dual T^*. We shall see in a moment that the invariant to the dual is translated to T and T^* being metrically identical. In the simpleset case of $T = T^*$, it is evident why this is the case. If $T = T^*$, we can deepen our intuition with complex numbers: if we consider $T : \mathbb{C} \to \mathbb{C}$ where $T(z) = z$, then $T^*(z) = \overline{z}$. Hence, $T = T^*$ if and only if $T(\mathbb{C}) \subseteq \mathbb{R}$, since only the real number are invariant under conjugation.

Definition 5.2.8: Normal Operator

Let $T \in \text{End}_k(\mathbb{H})$. Then T is said to be *normal* if it commutes with its adjoint, i.e. if $T^*T = TT^*$

As an immedaite consequence, we have

$$\|T(x)\| = \langle T^* \circ T(x), x\rangle^{1/2} = \langle T \circ T^*(x), x\rangle^{1/2} = \|T^* x\|$$

showing that T and T^* are metrically identical. In fact, by the palalalogram law (also known as polarization identity) if $T \in \text{End}_k(\mathbb{H})$ such that T and T^* ar emetrically identical, we can show $\langle T^* \circ T(x), y\rangle = \langle T \circ T^*(x), y\rangle$ for all x, y and so T is normal by lemma 5.2.1. By proposition 5.2.6, an operator is normal if and only if it is bounded away from zero.

Positive Operators

Definition 5.2.9: Positive Operator

Let $T \in \text{End}_k(\mathbb{H})$. Then T is said to be *positive* if $T = T^*$ and $\langle T(x), x\rangle \geq 0$ for all $x \in \mathbb{H}$. If $k = \mathbb{C}$, the positivey condition implies self-adjointness by the polarization identity.

Certainly, the sum of two positive operators is positive (geometrically, positive operatros form a cone in $\text{End}_k(\mathbb{H})$). However the product is not positive, and need not be self-adjoint.

Example 5.2.10: Positive And Non-Positive Operators
here

However, if the opeartors commute (if the product is self-adjoint), we see that there is a product that is positive since ($ST = S^{1/2}TS^{1/2} \geq 0$ by what we'll show in the next proposition)

Let $B(\mathbb{H})_{sa} \subseteq \text{End}_{\mathbb{R}}(\mathbb{H})$ be the collection of self-adjoint operators. Then this is a closed subspace, this subspace intersected with the conte of positive operators defines an ordder $S \leq T$ if $T - S \geq 0$. By taking the case of 2 dimensional real subspaces and $\text{End}_{\mathbb{R}}(\mathbb{R}^2)$, this order is not total, not even a lattice order.

We require the following lemma to gain some important results (which will be an immedate consequence of the spectral theorem in ref:HERE)

Lemma 5.2.11: Square Root Lemma

If $S \leq T$ in $B(\mathbb{H})_{sa}$, then $A^*SA \leq A^*TA$ for every $A \in \text{End}_k(\mathbb{H})$. If morever $) \leq S$, then $\|S\| \leq \|T\|$

***Proof*:**
Since $S \leq T$, we know that $\langle (T - S)(y), y\rangle \geq 0$ for evey $y \in \mathbb{H}$. Replacing y with $A(x)$ for appropraite $x \in \mathbb{H}$, it follows that $A^*(T - S)A \geq 0$, so $A^*SA \leq A^*TA$.

If $0 \leq S \leq T$, by the Cauchy-Schwarz inequality on the positive sesquilinear form $\langle S(\cdot), \cdot\rangle$, we get that for every pair of unit vectors $x, y \in \mathbb{H}$

$$|\langle S(x), y\rangle|^2 \leq \langle S(x), x\rangle\langle S(y), y\rangle \leq \langle T(x), x\rangle\langle T(y), y\rangle \leq \|T\|^2$$

Hence, $\|S\|^2 \leq \|T\|^2$ by lemma 5.2.1.

Lemma 5.2.12: Approximation Lemma

There is a sequence $(p_n) \subseteq \mathbb{R}[x]$ of polynomials with positive coefficients such that $\sum p_n$ converges uniformally on $[0, 1]$ to

$$t \mapsto 1 - (1 - t)^{1/2}$$

***Proof*:**
This is immediate from real analysis, but we can also construct it explicitly.If $q_0 = 0$ and $q_n(t) = \frac{1}{2}(t + q_{n-1}(t)^2)$, then we can see that each coefficent of q_n is positive and that $q_n(t) \leq 1$ for every $t \in [0, 1]$. Morever, since

$$2(q_{n+1} - q_n) = q_n^2 - q_{n-1}^2 = (q_n - q_{n-1})(q_n + q_{n-1})$$

it follows by induction that each polynomial $p_n = q_n - q_{n-1}]$ has positive coefficients. Taking the q_n as elements of $C([0, 1])$, we see they converge pointwise to $[0, 1]$ to a function q such that $2q(t) = t + q(t)^2$. Thus, $q(t) = 1 - (1 - t)^{1/2}$. Sicne $q \in C([0, 1])$ and $[0, 1]$ is compact, by the monotone ocnvergence $q_n \nearrow q$ si in fact uniform by Dini's lemma, hence

$$\sum p_n = \lim q_n = q$$

completing the proof.

Proposition 5.2.13: Square-Rooting Positive Operator

Let $T \in \mathrm{End}_k(\mathbb{H})$ be a positive operator. Then there is a unique positive operator $T^{1/2}$ such that $(T^{1/2})^2 = T$. Moreover, $T^{1/2}$ commutes with every operator commuting with T.

***Proof*:**
Analysis Now p.92

Proposition 5.2.14: Invertible Positive Operator

A positive operator T is invertible in $\mathrm{End}_k(\mathbb{H})$ if and only if $T \geq \epsilon I$ for some $\epsilon > 0$. I that case, $T^-1 \geq$ and $T^{1/2}$ is invertib,e and $(T^{-1})^{1/2} = (T^{1/2})^{-1}$. If $T \leq S$, then $S^{-1} \leq T^{-1}$

***Proof*:**
Analayis now p.94

Projection

Let $W \subseteq \mathbb{H}$ be a closed subspace of a hilbert space H. Then we know that $\mathbb{H} = W + W^{\perp}$. In particular, for eah $y \in \mathbb{H}$, we have $y = w + w^{\perp}$, $w \in W$ and $w^{\perp} \in W^{\perp}$. Since

$$\alpha y_1 + \beta y_2 = (\alpha x_1 + \beta x_2) + (\alpha x_1^{\perp} + \beta x_2^{\perp})$$

is the decomposition of a linear combination, it follows that $P : \mathbb{H} \to \mathbb{H}$ given by $P(y) = x$ is an operator in $\mathrm{End}_k(\mathbb{H})$ with $\|P\| \leq 1$. Note P is idempotent ($P^2 = P$), self-adjoint, and positive since

$$\langle P(y_1), y_2 \rangle = \langle x_1 < x_2 + x_2^{\perp} \rangle = \langle x_1, x_2 \rangle = \langle x_1 + x_1^{\perp}, x_2 \rangle = \langle y_1, P(y_2) \rangle$$

and

$$\langle P(y), y \rangle = \langle x, x + x^{\perp} \rangle = \|x\|^2 \geq 0$$

Definition 5.2.15: (Orthogonal) Projection

The map P in the above is called the *(orthogonal) projection.*

If P is a self-adjoint and idempotent in $\mathrm{End}_k(\mathbb{H})$, then

$$W = \{x \in \mathbb{H} \mid P(x) = x\} = P(\mathbb{H})$$

is a closed subsapce of $\mathbb{H}$. If $w^{\perp} \in W^{\perp}$, we have (since $P = P^*$) that

$$\|P(w^{\perp})\|^2 = \langle w^{\perp}, p^2(w^{\perp}) \rangle = 0$$

Thus, P is an orthogonal projection of $\mathbb{H}$ onto W. Note that $I - P$ is the projection on $W^{\perp}$.

Projections are some of the east operators we may construct, and can be thought as the "characteristic function" from real analysis. The analogue of simple function is then obtained by decomposing $\mathbb{H}$ into a finite orthogonal sum

$$\mathbb{H} = W_1 \oplus W_2 \oplus \cdots \oplus W_n$$

corresponding to the projections $P_1, ..., P_n$ which are pairwise orthogonal ($P_i P_j = 0$ for $i \neq j$ satisfying $\sum P_i = I$). We may then consider

$$T = \sum \lambda_n P_n$$

for scalars $\lambda_1, \lambda_2, ..., \lambda_n \in k$. This may look familiar to the notion of *diagonalization.* Indeed, we may generalize the concept to infintie dimensions like so:

Definition 5.2.16: Diagonalizable

Let $T \in \mathrm{End}_k(\mathbb{H})$ be an operator. Then T is said to be *diagonalizable* if there is an orthonormal basis $\{e_i \mid i \in I\}$ for $\mathbb{H}$ and a bounded set $\{\lambda_i \mid i \in I\} \subseteq k$ such that

$$T(x) = \sum \lambda_i \langle x, e_i \rangle e_i$$

The number $\langle x, e_i \rangle$ are the coordiantes for x in the basis $\{e_i\}$ and each λ_i is an *eigenvalue* for T corresponding to the eignevector $e_i >$ Denotign P_i to be the projecti nofo $\mathbb{H}$ onto ke_i, we have $P_i(x) = \langle x, e_i \rangle e_i$, so we may write the operator as:

$$T = \sum \lambda_i P_i$$

Note that the above sum is not necessarily convergent in the norm topology on $\mathrm{End}_k(\mathbb{H})$, but only pointwise convergent (i.e. in convergent in the strong operator topology). If T is diagonalizable, then T^* is diagonalizable with

$$T^*(x) = \sum \overline{\lambda_i}\langle x, e_i\rangle e_i$$

with eigenvalues $\{\overline{\lambda_i}\}$. Furthermore, if T is diagonalizable, $T^*T = TT^*$ (with eigenvalues $|\lambda_i|^2$ with basis $\{e_i\}$), and hence T is normal. T is self-adjoint if and only if all λ_i are real and T is positive if and only if $\lambda_i \geq 0$ for all i. If $\mathbb{H}$ is finite-dimensional, every normal operator is diagonalizable. This is not generally true for infinite dimensions. In general, most "interesting" (normal) operators on infinite-dimensional hilbert sapces are nondiagonalizable, and must be handled with a continuous analogue of the concept of a basis (this will be the spectrum, see section 5.3 and chapter 6).

Unitary Operators

Recall $T : \mathbb{H} \to \mathbb{H}$ is unitary if T is an isometric isomorphism from $\mathbb{H}$ onto itself (if $k = \mathbb{R}$ we say it is an *orthogonal operator*). By the polarization identity, we see that $U^*U = UU^* = I$, so that U is normal and invertible with $U^{-1} = U^*$. Conversely, if $U \in \mathrm{End}_k(\mathbb{H})$ such that $U^{-1} = u^*$ then U is unitary since

$$\|U(x)\|^2 = \langle U^*U(x), x\rangle = \|x\|^2$$

So U is an isometry and is surjective since U is invertible. If $\{e_i \mid i \in I\}$ is an orthonormal basis for $\mathbb{H}$ and U is unitary, then $\{Ue_i \mid i \in I\}$ is a basis for $\mathbb{H}$. Conversely, we saw that every transition between orthonormal bases is given by a unitary operator. For future reference the denote the group of unitary operators $U(\mathbb{H}) \subseteq \mathrm{End}_k(\mathbb{H})$ (it should be evident it forms a group).

Definition 5.2.17: Unitary Equivalent

Let $S, T \in \mathrm{End}_k(\mathbb{H})$. Then we say that S and T are *unitarily equivalnet* of $S = UTU^*$ for some unitary operator U.

It should be clear that unitary equivalent preserves norm, self-adjointness, normality, diagonalizability, and unitarity.

Definition 5.2.18: Partial Isometry

Let $U \in \mathrm{End}_k(\mathbb{H})$. Then U is said to be a *partial isometry* if there is a closed subpspace $X \subseteq \mathbb{H}$ such that $U|_X$ is isometric and $U|_{X^\perp} = 0$

This immediately implies that $X^\perp = \ker(U)$, hence $X = \bar{(}U^*(\mathbb{H}))$. Setting $P = U^*U$, then

$$\langle P(x), x\rangle = \|U(x)^2 = \|x\|^2$$

for every $x \in X$, hence $P(x) = x$ by the Cauchy-Schwarz inequality. Furthermore, $P(x^\perp) = 0$ for every $x^\perp \in X^\perp$, so P is a projetion of $\mathbb{H}$ onto X. Conversely, if $U \in \mathrm{End}_k(\mathbb{H})$ such that $U^*U = P$ for some projeciotn P, taking $X = P(\mathbb{H})$ and see from the equation $\|U(x)$
$^2 = \langle p(x), x\rangle$ that U is isometric on X and 0 on $X^\perp$, and so U is a partial isometry. Using this, we can also show that U^* is a partial isometry. Since $U(I - P) = 0$,

$$(UU^*)^2 = UU^*UU^* = UPU^* = UU^*$$

so UU^* is self-adjoint idempodent, i.e. a projection. Note that U^* is the projectio nof $U(\mathbb{H})$ back onto $U^*(\mathbb{H})$ so that U^* is a *partial inverse* of U.

Be weary that there exist isometries that are not unitary (since they need not be surjective). For example, if $\mathbb{H}$ is a separable and has countable basis $\{e_n \mid n \in \mathbb{N}\}$, we have

$$S\left(\sum \alpha_n e_n\right) = \sum \alpha_n e_{n+1}$$

Then S (the unilateral shift operator) is an isometry of $\mathbb{H}$ ont othe subspace $\{e_1\}^{|}perp$ and consequently S^* is a patial isometry of $\{e_1\}^{\perp}$ onto $\mathbb{H}$. In particular, $S^*S = I$, wheas SS^* is the projection onto $\{e_1\}^{\perp}$.

Theorem 5.2.19: Polar Decomposition of T

Let $T \in \mathrm{End}_k(\mathbb{H})$. Then there is a unique positive operator $|T| \in \mathrm{End}_k(\mathbb{H})$ such that

$$\|T(x)\| = \||T|(x)\|$$

and $|T| = (T^*T)^{1/2}$. Moreover, there is a unique partial isometry U with kernel $\ker(U) = \ker(T)$ and $U \circ |T| = T$. In particular

$$U^*U|T| = |T| \qquad U^*T = |T| \qquad UU^*T = T$$

***Proof*:**
Analysis Now p.96

We may think of U as the generalized "sign" of T and $|T|$ as the generalized "absolute value" of T. By the formula

$$T^* = |T|U^* = U^*(U|T|U^*)$$

and the unique of polar decomposition, we see that U^* is the sign of T^*, whereas $U|T|U^*$ is the absolute value fo T^*. In general, the absolute value of a sum or product of noncommuting operators doens't have much relation to the sum or product of the absolute values. Similarly, the sign (in the finite dimensional case), can't always be chosen to be unitary (the unilateral shift is an immedaite counter example). For T to have the form $U|T|$ for some unitary operator U, it is necessary and sufficient that the cloesd subspaces $\ker(T)$ and $\ker(T^*)$ have the same dimension (finite or infinite). Easy cases of this general theorem are estalblisehd in the next result

Proposition 5.2.20: Polar Decomposition For Isomorphisms

Let $T \in \mathrm{End}_k(\mathbb{H})$ be invertible. Then the partial isometry in its polar decomposition is unitary

***Proof*:**
Analysis Now p.97

Proposition 5.2.21: Polar Decomposition For Normal Operators

Let $T \in \mathrm{End}_k(\mathbb{H})$ be normal. Then there is a unitary operator W commuting with T, T^*, and $|T|$ such that $T = W|T|$

***Proof*:**
Analysis Now p.97

I want to add one more thing, I just don't konw where: for any $T \in B(\mathbb{H})$, $T = T_1 + iT_2$ for self-adjoint operators T_1, T_2. This lets us generalize the spectral theorem for noraml operators, since we will konw it for self-adjoint operator.

Lemma 5.2.22: lemma I

If $T = T^*$ and $\|T\| \leq 1$, the operator $U = T + i(I - T^2)^{1/2}$ is unitary and $T = \frac{1}{2}(U + U^*)$

***Proof*:**
Analysis Now p.97

Lemma 5.2.23: Lemma II

Let $S \in \mathrm{End}_k(\mathbb{H})$ with $\|S\| < 1$. Then for every unitary U, there are unitaries U_1, V_1 such that

$$S + U = U_1 + V_1$$

***Proof*:**
Analysis Now p.98

Theorem 5.2.24: Russo-Dye-Gardner Theorem

Let $T \in \mathrm{End}_k(\mathbb{H})$ with

$$\|T\| < 1 - \frac{2}{n} \qquad n > 2$$

Then there are unitary operators $U_1, U_2, ..., U_n$ such that

$$T = \frac{1}{n}(U_1 + U_2 + \cdots + U_n)$$

***Proof*:**
Analysis Now p.98

Since the self-adjoint elements span $\mathrm{End}_k(\mathbb{H})$, we see from lemma 5.2.22 that every $T \in \mathrm{End}_k(\mathbb{H})$ is the linear combination of (at least 4) unitary operators The estimate in the Russo-Dye-Gardner Theorem is much more precise with respect to the norm, and shows that every element in the open unit ball of $\mathrm{End}_k(\mathbb{H})$ is a convex combination (indeed, a mean) of unitary operators. On the surface of the ball this need not be the case, in fact the unilateral shift cannot be expressed as a convex combination of unitaries (exercise 3.2.8 in Gert K. Pederson Analysis Now).

Proposition 5.2.25: Numerical Radius Of Operator

Let $\mathbb{H}$ be a complex hilbert space and $T \in \mathrm{End}_{\mathbb{C}}(\mathbb{H})$. Then we deifn eth *numerical radius* of T as

$$|||T||| = \sup\{\langle T(x), x\rangle \mid x \in \mathbb{H}, \|x\| \leq 1\}$$

Then $\|\cdot\|$ is a norm on $\mathrm{End}_{\mathbb{C}}(\mathbb{H})$ satisfying $\frac{1}{2}\|\cdot\| \leq |||\cdot||| \leq \|\cdot\|$. More over, $|||T^2||| \leq |||T|||^2$ for every T and $|||T||| = \|T$
if T is normal

***Proof*:**
Analysis Now p.99

5.3 Compact Operators

Throughout this section, let $\mathbb{H}$ be a hilbert space over $k \in \{\mathbb{R}, \mathbb{C}\}$ unless otherwise specified.

Definition 5.3.1: Finite Rank Operator

Let $T \in \mathrm{End}_k(\mathbb{H})$. Then T is said to have *finite rank* if $T(\mathbb{H})$ is finite-dimensional subspace of $\mathbb{H}$. Denote by $B_f(\mathbb{H}) \subseteq \mathrm{End}_k(\mathbb{H})$ the set of all finite rank opeartors.

Clearly, $B_f(\mathbb{H})$ is a subspace, even a subalgebra, even an ideal of $\mathrm{End}_k(\mathbb{H})$. If $T \in B_f(\mathbb{H})$, by proposition 5.2.4, we get an orthogonal decomposition $\mathbb{H} = T(\mathbb{H}) \oplus \ker(T^*)$, which shows that $T^*(\mathbb{H}) - T^*T(\mathbb{H})$, so T^* has finite rank. Thus, $B_f(\mathbb{H})$ is *self-adjoint* ideal in $\mathrm{End}_k(\mathbb{H})$ ($(B_f(\mathbb{H}))^* = B_f(\mathbb{H})$ as a set). The subset $B_f(\mathbb{H}) \subseteq \mathrm{End}_k(\mathbb{H})$ is similar in relation to $C_c(X) \subseteq C_b(X)$ when X is a locally compact Hausdorff space. In particular, these classes describe the local phenomena on $\mathbb{H}$ and on X. Passing to a limit in norm may destroy the "locality". but enogh structure is preservd to make these "quasilocal" operators and functions very useful and interesting. We shall stud the closure of $B_f(\mathbb{H})$ in this section as a noncommutative analogue of $C_0(X)$ in function theory.

Lemma 5.3.2: Net And Compact Operators

There is a net $(P_\lambda)_{\lambda\in\Lambda}$ of projection in $B_f(\mathbb{H})$ such that $\|P_\lambda(x) - x\| \to 0$ for each $x \in \mathbb{H}$.

that is, the projection are "dense" in $B_f(\mathbb{H})$.

***Proof*:**
We want to find a next P_n such that $\|P_n(x) - x\| \to 0$. Let $\{e_i \mid i \in I\}$ be an orthonormal basis for $\mathbb{H}$ and let N be a net of finite subsets of I ordered by inclusion. For each $n \in N$, let P_n denote the projection of $\mathbb{H}$ onto $\operatorname{span}\{e_i \mid i \in n\}$ so that $(P_n)_{n\in N}$ is a net in $B_f(\mathbb{H})$.

Now, if $x \in \mathbb{H}$, then $x = \sum a_i e_i$, thus

$$\|P_n(x) - x\|^2 = \sum_{i \notin n} |a_i|^2$$

By Parseval's identity, this tends to zero, completing the proof.

Theorem 5.3.3: Closed Unit Balls In Hilbert Spaces

Let $B \subseteq \mathbb{H}$ be a closed unit ball in a hilbert space $\mathbb{H}$. Then the following conditions on an operator $T \in \mathrm{End}_k(\mathbb{H})$ are equivalent

1. $T \in \overline{B_f(\mathbb{H})}$
2. $T|_B$ is a weak-norm continuous function from B into $\mathbb{H}$
3. $T(B)$ is compact in $\mathbb{H}$
4. $\overline{T(B)}$ is compact in $\mathbb{H}$
5. each net in B has a subnet whose iamge under T converges in $\mathbb{H}$

***Proof*:**
Analysis Now p.105

Definition 5.3.4: Compact Operators

Let $T \in B(\mathbb{H})$. Then T is compact if $T(B)$ is compact. The set of compact operators is denoted by $B_0(\mathbb{H})$ to show that these operators "vanish at infinity".

Lemma 5.3.5: Diagonalizable Operator Is Compact

Let $T \in \mathrm{End}_k(\mathbb{H})$. Then T is comapct if and only if its eigenvalues $\{\lambda_i \mid i \in I\}$ belongs to $c_0(I)$

***Proof*:**
Let $T(x) = \sum \lambda_i \langle x, e_i \rangle e_i$. If $T \in B_0(\mathbb{H})$, then for any $\epsilon > 0$, define

$$I_\epsilon = \{i \in I \mid |\lambda_i| \geq \epsilon\}$$

If I_ϵ is infinite, the net $(e_i)_{i \in I_\epsilon}$ converges weakly to zero for any well-ordering of I_ϵ since $\langle e_i, x \rangle \to 0$ by Parsaval's identity. Sicne $\|T(e_i)\| = |\lambda_i| \geq \epsilon$ for $i \in I_\epsilon <$ this contradicts theorem 5.3.3(3). Thus, I_ϵ must be finite for each $\epsilon > 0$, hence λ_i must vanish at infinity.

Conversely, if I_ϵ is finite for all $\epsilon > 0$. et

$$T_\epsilon = \sum_{i \in I_\epsilon} \lambda_i \langle \cdot, e_i \rangle e_i$$

Then T_ϵ has finite rank, and

$$\begin{aligned}\|(T-T_\epsilon)(x)\|^2 &= \left\|\sum_{i\notin I_\epsilon}\lambda_i\langle x,e_i\rangle e_i\right\|^2\\ &= \sum_{i\notin I_\epsilon}|\lambda_i|^2|\langle x,e_i\rangle|^2\\ &\leq \epsilon^2\|x\|^2\end{aligned}$$

hence, $\|T-T_\epsilon\|\leq\epsilon$, thus $T\in B_0(\mathbb{H})$ by theorem 5.3.3(1), completing the proof.

Lemma 5.3.6: Eigen Vectors Between Adjoint

Let x be an eigenvector for a normal operator $T\in\mathrm{End}_k(\mathbb{H})$ corresponding to an eigenvalue λ. Then x is an eigenvector for T^* corresponding to the eigenvalue $\overline{\lambda}$. Eigenvectors for T correspondign to different eigenvalues are orthogonal.

***Proof*:**
Note that the operator $T-\lambda I$ is normal and it's adjoitn is $T^*-\overline{\lambda}I$. COnversely,

$$\|(T^*-\overline{\lambda}I)(x)\| = \|(T-\lambda I)(x)\| = 0$$

If $T(y)=\gamma y$ where $\gamma\neq\lambda$, then we may certainly assume $\lambda\neq 0$ so that

$$\begin{aligned}\langle x,y\rangle &= \lambda^{-1}\langle T(x),y\rangle\\ &= \lambda^{-1}\langle x,T^*(y)\rangle\\ &= \lambda^{-1}\langle x,\overline{\gamma}y\rangle\\ &= \lambda^{-1}\gamma\langle x,y\rangle\end{aligned}$$

Hence, $\langle x,y\rangle = 0$, showing they are different eigenvectors, completing the proof.

Lemma 5.3.7: Normal Compact Operator, then Eivenvalue Exists

Every normal, compact operator T on a complex Hilbert space $\mathbb{H}$ has an eigenvalue λ with $|\lambda|=\|T\|$

***Proof*:**
Analysis now p. 107

Theorem 5.3.8: Normal Compact, Then Diagonalizable

Every normal compact operator $T\in\mathrm{End}_k(\mathbb{H})$ is diagonalizable and its eigenvalues (counted-with multiplicity) vanish at infinite. Conversely, each such operator is normal and compact.

***Proof*:**
Analysis now p.108

It will be convenient for future sections to introduce a new notation: Let $x \odot y$ represent the rank one operator in $\mathrm{End}_k(\mathbb{H})$ determined by the vectors $x, y \in \mathbb{H}$ and the formula

$$(x \odot y)(z) = \langle z, y \rangle x$$

The map $(x, y) \mapsto x \odot y$ is sesquilinear map of $\mathbb{H} \times \mathbb{H}$ into $B_f(\mathbb{H})$. If $\|e\| = 1$ then $e \odot e$ is the one-dimensional projection of $\mathbb{H}$ on $\mathbb{C}e$. Every normal compact operator $\mathbb{H}$ can now be written by the above theorem in the form

$$T = \sum \lambda_i e_i \odot e_j$$

For suitable orthonormal basis $\{e_i \mid i \in I\}$. The sum will converge in norm, since either the set $I_0 = \{i \in I \mid \lambda_i \neq 0\}$ is finite (so that $T \in B_f(\mathbb{H})$) or else countably infinite in which case the sequence $\{\lambda_i \mid i \in I_0\}$ converges to zero.

Definition 5.3.9: Spectrum

We say that the compact set

$$^{(}T) = \{\lambda_i \mid i \in I_0\} \cup \{0\}$$

is the *spectrum* of T.

For every continuous function f on $^{(}T)$, define

$$f(T) = \sum f(\lambda_i) e_i \odot e_i$$

Then $f(T)$ is compact if and only if $f(0) = 0$ and the map $f \mapsto f(T)$ Is an isometric $*$-preserving homomorhpism of $C(^{(}T))$ into $\mathrm{End}_k(\mathbb{H})$. If $f(z) = \sum \alpha_{nm} z^n \overline{z}^m$ is a polynomial of two commuting variables z and $\overline{z}$, then $f(T) = \sum \alpha_{nm} T^n (T^*)^m$. This result is the *spectrial (mapping) theorem* for normal compact operators. In the next chapter, we shall generalize this to any normal operator.

Calkin Algebra and Index

Since $B_0(\mathbb{H}) \subseteq B(\mathbb{H})$ is a closed ideal, we may define $B(\mathbb{H})/B_0(\mathbb{H})$ with the quotient norm[7]. This quotient is given a name

Definition 5.3.10: Calkin Algebra

Let $B_0(\mathbb{H}) \subseteq B(\mathbb{H})$. Then the algebra

$$B(\mathbb{H})/B_0(\mathbb{H})$$

is called the *Calkin Algebra* of $\mathbb{H}$

We shall see that many properties of $B(\mathbb{H})$ are conveniently expressed in terms of the Calkin algebra.

[7]This is even a C^*-algebra

Definition 5.3.11: Compact Pertubation

If S and T are elements in $B(\mathbb{H})$ and $S - T \in B_0(\mathbb{H})$, we say that S is a *compact pertubation* of T.

This is antoher way of saying that S and T hav the same image in the Calkin algebra. Such "local" pertubation happen frequently in application and properties of an operator that are invariatn under compact pertubations are highly used. The best known of these we shall study know is known as the *index*

Theorem 5.3.12: Atkinson's Theorem

Let $T \in B(\mathbb{H})$. Then the following are equivalent:

1. There is a unique operator $S \in B(\mathbb{H})$ such that ST and TS are the projections on $(\ker(T))^\perp$ and $\ker(T^*)^\perp$, respectively, and both projections have finite co-rank
2. For some operator $S \in B(\mathbb{H})$ both operators $ST - I$ and $TS - I$ are compact
3. the image of T is invertible in the Calkin algebra $B(\mathbb{H})/B_0(\mathbb{H})$
4. Both $\ker(T)$ and $\ker(T^*)$ are finite-dimensional subsapces and $T(\mathbb{H})$ is closed

***Proof*:**
Analysis now p.109

In essence, Fredholm operators are operators that are are almost invertible up to some finite-dimension beign ignored, in particular they are invertible modulo compact operators.

Definition 5.3.13: Fredholm Operator

The operator satisfying the condition's in Atkinson's theorem are called *Fredholm operators*, and the class of them is dneoted $F(\mathbb{H})$.

Definition 5.3.14: Index

For each $T \in F(\mathbb{H})$, define the *index* of T to be

$$\text{index}(T) = \dim(\ker(T)) - \dim(\ker(T^*))$$

If we shooe S for T as in (1) of Atkinson's theorem and write $suchthat = I - P$ and $TS = I - Q$ with P and Q projection of rank, then another way of writing this would be

$$\text{index}(T) = \text{rank}(P) - \text{rank}(Q)$$

By (3) of Atkinson's Theorem, the product of Fredholm operators is again a Fredholm operator. In particular, every product $RT \in F(\mathbb{H})$ if $T \in F(\mathbb{H})$ and R is invertible in $B(\mathbb{H})$, and in this case

$$\text{index}(RT) = \text{index}(TR) = \text{index}(T)$$

since R is bijective. It should also be clear that $T^* \in F(\mathbb{H})$ if $T \in F(\mathbb{H})$ with index $\operatorname{index}(T^*) = -\operatorname{index}(T)$. Finally, the partial inverse S to T in (1) of Atkinson's theorem is a Fredholm operator with index $S) = -T)$. F

Definition 5.3.15: Indexed Fredholm Operators

For each $n \in \mathbb{Z}$ define
$$F_n(\mathbb{H}) = \{T \in F(\mathbb{H}) \mid T) = n\}$$

All of these subsets are nonemtpy, since if S is the unilateral shift, then $S^n \in F_{-n}(\mathbb{H})$ and $(S^*)^n \in F_n(\mathbb{H})$ for every $n > 0$.

Lemma 5.3.16: Finite Rank And 0 Index

If $A \in B_f(\mathbb{H})$, then $I + A \in F_0(\mathbb{H})$

***Proof*:**
Analysis now p.110

Corollary 5.3.17: Fredholm Alternative

If $A \in B_0(\mathbb{H})$ and $\lambda \in \mathbb{C} \setminus \{9\}$, then either $\lambda I - A$ is invertible in $B(\mathbb{H})$ or λ is an eigenvalue of A with finite multiplicity, in which case $\overline{\lambda}$ is an eigenvalue for A^* with the same multiplicity

***Proof*:**
Let $T + I - \lambda^{-1}A$. Since $T \in F_0(\mathbb{H})$, by lemma 5.3.16 we know that $T(\mathbb{H})$ is closed and that
$$\dim(\ker(T)) = \dim(\ker(T^*)) < \infty$$
If these dimension are 0, then $T(\mathbb{H}) = (\ker(T^*))^\perp = \mathbb{H}$ and $\ker(T) = \{0\}$, hence T is invertible by proposition 5.2.6, completing the proof.

This lemma implies that the spectrum of a compact operator (i.e. those λ such that $\lambda I - T$ is not invertible, as we'll see in the next chapter) consists of $\{0\}$ and the eigenvalues for T. IN particualr, it is a countable subset of $\mathbb{C}$ with 0 as the only possible accumulatpion point.

Theorem 5.3.18: Index Invariant Under Compact Operator

Let $T \in F(\mathbb{H})$ and A be a compact operator. Then
$$T + A) = T)$$

***Proof*:**
Analysis now p.111

Proposition 5.3.19: Fredholm Class Is Open

Each Fredholm class $F_n(\mathbb{H})$ for each $n \in \mathbb{H}$ is open in $B(\mathbb{H})$

***Proof*:**
analysis now p.112

Proposition 5.3.20: Grading Structure Of Freidholm Operators

If $T_1 \in F_n(\mathbb{H})$ and $T_2 \in F_m(\mathbb{H})$ then $T_1 T_2 \in F_{n+m}(\mathbb{H})$

***Proof*:**
Analysis now p. 112

We close with an interesting topological fact that brings some light to the index .Let G denote the group of invertible elements of the Calkin algebra $B(\mathbb{H})/B_0(\mathbb{H})$. and G_0 is the connected component of the identity (which is the group generated by the elements of the form $\exp(A)$ for some $A \in B(\mathbb{H})/B_0(\mathbb{H})$). Then G_0 is an open and closed subgroup of G and G/G_0 is a discrete group that labels the connected components in G. In this case, G/G_0 is isomorphic to $\mathbb{Z}$, and the index of a Fredholm operator T is the image of T under the compsoed quotinet maps from $F(\mathbb{H})$ to G and from G to G/G_0 $(= \mathbb{Z})$

5.4 Trace

5.5 *Intuition to add

This is a conversation I had with Gemini that I really want to incorporate somewhere as it gave some good insights. I will save it here for now:

Zak Katz: Let $F(x_1, \ldots, x_n, t)$ represent a PDE w.r.t time. Let's say $u(x,t)$ represent a solution and $f_t(x) = u(x,t)$ is a "state" at time 0. Let S_t be the "time-shift" operator so that $S_t(f_a) = f_{a+t}$ for some state f_a at time a.

My question is: Is S_t always self-adjoint? If yes, can you prove it, and if not, can you give a counter-example?

Raycast AI:

`Tool: web_search`

No, the time-shift operator S_t is not always self-adjoint.

In fact, for many common PDEs (like the wave equation, transport equation, or Schrödinger equation), S_t is almost never self-adjoint. It is usually unitary (energy-conserving) or a contraction semi-group (dissipative).

6

Spectral Theory

In short, Spectral theory generalizes the notion of eigen vectors and values to infinite dimensional operators. In the previous chapter, we saw how there is a notion of diagonalizability for compact normal operators. In this chapter, we extend this result to normal operators. We also along the way learn some important dualities such as the Gelfand duality which has some useful applications, and is an intersting motivator for classical algebraic geometry.

6.1 Banach Algebras

Recall the definition of a banach algebra

Definition 6.1.1: Banach Algebra Revisited

A *Banach Alagebra* is an algebra A with a submultiplicative norm (so that it is continuous) st the unerlying vector space is a banach space.

All the same applies for Banach Algebras as we expect. Note that for A/I to be a Banach algebra for an ideal I, I has to be closed, just like for subspaces. If A is unital and $A \neq \{0\}$, then it is certainly unique and $\|I\| \geq 1$ (why not =?? Analysis now p.128) If A has no units, we may try to isometrically embed it into a larger Banach Algebra $\widehat{A}$ such that $A \subseteq \widehat{A}$ becomes an ideal such that every nonzero ideal of $\widehat{A}$ has a nonzero intersection with A (such an ideal is called an *essential ideal*). This can be thought as the algebraic counterpart of the notion of compactification of a topological space. For classical Banach algebra, there is usually a natural way of *adjoining a unit*, but there is always a general method of doing so that is the counter-part of one-point compactification. Take the direct sum $A \oplus k = \widehat{A}$ along with the product

$$(x, \alpha)(y, \beta) = (xy + \alpha y + \beta x, \alpha\beta)$$

and the submultiplicative 1-norm. Then $A \oplus k$ is a unital Banach algebra with $I(0,1)$ and contains A as an ideal of co-dimension 1.

Definition 6.1.2: Regular Representation

Let A be a Banch Algebra. Then the (left) *regular representation* $\rho : A \to B(A)$ by $\rho(x)(y) = xy$ for $x, y \in A$.

It should be verified that ρ is a norm decreasing algebra homomorphism, and if A is unital, the inequalities

$$\|A\| \leq \|\rho(A)\|\|I\| \leq \|A\|\|I\|$$

show that ρ is a homeomrophism. Renorming A by using the equivalent norm from $B(A)$, we may assume that $\|I\| = 1$ (if $A \neq \{0\}$), which we do.

Definition 6.1.3: Approximate Unit

A net $(E_\lambda)_{\lambda\in\Lambda}$ in the unit ball of a Banah algebra A is an *approximate unit* if

$$\lim E_\lambda x = \lim xE_\lambda = x$$

for all $x \in A$.

Example 6.1.4: Approximate Units

Can you think of approximate units for the non-unital Banach algebra $C_0(X)$, $B_0(\mathbb{H})$ and $L^1(\mathbb{R}^n)$? Show that the existence of an approximate unit guarantees that the regular representation ρ is an isometry

We shall denote $A^\times = \mathrm{GL}(A)$ the set of all invertible elements of A (in algebra, we'd usually call these elements unital, but the term unit in functional analysis is more commonly reserved for the identity)

Lemma 6.1.5: Almost Invertible Elements

Let $x \in A$ where A is a unital banach algebra with $\|A\| < 1$. Then $I - A \in \mathrm{GL}(A)$ and

$$(I - A)^{-1} = \sum_{n=0}^{\infty} A^n$$

***Proof*:**

Since the norm is submultiplicative $\|x^n\| \leq \|x\|^n$, thus the series $\sum x^n$ converges in x to an element y. Sicne $xy = yx = y - I$ by manipulating the above power series, it follwos that $y = (I - x)^{-1}$, completing the proof.

Proposition 6.1.6: Properties Of $\mathrm{GL}(A)$

Let A be a unital algebra and $\mathrm{GL}(A)$ be it's multiplicative group. Then $\mathrm{GL}(A)$ is an open subset of A, an the map $x \mapsto x^{-1}$ is a homeomorphism of $\mathrm{GL}(A)$

***Proof*:**
Let $x \in \mathrm{GL}(A)$ and $y \in A$, Then by lemma 6.1.5

$$y = x - (x - y) = x(I - x^{-1}(x - y)) \in \mathrm{GL}(A)$$

provided that $\|A^{-1}(A - B)\| < 1$. In particular, the ball $B_\epsilon(x)$ is contained in $\mathrm{GL}(A)$ for $\epsilon < \|x^{-1}\|^{-1}$, which proves that $\mathrm{GL}(A)$ is open. If $y \to a$, we see from the above equation and lemma 6.1.5 that $y^{-1} \mapsto a^{-1}$, thus inverting is a continuous process and therefore a homeomorphism since $(A^{-1})^{-1} = A$, completing the proof.

From ehre on out, all Banach algebra will be over $\mathbb{C}$, i.e. all Banach algebras will be *complex*. See exercise ref:HERE on the complexicication of general real Banach algebras (or E.2.1.13 in Analysis Now).

Definition 6.1.7: Spectrum

Let A be a complex unital Banach algebra. Then for every $x \in A$, define the *spectrum* of x t obe the set

$${}^{(}x) = \{\lambda \in \mathbb{C} \mid \lambda I - x \notin \mathrm{GL}(A)\}$$

The comliment of ${}^{(}x)$ is the *resolvent set*.

On the resolvent set, we can define the resolvent function

$$R(x, \lambda) = (\lambda I - A)^{-1}$$

Definition 6.1.8: Spectral Radius

Let A be a complex unital Banach algera, $x \in A$, and ${}^{(}x)$ be its spectrum. Then the smalleest $r \geq 0$ such that ${}^{(}x) \subseteq B_r(0)$ is called the *spectral radius* of x, and is denoted $r(x)$. THus

$$r(x) = \sup\left\{|\lambda| \;\middle|\; \lambda \in^{(} x)\right\}$$

Lemma 6.1.9: Build-Up Lemma I

Let $x \in A$ and $f(z) = \sum \alpha_n z^n$ be a holomorphic function in a region contined in the closed disk $B_r(0)$ with $\|x\| \leq r$. then we can define $f(x) = \sum \alpha_n x^n$ in A. Morever, if $\lambda \in^{(} x)$ and $|\lambda| \leq r$, then $f(\lambda) \in^{(} f(x))$

***Proof*:**
The series $f(x)$ is absolutely convergence ($\sum |\alpha_n| \|x\|^n < \infty$), so certainly $f(x) \in A$. Moerover

$$\begin{aligned} f(\lambda)I - f(x) &= \sum \alpha_n(\lambda^n I - x^n) \\ &= (\lambda I - x) \sum \alpha P_{n-1}(\lambda, x) \\ &= (\lambda I - x)y \end{aligned}$$

where $P_{n-1}(\lambda, x) = \sum_{k=0}^{n-1} \lambda^k A^{n-k-1}$ so that $\|P_{n-1}(\lambda, x)\| \leq nr^{n-1}$, which implies the series of polynomials converges in A to an element y commuting with x. We see form this computation that if $f(\lambda)I - f(x) \in \mathrm{GL}(A)$ with inverse v, then yv will be the inverse of $\lambda I - x$, so tha $\lambda I - x \in \mathrm{GL}(A)$

Lemma 6.1.10: Build-Up Lemma II

For every $x\ inA$, we have

$$r(x) \leq \inf \|A^n\|^{1/n}$$

***Proof*:**
If $|\lambda| > \|A\|$, thn by lemma 6.1.5

$$(\lambda I - x)^{-1} = \lambda^{-1}(I - \lambda^{-1}x)^{-1} = \sum \lambda^{-n-1} x^n$$

which shows that $r(x) \leq \|x\|$. Now if $\lambda \in^(x)$, then $\lambda^n \in^(x^n)$ by lemma 6.1.9. Thus $|\lambda^n| \leq \|x^n\|$ from what we've just shown. It follows that $r(x) \leq \|x^n\|^{1/n}$ for every n, completing the proof.

Theorem 6.1.11: Spectrum Is Compact

Let $x \in A$ be an element of a complex unital Banach algebra A. Then the spectrum of x, $^(x)$ is a compact nonempty subset of $\mathbb{C}$, and the spectral radius of x is the limit of the convergent sequence $\|x^n\|^{1/n}$

***Proof*:**
Compactness. If $|\lambda| > \|x\|$ then $\lambda I - x = \lambda(I - \lambda^{-1}x)$ is invertible, its inverse being the norm-convergent Neumann series $\sum_{n\geq 0} \lambda^{-n-1}x^n$; hence $^(x) \subseteq \{\lambda : |\lambda| \leq \|x\|\}$ is bounded. The map $\lambda \mapsto \lambda I - x$ is continuous and the group $\mathrm{GL}(A)$ of invertibles is open, so the resolvent set $\rho(x) = \mathbb{C} \setminus^(x)$ is open and $^(x)$ is closed. A closed bounded subset of $\mathbb{C}$ is compact.

Nonemptiness. Write $R(\lambda) = (\lambda I - x)^{-1}$ for the resolvent on $\rho(x)$. Near any $\lambda_0 \in \rho(x)$ the factorization $\lambda I - x = (\lambda_0 I - x)\big(I + (\lambda - \lambda_0)R(\lambda_0)\big)$ gives the norm-convergent expansion $R(\lambda) = \sum_{n\geq 0}(\lambda_0 - \lambda)^n R(\lambda_0)^{n+1}$, so R is an A-valued analytic function on $\rho(x)$. Suppose $^(x) = \varnothing$, so $\rho(x) = \mathbb{C}$ and R is entire. From $R(\lambda) = \lambda^{-1}(I - \lambda^{-1}x)^{-1}$ we get $\|R(\lambda)\| \to 0$ as $|\lambda| \to \infty$. For any φ in the dual A', the scalar function $\varphi \circ R$ is then entire and vanishes at infinity, hence is bounded, hence constant by Liouville's theorem ([9]); vanishing at infinity forces that constant to be 0. As $\varphi \in A'$ was arbitrary, proposition 1.3.6 gives $R(\lambda) = 0$ for every λ, contradicting the invertibility of $\lambda I - x$. Hence $^{(}x) \neq \varnothing$.

The spectral radius formula. The estimate $r(x) \leq \|x^n\|^{1/n}$ for every n was shown above (from $\lambda^n \in^{(} x^n)$, lemma 6.1.9), so $r(x) \leq \liminf_n \|x^n\|^{1/n}$. For the reverse inequality fix $\varphi \in A'$. On $\{|\lambda| > \|x\|\}$ the Neumann series gives the Laurent expansion $\varphi(R(\lambda)) = \sum_{n\geq 0} \varphi(x^n)\lambda^{-n-1}$; since $\varphi \circ R$ is analytic on the larger region $\{|\lambda| > r(x)\} \subseteq \rho(x)$, uniqueness of Laurent coefficients extends this expansion to all $|\lambda| > r(x)$. For each such λ the terms $\varphi(x^n)\lambda^{-n-1}$ are then bounded in n, so $\sup_n |\varphi(\lambda^{-n}x^n)| < \infty$. This holds for every $\varphi \in A'$, so by the principle of uniform boundedness (theorem 1.2.11) the sequence $(\lambda^{-n}x^n)_n$ is norm-bounded, say $\|x^n\| \leq C_\lambda |\lambda|^n$; hence $\limsup_n \|x^n\|^{1/n} \leq |\lambda|$. Letting $|\lambda| \downarrow r(x)$ yields $\limsup_n \|x^n\|^{1/n} \leq r(x)$. Combining, $\lim_n \|x^n\|^{1/n} = r(x)$.

An immediate consequence is the classical Gelfand-Mazur theorem, which pins down the only complex Banach division algebra:

Corollary 6.1.12: Gelfand-Mazur

Let A be a unital Banach algebra over $\mathbb{C}$. Then if A is a division ring, $A = \mathbb{C}$

***Proof*:**

For each $x \in A$, there is some $\lambda \in^{(} x)$ by the theorem. Thus, $\lambda I - x \notin \mathrm{GL}(A)$, hence $x = \lambda I$.

6.1.1 The Gelfand Transform

Gelfand-Mazur pins down $\mathbb{C}$ as the only complex Banach division algebra. For a *commutative* Banach algebra this one fact reorganizes the whole algebra into a space of functions: every maximal ideal has quotient $\mathbb{C}$, so it is the kernel of a homomorphism onto $\mathbb{C}$, and evaluating an element against all such homomorphisms turns it into a function on the set of homomorphisms. This assignment is the Gelfand transform, and it is the mechanism behind every functional calculus.

Definition 6.1.13: Character and Character Space

Let A be a commutative Banach algebra over $\mathbb{C}$. A *character* (or *multiplicative functional*) of A is a nonzero algebra homomorphism $\varphi\colon A \to \mathbb{C}$: a linear map with $\varphi(xy) = \varphi(x)\varphi(y)$ and $\varphi \neq 0$. The set of all characters is the *character space* (equivalently the *Gelfand spectrum* or *maximal ideal space*) of A, written $\Delta(A)$ and topologized as a subspace of $(A', \text{weak-}*)$; this subspace topology is the *Gelfand topology*.

If A is unital then $\varphi(1) = 1$ for every character (apply φ to $1 \cdot 1 = 1$ and cancel, using $\varphi \neq 0$), and φ cannot vanish on an invertible element (from $\varphi(x)\varphi(x^{-1}) = 1$). The characters turn out to be exactly the quotient maps of A onto $\mathbb{C}$.

Proposition 6.1.14: Characters, Maximal Ideals, and Compactness

Let A be a commutative Banach algebra over $\mathbb{C}$.

1. Every character is contractive: $\|\varphi\| \leq 1$. Thus $\Delta(A)$ lies in the closed unit ball of A'.
2. If A is unital, then $\varphi \mapsto \ker\varphi$ is a bijection of $\Delta(A)$ onto the set of maximal ideals of A, and $\Delta(A)$ is weak-$*$ compact and Hausdorff.
3. If A is non-unital, then $\Delta(A) \cup \{0\}$ is weak-$*$ compact, so $\Delta(A)$ is locally compact and Hausdorff.

***Proof*:**

(1) Adjoin a unit: on $\widetilde{A} = A \oplus \mathbb{C}$ (definition 6.1.1) the map $\widetilde{\varphi}(x, \lambda) = \varphi(x) + \lambda$ is a unital character extending φ. If $|\varphi(x)| > \|x\|$ for some $x \in A$, put $\lambda = \varphi(x) \neq 0$; then $\|x/\lambda\| < 1$, so $1 - x/\lambda$ is invertible in $\widetilde{A}$ by lemma 6.1.5, yet $\widetilde{\varphi}(1 - x/\lambda) = 1 - \varphi(x)/\lambda = 0$ - a character vanishing on an invertible, which is impossible. Hence $|\varphi(x)| \leq \|x\|$.

(2) The kernel of a character has codimension 1 (since $A/\ker\varphi \cong \mathbb{C}$), so it is a maximal ideal. Conversely let M be a maximal ideal. It is closed: $\mathrm{GL}(A)$ is open (proposition 6.1.6) and disjoint from M, so the unit has a neighbourhood missing M, whence the closure $\overline{M}$ is a proper ideal containing M, forcing $\overline{M} = M$. Then A/M is a unital Banach algebra and a field, hence a Banach division algebra, so $A/M \cong \mathbb{C}$ by Gelfand-Mazur (corollary 6.1.12); the quotient map is a character with kernel M. The two assignments are mutually inverse. For compactness, $\Delta(A)$ is the set of φ in the closed unit ball of A' meeting the conditions $\varphi(xy) = \varphi(x)\varphi(y)$ for all x, y and $\varphi(1) = 1$; each is weak-$*$ closed (the maps $\varphi \mapsto \varphi(xy)$ and $\varphi \mapsto \varphi(x)\varphi(y)$ are weak-$*$ continuous), so $\Delta(A)$ is weak-$*$ closed in the ball, which is weak-$*$ compact by Alaoglu's theorem (theorem 1.5.7).

(3) Characters of A correspond to the unital characters of $\widetilde{A}$ other than the augmentation $(x, \lambda) \mapsto \lambda$; that is, $\Delta(\widetilde{A}) = \Delta(A) \cup \{0\}$, writing 0 for the character that vanishes on A. By (2) applied to $\widetilde{A}$ this set is weak-$*$ compact, and $\Delta(A)$ is its complement of the single point 0, hence open in a compact Hausdorff space and so locally compact Hausdorff.

Evaluation now turns each element of A into a function on $\Delta(A)$.

Theorem 6.1.15: The Gelfand Transform

Let A be a commutative Banach algebra over $\mathbb{C}$. For $x \in A$ define $\widehat{x} \colon \Delta(A) \to \mathbb{C}$ by $\widehat{x}(\varphi) = \varphi(x)$. Then:

1. $\widehat{x} \in C_0(\Delta(A))$, and the *Gelfand transform* $\Gamma \colon A \to C_0(\Delta(A))$, $x \mapsto \widehat{x}$, is a norm-decreasing algebra homomorphism.
2. If A is unital, then for every $x \in A$ the range of $\widehat{x}$ is exactly the spectrum,

$$^{(}x) = \widehat{x}(\Delta(A)) = \{\varphi(x) \mid \varphi \in \Delta(A)\}$$

   and in particular $\|\widehat{x}\|_\infty = r(x) \leq \|x\|$.

***Proof*:**
Continuity of $\widehat{x}$ is the definition of the weak-$*$ topology, and Γ is linear and multiplicative because each φ is. For vanishing at infinity, pass to $\widetilde{A}$: there $\Delta(\widetilde{A}) = \Delta(A) \cup \{0\}$ is compact Hausdorff (proposition 6.1.14), the extended function $\widehat{x}$ is continuous on it, and it takes the value 0 at the added character; continuity at that point is precisely $\widehat{x} \in C_0(\Delta(A))$.

For (2), with A unital: $\lambda \in^{(} x)$ iff $x - \lambda$ is not invertible, iff $x - \lambda$ generates a proper ideal and so lies in some maximal ideal M (Zorn), iff by proposition 6.1.14(2) there is a character φ with $\ker \varphi = M$, that is $\varphi(x - \lambda) = 0$, i.e. $\varphi(x) = \lambda$. Hence $^{(}x) = \{\varphi(x) \mid \varphi \in \Delta(A)\}$, and $\|\widehat{x}\|_\infty = \sup \left\{ |\lambda| \;\middle|\; \lambda \in^{(} x) \right\} = r(x) \leq \|x\|$ by theorem 6.1.11.

For a general commutative Banach algebra Γ is neither injective nor isometric: its kernel is the *radical* $\bigcap_{\varphi \in \Delta(A)} \ker \varphi$, and the norm it induces is the spectral radius $r(x)$, not $\|x\|$. Both defects vanish for a commutative C^*-algebra, where the C^*-identity $\|A^*A\| = \|A\|^2$ forces $r = \|\cdot\|$ and Γ becomes an isometric $*$-isomorphism onto $C_0(\Delta(A))$ - the commutative Gelfand-Naimark theorem, taken up in the C^*-algebras companion. The next section is the first payoff: the singly generated algebra $C^*(T)$ is commutative, and identifying it with $C(^{T})$ via Γ is exactly what upgrades the polynomial calculus to a continuous one.

6.2 The Continuous Functional Calculus

For a self-adjoint operator $T \in B(\mathbb{H})$, the polynomial $p(T) = \alpha_0 I + \alpha_1 T + \cdots + \alpha_n T^n$ is unambiguous: it is built from T by the algebra operations alone. The question this section answers is how far the assignment $p \mapsto p(T)$ can be pushed. If T were a diagonalizable matrix with eigenvalues $\lambda_1, \ldots, \lambda_k$, then $p(T)$ would act as multiplication by $p(\lambda_i)$ on the ith eigenspace, and this description would make sense for *any* function f on the eigenvalues, not just polynomials: $f(T)$ multiplies the ith eigenspace by $f(\lambda_i)$. The eigenvalues are the spectrum, so the natural domain of f is $\sigma(T)$. The continuous functional calculus makes this precise in infinite dimensions, where T need not have eigenvectors: it produces an operator $f(T)$ for every continuous f on $\sigma(T)$, compatible with the algebra operations and with complex conjugation, and it does so isometrically. The mechanism is a single inequality, $\|p(T)\| = \sup_{\sigma(T)} |p|$, which lets the discrete algebra of polynomials be completed to the continuous algebra $C(\sigma(T))$ by density.

Two facts about the spectrum of a self-adjoint operator drive everything. First, its spectrum is real, so $\sigma(T)$ is a compact subset of $\mathbb{R}$ and $C(\sigma(T))$ is an algebra of functions of a real variable. Second, its spectral radius equals its norm, so the spectrum "sees" the full size of T. Both follow from the single algebraic identity $\|T^2\| = \|T\|^2$, which is the operator-norm signature of self-adjointness.

Theorem 6.2.1: Spectrum of a Self-Adjoint Operator

Let $T \in B(\mathbb{H})$ be self-adjoint. Then $\sigma(T) \subseteq \mathbb{R}$, and the spectral radius equals the norm:

$$r(T) = \|T\|$$

***Proof*:**
Step 1: The spectral radius equals the norm. For self-adjoint T the adjoint identity $\|S^*S\| = \|S\|^2$ (proposition 5.2.7) applied to $S = T$ gives $\|T^2\| = \|T^*T\| = \|T\|^2$. Since T^2 is again self-adjoint ($(T^2)^* = (T^*)^2 = T^2$), the identity iterates: replacing T by T^2 gives $\|T^4\| = \|T^2\|^2 = \|T\|^4$, and by induction

$$\|T^{2^k}\| = \|T\|^{2^k}$$

for every $k \geq 0$. Taking 2^kth roots, $\|T^{2^k}\|^{1/2^k} = \|T\|$ for all k. The spectral radius is the limit of the convergent sequence $\|T^n\|^{1/n}$ (theorem 6.1.11), so its value is read off from any subsequence:

$$r(T) = \lim_{k\to\infty} \|T^{2^k}\|^{1/2^k} = \|T\|$$

Step 2: The spectrum is real. Fix $\lambda = a + ib$ with $a, b \in \mathbb{R}$ and $b \neq 0$; the claim is that $\lambda I - T$ is invertible, so $\lambda \notin \sigma(T)$. For any $x \in \mathbb{H}$,

$$\|(\lambda I - T)x\|^2 = \langle (\lambda I - T)x, (\lambda I - T)x\rangle$$

Expand using $T = T^*$ and $\langle Tx, x\rangle = \langle x, Tx\rangle = \overline{\langle Tx, x\rangle}$, so that $\langle Tx, x\rangle$ is real. Writing $\lambda I - T = (aI - T) + ibI$ and using that $aI - T$ is self-adjoint,

$$\|(\lambda I - T)x\|^2 = \|(aI - T)x\|^2 + b^2\|x\|^2 + 2\,\mathrm{Re}\langle (aI - T)x, ibx\rangle$$

The cross term is $2\,\mathrm{Re}\big(\overline{(ib)}\,\langle (aI - T)x, x\rangle\big) = 2\,\mathrm{Re}\big(-ib\,\langle (aI - T)x, x\rangle\big)$, and $\langle (aI - T)x, x\rangle$ is real, so the whole term is $2\,\mathrm{Re}(-ib \cdot (\text{real})) = 0$. Hence

$$\|(\lambda I - T)x\|^2 = \|(aI - T)x\|^2 + b^2\|x\|^2 \geq b^2\|x\|^2 \tag{6.1}$$

so $\|(\lambda I - T)x\| \geq |b|\,\|x\|$. Thus $\lambda I - T$ is bounded below, hence injective with closed range (proposition 5.2.6). The same bound applies to $(\lambda I - T)^* = \overline{\lambda} I - T$, since $\overline{\lambda} = a - ib$ has the same nonzero imaginary part in modulus; so $\overline{\lambda} I - T$ is injective as well. By proposition 5.2.4 the range of $\lambda I - T$ is dense: $\overline{\mathrm{ran}(\lambda I - T)} = \ker((\lambda I - T)^*)^\perp = \ker(\overline{\lambda} I - T)^\perp = \{0\}^\perp = \mathbb{H}$. A dense closed subspace is everything, so $\lambda I - T$ is a bijection, invertible by the open mapping theorem. Therefore no λ with nonzero imaginary part lies in $\sigma(T)$, giving $\sigma(T) \subseteq \mathbb{R}$, as we sought to show.

The two conclusions play different roles. Reality of the spectrum is what makes $C(\sigma(T))$ an algebra of functions on a compact subset of the real line, so that a function like $\sqrt{\cdot}$ or $|\cdot|$ can be applied to T whenever the spectrum lies where the function is continuous. The equality $r(T) = \|T\|$ is the quantitative heart: it says the operator norm is recovered from the spectrum alone, $\|T\| = \sup\{|\lambda| : \lambda \in \sigma(T)\}$, so nothing about the size of T is invisible to its spectrum. For a general (non-self-adjoint) operator this fails, as the lower bound (7.6) exhibits directly: a nilpotent matrix has spectrum $\{0\}$ yet nonzero norm, so $r < \|\cdot\|$ strictly.

Example 6.2.2: A Multiplication Operator
Let $\mathbb{H} = L^2([0,1])$ and let M_g be multiplication by a real-valued $g \in C([0,1])$: $(M_g f)(t) = g(t)f(t)$. This is self-adjoint, since $\langle M_g f, h\rangle = \int_0^1 g(t)f(t)\overline{h(t)}\,dt = \langle f, M_g h\rangle$ using that g is real. Its spectrum is the range of g: for $\lambda \in \mathbb{C}$, the operator $\lambda I - M_g$ is multiplication by $\lambda - g$, invertible exactly when $\lambda - g$ is bounded away from zero, that is when $\lambda \notin g([0,1])$. Hence

$$\sigma(M_g) = g([0,1]) \subseteq \mathbb{R}$$

which is a compact real interval (or union of intervals), consistent with theorem 6.2.1. The norm is $\|M_g\| = \sup_t |g(t)| = \max_{\lambda \in \sigma(M_g)} |\lambda| = r(M_g)$, confirming $r = \|\cdot\|$ concretely. When $g(t) = t$, the spectrum is the full interval $[0,1]$ and M_g has no eigenvalues at all: $\lambda f = tf$ forces $f = 0$ almost everywhere. This is the model to keep in mind. The spectrum can be a solid interval with no eigenvectors, and the functional calculus below is exactly what replaces the missing eigenspace decomposition.

With reality and $r(T) = \|T\|$ in hand, the polynomial calculus can be normed. A polynomial p with complex coefficients defines $p(T) \in B(\mathbb{H})$ by substitution; the point is that its operator norm is computed by evaluating $|p|$ on the spectrum. This is where the real spectrum matters: to make $p(T)$ self-adjoint (so that Step 1 of the previous proof applies to it) one restricts to polynomials with *real* coefficients, or more usefully passes to $\overline{p}p$, whose value on the real spectrum is $|p|^2$.

Proposition 6.2.3: The Polynomial Functional Calculus

Let $T \in B(\mathbb{H})$ be self-adjoint. For a polynomial $p(z) = \sum_{k=0}^{n} \alpha_k z^k$ with $\alpha_k \in \mathbb{C}$, set $p(T) = \sum_{k=0}^{n} \alpha_k T^k$. Then $p \mapsto p(T)$ is an algebra homomorphism from $\mathbb{C}[z]$ into $B(\mathbb{H})$ with $(\overline{p})(T) = p(T)^*$, where $\overline{p}$ has the conjugate coefficients, and

$$\|p(T)\| = \sup_{\lambda \in \sigma(T)} |p(\lambda)| \tag{6.2}$$

***Proof*:**
That $p \mapsto p(T)$ respects sums and products is immediate from the fact that powers of T commute: $(pq)(T) = p(T)q(T)$ because both sides are the operator obtained by multiplying out the polynomial pq and substituting T. Taking adjoints term by term, $p(T)^* = \sum_k \overline{\alpha_k}(T^*)^k = \sum_k \overline{\alpha_k} T^k = (\overline{p})(T)$, using $T = T^*$.

For (6.2), consider the operator $\overline{p}(T)\, p(T) = (\overline{p}p)(T)$. It is self-adjoint, since $((\overline{p}p)(T))^* = (\overline{\overline{p}p})(T) = (\overline{p}p)(T)$ as $\overline{p}p$ has real values on $\mathbb{R}$ and its coefficient conjugate is itself. By the C^*-identity proposition 5.2.7,

$$\|p(T)\|^2 = \|p(T)^* p(T)\| = \|(\overline{p}p)(T)\|$$

Since $\overline{p}p$ is a self-adjoint operator's defining polynomial, theorem 6.2.1 gives $\|(\overline{p}p)(T)\| = r\big((\overline{p}p)(T)\big)$. Now apply the polynomial spectral mapping theorem: for the polynomial $q = \overline{p}p$,

$$\sigma(q(T)) = q(\sigma(T)) = \{q(\lambda) : \lambda \in \sigma(T)\} \tag{6.3}$$

This is the two-sided refinement of lemma 6.1.9, which supplies the inclusion $q(\sigma(T)) \subseteq \sigma(q(T))$; the reverse inclusion holds because for $\mu \notin q(\sigma(T))$ one factors $q(z) - \mu = c \prod_j (z - \mu_j)$ over $\mathbb{C}$, and each root μ_j satisfies $q(\mu_j) = \mu$, so no μ_j lies in $\sigma(T)$, whence each $\mu_j I - T$ is invertible and $q(T) - \mu I = c \prod_j (\mu_j I - T)$ is a product of invertibles, invertible. Thus $\mu \notin \sigma(q(T))$, proving (6.3). Taking suprema of $|\cdot|$ over both sides of (6.3) for $q = \overline{p}p$, and using that $q(\lambda) = |p(\lambda)|^2$ for real λ (since $\lambda \in \sigma(T) \subseteq \mathbb{R}$ makes $\overline{p}(\lambda) = \overline{p(\lambda)}$),

$$r\big((\overline{p}p)(T)\big) = \sup_{\mu \in \sigma(q(T))} |\mu| = \sup_{\lambda \in \sigma(T)} |q(\lambda)| = \sup_{\lambda \in \sigma(T)} |p(\lambda)|^2$$

Chaining the equalities, $\|p(T)\|^2 = \sup_{\lambda\in\sigma(T)} |p(\lambda)|^2$, and taking square roots gives (6.2), completing the proof.

Equation (6.2) is the linchpin. Written in the sup norm on $C(\sigma(T))$, it reads $\|p(T)\|_{B(\mathbb{H})} = \|p\|_\infty$, where $\|p\|_\infty = \sup_{\sigma(T)} |p|$: the assignment $p \mapsto p(T)$ is an *isometry* from the polynomials, viewed as a subspace of $C(\sigma(T))$, into $B(\mathbb{H})$. An isometry defined on a dense subspace of a complete space extends uniquely to the whole space, and the polynomials are dense in $C(\sigma(T))$ by Stone-Weierstrass. This is the entire content of the next theorem: the extension exists because the norm is already controlled on the dense subspace.

One subtlety deserves attention before the extension. In (6.2) the norm of $p(T)$ depends on p only through its restriction to $\sigma(T)$: if two polynomials agree on the spectrum, they produce the same operator. This is forced by isometry, $\|(p-q)(T)\| = \sup_{\sigma(T)} |p-q| = 0$ when $p = q$ on $\sigma(T)$, and it is exactly what makes $C(\sigma(T))$, rather than $\mathbb{C}[z]$, the correct domain: the calculus factors through evaluation on the spectrum.

Theorem 6.2.4: The Continuous Functional Calculus

Let $T \in B(\mathbb{H})$ be self-adjoint. There is a unique map

$$\Phi\colon C(\sigma(T)) \to B(\mathbb{H})$$

that is an isometric $*$-homomorphism with $\Phi(1) = I$ and $\Phi(\mathrm{id}) = T$, where 1 is the constant function and $\mathrm{id}(\lambda) = \lambda$. Explicitly, Φ is linear and multiplicative, satisfies $\Phi(\overline{f}) = \Phi(f)^*$, and

$$\|\Phi(f)\| = \|f\|_\infty = \sup_{\lambda\in\sigma(T)} |f(\lambda)|$$

for every $f \in C(\sigma(T))$. One writes $f(T) := \Phi(f)$.

***Proof*:**
Step 1: Density of polynomials. The spectrum $\sigma(T)$ is a compact subset of $\mathbb{R}$ (theorem 6.2.1). The polynomials in the real variable λ form a subalgebra of $C(\sigma(T))$ that contains the constants, separates points (the function id already does), and is closed under complex conjugation (the conjugate of a polynomial is the polynomial with conjugated coefficients). By the Stone-Weierstrass theorem this subalgebra is dense in $C(\sigma(T))$ for the sup norm. Every $f \in C(\sigma(T))$ is thus a uniform limit $f = \lim_m p_m$ of polynomials.

Step 2: Definition of Φ by extension. On the dense subalgebra of polynomials, define $\Phi(p) = p(T)$. By (6.2) this is an isometry into $B(\mathbb{H})$: $\|\Phi(p) - \Phi(q)\| = \|(p-q)(T)\| = \|p-q\|_\infty$. Given $f \in C(\sigma(T))$ and polynomials $p_m \to f$ uniformly, the sequence (p_m) is Cauchy in $\|\cdot\|_\infty$, so $(\Phi(p_m)) = (p_m(T))$ is Cauchy in $B(\mathbb{H})$; since $B(\mathbb{H})$ is complete (it is a Banach space, $\mathbb{H}$ being complete), the limit

$$\Phi(f) := \lim_{m\to\infty} p_m(T)$$

exists in $B(\mathbb{H})$. The limit is independent of the approximating sequence: if $p_m \to f$ and $q_m \to f$, then $\|p_m(T) - q_m(T)\| = \|p_m - q_m\|_\infty \to 0$, so the two limits coincide. Thus Φ is well defined on all of $C(\sigma(T))$.

Step 3: Φ is an isometric $*$-homomorphism. Each property holds for polynomials and passes to the limit because all the operations involved are continuous. For $f = \lim p_m$ and $g = \lim q_m$:

- *Linearity.* $p_m + \alpha q_m \to f + \alpha g$ uniformly, and $\Phi(p_m + \alpha q_m) = p_m(T) + \alpha\, q_m(T) \to \Phi(f) + \alpha\Phi(g)$, so $\Phi(f + \alpha g) = \Phi(f) + \alpha\Phi(g)$.
- *Isometry.* The norm is continuous, so $\|\Phi(f)\| = \lim \|p_m(T)\| = \lim \|p_m\|_\infty = \|f\|_\infty$, using $\|p_m\|_\infty \to \|f\|_\infty$.
- *Multiplicativity.* $p_m q_m \to fg$ uniformly (uniform convergence is stable under products of uniformly bounded sequences, and $\|p_m\|_\infty, \|q_m\|_\infty$ are bounded as convergent sequences). Multiplication in $B(\mathbb{H})$ is jointly continuous on bounded sets: $\|p_m(T)q_m(T) - \Phi(f)\Phi(g)\| \le \|p_m(T)\|\,\|q_m(T) - \Phi(g)\| + \|p_m(T) - \Phi(f)\|\,\|\Phi(g)\| \to 0$. Hence $\Phi(fg) = \lim(p_m q_m)(T) = \lim p_m(T)q_m(T) = \Phi(f)\Phi(g)$.
- *$*$-preservation.* $\overline{p_m} \to \overline{f}$ uniformly and adjunction is a (conjugate-linear) isometry, hence continuous, so $\Phi(\overline{f}) = \lim \overline{p_m}(T) = \lim p_m(T)^* = (\lim p_m(T))^* = \Phi(f)^*$, using $(\overline{p})(T) = p(T)^*$ from proposition 6.2.3.

The normalizations $\Phi(1) = I$ and $\Phi(\mathrm{id}) = T$ hold already at the polynomial level, since the constant 1 and id are polynomials with $1(T) = I$ and $\mathrm{id}(T) = T$.

Step 4: Uniqueness. Suppose $\Psi\colon C(\sigma(T)) \to B(\mathbb{H})$ is any isometric $*$-homomorphism with $\Psi(1) = I$ and $\Psi(\mathrm{id}) = T$. Multiplicativity and linearity force $\Psi(\lambda^k) = T^k$ for every k, so $\Psi(p) = p(T) = \Phi(p)$ on every polynomial p. Two continuous maps that agree on a dense set agree everywhere; both Ψ and Φ are continuous, being isometries, and the polynomials are dense by Step 1. Hence $\Psi = \Phi$, completing the proof.

The theorem converts the algebra $C(\sigma(T))$ into an algebra of operators, faithfully and without distortion of norms. Three features of the map $f \mapsto f(T)$ are worth isolating. It is an *algebra* map, so composition of the calculus with algebra operations is transparent: $(f+g)(T) = f(T) + g(T)$, $(fg)(T) = f(T)g(T)$, and in particular any two operators $f(T), g(T)$ produced by the calculus commute, because $C(\sigma(T))$ is commutative. It is a *$*$-map*, so real-valued f give self-adjoint $f(T)$ and $|f| = 1$ on the spectrum gives unitary $f(T)$; the calculus respects the involution as well as the multiplication. And it is *isometric*, so estimates on operators reduce to estimates on scalar functions, $\|f(T)\| = \|f\|_\infty$, which is the single most useful computational consequence. Together these say that studying the self-adjoint T through functions of it is the same as studying scalar functions on $\sigma(T)$, with no loss.

The construction is a completion argument, so it is worth naming what it buys over the polynomial calculus: functions that are continuous but not polynomial. The most consequential of these is the modulus.

Example 6.2.5: The Absolute Value and the Positive Part

Let $T \in B(\mathbb{H})$ be self-adjoint, so $\sigma(T) \subseteq \mathbb{R}$. The function $f(\lambda) = |\lambda|$ is continuous on $\sigma(T)$ but is not a polynomial; the calculus produces an operator

$$|T| := f(T)$$

the *absolute value* of T. By the $*$-property, $|T|$ is self-adjoint (since f is real-valued), and by multiplicativity $|T|^2 = (f^2)(T)$ where $f(\lambda)^2 = \lambda^2$, so $|T|^2 = T^2$: the operator $|T|$ is a square root of T^2. It is a positive operator, since $|T| = g(T)^* g(T)$ for $g(\lambda) = \sqrt{|\lambda|}$ (continuous on $\sigma(T)$), giving $\langle |T|x, x\rangle = \|g(T)x\|^2 \geq 0$. Likewise the positive and negative parts $\lambda^+ = \max(\lambda, 0)$ and $\lambda^- = \max(-\lambda, 0)$ are continuous on $\sigma(T)$ and yield positive operators $T^+ = f^+(T)$, $T^- = f^-(T)$ with

$$T = T^+ - T^-, \qquad |T| = T^+ + T^-, \qquad T^+T^- = 0$$

the last because $\lambda^+\lambda^- = 0$ identically, so multiplicativity forces $T^+T^- = (f^+f^-)(T) = 0$. These identities decompose a self-adjoint operator into its positive and negative parts exactly as one decomposes a real function, and none of $|T|$, T^+, T^- is reachable by polynomials in T: the continuous calculus is what makes them available. For the multiplication operator M_g of example 6.2.2 with g real-valued, $|M_g| = M_{|g|}$ and $M_g^{\pm} = M_{g^{\pm}}$, the calculus acting by composition on the multiplier.

The isometry (6.2) was the input to the construction; the same identity, now for arbitrary continuous f, is the output, and it upgrades to a full description of the spectrum of $f(T)$. The polynomial spectral mapping theorem used inside the proof of proposition 6.2.3 extends verbatim to the continuous calculus: the spectrum of $f(T)$ is the image of $\sigma(T)$ under f.

Proposition 6.2.6: Continuous Spectral Mapping

Let $T \in B(\mathbb{H})$ be self-adjoint and $f \in C(\sigma(T))$. Then

$$\sigma(f(T)) = f(\sigma(T)) = \{f(\lambda) : \lambda \in \sigma(T)\}$$

***Proof*:**
Both sides are compact subsets of $\mathbb{C}$: $f(\sigma(T))$ is the continuous image of the compact set $\sigma(T)$, and $\sigma(f(T))$ is compact by theorem 6.1.11. It suffices to prove the two inclusions.

Step 1: $f(\sigma(T)) \subseteq \sigma(f(T))$. Fix $\lambda_0 \in \sigma(T)$ and set $\mu = f(\lambda_0)$; the claim is $\mu \in \sigma(f(T))$, that is $f(T) - \mu I$ is not invertible. Suppose it were, with inverse S. Choose polynomials $p_m \to f$ uniformly on $\sigma(T)$ (theorem 6.2.4, Step 1). Then $p_m(T) \to f(T)$ in norm, so for large m the operator $p_m(T) - \mu I$ is invertible (the invertibles are open, proposition 6.1.6), with inverses bounded uniformly near S. But $p_m(\lambda_0) \to f(\lambda_0) = \mu$, and by the polynomial spectral mapping (6.3), $p_m(\lambda_0) \in p_m(\sigma(T)) = \sigma(p_m(T))$, so $p_m(T) - p_m(\lambda_0)I$ is *not* invertible. Since $p_m(\lambda_0) \to \mu$ and inversion is continuous where defined, the non-invertible elements $p_m(T) - p_m(\lambda_0)I$ converge to $f(T) - \mu I$; the non-invertibles form a closed set (the complement of the open group of invertibles), so the limit $f(T) - \mu I$ is not invertible. This contradicts the supposition, so $\mu \in \sigma(f(T))$.

Step 2: $\sigma(f(T)) \subseteq f(\sigma(T))$. Let $\mu \notin f(\sigma(T))$; the claim is $\mu \notin \sigma(f(T))$, that is $f(T) - \mu I$ is invertible. Because μ is not in the compact set $f(\sigma(T))$, the function

$$g(\lambda) = \frac{1}{f(\lambda) - \mu}$$

is continuous on $\sigma(T)$: its denominator $f(\lambda) - \mu$ never vanishes on $\sigma(T)$, so $g \in C(\sigma(T))$. Applying the calculus and its multiplicativity (theorem 6.2.4) to the identity $g \cdot (f - \mu) = 1$ in $C(\sigma(T))$,

$$g(T)\,(f(T) - \mu I) = (f(T) - \mu I)\,g(T) = I$$

so $g(T)$ is a two-sided inverse of $f(T) - \mu I$. Hence $f(T) - \mu I$ is invertible and $\mu \notin \sigma(f(T))$, as we sought to show.

The spectral mapping theorem is the statement that the calculus is compatible with the spectrum in the strongest possible sense: applying f to T applies f to its spectrum. It repackages the isometry $\|f(T)\| = \|f\|_\infty$ (take suprema of $|\cdot|$ over $\sigma(f(T)) = f(\sigma(T))$ and use $r(f(T)) = \|f(T)\|$ when f is real-valued) and it makes the calculus *internally consistent*: applying a second function $h \in C(\sigma(f(T)))$ to $f(T)$ agrees with applying the composite $h \circ f \in C(\sigma(T))$ to T, since both are computed by uniform approximation and the spectra match. For the multiplication operator M_g of example 6.2.2, $\sigma(f(M_g)) = f(\sigma(M_g)) = f(g([0,1])) = (f \circ g)([0,1])$, which is $\sigma(M_{f\circ g})$, recovering $f(M_g) = M_{f\circ g}$ and matching the concrete calculation there. The one gap the continuous calculus leaves open is discontinuous f such as indicators of subsets of the spectrum, which are exactly the functions that would carve $\mathbb{H}$ into spectral subspaces; supplying those is the task of the Borel calculus in the next section.

6.3 The Borel Functional Calculus and Spectral Projections

The continuous functional calculus (theorem 6.2.4) lets a continuous function f act on a self-adjoint operator T, but continuity is a severe restriction on the functions available. To slice the operator apart the way a diagonal matrix is sliced by its eigenprojections, the indicator χ_Ω of a Borel set $\Omega \subseteq \sigma(T)$ must be admissible, and χ_Ω is almost never continuous. The task of this section is to enlarge the calculus $f \mapsto f(T)$ from $C(\sigma(T))$ to the bounded Borel functions $B_b(\sigma(T))$. The enlargement is forced: each pair of vectors $x, y \in \mathbb{H}$ turns the continuous calculus into a bounded functional on $C(\sigma(T))$, and the Riesz representation theorem (theorem 2.3.3) converts that functional into a complex measure against which any bounded Borel function may be integrated. The indicators χ_Ω then produce the *spectral projections* $E(\Omega)$, the objects that will assemble into the projection-valued measure of the spectral theorem.

Throughout, $T \in B(\mathbb{H})$ is self-adjoint, so $\sigma(T) \subseteq \mathbb{R}$ is a compact subset of the line (theorem 6.2.1), and $f \mapsto f(T)$ denotes the isometric $*$-homomorphism $C(\sigma(T)) \to B(\mathbb{H})$ of theorem 6.2.4. Write $B_b(\sigma(T))$ for the bounded Borel functions on $\sigma(T)$, a commutative C^*-algebra under pointwise operations, complex conjugation, and the supremum norm $\|f\|_\infty = \sup_{\lambda \in \sigma(T)} |f(\lambda)|$; it contains $C(\sigma(T))$ as a subalgebra.

The first step is to attach a measure to each pair of vectors. Fix $x, y \in \mathbb{H}$. The continuous calculus already gives a map $C(\sigma(T)) \to \mathbb{C}$ by

$$f \longmapsto \langle f(T)x, y\rangle$$

This is linear in f (the calculus is an algebra homomorphism, hence linear), and it is bounded, because the calculus is isometric:

$$|\langle f(T)x, y\rangle| \leq \|f(T)x\| \, \|y\| \leq \|f(T)\| \, \|x\| \, \|y\| = \|f\|_\infty \|x\| \, \|y\|$$

So $f \mapsto \langle f(T)x, y\rangle$ is a bounded linear functional on $C(\sigma(T))$ of norm at most $\|x\| \, \|y\|$. Since $\sigma(T)$ is compact, $C(\sigma(T)) = C_0(\sigma(T))$, and theorem 2.3.3 identifies its dual with the complex regular Borel measures on $\sigma(T)$. There is therefore a unique such measure, which we call $\mu_{x,y}$, representing the functional.

Definition 6.3.1: Spectral Measures $\mu_{x,y}$

Let $T \in B(\mathbb{H})$ be self-adjoint and $x, y \in \mathbb{H}$. The *spectral measure* associated to the pair (x, y) is the unique complex regular Borel measure $\mu_{x,y}$ on $\sigma(T)$ furnished by theorem 2.3.3 for the bounded functional $f \mapsto \langle f(T)x, y\rangle$ on $C(\sigma(T))$; that is,

$$\langle f(T)x, y\rangle = \int_{\sigma(T)} f \, d\mu_{x,y} \qquad (f \in C(\sigma(T)))$$

Its total variation satisfies $\|\mu_{x,y}\| \leq \|x\| \, \|y\|$.

The bound $\|\mu_{x,y}\| \leq \|x\| \, \|y\|$ is exactly the operator norm of the functional, which theorem 2.3.3 equates with the total variation $|\mu_{x,y}|(\sigma(T))$. The family $\{\mu_{x,y}\}$ inherits the sesquilinearity of the inner product: for fixed y the map $x \mapsto \mu_{x,y}$ is linear, and for fixed x the map $y \mapsto \mu_{x,y}$ is conjugate-linear, since $\langle f(T)x, y\rangle$ has these properties in x and y and the representing measure is unique. This is the raw material for the extension: any bounded Borel f can be integrated against $\mu_{x,y}$, and the resulting number will play the role of $\langle f(T)x, y\rangle$.

Example 6.3.2: Spectral Measures for a Diagonal Operator

Let $\mathbb{H} = \ell^2(\mathbb{N})$ and let T be the diagonal operator $Te_n = \lambda_n e_n$ on the standard orthonormal basis (e_n), where (λ_n) is a bounded real sequence; T is self-adjoint and $\sigma(T) = \overline{\{\lambda_n \mid n \in \mathbb{N}\}}$. For $f \in C(\sigma(T))$ the continuous calculus acts diagonally, $f(T)e_n = f(\lambda_n)e_n$, so for $x = \sum_n a_n e_n$ and $y = \sum_n b_n e_n$,

$$\langle f(T)x, y\rangle = \sum_n f(\lambda_n) \, a_n \overline{b_n} = \int_{\sigma(T)} f \, d\left(\sum_n a_n \overline{b_n} \, \delta_{\lambda_n}\right)$$

where δ_{λ_n} is the point mass at λ_n. Thus $\mu_{x,y} = \sum_n a_n \overline{b_n} \, \delta_{\lambda_n}$, a purely atomic measure supported on the eigenvalues, with total variation $\sum_n |a_n| \, |b_n| \leq \|x\| \, \|y\|$ by Cauchy-Schwarz. Taking $x = y = e_k$ gives $\mu_{e_k,e_k} = \delta_{\lambda_k}$: the diagonal spectral measure at a basis vector is the point mass at its eigenvalue. Integrating $\chi_{\{\lambda_k\}}$ against it will recover the eigenprojection onto $\mathbb{C}e_k$.

The measures $\mu_{x,y}$ now drive the extension of the calculus to bounded Borel functions. The definition $\langle f(T)x, y\rangle = \int f \, d\mu_{x,y}$ for Borel f must be shown to determine a genuine bounded operator $f(T)$, and then to respect the algebra operations. Boundedness comes from the estimate on the total variation; the operator is produced by the correspondence between bounded sesquilinear forms and bounded operators.

Theorem 6.3.3: The Borel Functional Calculus

Let $T \in B(\mathbb{H})$ be self-adjoint. There is a unique linear map

$$\Phi_b \colon B_b(\sigma(T)) \longrightarrow B(\mathbb{H}), \qquad \Phi_b(f) =: f(T)$$

extending the continuous calculus of theorem 6.2.4 and characterized by

$$\langle f(T)x, y\rangle = \int_{\sigma(T)} f \, d\mu_{x,y} \qquad (x, y \in \mathbb{H},\ f \in B_b(\sigma(T))) \tag{6.4}$$

It has the following properties.

1. Φ_b is a $*$-homomorphism: $(fg)(T) = f(T)g(T)$ and $\overline{f}(T) = f(T)^*$, with $1(T) = I$.
2. Φ_b is a contraction: $\|f(T)\| \leq \|f\|_\infty$.
3. Φ_b is boundedly convergent: if $(f_n) \subseteq B_b(\sigma(T))$ satisfies $\sup_n \|f_n\|_\infty < \infty$ and $f_n \to f$ pointwise, then $f_n(T)x \to f(T)x$ for every $x \in \mathbb{H}$ (convergence in the strong operator topology).
4. If $f \geq 0$ then $f(T) \geq 0$; consequently $f(T)$ is self-adjoint when f is real-valued.

***Proof*:**
The proof runs in five steps: build the operator from the sesquilinear form and prove the norm bound (Step 1), establish the adjoint identity (Step 2), promote bounded convergence from the continuous calculus to Borel functions weakly (Step 3), deduce multiplicativity from that weak convergence by a monotone-class argument (Step 4), and finally upgrade the convergence to the strong topology (Step 5). Positivity and uniqueness follow.

Step 1: $f(T)$ is a well-defined bounded operator. Fix $f \in B_b(\sigma(T))$ and define

$$B(x, y) = \int_{\sigma(T)} f \, d\mu_{x,y}$$

The assignment $(x, y) \mapsto \mu_{x,y}$ is linear in x and conjugate-linear in y, and f is fixed, so B is sesquilinear (linear in x, conjugate-linear in y). It is bounded:

$$|B(x, y)| = \left| \int f \, d\mu_{x,y} \right| \leq \|f\|_\infty \, |\mu_{x,y}|(\sigma(T)) = \|f\|_\infty \|\mu_{x,y}\| \leq \|f\|_\infty \|x\| \, \|y\|$$

By the correspondence between bounded sesquilinear forms and bounded operators on a Hilbert space, there is a unique $f(T) \in B(\mathbb{H})$ with $\langle f(T)x, y\rangle = B(x, y)$ for all x, y, and $\|f(T)\| = \sup_{\|x\|,\|y\| \leq 1} |B(x, y)| \leq \|f\|_\infty$. This gives (6.4), property (2), and, by uniqueness of the representing operator, linearity of Φ_b in f. When $f \in C(\sigma(T))$ the defining property of $\mu_{x,y}$ makes $B(x, y) = \langle f(T)x, y\rangle$ for the *continuous*-calculus $f(T)$, so Φ_b agrees with theorem 6.2.4 there and is a genuine extension.

Step 2: the adjoint identity. For $f \in B_b(\sigma(T))$ and any x, y,

$$\langle \overline{f}(T)x, y\rangle = \int \overline{f} \, d\mu_{x,y} = \overline{\int f \, d\overline{\mu_{x,y}}}$$

The measure $\overline{\mu_{x,y}}$ (conjugate of each complex value) represents $g \mapsto \int g\, d\mu_{y,x}$ over continuous g: for such g, $\overline{\langle \overline{g}(T)x, y\rangle} = \langle y, \overline{g}(T)x\rangle = \langle g(T)y, x\rangle = \int g\, d\mu_{y,x}$, where the middle step uses $\overline{g}(T)^* = g(T)$ from the continuous calculus. By uniqueness of the representing measure, $\overline{\mu_{x,y}} = \mu_{y,x}$. Hence $\langle \overline{f}(T)x, y\rangle = \overline{\int f\, d\mu_{y,x}} = \overline{\langle f(T)y, x\rangle} = \langle x, f(T)y\rangle = \langle f(T)^*x, y\rangle$ for all x, y, so $\overline{f}(T) = f(T)^*$. Applying this to a real f gives $f(T) = f(T)^*$, the self-adjointness in (4).

Step 3: bounded convergence, weakly. Let (f_n) be uniformly bounded, $\|f_n\|_\infty \le M$, with $f_n \to f$ pointwise on $\sigma(T)$. For fixed x, y the sequence integrates against the finite measure $\mu_{x,y}$: each $|f_n| \le M$ is dominated by the constant $M \in L^1(|\mu_{x,y}|)$ (finite since $|\mu_{x,y}|(\sigma(T)) \le \|x\|\,\|y\|$), so dominated convergence gives

$$\langle f_n(T)x, y\rangle = \int f_n\, d\mu_{x,y} \longrightarrow \int f\, d\mu_{x,y} = \langle f(T)x, y\rangle$$

Thus $f_n(T) \to f(T)$ in the weak operator topology whenever $\|f_n\|_\infty$ is bounded and $f_n \to f$ pointwise. (Strong convergence is deferred to Step 5, once multiplicativity is available.)

Step 4: multiplicativity, by a bounded-convergence / monotone-class argument. Multiplicativity $(fg)(T) = f(T)g(T)$ holds on $C(\sigma(T))$ by theorem 6.2.4. Extend it in two stages, each using only the weak convergence of Step 3.

First fix $g \in C(\sigma(T))$ and let

$$\mathcal{M}_g = \{f \in B_b(\sigma(T)) \mid (fg)(T) = f(T)g(T)\}$$

This is a linear subspace containing $C(\sigma(T))$, and it is closed under uniformly bounded pointwise limits: if $f_n \in \mathcal{M}_g$ with $\sup_n \|f_n\|_\infty < \infty$ and $f_n \to f$ pointwise, then $f_n g \to fg$ pointwise and boundedly, so by Step 3 both $(f_n g)(T) \to (fg)(T)$ and $f_n(T) \to f(T)$ weakly. Testing against x, y and using that $g(T)$ is bounded,

$$\langle (fg)(T)x, y\rangle = \lim_n \langle (f_n g)(T)x, y\rangle = \lim_n \langle f_n(T)\, g(T)x, y\rangle = \langle f(T)g(T)x, y\rangle$$

the last step applying weak convergence of $f_n(T)$ to the fixed vector $g(T)x$; so $f \in \mathcal{M}_g$. A linear class of bounded functions that contains $C(\sigma(T))$ and is closed under uniformly bounded pointwise limits contains every bounded Borel function. Indeed, for an open $U \subseteq \sigma(T)$ the Urysohn functions $g_n \in C(\sigma(T))$ with $0 \le g_n \uparrow \chi_U$ are uniformly bounded and converge pointwise to χ_U, so $\chi_U \in \mathcal{M}_g$; the collection $\mathcal{D} = \{\Omega \subseteq \sigma(T) \mid \chi_\Omega \in \mathcal{M}_g\}$ then contains the open sets, contains $\sigma(T)$ (as $1 \in C(\sigma(T)) \subseteq \mathcal{M}_g$), is closed under complements ($\chi_{\sigma(T)\setminus\Omega} = 1 - \chi_\Omega \in \mathcal{M}_g$ by linearity), and is closed under countable disjoint unions (for pairwise disjoint Ω_n the indicators $\chi_{\bigcup_{n\le N}\Omega_n} = \sum_{n \le N} \chi_{\Omega_n}$ are uniformly bounded and increase pointwise to $\chi_{\bigcup_n \Omega_n}$, a bounded pointwise limit). Thus $\mathcal{D}$ is a Dynkin system. The open sets are closed under finite intersection, so they form a π-system generating the Borel σ-algebra; by Dynkin's π-λ theorem $\mathcal{D}$ contains the σ-algebra they generate, that is, every Borel set. (Closure under complements and disjoint unions alone gives only a Dynkin system, not a σ-algebra; the π-λ theorem supplies the passage to all Borel sets, since products $\chi_A\chi_B$ are not yet known to lie in $\mathcal{M}_g$.) Every bounded Borel function is a uniformly bounded pointwise limit of simple functions built from these indicators, so $\mathcal{M}_g = B_b(\sigma(T))$. Thus $(fg)(T) = f(T)g(T)$ for all $f \in B_b(\sigma(T))$ and $g \in C(\sigma(T))$.

Now fix $f \in B_b(\sigma(T))$ and repeat with $\mathcal{N}_f = \{g \in B_b(\sigma(T)) \mid (fg)(T) = f(T)g(T)\}$. The stage just completed shows $C(\sigma(T)) \subseteq \mathcal{N}_f$, and the identical monotone-class argument shows $\mathcal{N}_f$ closed under uniformly bounded pointwise limits, so $\mathcal{N}_f = B_b(\sigma(T))$. Hence $(fg)(T) = f(T)g(T)$

for all bounded Borel f, g, which with Step 2 and $1(T) = I$ (inherited from the continuous calculus) makes Φ_b a $*$-homomorphism, property (1).

Positivity. If $f \geq 0$ is bounded Borel, write $f = h^2$ with $h = \sqrt{f} \in B_b(\sigma(T))$ real; then $f(T) = h(T)^2 = h(T)^* h(T)$ by property (1) and Step 2, so $\langle f(T)x, x\rangle = \|h(T)x\|^2 \geq 0$, giving property (4).

Step 5: upgrading to strong convergence. With multiplicativity in hand, the weak convergence of Step 3 improves to strong. Keep $\|f_n\|_\infty \leq M$ and $f_n \to f$ pointwise, and fix x. The scalar measure $\mu_{x,x}$ is finite and positive with total mass $\|x\|^2$: for real Borel $u \geq 0$, writing $u = w^2$ with $w = \sqrt{u}$ real Borel gives $\int u \, d\mu_{x,x} = \langle u(T)x, x\rangle = \langle w(T)^* w(T)x, x\rangle = \|w(T)x\|^2 \geq 0$ by property (1) and Step 2, and $\mu_{x,x}(\sigma(T)) = \langle 1(T)x, x\rangle = \|x\|^2$. Applying the $*$-homomorphism to $|f_n - f|^2 = \overline{(f_n - f)}(f_n - f)$,

$$\|f_n(T)x - f(T)x\|^2 = \langle (f_n - f)(T)^*(f_n - f)(T)x, x\rangle = \int |f_n - f|^2 \, d\mu_{x,x}$$

The integrand is dominated by the constant $(2M)^2 \in L^1(\mu_{x,x})$ and tends to 0 pointwise, so dominated convergence sends the integral to 0. Hence $f_n(T)x \to f(T)x$ in norm for every x, which is property (3).

Uniqueness. Any linear extension satisfying (6.4) is determined on every pair (x, y) by the measure $\mu_{x,y}$, hence is unique, completing the proof.

The extension trades one property for another. The continuous calculus is an *isometry*; the Borel calculus is only a *contraction* (property (2)), because a discontinuous f may have $\|f(T)\| < \|f\|_\infty$ once f is altered off the "mass" of the measures $\mu_{x,x}$ (a function supported where T has no spectrum in the measure-theoretic sense contributes nothing). What is gained is closure under bounded pointwise limits (property (3)): the calculus now commutes with the limit operations of measure theory, not merely with uniform limits. That single gain is what admits the indicators χ_Ω, and with them the entire machinery of projections and integration developed next. The $*$-homomorphism property (1) says the algebraic structure survives intact: products, adjoints, and the unit all transport from functions to operators exactly as for continuous f.

Example 6.3.4: Borel Calculus on a Diagonal Operator

Continue with the diagonal $Te_n = \lambda_n e_n$ on $\ell^2(\mathbb{N})$ from example 6.3.2. For a bounded Borel f on $\sigma(T)$, formula (6.4) with $\mu_{e_n,e_m} = \delta_{\lambda_n}$ if $n = m$ and $\mu_{e_n,e_m} = 0$ for $n \neq m$ gives

$$\langle f(T)e_n, e_m\rangle = \int_{\sigma(T)} f \, d\mu_{e_n,e_m} = f(\lambda_n)\, \delta_{nm}$$

so $f(T)e_n = f(\lambda_n)e_n$: the Borel calculus still acts diagonally, now for *any* bounded Borel f, not just continuous f. Taking $f = \chi_{\{\lambda_k\}}$ (the indicator of a single eigenvalue, a genuinely discontinuous function when λ_k is a limit of other eigenvalues) yields the operator $e_n \mapsto \chi_{\{\lambda_k\}}(\lambda_n)e_n$, which annihilates every e_n with $\lambda_n \neq \lambda_k$ and fixes those with $\lambda_n = \lambda_k$. This is precisely the orthogonal projection onto the eigenspace $\overline{\operatorname{span}}\{e_n \mid \lambda_n = \lambda_k\}$. The isometry defect is visible here: if some λ_k is isolated with a one-dimensional eigenspace, $\|\chi_{\{\lambda_k\}}\|_\infty = 1 = \|\chi_{\{\lambda_k\}}(T)\|$, but χ_Ω for a Borel Ω disjoint from $\{\lambda_n \mid n\} \cup \{0\}$ has $\|\chi_\Omega\|_\infty = 1$ while $\chi_\Omega(T) = 0$.

With the calculus in hand, the indicators are now legitimate arguments, and their operators are the spectral projections. These are the building blocks that a general $f(T)$ integrates against, the operator analogue of the sets a scalar integral integrates over.

Definition 6.3.5: Spectral Projection

Let $T \in B(\mathbb{H})$ be self-adjoint. For a Borel set $\Omega \subseteq \sigma(T)$, the *spectral projection* of T over Ω is the operator

$$E(\Omega) := \chi_\Omega(T)$$

the Borel calculus (theorem 6.3.3) applied to the indicator function χ_Ω.

Each $E(\Omega)$ deserves its name: it is an orthogonal projection in the sense of definition 5.2.15.

Proposition 6.3.6: Spectral Projections Are Orthogonal Projections

For every Borel $\Omega \subseteq \sigma(T)$, the operator $E(\Omega) = \chi_\Omega(T)$ is an orthogonal projection: $E(\Omega) = E(\Omega)^* = E(\Omega)^2$.

***Proof*:**
The indicator satisfies $\chi_\Omega^2 = \chi_\Omega$ and $\overline{\chi_\Omega} = \chi_\Omega$ (it is real-valued and idempotent). Applying the $*$-homomorphism Φ_b of theorem 6.3.3, part (1),

$$E(\Omega)^2 = \chi_\Omega(T)^2 = (\chi_\Omega^2)(T) = \chi_\Omega(T) = E(\Omega), \qquad E(\Omega)^* = \overline{\chi_\Omega}(T) = \chi_\Omega(T) = E(\Omega)$$

so $E(\Omega)$ is self-adjoint and idempotent, hence an orthogonal projection by definition 5.2.15, as we sought to show.

An orthogonal projection is completely described by its range, a closed subspace of $\mathbb{H}$, and $I - E(\Omega) = \chi_{\sigma(T)\setminus\Omega}(T) = E(\sigma(T) \setminus \Omega)$ is the projection onto the orthogonal complement. The range of $E(\Omega)$ is the "part of $\mathbb{H}$ that T places over Ω", made precise in example 6.3.7 for the diagonal case and, in general, by the spectral theorem of the next section. When $\Omega = \{\lambda\}$ is a single point and λ is an eigenvalue, $E(\{\lambda\})$ is the orthogonal projection onto the eigenspace $\ker(T - \lambda I)$; when λ is in the spectrum but not an eigenvalue (continuous spectrum), $E(\{\lambda\}) = 0$ even though small neighborhoods of λ carry nonzero projections. This is the mechanism by which the calculus handles operators with no eigenvectors at all.

Example 6.3.7: Spectral Projections of a Diagonal Operator
For the diagonal $Te_n = \lambda_n e_n$ on $\ell^2(\mathbb{N})$, example 6.3.4 gives $E(\Omega)e_n = \chi_\Omega(\lambda_n)e_n$, so $E(\Omega)$ is the orthogonal projection onto the closed span of the basis vectors whose eigenvalue lies in Ω:

$$\operatorname{ran} E(\Omega) = \overline{\operatorname{span}}\,\{e_n \mid \lambda_n \in \Omega\}$$

Two special cases fix the intuition. With $\Omega = \sigma(T)$ every $\lambda_n \in \Omega$, so $E(\sigma(T)) = I$; with $\Omega = \varnothing$ no eigenvalue qualifies, so $E(\varnothing) = 0$. For $\Omega = [0, \infty)$ the projection $E([0, \infty))$ cuts $\mathbb{H}$ into the closed span of the eigenvectors with nonnegative eigenvalue and its complement, the "positive part" of the operator. These are the countable analogue of grouping the diagonal entries of a matrix by which subset of the spectrum they fall in and reading off the corresponding coordinate projection.

Assembled over all Borel sets, these projections form a set function. The map $\Omega \mapsto E(\Omega)$ carries the Boolean algebra of Borel sets, with intersection and complement, to a commuting family of projections, with operator product and orthogonal complement. The correspondence is exact, and its countable additivity in the strong topology is what makes E a measure valued in projections rather than a mere finitely additive assignment.

Proposition 6.3.8: Properties of the Spectral Projections

Let $T \in B(\mathbb{H})$ be self-adjoint and $\Omega, \Omega', (\Omega_n)_{n\in\mathbb{N}}$ Borel subsets of $\sigma(T)$. The spectral projections satisfy:

1. $E(\varnothing) = 0$ and $E(\sigma(T)) = I$;
2. $E(\Omega \cap \Omega') = E(\Omega)E(\Omega')$; in particular the projections commute, and if $\Omega \cap \Omega' = \varnothing$ then $E(\Omega)E(\Omega') = 0$ (their ranges are orthogonal);
3. (countable strong additivity) if the Ω_n are pairwise disjoint with union $\Omega = \bigcup_n \Omega_n$, then
$$E(\Omega)x = \sum_{n=1}^{\infty} E(\Omega_n)x \qquad (x \in \mathbb{H})$$
the series converging in the norm of $\mathbb{H}$.

***Proof*:**
Part (1). The empty set has $\chi_\varnothing = 0$, so $E(\varnothing) = 0(T) = 0$; the full set has $\chi_{\sigma(T)} = 1$, so $E(\sigma(T)) = 1(T) = I$ by the unit property in theorem 6.3.3(1).

Part (2). Indicators multiply as $\chi_\Omega\chi_{\Omega'} = \chi_{\Omega\cap\Omega'}$ pointwise. Applying the multiplicativity of theorem 6.3.3(1),

$$E(\Omega)E(\Omega') = \chi_\Omega(T)\chi_{\Omega'}(T) = (\chi_\Omega\chi_{\Omega'})(T) = \chi_{\Omega\cap\Omega'}(T) = E(\Omega \cap \Omega')$$

Since $\Omega\cap\Omega' = \Omega'\cap\Omega$, this operator equals both $E(\Omega)E(\Omega')$ and $E(\Omega')E(\Omega)$, so the two projections commute. When $\Omega \cap \Omega' = \varnothing$ the product is $E(\varnothing) = 0$; then for any x, y, $\langle E(\Omega)x, E(\Omega')y\rangle = \langle E(\Omega')^*E(\Omega)x, y\rangle = \langle E(\Omega')E(\Omega)x, y\rangle = 0$, so the ranges are orthogonal.

Part (3). Set $S_N = \bigcup_{n=1}^N \Omega_n$, a Borel set, so $\chi_{S_N} = \sum_{n=1}^N \chi_{\Omega_n}$ (the Ω_n are disjoint) and by linearity of the calculus $E(S_N) = \sum_{n=1}^N E(\Omega_n)$. As $N \to \infty$ the indicators converge pointwise and monotonically to χ_Ω,
$$\chi_{S_N} \uparrow \chi_\Omega, \qquad 0 \le \chi_{S_N} \le 1$$
so the family (χ_{S_N}) is uniformly bounded (by 1) and converges pointwise to χ_Ω. The bounded convergence property theorem 6.3.3(3) gives $E(S_N)x \to E(\Omega)x$ in norm for every $x \in \mathbb{H}$, that is,

$$\sum_{n=1}^{N} E(\Omega_n)x = E(S_N)x \longrightarrow E(\Omega)x$$

which is the asserted norm-convergent series. (The convergence is at the level of vectors, the strong operator topology, not in operator norm: for disjoint Ω_n with nonzero projections the partial sums are projections onto growing subspaces and cannot be operator-norm Cauchy.) This completes the proof.

Part (3) is the reason E is called a projection-valued *measure*: it is countably additive on disjoint unions, in the strong operator topology. The failure of norm convergence noted at the end is not a defect but the correct behavior; it mirrors the fact that in scalar measure theory a countable partition of the space gives a series of measures $\mu(\Omega_n)$ summing to $\mu(\Omega)$, and the operator statement upgrades numbers to orthogonal projections onto orthogonal subspaces. The Pythagorean identity is exact here: for disjoint Ω_n, part (2) makes the ranges of $E(\Omega_n)$ mutually orthogonal, so

$$\|E(\Omega)x\|^2 = \left\|\sum_n E(\Omega_n)x\right\|^2 = \sum_n \|E(\Omega_n)x\|^2$$

and applying this to $\Omega = \sigma(T)$ with any partition decomposes $\|x\|^2 = \|E(\sigma(T))x\|^2$ into the masses the vector places over the pieces of the spectrum. The scalar measure $\Omega \mapsto \langle E(\Omega)x, x\rangle = \|E(\Omega)x\|^2$ is then a genuine finite positive Borel measure of total mass $\|x\|^2$, the diagonal spectral measure $\mu_{x,x}$ of definition 6.3.1 recovered from the projections. This positive measure is the object the spectral theorem integrates λ against to write $T = \int \lambda \, dE(\lambda)$, the construction taken up in section 6.4.

6.4 The Spectral Theorem for Bounded Self-Adjoint Operators

The Borel calculus of the previous section produced, for each self-adjoint $T \in B(\mathbb{H})$ and each Borel set $\Omega \subseteq \sigma(T)$, an orthogonal projection $E(\Omega) = \chi_\Omega(T)$, and recorded its arithmetic (proposition 6.3.8): the projections multiply the way the sets intersect, and they add up strongly over disjoint unions. This section reads that family the other way around. Packaged as a single object, the assignment $\Omega \mapsto E(\Omega)$ is a measure whose values are projections; against it one integrates a bounded Borel function to recover $f(T)$, and integrating the coordinate function λ itself returns T. The identity $T = \int_{\sigma(T)} \lambda \, dE(\lambda)$ is the resolution of the identity, and it is the exact infinite-dimensional replacement for writing a Hermitian matrix as $\sum_i \lambda_i P_i$ over its eigenvalues. A second reading of the same data trades the abstract projections for concrete multiplication operators: every bounded self-adjoint operator is, up to a unitary change of coordinates, multiplication by the independent variable on a direct sum of L^2 spaces.

The projection family from proposition 6.3.8 has exactly the structure that makes it integrable. Abstracting that structure gives the central object of the section.

Definition 6.4.1: Projection-Valued Measure

Let $K \subseteq \mathbb{R}$ (or $K \subseteq \mathbb{C}$) be compact and let $\mathcal{B}(K)$ denote its Borel σ-algebra. A *projection-valued measure* on K is a map $E\colon \mathcal{B}(K) \to B(\mathbb{H})$ assigning to each Borel set Ω an orthogonal projection $E(\Omega)$ (definition 5.2.15), such that

1. $E(\varnothing) = 0$ and $E(K) = I$;
2. $E(\Omega \cap \Omega') = E(\Omega)E(\Omega')$ for all Borel Ω, Ω';
3. for every countable family $\{\Omega_n\}$ of pairwise disjoint Borel sets and every $x \in \mathbb{H}$,

$$E\Big(\bigcup_n \Omega_n\Big)x = \sum_n E(\Omega_n)x$$

the series converging in the norm of $\mathbb{H}$.

Condition (2) with $\Omega = \Omega'$ recovers idempotence $E(\Omega)^2 = E(\Omega)$, which is built into "projection" already; its content is that for disjoint Ω, Ω' the ranges are orthogonal, since $E(\Omega)E(\Omega') = E(\varnothing) = 0$. The convergence in (3) is strong, not in operator norm: for a diagonal operator the partial sums $\sum_{n \le N} E(\Omega_n)$ are projections onto growing subspaces and never approach $E(\bigcup_n \Omega_n)$ uniformly unless the tail is eventually trivial. So a projection-valued measure behaves like an ordinary measure evaluated one vector at a time, and it is exactly this vectorwise reading that the integral below exploits. The spectral projections of definition 6.3.5 are the motivating instance.

Example 6.4.2: The Spectral Projections of a Self-Adjoint Operator

Let $T \in B(\mathbb{H})$ be self-adjoint and set $K = \sigma(T) \subseteq \mathbb{R}$ (theorem 6.2.1). Define $E(\Omega) = \chi_\Omega(T)$ for Borel $\Omega \subseteq K$, the spectral projection of definition 6.3.5. Then E is a projection-valued measure on K: each $E(\Omega)$ is an orthogonal projection, and conditions (1)-(3) are precisely the properties collected in proposition 6.3.8 ($E(\varnothing) = \chi_\varnothing(T) = 0$, $E(\sigma(T)) = \chi_{\sigma(T)}(T) = I$, multiplicativity from $\chi_{\Omega\cap\Omega'} = \chi_\Omega \chi_{\Omega'}$, and strong countable additivity). For a self-adjoint operator on $\mathbb{C}^n$ with distinct eigenvalues $\lambda_1, \dots, \lambda_m$ and eigenprojections $P_1, \dots, P_m$, the spectrum is the finite set $\{\lambda_1, \dots, \lambda_m\}$ and $E(\Omega) = \sum_{\lambda_i \in \Omega} P_i$: the projection-valued measure simply sums the eigenprojections whose eigenvalues land in Ω. Every projection-valued measure below is modelled on this example, and theorem 6.4.6 shows that this example is the only one that arises.

To integrate against E one first turns it back into ordinary scalar measures by pairing with vectors, exactly as in the construction of the Borel calculus. Fix $x, y \in \mathbb{H}$ and set

$$E_{x,y}(\Omega) := \langle E(\Omega)x, y\rangle \tag{6.5}$$

Additivity of E in the strong sense makes $\Omega \mapsto E_{x,y}(\Omega)$ countably additive, so $E_{x,y}$ is a complex Borel measure on K; its total variation is at most $\|x\|\,\|y\|$. Bounding each term $|E_{x,y}(\Omega_i)| \le \|x\|\,\|y\|$ would not control the sum, since a countable partition carries infinitely many terms; the bound instead uses the orthogonality of the ranges. For any Borel partition $\{\Omega_i\}$ of K, choose unimodular scalars α_i with $\alpha_i E_{x,y}(\Omega_i) = |E_{x,y}(\Omega_i)|$; then, because the projections $E(\Omega_i)$ have pairwise orthogonal ranges,

$$\sum_i |E_{x,y}(\Omega_i)| = \Big\langle \sum_i \alpha_i E(\Omega_i)x, y\Big\rangle \le \Big\|\sum_i \alpha_i E(\Omega_i)x\Big\|\,\|y\| = \Big(\sum_i \|E(\Omega_i)x\|^2\Big)^{1/2}\|y\| \le \|x\|\,\|y\|$$

using $|\alpha_i| = 1$ and $\sum_i \|E(\Omega_i)x\|^2 = \|x\|^2$. Taking the supremum over partitions gives $|E_{x,y}|(K) \leq \|x\|\,\|y\|$. When $y = x$ the measure

$$E_{x,x}(\Omega) = \langle E(\Omega)x, x\rangle = \langle E(\Omega)^2 x, x\rangle = \|E(\Omega)x\|^2 \geq 0$$

is a positive measure of total mass $E_{x,x}(K) = \langle E(K)x, x\rangle = \|x\|^2$, using $E(\Omega) = E(\Omega)^* = E(\Omega)^2$. These scalar measures are the raw material for the integral.

Definition 6.4.3: Integral against a Projection-Valued Measure

Let E be a projection-valued measure on a compact $K \subseteq \mathbb{C}$ and let $f \in B_b(K)$ be a bounded Borel function. The *integral of f against E* is the operator $\int_K f\,dE \in B(\mathbb{H})$ determined by

$$\Big\langle \Big(\int_K f\,dE\Big)x, y\Big\rangle = \int_K f\,dE_{x,y} \qquad (x, y \in \mathbb{H}) \tag{6.6}$$

where $E_{x,y}$ is the complex measure of (6.5).

The definition asserts that a single operator is pinned down by (6.6); the following proposition supplies the operator and its basic estimate.

Proposition 6.4.4: The Integral against a Projection-Valued Measure

Let E be a projection-valued measure on a compact $K \subseteq \mathbb{C}$. For each $f \in B_b(K)$ there is a unique operator $\int_K f\,dE \in B(\mathbb{H})$ satisfying (6.6), and

$$\Big\| \int_K f\,dE \Big\| \leq \|f\|_\infty$$

Moreover $f \mapsto \int_K f\,dE$ is linear, and $\int_K \chi_\Omega\,dE = E(\Omega)$ for every Borel set $\Omega \subseteq K$.

***Proof*:**
Fix $f \in B_b(K)$ and consider the map $B(x, y) := \int_K f\,dE_{x,y}$ on $\mathbb{H} \times \mathbb{H}$. In the first argument $x \mapsto E(\Omega)x$ is linear for each Ω, so $x \mapsto E_{x,y}$ is linear as a map into complex measures and hence $x \mapsto B(x, y)$ is linear; in the second argument the conjugate-linearity of $\langle\cdot,\cdot\rangle$ makes $y \mapsto B(x, y)$ conjugate-linear. Thus B is a sesquilinear form. It is bounded, because

$$|B(x, y)| = \Big| \int_K f\,dE_{x,y} \Big| \leq \|f\|_\infty\,|E_{x,y}|(K) \leq \|f\|_\infty\,\|x\|\,\|y\|$$

using the total-variation bound $|E_{x,y}|(K) \leq \|x\|\,\|y\|$ noted above. A bounded sesquilinear form is represented by a unique bounded operator: by the Riesz representation of the Hilbert space (theorem 5.1.13), for each fixed x the conjugate-linear functional $y \mapsto \overline{B(x, y)}$ is $y \mapsto \langle y, Sx\rangle$ for a unique vector Sx, and $x \mapsto Sx$ is linear with $\|Sx\| \leq \|f\|_\infty \|x\|$. Writing $S = \int_K f\,dE$ gives $\langle Sx, y\rangle = B(x, y)$, which is (6.6), and the operator norm bound $\|S\| \leq \|f\|_\infty$. Uniqueness is immediate: an operator is determined by all inner products $\langle Sx, y\rangle$. Linearity in f follows from linearity of $f \mapsto \int_K f\,dE_{x,y}$ for the fixed measures $E_{x,y}$. Finally, for $f = \chi_\Omega$ the defining identity reads $\langle(\int_K \chi_\Omega\,dE)x, y\rangle = E_{x,y}(\Omega) = \langle E(\Omega)x, y\rangle$ for all x, y, so $\int_K \chi_\Omega\,dE = E(\Omega)$, completing the proof.

The integral turns the algebra of bounded Borel functions into an algebra of operators, and it does so multiplicatively. This is the property that makes it a functional calculus rather than a mere assignment of operators to functions.

Proposition 6.4.5: Multiplicativity of the Projection-Valued Integral

Let E be a projection-valued measure on a compact $K \subseteq \mathbb{C}$. For $f, g \in B_b(K)$,

$$\int_K fg\, dE = \Big(\int_K f\, dE\Big)\Big(\int_K g\, dE\Big)$$

and $\int_K \overline{f}\, dE = \big(\int_K f\, dE\big)^*$. In particular $f \mapsto \int_K f\, dE$ is a $*$-homomorphism from $B_b(K)$ into $B(\mathbb{H})$.

***Proof*:**
Step 1: Indicators. For Borel sets Ω, Ω' the product $\chi_\Omega \chi_{\Omega'} = \chi_{\Omega\cap\Omega'}$, so by proposition 6.4.4 and the multiplicativity of E (definition 6.4.1(2)),

$$\int_K \chi_\Omega \chi_{\Omega'}\, dE = E(\Omega\cap\Omega') = E(\Omega)E(\Omega') = \Big(\int_K \chi_\Omega\, dE\Big)\Big(\int_K \chi_{\Omega'}\, dE\Big)$$

By linearity in each factor, the product identity extends to all pairs of simple Borel functions (finite linear combinations of indicators).

Step 2: Passage to bounded Borel functions. Every $f \in B_b(K)$ is a uniform limit of simple functions s_n with $\|s_n\|_\infty \le \|f\|_\infty$: partition the bounded range of f into finitely many small cells and let s_n take a sample value on each preimage. By the norm bound of proposition 6.4.4, $\int_K s_n\, dE \to \int_K f\, dE$ in operator norm, and likewise $\int_K t_n\, dE \to \int_K g\, dE$ for simple $t_n \to g$ uniformly. Since multiplication is jointly norm-continuous on bounded sets and $s_n t_n \to fg$ uniformly,

$$\int_K fg\, dE = \lim_n \int_K s_n t_n\, dE = \lim_n \Big(\int_K s_n\, dE\Big)\Big(\int_K t_n\, dE\Big) = \Big(\int_K f\, dE\Big)\Big(\int_K g\, dE\Big)$$

where the middle equality is Step 1 applied to the simple functions s_n, t_n.

Step 3: The involution. Each $E(\Omega)$ is self-adjoint, so for a simple $s = \sum_j c_j \chi_{\Omega_j}$ one has $(\int_K s\, dE)^* = \sum_j \overline{c_j} E(\Omega_j) = \int_K \overline{s}\, dE$. Passing to the uniform limit and using continuity of the adjoint gives $\int_K \overline{f}\, dE = (\int_K f\, dE)^*$ for all $f \in B_b(K)$. Together with linearity and $\int_K 1\, dE = E(K) = I$, the map $f \mapsto \int_K f\, dE$ preserves products, adjoints, and the unit, hence is a $*$-homomorphism, as we sought to show.

The projection-valued integral is now available for any projection-valued measure. Applied to the spectral projections of a self-adjoint operator, it reconstructs both the operator and its entire functional calculus, and this is the spectral theorem. The measure E carries all the information in T, distributed across the Borel sets of the spectrum, and integration reassembles it.

Theorem 6.4.6: The Spectral Theorem (Resolution of the Identity)

Let $T \in B(\mathbb{H})$ be self-adjoint and let $E(\Omega) = \chi_\Omega(T)$ be its spectral projection-valued measure on $\sigma(T) \subseteq \mathbb{R}$. Then, with all integrals taken against E over $\sigma(T)$:

1. For every bounded Borel function $f \in B_b(\sigma(T))$, the Borel calculus of theorem 6.3.3 agrees with integration against E,
$$f(T) = \int_{\sigma(T)} f\,dE$$
2. Taking $f(\lambda) = \lambda$ gives the resolution of the identity
$$T = \int_{\sigma(T)} \lambda\,dE(\lambda)$$
3. For every $x \in \mathbb{H}$,
$$\|x\|^2 = \int_{\sigma(T)} d\langle E(\lambda)x, x\rangle$$

The projection-valued measure E with these properties is unique.

***Proof*:**
Step 1: The calculus is integration against E. Both $f \mapsto f(T)$ (the Borel calculus of theorem 6.3.3) and $f \mapsto \int_{\sigma(T)} f\,dE$ are linear maps $B_b(\sigma(T)) \to B(\mathbb{H})$. They agree on indicators: $\chi_\Omega(T) = E(\Omega)$ by the definition of the spectral projection (definition 6.3.5), while $\int_{\sigma(T)} \chi_\Omega\,dE = E(\Omega)$ by proposition 6.4.4. By linearity they agree on simple Borel functions. For general $f \in B_b(\sigma(T))$, choose simple $s_n \to f$ uniformly with $\|s_n\|_\infty \leq \|f\|_\infty$. The Borel calculus satisfies $\|s_n(T) - f(T)\| \leq \|s_n - f\|_\infty \to 0$ (theorem 6.3.3), and the projection-valued integral satisfies $\|\int s_n\,dE - \int f\,dE\| \leq \|s_n - f\|_\infty \to 0$ (proposition 6.4.4); as the two limits agree on the s_n, they agree on f. This proves (1).

Step 2: Recovering T. Apply (1) to the coordinate function $\iota(\lambda) = \lambda$, which is continuous, hence in $B_b(\sigma(T))$. The continuous calculus of theorem 6.2.4 sends ι to T (it sends id to T by construction), and the Borel calculus extends the continuous one, so $\iota(T) = T$. Thus
$$T = \iota(T) = \int_{\sigma(T)} \iota\,dE = \int_{\sigma(T)} \lambda\,dE(\lambda)$$
which is (2).

Step 3: The norm identity. Apply (1) to $f = 1$, the constant function. Then $1(T) = I$ by the continuous calculus, so $\int_{\sigma(T)} 1\,dE = I$. Pairing with x and unwinding the defining identity (6.6),
$$\|x\|^2 = \langle Ix, x\rangle = \Big\langle \Big(\int_{\sigma(T)} 1\,dE\Big)x, x\Big\rangle = \int_{\sigma(T)} 1\,dE_{x,x} = \int_{\sigma(T)} d\langle E(\lambda)x, x\rangle$$
which is (3); the integrand 1 integrates the total mass of the positive measure $E_{x,x}$, already seen to equal $\|x\|^2$.

Step 4: Uniqueness. Suppose E' is a projection-valued measure on $\sigma(T)$ with $\int \lambda \, dE' = T$. Then $\int p \, dE' = p(T)$ for every polynomial p: by proposition 6.4.5 the map $f \mapsto \int f \, dE'$ is a $*$-homomorphism sending $\lambda \mapsto T$ and $1 \mapsto I$, so it agrees with the polynomial calculus (proposition 6.2.3) on polynomials. By Stone-Weierstrass, polynomials in λ (real spectrum, so $\overline{\lambda} = \lambda$ and complex conjugates add nothing) are uniformly dense in $C(\sigma(T))$, and both $\int(\cdot) \, dE'$ and the continuous calculus are norm-continuous, so $\int f \, dE' = f(T)$ for all $f \in C(\sigma(T))$. Fixing x, y, the two complex measures $E'_{x,y}$ and $E_{x,y}$ therefore satisfy $\int f \, dE'_{x,y} = \langle f(T)x, y\rangle = \int f \, dE_{x,y}$ for every $f \in C(\sigma(T))$. A complex Borel measure on a compact set is determined by its integrals against continuous functions (theorem 2.3.3), so $E'_{x,y} = E_{x,y}$ for all x, y, whence $E'(\Omega) = E(\Omega)$ for every Borel Ω. Thus E is unique, completing the proof.

The three statements are three readings of one fact. Statement (1) says the projection-valued measure is a repackaging of the Borel calculus with no loss: every $f(T)$, and in particular every continuous function of T, is an integral against E. Statement (2) is the exact analogue of the finite-dimensional $T = \sum_i \lambda_i P_i$: the eigenprojections have become the projection-valued measure, and the finite sum over eigenvalues has become an integral over the spectrum. The passage from sum to integral is what accommodates operators with no eigenvalues at all, such as multiplication by x on $L^2[0,1]$, whose spectrum $[0,1]$ is entirely continuous and carries no point masses. Statement (3) records that the positive measure $\lambda \mapsto \langle E(\lambda)x, x\rangle$ has total mass $\|x\|^2$; it is the operator-theoretic Parseval identity, the continuous replacement for $\|x\|^2 = \sum_i \|P_i x\|^2$ (theorem 5.1.15), distributing the squared length of x over the spectrum. Uniqueness is what earns the phrase "the" resolution of the identity: the data of T and the data of E are equivalent.

Example 6.4.7: The Multiplication Operator on $L^2[0,1]$

Let $\mathbb{H} = L^2[0,1]$ and let T be multiplication by the independent variable, $(Tf)(t) = tf(t)$. Then T is self-adjoint (multiplication by a real bounded function equals its own adjoint) with $\|T\| = 1$, and $\sigma(T) = [0,1]$: for $\lambda \notin [0,1]$ the function $1/(t-\lambda)$ is bounded on $[0,1]$ and multiplication by it inverts $T - \lambda I$, while for $\lambda \in [0,1]$ the operator $T - \lambda I$ has dense but non-closed range and no bounded inverse. The spectral projection $E(\Omega)$ is multiplication by χ_Ω, the indicator of $\Omega \subseteq [0,1]$: one checks $\chi_\Omega(T)f = \chi_\Omega \cdot f$ directly, since for continuous g the operator $g(T)$ is multiplication by g, and the Borel calculus extends this. The scalar measures are then $\langle E(\Omega)f, f\rangle = \int_\Omega |f(t)|^2 \, dt$, so $\langle E(\cdot)f, f\rangle$ is the measure $|f(t)|^2 \, dt$ on $[0,1]$. The resolution of the identity reads

$$\langle Tf, f\rangle = \int_0^1 t \, |f(t)|^2 \, dt = \int_{[0,1]} \lambda \, d\langle E(\lambda)f, f\rangle$$

which is just the statement that T acts as multiplication by t. This operator has no eigenvalues: $Tf = \lambda f$ forces $(t-\lambda)f(t) = 0$ almost everywhere, hence $f = 0$ in L^2. The spectral theorem thus captures an operator that the eigenvalue picture cannot reach, and the projection-valued measure E has no atoms.

The last example is not a special case but the general shape. Multiplication by the coordinate on an L^2 space is what every bounded self-adjoint operator looks like once the right coordinates are chosen. Making this precise requires decomposing $\mathbb{H}$ into pieces on which T has a single generating vector, and on each such piece the Borel calculus produces an isometry onto an L^2 space that conjugates T into multiplication.

The building block is the closed subspace generated by one vector under T.

Definition 6.4.8: Cyclic Vector and Cyclic Subspace

Let $T \in B(\mathbb{H})$ be self-adjoint. For $x \in \mathbb{H}$, the *cyclic subspace* generated by x is the closed linear span

$$\mathbb{H}_x := \overline{\operatorname{span}} \{p(T)x \mid p \text{ a polynomial}\}$$

A vector x is a *cyclic vector* for T if $\mathbb{H}_x = \mathbb{H}$, and T is a *cyclic operator* if it has a cyclic vector.

Because T is self-adjoint, $p(T)^* = \overline{p}(T)$ is again a polynomial in T, so $\mathbb{H}_x$ is invariant under both T and T^*; its orthogonal complement is then invariant as well, which is what lets the space be split off cleanly. On a cyclic subspace the operator is as simple as possible, and the following lemma identifies that simple model.

Lemma 6.4.9: Cyclic Operators are Multiplication Operators

Let $T \in B(\mathbb{H})$ be self-adjoint with a cyclic vector x. Then there is a finite positive Borel measure μ on $\sigma(T)$ and a unitary operator $U \colon \mathbb{H} \to L^2(\sigma(T), \mu)$ with $Ux = 1$ (the constant function) such that

$$(UTU^{-1}g)(\lambda) = \lambda\, g(\lambda) \qquad (g \in L^2(\sigma(T), \mu))$$

so T is unitarily equivalent to multiplication by λ on $L^2(\sigma(T), \mu)$.

***Proof*:**
Step 1: The measure. Let $\mu := E_{x,x}$ be the positive Borel measure $\mu(\Omega) = \langle E(\Omega)x, x\rangle = \|E(\Omega)x\|^2$ on $\sigma(T)$, of total mass $\|x\|^2$ (theorem 6.4.6(3)). For a continuous function $f \in C(\sigma(T))$, the Borel calculus gives

$$\|f(T)x\|^2 = \langle f(T)^* f(T)x, x\rangle = \langle |f|^2(T)x, x\rangle = \int_{\sigma(T)} |f|^2 \, dE_{x,x} = \int_{\sigma(T)} |f|^2 \, d\mu = \|f\|^2_{L^2(\mu)} \tag{6.7}$$

using $f(T)^* f(T) = \overline{f}(T) f(T) = |f|^2(T)$ (theorem 6.2.4) and the defining identity (6.6) for the measure $E_{x,x}$.

Step 2: The isometry on a dense set. Define $V \colon C(\sigma(T)) \to \mathbb{H}$ by $Vf = f(T)x$. Its range $\{f(T)x \mid f \in C(\sigma(T))\}$ contains all $p(T)x$ for polynomials p, hence is dense in $\mathbb{H}_x = \mathbb{H}$. By (6.7), V is an isometry from $(C(\sigma(T)), \|\cdot\|_{L^2(\mu)})$ into $\mathbb{H}$. Since $C(\sigma(T))$ is dense in $L^2(\sigma(T), \mu)$ (continuous functions are dense in L^2 of a finite Borel measure on a compact metric space) and V is isometric, V extends uniquely to an isometry $W \colon L^2(\sigma(T), \mu) \to \mathbb{H}$. Its range is closed (an isometry has closed range) and dense (it contains the dense set $\{f(T)x \mid f \in C(\sigma(T))\}$), hence all of $\mathbb{H}$; so W is unitary. Set $U := W^{-1} \colon \mathbb{H} \to L^2(\sigma(T), \mu)$. The constant function $1 \in C(\sigma(T))$ maps to $W1 = 1(T)x = x$, so $Ux = 1$.

Step 3: Conjugating T into multiplication. It suffices to check UTU^{-1} on the dense subspace $C(\sigma(T)) \subseteq L^2(\sigma(T), \mu)$. For $f \in C(\sigma(T))$,

$$UTU^{-1}f = UT(f(T)x) = U\big((\iota f)(T)x\big) = \iota f$$

where $\iota(\lambda) = \lambda$ is the coordinate function, using $Tf(T) = (\iota f)(T)$ by multiplicativity of the calculus (theorem 6.2.4) and $U(g(T)x) = g$ for $g \in C(\sigma(T))$. Thus UTU^{-1} acts as $f \mapsto \iota f$, multiplication by λ, on a dense subspace, and by continuity on all of $L^2(\sigma(T), \mu)$, as we sought to show.

A general self-adjoint operator need not be cyclic, but the whole space breaks into an orthogonal sum of cyclic pieces. Assembling the models from each piece gives the multiplication form of the spectral theorem. The direct sum is indexed by however many cyclic subspaces are needed to exhaust $\mathbb{H}$, which for a separable space is at most countable.

Theorem 6.4.10: The Spectral Theorem (Multiplication-Operator Form)

Let $T \in B(\mathbb{H})$ be self-adjoint. Then there is a family of finite positive Borel measures $\{\mu_n\}_{n \in N}$ on $\sigma(T)$ and a unitary operator

$$U : \mathbb{H} \to \bigoplus_{n \in N} L^2(\sigma(T), \mu_n)$$

such that UTU^{-1} is multiplication by λ,

$$\big(UTU^{-1}(g_n)_n\big)(\lambda) = (\lambda\, g_n(\lambda))_n$$

If $\mathbb{H}$ is separable, the index set N may be taken finite or countably infinite. In particular every bounded self-adjoint operator is unitarily equivalent to multiplication by the independent variable on a direct sum of L^2 spaces.

***Proof*:**
Step 1: An orthogonal cyclic decomposition. Consider the collection of sets $\{\mathbb{H}_{x_i}\}_{i \in I}$ of mutually orthogonal nonzero cyclic subspaces (definition 6.4.8), ordered by inclusion of the index families. If $\mathbb{H} = \{0\}$ the statement is vacuous, so assume $\mathbb{H} \neq \{0\}$; then a single nonzero cyclic subspace exists, so the collection is nonempty. Any chain has an upper bound (its union), so by Zorn's lemma there is a maximal family $\{\mathbb{H}_{x_n}\}_{n \in N}$ of pairwise orthogonal cyclic subspaces. Let $K = \overline{\bigoplus_n \mathbb{H}_{x_n}}$ be the closure of their orthogonal sum. If $K \neq \mathbb{H}$, pick a nonzero $y \in K^\perp$. Each $\mathbb{H}_{x_n}$ is invariant under T and $T^* = T$, so K is invariant under T, hence $K^\perp$ is invariant under $T^* = T$; then $\mathbb{H}_y \subseteq K^\perp$ is a nonzero cyclic subspace orthogonal to every $\mathbb{H}_{x_n}$, contradicting maximality. Therefore $K = \mathbb{H}$ and

$$\mathbb{H} = \bigoplus_{n \in N} \mathbb{H}_{x_n}$$

an orthogonal direct sum of cyclic subspaces.

Step 2: Model each summand. For each n, the restriction $T|_{\mathbb{H}_{x_n}}$ is self-adjoint on $\mathbb{H}_{x_n}$ with cyclic vector x_n, so by lemma 6.4.9 there is a finite positive Borel measure μ_n on $\sigma(T|_{\mathbb{H}_{x_n}}) \subseteq \sigma(T)$ (the inclusion holds because $\mathbb{H}_{x_n}$ reduces T, so $T = T|_{\mathbb{H}_{x_n}} \oplus T|_{\mathbb{H}_{x_n}^\perp}$ splits the spectrum as a union; extend μ_n by zero to a measure on $\sigma(T)$) and a unitary $U_n : \mathbb{H}_{x_n} \to L^2(\sigma(T), \mu_n)$ with $U_n(T|_{\mathbb{H}_{x_n}})U_n^{-1}$ equal to multiplication by λ.

Step 3: Assemble. Define $U := \bigoplus_n U_n : \mathbb{H} \to \bigoplus_n L^2(\sigma(T), \mu_n)$ by acting as U_n on the summand $\mathbb{H}_{x_n}$. As an orthogonal sum of unitaries between the summands it is unitary, and since T preserves each $\mathbb{H}_{x_n}$,

$$UTU^{-1}(g_n)_n = \big(U_n T|_{\mathbb{H}_{x_n}} U_n^{-1} g_n\big)_n = (\lambda\, g_n)_n$$

which is multiplication by λ on the direct sum. When $\mathbb{H}$ is separable, each $\mathbb{H}_{x_n}$ is a nonzero closed subspace, and mutually orthogonal nonzero subspaces of a separable space are at most countable in number (each contributes an independent direction to a countable orthonormal basis), so N is finite or countably infinite. This completes the proof.

The two forms of the spectral theorem are the same statement dressed differently. The resolution of the identity is coordinate-free and intrinsic, built from projections attached to the operator; the multiplication form is a concrete normal form, exhibiting T as an operator one can compute with directly. The measures μ_n record how the spectrum is "filled in" along each cyclic direction: a point mass at λ signals a genuine eigenvector, while a diffuse part signals continuous spectrum with no eigenvector, as in example 6.4.7. On a finite-dimensional space each μ_n is a finite sum of point masses and the direct sum recovers the diagonalization $T = \sum_i \lambda_i P_i$; the theorem is the statement that the same picture survives verbatim into infinite dimensions once "sum over eigenvalues" is read as "integral over the spectrum."

The self-adjointness of T entered only through two facts: its spectrum is real, and the functional calculus is a $*$-homomorphism on which conjugation of functions matches the adjoint of operators. Both survive for a normal operator, at the cost of moving from the real line to the complex plane. A normal T (definition 5.2.8) has $\sigma(T) \subseteq \mathbb{C}$ compact, and the unital C^*-subalgebra $C^*(T, T^*) \subseteq B(\mathbb{H})$ generated by T and T^* is commutative. The Gelfand transform (theorem 6.1.15) identifies its character space with $\sigma(T)$ and maps it onto $C(\sigma(T))$; this transform is isometric because every self-adjoint element $S = S^*$ of the commutative algebra $C^*(T, T^*)$ has $r(S) = \|S\|$ (the argument of theorem 6.2.1 applies to S inside the algebra), and the C^*-identity $\|A^*A\| = \|A\|^2$ propagates the equality to every element A. The transform therefore gives $C^*(T, T^*) \cong C(\sigma(T))$ carrying T to the coordinate function $z \mapsto z$ on $\sigma(T) \subseteq \mathbb{C}$. This continuous calculus over $\mathbb{C}$ replaces theorem 6.2.4, and the Riesz extension of theorem 6.3.3 produces a projection-valued measure E on the compact set $\sigma(T) \subseteq \mathbb{C}$ verbatim.

Theorem 6.4.11: The Spectral Theorem for Normal Operators

Let $T \in B(\mathbb{H})$ be normal. Then there is a unique projection-valued measure E on the compact set $\sigma(T) \subseteq \mathbb{C}$ such that

$$T = \int_{\sigma(T)} \lambda \, dE(\lambda)$$

and $f(T) = \int_{\sigma(T)} f \, dE$ for every $f \in B_b(\sigma(T))$, where $f \mapsto f(T)$ is the Borel calculus over $C^*(T, T^*) \cong C(\sigma(T))$. Equivalently, T is unitarily equivalent to multiplication by λ on a direct sum $\bigoplus_n L^2(\sigma(T), \mu_n)$ of L^2 spaces over $\sigma(T) \subseteq \mathbb{C}$.

***Proof Key ideas*:**
The self-adjoint argument transposes without change once the continuous calculus is supplied over $\mathbb{C}$. The commutative C^*-algebra $C^*(T, T^*)$ has, by the Gelfand transform (theorem 6.1.15), character space homeomorphic to $\sigma(T) \subseteq \mathbb{C}$, with Gelfand transform a unital $*$-homomorphism onto $C(\sigma(T))$ sending T to z and T^* to $\overline{z}$. The transform is isometric: every self-adjoint $S = S^*$ in the algebra satisfies $r(S) = \|S\|$ (the $\|S^2\| = \|S\|^2$ argument of theorem 6.2.1, applied to S inside $C^*(T, T^*)$), and the C^*-identity $\|A^*A\| = \|A\|^2$ upgrades this to $\|A\|^2 = \|A^*A\| = r(A^*A)$ for every element A, matching the algebra norm to the sup norm on $\sigma(T)$, so the transform is an isometric $*$-isomorphism onto $C(\sigma(T))$. This is the input played by theorem 6.2.4 in the self-adjoint case; the polynomial calculus is now the calculus of polynomials in the two

commuting variables $z, \overline{z}$. From here the Riesz-measure extension to $B_b(\sigma(T))$ (theorem 6.3.3), the projection-valued integral (proposition 6.4.4, proposition 6.4.5), the resolution of the identity and its uniqueness (theorem 6.4.6), and the cyclic decomposition (theorem 6.4.10) all repeat verbatim, reading $\sigma(T) \subseteq \mathbb{C}$ in place of $\sigma(T) \subseteq \mathbb{R}$ throughout, completing the proof.

The normal case is where the theory reaches its natural boundary. A self-adjoint operator is normal with real spectrum; a unitary operator is normal with spectrum on the unit circle (definition 5.2.8); both are now handled by one theorem, integration of λ over a compact subset of $\mathbb{C}$ against a projection-valued measure. The engine driving the generalization is the identification of the commutative C^*-algebra $C^*(T, T^*)$ with $C(\sigma(T))$, carried out here by the Banach-algebra Gelfand theory of the first section together with the C^*-identity. This identification is the single-operator instance of the commutative Gelfand-Naimark duality, which the abstract theory of C^*-algebras develops in full generality. Pushing past bounded operators, to the unbounded self-adjoint operators that carry the differential operators of quantum mechanics and of partial differential equations, requires domains, closedness, and a projection-valued measure supported on an unbounded subset of $\mathbb{R}$; that extension is the subject of the next chapter, and the resolution of the identity proved here is the template it generalizes.

7

Unbounded Operators and Semigroups

The operators of analysis are rarely bounded. Differentiation, the Laplacian, and the position and momentum operators of quantum mechanics are each defined only on a dense subspace of a Hilbert space and are discontinuous there, so the bounded theory of the preceding chapters does not apply to them directly. Its central result survives the extension, however: the spectral theorem holds for unbounded self-adjoint operators and remains the organizing tool of the subject. This chapter builds the framework around it and then reads off the two ways a self-adjoint or dissipative operator generates a dynamics, one unitary and one contractive.

The chapter opens with the vector-valued integration it will need: a solution of an evolution equation is a curve $t \mapsto u(t)$ in a Banach space, and the formulas that produce and analyze it integrate such curves, which the Bochner integral makes precise. On that base we develop unbounded operators and their adjoints, carry the spectral theorem of chapter 6 to the unbounded self-adjoint case through the Cayley transform, and prove the two generation theorems: Stone's theorem, that the strongly continuous one-parameter unitary groups are exactly the e^{itA} for self-adjoint A, and the Hille-Yosida theorem, which characterizes the generators of C_0-semigroups and underwrites the parabolic evolution equations of the later volumes.

7.1 Bochner Integration

This section builds the integral of a Banach-space-valued function of a real variable, records the criterion that decides when it exists, and proves the two properties the later sections lean on: that a closed operator passes through the integral, and that integrating a continuous curve and then differentiating returns the curve.

Throughout, (Ω, Σ, μ) is a measure space and X a Banach space; for the applications Ω is an interval of $\mathbb{R}$ with Lebesgue measure and X is the Hilbert space $\mathbb{H}$ or the Banach space carrying a semigroup.

The Lebesgue integral is built from simple functions, and the vector-valued theory begins the same way. A *simple function* $s\colon \Omega \to X$ takes finitely many values $x_1, \dots, x_n \in X$, each on a measurable set $A_k = s^{-1}(x_k)$; it is written $s = \sum_{k=1}^{n} x_k \mathbf{1}_{A_k}$. The subtlety absent in the scalar case is that a general X-valued function need not be an everywhere limit of simple functions unless X is separable along the range: a function into a nonseparable space can fail to be approximable even when each scalar coordinate is measurable. The right notion of measurability therefore builds separability into its definition.

Definition 7.1.1: Strongly Measurable Function

Let (Ω, Σ, μ) be a measure space and X a Banach space. A function $f\colon \Omega \to X$ is *strongly measurable* if there is a sequence of simple functions $s_n\colon \Omega \to X$ with

$$\|s_n(\omega) - f(\omega)\| \to 0 \quad \text{for } \mu\text{-almost every } \omega \in \Omega$$

The function f is *weakly measurable* if $\omega \mapsto \langle f(\omega), \varphi\rangle$ is measurable for every φ in the dual X^*, and *essentially separably valued* if there is a μ-null set N with $f(\Omega \setminus N)$ separable in X.

Strong measurability is the exact hypothesis under which the norm-limit construction of the integral will go through: it supplies the approximating simple functions directly. The two weaker notions in the definition matter because strong measurability is often awkward to check by hand, whereas weak measurability reduces to the scalar theory one functional at a time, and essential separable-valuedness is frequently visible from the range. Pettis's theorem, below, says these two together are equivalent to the strong notion, so one may verify strong measurability by the two convenient conditions.

A first consequence records how the norm interacts with strong measurability: if f is strongly measurable then $\omega \mapsto \|f(\omega)\|$ is a measurable scalar function, being the almost-everywhere limit of the measurable functions $\omega \mapsto \|s_n(\omega)\|$. This is what makes the phrase "$\|f(\cdot)\| \in L^1$" meaningful in the integrability criterion.

Example 7.1.2: A Step-Modulated Operator Orbit

Fix a Banach space X, an operator $B \in B(X)$ with $\|B\| \le 1$, and a vector $x \in X$. On $\Omega = [0, \infty)$ with Lebesgue measure, define $f\colon [0,\infty) \to X$ by $f(t) = e^{-t} B^{\lfloor t \rfloor} x$, where $\lfloor t \rfloor$ is the integer part. To exhibit approximating simple functions, partition $[0, \infty)$ at the dyadic points $t_{n,k} = n + k2^{-N}$ for integers $n \ge 0$ and $0 \le k < 2^N$, and let s_N take on each half-open interval $[t_{n,k}, t_{n,k+1})$ the constant value $e^{-t_{n,k}} B^n x$; on $[N, \infty)$ set $s_N = 0$. Each s_N is a simple function. For $t < N$, writing $t \in [t_{n,k}, t_{n,k+1})$ with $n = \lfloor t \rfloor$,

$$\|f(t) - s_N(t)\| = |e^{-t} - e^{-t_{n,k}}|\, \|B^n x\| \le (e^{-t_{n,k}} - e^{-t_{n,k+1}})\|x\| \le 2^{-N}\|x\|$$

using $\|B^n\| \le 1$ and the mean value theorem on e^{-t}. Hence $s_N(t) \to f(t)$ at every t, so f is strongly measurable. Its range lies in the separable subspace spanned by $\{B^n x \mid n \ge 0\}$, which is the separability that Pettis's theorem will read off directly.

The equivalence promised above is Pettis's measurability theorem. It is the vector-valued analogue of the fact that a scalar function is measurable iff its preimages of intervals are measurable: it trades the hard-to-check pointwise-limit condition for two conditions that decompose along the two structures of X, its linear functionals and its separable subspaces.

Theorem 7.1.3: Pettis's Measurability Theorem

Let X be a Banach space and $f\colon \Omega \to X$. Then f is strongly measurable if and only if f is weakly measurable and essentially separably valued.

***Proof*:**
Necessity. Suppose f is strongly measurable, with simple functions $s_n \to f$ almost everywhere. For each $\varphi \in X^*$ the map $\omega \mapsto \langle s_n(\omega), \varphi\rangle$ is a scalar simple function, and $\langle s_n(\omega), \varphi\rangle \to \langle f(\omega), \varphi\rangle$ almost everywhere by continuity of φ, so the limit $\omega \mapsto \langle f(\omega), \varphi\rangle$ is measurable; thus f is weakly measurable. For essential separable-valuedness, let N be the null set off which $s_n \to f$. Each s_n takes finitely many values, so the countable set D of all values taken by all s_n is a countable subset of X, and for $\omega \notin N$ the vector $f(\omega) = \lim_n s_n(\omega)$ lies in the closure $\overline{D}$. Hence $f(\Omega \setminus N)$ is contained in the separable set $\overline{D}$, so f is essentially separably valued.

Sufficiency. Suppose f is weakly measurable and essentially separably valued. Discarding a null set, assume $f(\Omega)$ lies in a separable closed subspace $X_0 \subseteq X$; it suffices to produce the approximating simple functions with values in X_0. Let $\{d_j \mid j \geq 1\}$ be a countable dense subset of X_0.

Step 1: a countable norming set. The separable space X_0 carries a countable set $\{\varphi_i \mid i \geq 1\} \subseteq X^*$ with $\|\varphi_i\| \leq 1$ that is *norming*, meaning $\|z\| = \sup_i |\langle z, \varphi_i\rangle|$ for every $z \in X_0$. To construct it, enumerate a countable dense subset $\{d_j \mid j \geq 1\}$ of X_0 and, for each j with $d_j \neq 0$, use the Hahn-Banach theorem to pick $\varphi_j \in X^*$ with $\|\varphi_j\| = 1$ and $\langle d_j, \varphi_j\rangle = \|d_j\|$. For $z \in X_0$ and $\varepsilon > 0$, choosing d_j with $\|z - d_j\| < \varepsilon$ gives $|\langle z, \varphi_j\rangle| \geq \|d_j\| - \varepsilon \geq \|z\| - 2\varepsilon$, while $|\langle z, \varphi_i\rangle| \leq \|z\|$ for every i; letting $\varepsilon \to 0$ yields $\sup_i |\langle z, \varphi_i\rangle| = \|z\|$.

The norm to a fixed vector is measurable. Fix $x \in X_0$. Applying the norming identity to $z = f(\omega) - x \in X_0$ gives

$$\|f(\omega) - x\| = \sup_i |\langle f(\omega), \varphi_i\rangle - \langle x, \varphi_i\rangle|$$

Each $\omega \mapsto \langle f(\omega), \varphi_i\rangle$ is measurable by weak measurability, so $\omega \mapsto \|f(\omega) - x\|$ is a countable supremum of measurable functions, hence measurable.

Step 2: nearest-point simple approximants. Fix $n \geq 1$. For each ω, at least one of the first n dense points $d_1, \ldots, d_n$ lies within $\inf_{1\leq j\leq n} \|f(\omega) - d_j\| + 1/n$ of $f(\omega)$; let $j_n(\omega)$ be the least index $j \leq n$ achieving this, and set $s_n(\omega) = d_{j_n(\omega)}$. The sets $\{\omega \mid j_n(\omega) = j\}$ are measurable by Step 1 (they are defined by finitely many inequalities among the measurable functions $\omega \mapsto \|f(\omega) - d_j\|$), so s_n is a simple function with values in $\{d_1, \ldots, d_n\}$.

Step 3: convergence. Fix ω and $\varepsilon > 0$. By density of $\{d_j \mid j\}$ in X_0 there is a j_0 with $\|f(\omega) - d_{j_0}\| < \varepsilon/2$. For $n \geq \max(j_0, 2/\varepsilon)$ the choice in Step 2 gives

$$\|f(\omega) - s_n(\omega)\| \leq \inf_{1\leq j\leq n} \|f(\omega) - d_j\| + \frac{1}{n} \leq \|f(\omega) - d_{j_0}\| + \frac{1}{n} < \varepsilon$$

Hence $s_n(\omega) \to f(\omega)$ for every ω, and f is strongly measurable, as we sought to show.

Pettis's theorem is the working criterion for strong measurability. In every application below the range of the function is visibly separable, either because X itself is separable or because the function is a norm-continuous curve $t \mapsto u(t)$ whose image is the continuous picture of an interval; weak measurability is then automatic, and the theorem delivers strong measurability with no further work.

With the class of integrands fixed, the integral is defined by the same two-stage passage as the Lebesgue integral: first on simple functions, where it is a finite sum, then extended by a norm-limit. The one point requiring care is that the limit of the integrals of approximating simple functions exists and does not depend on the approximating sequence.

For a simple function $s = \sum_{k=1}^{n} x_k \mathbf{1}_{A_k}$ with the A_k of finite measure and disjoint, the integral is the finite sum

$$\int_\Omega s\, d\mu = \sum_{k=1}^{n} \mu(A_k)\, x_k \in X$$

which is independent of the representation of s, by the same additivity computation as in the scalar case. Call a simple function *integrable* if each A_k with $x_k \neq 0$ has finite measure; then $\int_\Omega \|s\|\, d\mu = \sum_k \mu(A_k)\|x_k\|$ is finite, and the triangle inequality gives the basic estimate $\|\int_\Omega s\, d\mu\| \le \int_\Omega \|s\|\, d\mu$.

Definition 7.1.4: Bochner Integral

A strongly measurable function $f \colon \Omega \to X$ is *Bochner integrable* if there is a sequence of integrable simple functions $s_n \colon \Omega \to X$ with

$$\int_\Omega \|s_n - f\|\, d\mu \to 0$$

The *Bochner integral* of such an f is

$$\int_\Omega f\, d\mu = \lim_{n\to\infty} \int_\Omega s_n\, d\mu$$

the limit taken in the norm of X. For a measurable $E \subseteq \Omega$ one writes $\int_E f\, d\mu = \int_\Omega \mathbf{1}_E f\, d\mu$.

The defining condition asks the approximants to converge to f in the L^1-sense of the integrated norm, which is stronger than the almost-everywhere convergence of strong measurability. The estimate for simple functions shows the sequence of integrals is Cauchy, so the limit exists; that it is independent of the approximating sequence is the content of the following well-definedness check, after which the integral is an unambiguous element of X.

Proposition 7.1.5: The Bochner Integral is Well-Defined

Let $f \colon \Omega \to X$ be strongly measurable and Bochner integrable, with integrable simple functions (s_n) as in definition 7.1.4. Then $(\int_\Omega s_n\, d\mu)_n$ is a Cauchy sequence in X, its limit is independent of the choice of (s_n), and it agrees with $\int_\Omega s\, d\mu$ when $f = s$ is itself an integrable simple function.

***Proof*:**
Cauchy. For indices m, n, the function $s_m - s_n$ is an integrable simple function, so by the basic

estimate and the triangle inequality inside the integral,

$$\left\|\int_\Omega s_m\,d\mu-\int_\Omega s_n\,d\mu\right\|=\left\|\int_\Omega (s_m-s_n)\,d\mu\right\|\le\int_\Omega \|s_m-s_n\|\,d\mu\le\int_\Omega \|s_m-f\|\,d\mu+\int_\Omega \|f-s_n\|\,d\mu$$

Both terms on the right tend to 0 by the defining condition, so the sequence of integrals is Cauchy; since X is complete, it converges.

Independence of the sequence. Let (s_n) and (t_n) be two approximating sequences for the same f. Interleave them into the single sequence $s_1, t_1, s_2, t_2, \dots$, which also satisfies $\int_\Omega \|\cdot - f\|\,d\mu \to 0$ and hence, by the first part, has a convergent sequence of integrals. A convergent sequence has a unique limit, and both $(\int s_n)$ and $(\int t_n)$ are subsequences of the interleaved integrals, so they share that limit. The Bochner integral is therefore independent of the approximating sequence.

Consistency. If $f = s$ is an integrable simple function, the constant sequence $s_n = s$ approximates it, and $\int_\Omega f\,d\mu = \lim_n \int_\Omega s\,d\mu = \int_\Omega s\,d\mu$, so the two meanings of the symbol agree, completing the proof.

The construction is exactly the scalar one with $|\cdot|$ replaced by $\|\cdot\|$ and completeness of $\mathbb{R}$ replaced by completeness of X; every place the scalar argument used the absolute value, the norm serves in its stead. Completeness of X is what makes the Cauchy sequence converge, and it is the only property of X the definition uses beyond its linear and normed structure.

The definition presupposes that a suitable approximating sequence exists, and one wants a criterion deciding this from f alone. In the scalar theory a measurable function is integrable iff its absolute value has finite integral; the vector-valued statement is identical with the norm in place of the absolute value, and it is the criterion invoked whenever an integral is formed in the sequel.

Theorem 7.1.6: Bochner's Integrability Criterion

A strongly measurable function $f\colon \Omega \to X$ is Bochner integrable if and only if

$$\int_\Omega \|f(\omega)\|\,d\mu(\omega) < \infty$$

that is, $\|f(\cdot)\| \in L^1(\mu)$. In that case

$$\left\|\int_\Omega f\,d\mu\right\| \le \int_\Omega \|f\|\,d\mu$$

***Proof*:**
Necessity. Suppose f is Bochner integrable, with integrable simple functions s_n satisfying $\int_\Omega \|s_n - f\|\,d\mu \to 0$. Each $\|s_n\|$ is a nonnegative integrable simple function, and the reverse triangle inequality gives, pointwise, $\big|\|f\| - \|s_n\|\big| \le \|f - s_n\|$, so

$$\int_\Omega \big|\|f\| - \|s_n\|\big|\,d\mu \le \int_\Omega \|f - s_n\|\,d\mu \to 0$$

Thus $\|f\|$ is the L^1-limit of the integrable functions $\|s_n\|$, hence $\|f\| \in L^1(\mu)$, and $\int_\Omega \|f\|\,d\mu = \lim_n \int_\Omega \|s_n\|\,d\mu < \infty$.

Sufficiency. Suppose f is strongly measurable with $\int_\Omega \|f\|\,d\mu < \infty$. Let (σ_n) be simple functions with $\sigma_n \to f$ almost everywhere, as furnished by strong measurability. These need not be integrable, so they are truncated. Set

$$s_n(\omega) = \begin{cases} \sigma_n(\omega) & \text{if } \|\sigma_n(\omega)\| \le 2\|f(\omega)\| \\ 0 & \text{otherwise} \end{cases}$$

Each s_n is simple, since it takes each nonzero value of σ_n on the intersection of a measurable set with the measurable set $\{\omega \mid \|\sigma_n(\omega)\| \le 2\|f(\omega)\|\}$; and $\|s_n(\omega)\| \le 2\|f(\omega)\|$ everywhere, so $\int_\Omega \|s_n\|\, d\mu \le 2\int_\Omega \|f\|\, d\mu < \infty$, whence each s_n is integrable.

Pointwise convergence of s_n to f. Fix ω with $\sigma_n(\omega) \to f(\omega)$ (all but a null set of ω). If $f(\omega) \ne 0$, then $\|\sigma_n(\omega)\| \to \|f(\omega)\| < 2\|f(\omega)\|$, so $\|\sigma_n(\omega)\| \le 2\|f(\omega)\|$ for all large n, and thus $s_n(\omega) = \sigma_n(\omega) \to f(\omega)$. If $f(\omega) = 0$, then the truncation forces $\|s_n(\omega)\| \le 2\|f(\omega)\| = 0$, so $s_n(\omega) = 0 = f(\omega)$. In both cases $s_n(\omega) \to f(\omega)$, and moreover

$$\|s_n(\omega) - f(\omega)\| \le \|s_n(\omega)\| + \|f(\omega)\| \le 3\|f(\omega)\|$$

L^1 convergence via dominated convergence. The scalar functions $\omega \mapsto \|s_n(\omega) - f(\omega)\|$ converge to 0 pointwise almost everywhere and are dominated by the integrable function $3\|f\|$. By the dominated convergence theorem of the scalar theory,

$$\int_\Omega \|s_n - f\|\, d\mu \to 0$$

So the (s_n) witness Bochner integrability of f.

The norm estimate. With such (s_n), the basic estimate for simple functions and the continuity of the norm give

$$\left\|\int_\Omega f\, d\mu\right\| = \lim_n \left\|\int_\Omega s_n\, d\mu\right\| \le \lim_n \int_\Omega \|s_n\|\, d\mu = \int_\Omega \|f\|\, d\mu$$

the last equality by the necessity computation applied to the sequence (s_n), completing the proof.

The criterion reduces the existence of a vector-valued integral to a single scalar integrability question, which the reader already knows how to answer. In practice one checks that $t \mapsto \|f(t)\|$ is integrable, often by a crude bound such as continuity on a compact interval or an exponential decay estimate, and the Bochner integral then exists automatically. The accompanying norm inequality is the vector triangle inequality passed through the integral, controlling the size of an integrated curve by the integral of its length.

Example 7.1.7: The Laplace Transform of a Bounded Curve

Let X be a Banach space and $u\colon [0,\infty) \to X$ a norm-continuous curve with $\|u(t)\| \le Me^{\omega t}$ for constants $M \ge 0$ and $\omega \in \mathbb{R}$. For a scalar λ with $\operatorname{Re}\lambda > \omega$, consider the integrand $f(t) = e^{-\lambda t}u(t)$ on $[0,\infty)$. Continuity of u makes its range separable, so f is strongly measurable by Pettis's theorem (theorem 7.1.3); and

$$\int_0^\infty \|f(t)\|\, dt = \int_0^\infty e^{-(\operatorname{Re}\lambda)t}\|u(t)\|\, dt \le M\int_0^\infty e^{-(\operatorname{Re}\lambda - \omega)t}\, dt = \frac{M}{\operatorname{Re}\lambda - \omega}$$

is finite. By theorem 7.1.6, the Bochner integral $\int_0^\infty e^{-\lambda t}u(t)\,dt$ exists, and its norm is at most $M/(\operatorname{Re}\lambda-\omega)$. This is the Laplace transform of the curve u; section 7.5 identifies exactly this integral, taken against a semigroup orbit $u(t)=T(t)x$, with the resolvent $R(\lambda,A)x$ of the generator.

The integral would be of little use in operator theory if the unbounded operators of the chapter could not be moved across it. They can, provided the operator is closed: a closed operator applied to a Bochner integrable curve, whose image is again Bochner integrable, commutes with the integral. This is the vector-valued substitute for differentiation under the integral sign, and the semigroup theory derives its resolvent identities through it.

Proposition 7.1.8: Closed Operators Commute with the Bochner Integral

Let X and Y be Banach spaces and $T\colon \operatorname{dom}T\to Y$ a closed linear operator with domain $\operatorname{dom}T\subseteq X$. Let $f\colon\Omega\to X$ be Bochner integrable with $f(\omega)\in\operatorname{dom}T$ for almost every ω, and suppose the function $\omega\mapsto Tf(\omega)$ is Bochner integrable in Y. Then

$$\int_\Omega f\,d\mu\in\operatorname{dom}T \qquad\text{and}\qquad T\int_\Omega f\,d\mu=\int_\Omega Tf\,d\mu$$

***Proof*:**
Recall that T is closed means its graph $\Gamma(T)=\{(x,Tx)\mid x\in\operatorname{dom}T\}$ is a closed subspace of $X\oplus Y$, the direct sum carrying the norm $\|(x,y)\|=\|x\|+\|y\|$. The strategy is to integrate the graph curve $\omega\mapsto(f(\omega),Tf(\omega))$ into $X\oplus Y$ and read off that its integral lands in the closed subspace $\Gamma(T)$.

Step 1: the graph curve is Bochner integrable in $X\oplus Y$. Define $g\colon\Omega\to X\oplus Y$ by $g(\omega)=(f(\omega),Tf(\omega))$ where $f(\omega)\in\operatorname{dom}T$, and $g(\omega)=0$ on the null set where $f(\omega)\notin\operatorname{dom}T$. Both coordinate functions are strongly measurable, so g is strongly measurable, and

$$\int_\Omega\|g\|\,d\mu=\int_\Omega\big(\|f\|+\|Tf\|\big)\,d\mu=\int_\Omega\|f\|\,d\mu+\int_\Omega\|Tf\|\,d\mu<\infty$$

both terms finite by the integrability of f and of Tf and theorem 7.1.6. Hence g is Bochner integrable in $X\oplus Y$.

Step 2: the integral respects the two coordinate projections. The coordinate projections $\pi_X: X\oplus Y\to X$ and $\pi_Y: X\oplus Y\to Y$ are bounded linear maps. A bounded linear map S commutes with the Bochner integral: for integrable simple functions $s_n\to g$ in the L^1-norm one has $\int_\Omega Ss_n\,d\mu=S\int_\Omega s_n\,d\mu$ term by term, and passing to the limit through the bounded S gives $\int_\Omega Sg\,d\mu=S\int_\Omega g\,d\mu$. Applying this to π_X and π_Y,

$$\int_\Omega g\,d\mu=\left(\int_\Omega f\,d\mu,\ \int_\Omega Tf\,d\mu\right)$$

Step 3: the integral lies in the graph. Each value $g(\omega)$ lies in $\Gamma(T)$, which is a closed subspace of $X\oplus Y$. The integral of a $\Gamma(T)$-valued Bochner integrable function lies in $\Gamma(T)$, by the Hahn-Banach route through the annihilator. Let $\psi\in(X\oplus Y)^*$ annihilate $\Gamma(T)$, so $\psi(g(\omega))=0$ for every ω. Since ψ is a bounded linear functional, it passes through the integral by the bounded-map interchange of Step 2 (a functional is a bounded linear map into the scalars), giving

$$\psi\left(\int_\Omega g\,d\mu\right) = \int_\Omega \psi(g)\,d\mu = \int_\Omega 0\,d\mu = 0$$

Thus $\int_\Omega g\,d\mu$ is annihilated by every $\psi \in (X \oplus Y)^*$ that annihilates $\Gamma(T)$. A closed subspace equals the annihilator of its annihilator, by the Hahn-Banach separation theorem: any vector outside the closed subspace $\Gamma(T)$ is separated from it by some such ψ with $\psi \neq 0$ on that vector. Hence $\int_\Omega g\,d\mu \in \Gamma(T)$. Therefore

$$\left(\int_\Omega f\,d\mu,\ \int_\Omega Tf\,d\mu\right) = \int_\Omega g\,d\mu \in \Gamma(T)$$

By the definition of the graph, this says exactly that $\int_\Omega f\,d\mu \in \operatorname{dom} T$ and $T\int_\Omega f\,d\mu = \int_\Omega Tf\,d\mu$, as we sought to show.

The proposition is what lets an unbounded operator be handled by integrating its values rather than its action: to know T on the integral it suffices to integrate Tf. Closedness is essential and cannot be dropped, for a merely densely-defined operator can send a convergent sequence of integrals to a divergent one; the closed graph is precisely the hypothesis that survives the limit. Every resolvent formula of section 7.5, where T is the generator of a semigroup and f is a weighted orbit, rests on this interchange.

The last property is the one that makes the Bochner integral a genuine antiderivative. For a continuous curve on an interval, the integral from a fixed base point is differentiable in the upper limit, with the integrand as derivative. This is the fundamental theorem of calculus, and its vector-valued form is what turns the integral formulas of the semigroup theory into differential equations. The derivative here is the norm derivative: $t \mapsto v(t)$ has derivative w at t_0 if $\|(v(t) - v(t_0))/(t - t_0) - w\| \to 0$ as $t \to t_0$.

Theorem 7.1.9: Fundamental Theorem for the Bochner Integral

Let X be a Banach space and $g\colon [a,b] \to X$ continuous. Then the function

$$G\colon [a,b] \to X, \qquad G(t) = \int_a^t g(\tau)\,d\tau$$

is well-defined, continuously differentiable, and satisfies $G'(t) = g(t)$ for every $t \in [a,b]$.

***Proof*:**
Well-definedness. A continuous g on the compact interval $[a,b]$ has separable range, so it is strongly measurable by theorem 7.1.3, and it is bounded, say $\|g(\tau)\| \le M$, so $\int_a^t \|g\|\,d\tau \le M(b-a) < \infty$. By theorem 7.1.6 the Bochner integral defining $G(t)$ exists for each t.

The difference quotient. Fix $t_0 \in [a,b]$ and $t \neq t_0$ in $[a,b]$. By additivity of the integral over adjacent intervals,

$$G(t) - G(t_0) = \int_{t_0}^t g(\tau)\,d\tau$$

and since the constant function $g(t_0)$ integrates to $(t - t_0)g(t_0)$ over $[t_0, t]$,

$$\frac{G(t) - G(t_0)}{t - t_0} - g(t_0) = \frac{1}{t - t_0} \int_{t_0}^{t} \big(g(\tau) - g(t_0)\big)\, d\tau$$

Estimating by continuity. Let $\varepsilon > 0$. By continuity of g at t_0 there is a $\delta > 0$ with $\|g(\tau) - g(t_0)\| < \varepsilon$ whenever $|\tau - t_0| < \delta$. For $0 < |t - t_0| < \delta$ the integrand above satisfies $\|g(\tau) - g(t_0)\| < \varepsilon$ on the whole interval of integration, so by the norm estimate of theorem 7.1.6,

$$\left\| \frac{G(t) - G(t_0)}{t - t_0} - g(t_0) \right\| \le \frac{1}{|t - t_0|} \left| \int_{t_0}^{t} \|g(\tau) - g(t_0)\|\, d\tau \right| \le \frac{1}{|t - t_0|} \cdot \varepsilon\, |t - t_0| = \varepsilon$$

Hence the difference quotient converges to $g(t_0)$ in the norm of X as $t \to t_0$, so G is differentiable at t_0 with $G'(t_0) = g(t_0)$.

Continuity of the derivative. The derivative $G' = g$ is continuous by hypothesis, so G is continuously differentiable, completing the proof.

The theorem says the Bochner integral inverts differentiation exactly as in the scalar case: to solve $u'(t) = h(t)$ with a continuous right side, integrate h. The generation theorems of this chapter are read through this lens. A semigroup orbit $t \mapsto T(t)x$ has, for x in the domain of the generator, derivative $AT(t)x$, and the equation $\frac{d}{dt}T(t)x = AT(t)x$ that identifies A is exactly a statement that a Bochner integral of the right side recovers the orbit. The multiplication-operator and heat-semigroup models of the later sections will each be verified by differentiating an explicit Bochner integral of this kind.

7.2 Unbounded Operators and Their Adjoints

This section sets up the operators the spectral theorem will act on and locates the self-adjoint ones among them. An unbounded operator carries a domain as part of its data, and the two structural notions that make it manageable, closedness and the adjoint, both read cleanly off its graph. Symmetry is the naive analogue of self-adjointness and is strictly weaker; the section closes by identifying, through the ranges of $T \pm i$, exactly which symmetric operators are self-adjoint.

An operator on a Hilbert space $\mathbb{H}$ need not be defined on all of $\mathbb{H}$. The three operators named in the chapter opening make the point: momentum $-i\, d/dx$ is defined only on differentiable functions, and the value $-i\, f'$ of such a function need not be square-integrable even when f is. So an unbounded operator is a linear map defined on a subspace, and that subspace is not an afterthought but part of the operator's identity.

Definition 7.2.1: Densely-Defined Operator and Its Graph

An *operator* T on a Hilbert space $\mathbb{H}$ is a linear map $T\colon \operatorname{dom} T \to \mathbb{H}$ defined on a linear subspace $\operatorname{dom} T \subseteq \mathbb{H}$, its *domain*. The operator is *densely defined* if $\operatorname{dom} T$ is dense in $\mathbb{H}$. Its *range* is $\operatorname{ran} T = \{Tx \mid x \in \operatorname{dom} T\}$, and its *graph* is the subspace

$$\Gamma(T) = \{(x, Tx) \mid x \in \operatorname{dom} T\} \subseteq \mathbb{H} \oplus \mathbb{H}$$

Two operators are equal when they have the same domain and agree on it; equivalently, when their graphs coincide. Write $S \subseteq T$, and call T an *extension* of S, when $\Gamma(S) \subseteq \Gamma(T)$, that is, $\operatorname{dom} S \subseteq \operatorname{dom} T$ and T restricts to S on $\operatorname{dom} S$.

The graph encodes the operator completely, and the inclusion order on operators is the inclusion order on graphs. This is the reorganization that makes the theory work: statements about domains and values become statements about a single subspace of $\mathbb{H} \oplus \mathbb{H}$, on which the Hilbert-space geometry of the ambient product is available. The word "unbounded" is used loosely for an operator defined on a proper dense domain; a genuinely unbounded operator is discontinuous, but the domain, not the discontinuity, is what the definitions track.

Example 7.2.2: The Momentum Operator

Let $\mathbb{H} = L^2(\mathbb{R})$ and let $T = -i\,d/dx$ act on the domain $\operatorname{dom} T = C_c^\infty(\mathbb{R})$ of smooth compactly supported functions, so $Tf = -if'$. The domain is dense in $L^2(\mathbb{R})$, so T is densely defined. It is unbounded: with $f(x) = e^{-x^2/2}$ scaled to $f_\lambda(x) = f(\lambda x)$, so that $f_\lambda'(x) = \lambda f'(\lambda x)$, direct computation gives $\|f_\lambda'\|_2 = \lambda^{1/2}\|f'\|_2$ while $\|f_\lambda\|_2 = \lambda^{-1/2}\|f\|_2$, so the ratio $\|Tf_\lambda\|_2/\|f_\lambda\|_2$ grows like λ and is not bounded over the domain. (A smooth cutoff makes each f_λ compactly supported without changing the growth.) The factor $-i$ is chosen so that T is symmetric, as the next definitions will show: integration by parts moves the derivative across the inner product and the i absorbs the sign.

Example 7.2.3: The Position (Multiplication) Operator

Let $\mathbb{H} = L^2(\mathbb{R})$ and let M be multiplication by the coordinate, $(Mf)(x) = xf(x)$, on the natural domain $\operatorname{dom} M = \left\{f \in L^2(\mathbb{R}) \;\middle|\; xf(x) \in L^2(\mathbb{R})\right\}$, the largest domain on which the formula lands back in $\mathbb{H}$. It contains $C_c^\infty(\mathbb{R})$ and so is dense. The operator is unbounded, since for f supported near a large R the factor $x \approx R$ multiplies the norm by roughly R. This is the model unbounded operator: the multiplication operator by an unbounded measurable function is the concrete form to which every self-adjoint operator will reduce (theorem 7.3.6).

For the multiplication operator the maximal domain above is forced: it is the set of f for which Mf makes sense as an element of $\mathbb{H}$, and there is no room to enlarge it. For a differential operator the domain is a genuine choice, and different choices give different operators with different adjoints and different spectral behaviour. Managing that choice is what the rest of the section is about; the first requirement to impose is that the graph be closed.

Definition 7.2.4: Closed and Closable Operators

An operator T is *closed* if its graph $\Gamma(T)$ is a closed subspace of $\mathbb{H}\oplus\mathbb{H}$. Concretely, T is closed when for every sequence (x_n) in $\operatorname{dom} T$ with $x_n \to x$ and $Tx_n \to y$ in $\mathbb{H}$, one has $x \in \operatorname{dom} T$ and $Tx = y$. The operator T is *closable* if the closure $\overline{\Gamma(T)}$ of its graph in $\mathbb{H}\oplus\mathbb{H}$ is again the graph of an operator, that is, if $\overline{\Gamma(T)}$ contains no point $(0, y)$ with $y \neq 0$. In that case the operator whose graph is $\overline{\Gamma(T)}$ is the *closure* $\overline{T}$ of T, the smallest closed extension of T.

Closedness is the substitute for continuity available in the unbounded setting. A bounded operator extends by continuity to the closure of its domain, so a densely defined bounded operator is closed after that extension; the closed graph theorem says the two conditions coincide for operators defined on all of $\mathbb{H}$. For an unbounded operator continuity fails, but closedness can survive, and it is exactly enough for the arguments that matter: it lets one pass to limits inside the operator. The obstruction to closability is a subspace $\overline{\Gamma(T)}$ containing $(0, y)$ with $y \neq 0$, which would force $T\,0 = y \neq 0$ on any operator with that graph; closability rules this out, and then $\overline{T}$ is the operator read off the closed subspace $\overline{\Gamma(T)}$.

Example 7.2.5: The Momentum Operator Is Closable

Return to $T = -i\,d/dx$ on $C_c^\infty(\mathbb{R})$ from example 7.2.2. It is not closed: the limit of a sequence $f_n \in C_c^\infty(\mathbb{R})$ with $f_n \to f$ and $-if_n' \to g$ in L^2 need only have $g = -if'$ in the weak (distributional) sense, and such an f need not lie in $C_c^\infty(\mathbb{R})$. But T is closable. Suppose $f_n \to 0$ and $-if_n' \to g$ in L^2. For any $\varphi \in C_c^\infty(\mathbb{R})$, integration by parts on the compactly supported f_n gives $\langle -if_n', \varphi\rangle = \langle f_n, -i\varphi'\rangle$, and letting $n \to \infty$ the right side tends to $\langle 0, -i\varphi'\rangle = 0$ while the left tends to $\langle g, \varphi\rangle$. Thus g is orthogonal to the dense set $C_c^\infty(\mathbb{R})$, so $g = 0$. Hence $\overline{\Gamma(T)}$ contains no $(0, g)$ with $g \neq 0$, and T is closable. Its closure is $-i\,d/dx$ on the Sobolev space $H^1(\mathbb{R})$ (chapter 4), the functions with a square-integrable weak derivative.

Closability was checked by pairing against a dense set of test functions and pushing the derivative across the inner product. That maneuver is the adjoint, made into an operator. For a bounded T the adjoint is defined on all of $\mathbb{H}$ by the Riesz representation theorem (proposition 5.2.7); for an unbounded T the same defining relation still determines the values of the adjoint, but now it also determines the domain, since the relation can be satisfied only for those y that make $x \mapsto \langle Tx, y\rangle$ continuous.

Definition 7.2.6: Adjoint of a Densely-Defined Operator

Let T be densely defined on $\mathbb{H}$. The *adjoint* T^* is the operator with domain

$$\operatorname{dom} T^* = \{y \in \mathbb{H} \mid x \mapsto \langle Tx, y\rangle \text{ is continuous on } \operatorname{dom} T\}$$

defined as follows. For $y \in \operatorname{dom} T^*$ the functional $x \mapsto \langle Tx, y\rangle$, continuous on the dense subspace $\operatorname{dom} T$, extends uniquely to a bounded linear functional on $\mathbb{H}$, so by the Riesz representation theorem (theorem 5.1.13) there is a unique $z \in \mathbb{H}$ with $\langle Tx, y\rangle = \langle x, z\rangle$ for all $x \in \operatorname{dom} T$; set $T^*y = z$. Thus T^*y is characterized by

$$\langle Tx, y\rangle = \langle x, T^*y\rangle \qquad \text{for all } x \in \operatorname{dom} T \tag{7.1}$$

Density of $\operatorname{dom} T$ is what makes T^*y well defined: two vectors representing the same functional on a dense subspace agree on all of $\mathbb{H}$, so z is unique. Without density the value T^*y would be ambiguous, which is why the adjoint is defined only for densely defined operators. The domain of T^* is genuinely part of the construction, not a formality: a vector y belongs to it precisely when the pairing $\langle Tx, y\rangle$ is controlled by $\|x\|$ alone, uniformly over the domain, and how large this set is depends delicately on how large $\operatorname{dom} T$ was. Shrinking the domain of T enlarges the domain of T^*, since fewer constraints on x make continuity easier to achieve; this inverse relationship is the source of the gap between symmetric and self-adjoint operators below.

Example 7.2.7: The Adjoint of the Momentum Operator

The adjoint T^* of $T = -i\,d/dx$ on $C_c^\infty(\mathbb{R})$ is identified as follows. Let $y \in L^2(\mathbb{R})$ be such that $f \mapsto \langle -if', y\rangle$ is continuous on $C_c^\infty(\mathbb{R})$. Integration by parts gives, for $f \in C_c^\infty(\mathbb{R})$,

$$\langle Tf, y\rangle = \int_{\mathbb{R}} (-if'(x))\,\overline{y(x)}\,dx$$

and the requirement that this be a bounded functional of f in the L^2-norm says exactly that y has a weak derivative in L^2, that is, $y \in H^1(\mathbb{R})$, and then $\langle Tf, y\rangle = \langle f, -iy'\rangle$. Hence $\operatorname{dom} T^* = H^1(\mathbb{R})$ and $T^*y = -iy'$. The adjoint is the same differential expression on the strictly larger domain $H^1(\mathbb{R}) \supsetneq C_c^\infty(\mathbb{R})$: the operator is a proper restriction of its own adjoint, the exact situation the next definition names.

The momentum example shows the adjoint agreeing with the operator as a formula while acting on a larger domain. When the operator sits inside its adjoint this way the pairing $\langle Tx, y\rangle$ is symmetric on the original domain, and when the two operators coincide outright the spectral theorem will apply. These are the two conditions to name.

Definition 7.2.8: Symmetric, Self-Adjoint, Essentially Self-Adjoint

Let T be densely defined on $\mathbb{H}$.

1. T is *symmetric* if $T \subseteq T^*$, equivalently if $\langle Tx, y\rangle = \langle x, Ty\rangle$ for all $x, y \in \operatorname{dom} T$.
2. T is *self-adjoint* if $T = T^*$, that is, symmetric and with $\operatorname{dom} T^* = \operatorname{dom} T$.
3. T is *essentially self-adjoint* if it is symmetric and its closure $\overline{T}$ is self-adjoint.

The equivalence in (1) is immediate from (7.1): symmetry says $\langle Tx, y\rangle = \langle x, Ty\rangle$ on the domain, which is exactly the statement that every $y \in \operatorname{dom} T$ lies in $\operatorname{dom} T^*$ with $T^*y = Ty$, i.e. $T \subseteq T^*$. Self-adjointness demands the reverse inclusion as well, and by the inverse relationship noted above this is a real constraint on the domain, not on the formula. Essential self-adjointness is the practical middle ground: a symmetric operator defined on a convenient small domain (test functions, say) is rarely self-adjoint there, but if its closure is self-adjoint then the small domain already determines a unique self-adjoint operator, and the small domain is called a *core*. The distinction is not pedantry, as the following example shows: symmetry is genuinely weaker than self-adjointness, and the difference is measured by domains.

Example 7.2.9: Symmetric but Not Self-Adjoint

Let $\mathbb{H} = L^2([0,1])$ and let $T = -i\,d/dx$ act on

$$\operatorname{dom} T = \left\{ f \in H^1([0,1]) \;\middle|\; f(0) = f(1) = 0 \right\}$$

the smooth-enough functions vanishing at both endpoints. For $f, g \in \operatorname{dom} T$, integration by parts gives

$$\langle Tf, g\rangle = \int_0^1 (-if')\,\overline{g}\,dx = \big[-if\,\overline{g}\big]_0^1 + \int_0^1 f\,\overline{(-ig')}\,dx = \langle f, Tg\rangle$$

the boundary term vanishing because $f(0) = f(1) = 0$. So T is symmetric. But it is not self-adjoint. Computing T^*, the boundary term $[-if\,\overline{g}]_0^1$ vanishes for all f with $f(0) = f(1) = 0$ regardless of the values $g(0), g(1)$, so no boundary condition on g is forced: $\operatorname{dom} T^* = H^1([0,1])$ with $T^*g = -ig'$, the full first Sobolev space. Thus $\operatorname{dom} T^* \supsetneq \operatorname{dom} T$ and $T \subsetneq T^*$; the operator is symmetric but not self-adjoint. Imposing instead the periodic condition $f(0) = f(1)$ gives an operator T_{per} that is self-adjoint, since then the boundary term forces the matching condition on g too. The moral is that boundary conditions, encoded in the domain, decide self-adjointness; symmetry alone does not see them.

Two operators that agree as differential expressions can therefore differ as self-adjoint operators, and the difference is invisible to symmetry. To detect self-adjointness one needs an intrinsic criterion, and to state it one first needs the two structural facts about the adjoint: it is always closed, and its density governs whether T itself is closable. Both come from a single geometric observation about the graph.

The adjoint has a graph, and that graph is a rotation of the graph of T inside $\mathbb{H} \oplus \mathbb{H}$. Introduce the unitary

$$V : \mathbb{H} \oplus \mathbb{H} \to \mathbb{H} \oplus \mathbb{H}, \qquad V(x, y) = (-y, x) \tag{7.2}$$

a quarter-turn of the product space; it is unitary, with $V^2 = -I$ and $V^{-1}(x, y) = (y, -x)$. The defining relation (7.1) is an orthogonality statement between the two graphs under V, which is the content of the next proposition and the engine of everything after it.

Proposition 7.2.10: The Adjoint Is Closed; Closability via the Adjoint

Let T be densely defined on $\mathbb{H}$, and let V be the unitary of (7.2). Then:

1. $\Gamma(T^*) = \big(V\,\Gamma(T)\big)^\perp$ in $\mathbb{H} \oplus \mathbb{H}$. In particular T^* is closed.
2. T is closable if and only if T^* is densely defined, and then $T^{**} = \overline{T}$.

***Proof*:**

Step 1: The graph of T^* as an orthogonal complement. A pair $(y, z) \in \mathbb{H} \oplus \mathbb{H}$ lies in $\big(V\,\Gamma(T)\big)^\perp$ precisely when it is orthogonal to $V(x, Tx) = (-Tx, x)$ for every $x \in \operatorname{dom} T$, that is,

$$0 = \langle (y, z), (-Tx, x)\rangle = -\langle y, Tx\rangle + \langle z, x\rangle$$

for all $x \in \operatorname{dom} T$. Conjugating, this reads $\langle Tx, y\rangle = \langle x, z\rangle$ for all $x \in \operatorname{dom} T$. By (7.1) this holds exactly when $y \in \operatorname{dom} T^*$ and $z = T^*y$, i.e. when $(y, z) \in \Gamma(T^*)$. Hence $\Gamma(T^*) = (V\,\Gamma(T))^\perp$. An orthogonal complement is a closed subspace, so $\Gamma(T^*)$ is closed and T^* is a closed operator. This proves (1).

Step 2: V preserves orthogonal complements. Since V is unitary it carries closed subspaces to closed subspaces and commutes with taking orthogonal complements: $V(M^\perp) = (VM)^\perp$ for any subspace M. Moreover $V^2 = -I$ acts as multiplication by a scalar of modulus one on each factor, so $V^2M = M$ for every subspace M. Both facts are used below.

Step 3: Closability equals density of $\operatorname{dom} T^*$. The closure of $\Gamma(T)$ is recovered from (1) by taking complements twice. Since $M^{\perp\perp} = \overline{M}$ for any subspace M, and using Step 2,

$$\overline{\Gamma(T)} = \Gamma(T)^{\perp\perp} = \big(V^2\Gamma(T)\big)^{\perp\perp} = \big(V(V\Gamma(T))\big)^{\perp\perp}$$

Applying $V(M^\perp) = (VM)^\perp$ and part (1),

$$\overline{\Gamma(T)} = \big(V\,(V\Gamma(T))^\perp\big)^\perp = \big(V\,\Gamma(T^*)\big)^\perp$$

Now T is closable if and only if $\overline{\Gamma(T)}$ contains no $(0,w)$ with $w \neq 0$ (definition 7.2.4). A pair $(0,w)$ lies in $(V\,\Gamma(T^*))^\perp$ exactly when it is orthogonal to $V(y, T^*y) = (-T^*y, y)$ for every $y \in \operatorname{dom} T^*$, i.e. when $\langle w, y\rangle = 0$ for all $y \in \operatorname{dom} T^*$; equivalently $w \in (\operatorname{dom} T^*)^\perp$. Thus $\overline{\Gamma(T)}$ contains a nonzero $(0,w)$ if and only if $(\operatorname{dom} T^*)^\perp \neq \{0\}$, that is, if and only if $\operatorname{dom} T^*$ is *not* dense. Contrapositively, T is closable if and only if T^* is densely defined.

Step 4: $T^{**} = \overline{T}$. Suppose T is closable, so by Step 3 T^* is densely defined and $T^{**} = (T^*)^*$ makes sense. Applying part (1) to T^* in place of T,

$$\Gamma(T^{**}) = \big(V\,\Gamma(T^*)\big)^\perp$$

which is exactly $\overline{\Gamma(T)}$ by the displayed identity of Step 3. Two operators with the same graph are equal, so $T^{**} = \overline{T}$, completing the proof.

The proposition is the payoff of moving to graphs. The adjoint is the orthogonal complement of a rotated graph, so it is automatically closed, and taking the adjoint twice returns the closure of the original operator, provided the first adjoint is densely defined. For a symmetric operator $T \subseteq T^*$ this last proviso is free: T is densely defined and contained in T^*, so T^* is densely defined and T is closable, with $\overline{T} = T^{**} \subseteq T^*$. Every symmetric operator is thus automatically closable, and its closure is still symmetric. This is why one may always replace a symmetric operator by its closure without loss, and why essential self-adjointness is the right target for operators given on a small core.

For a self-adjoint operator the chain collapses: $T = T^*$ forces $T = T^* = T^{**} = \overline{T}$, so a self-adjoint operator is closed. A symmetric operator has the descending chain $T \subseteq \overline{T} = T^{**} \subseteq T^*$, and it is self-adjoint exactly when the whole chain collapses to a single operator. The criterion below measures the gap in the chain by the size of two subspaces and converts self-adjointness into a surjectivity statement one can check.

The idea is to test $T \pm i$ rather than T itself. Adding $\pm i$ to a symmetric operator makes it bounded below in a way that rules out kernel and forces closed range, so the only thing that can fail is surjectivity; and surjectivity of both $T + i$ and $T - i$ is precisely enough to close the gap between T and T^*. The two kernels $\ker(T^* \mp i)$, which measure the failure of surjectivity, are the deficiency spaces.

Theorem 7.2.11: Basic Criterion for Self-Adjointness

Let T be a symmetric operator on $\mathbb{H}$. The following are equivalent.

1. T is self-adjoint.
2. T is closed and $\ker(T^* - i) = \ker(T^* + i) = \{0\}$.
3. $\operatorname{ran}(T + i) = \mathbb{H}$ and $\operatorname{ran}(T - i) = \mathbb{H}$.

The dimensions $n_\pm = \dim\ker(T^* \mp i)$, the *deficiency indices* of T, thus measure the failure of self-adjointness: a closed symmetric operator is self-adjoint if and only if $n_+ = n_- = 0$.

***Proof*:**

Step 1: The fundamental estimate. For $x \in \operatorname{dom} T$ and the symmetry $\langle Tx, x\rangle = \langle x, Tx\rangle = \overline{\langle Tx, x\rangle}$, the number $\langle Tx, x\rangle$ is real. Hence

$$\|(T \pm i)x\|^2 = \|Tx\|^2 \pm \langle Tx, ix\rangle \pm \langle ix, Tx\rangle + \|x\|^2 = \|Tx\|^2 + \|x\|^2$$

the two cross terms $\mp i\langle Tx, x\rangle \pm i\overline{\langle Tx, x\rangle}$ cancelling because $\langle Tx, x\rangle$ is real. In particular

$$\|(T \pm i)x\| \geq \|x\| \qquad \text{for all } x \in \operatorname{dom} T \tag{7.3}$$

so $T \pm i$ is injective and bounded below. If moreover T is closed, then $T \pm i$ is closed (its graph is that of T sheared by the bounded map $x \mapsto \pm ix$), and a closed operator bounded below has closed range: if $(T \pm i)x_n \to w$, then (7.3) makes (x_n) Cauchy, so $x_n \to x$, and closedness gives $x \in \operatorname{dom} T$ with $(T \pm i)x = w$. Thus for closed symmetric T the ranges $\operatorname{ran}(T \pm i)$ are closed subspaces.

Step 2: The range-kernel duality. For any densely defined operator, a vector w is orthogonal to $\operatorname{ran}(T + i)$ if and only if $\langle (T + i)x, w\rangle = 0$ for all $x \in \operatorname{dom} T$, i.e. $\langle Tx, w\rangle = -\langle ix, w\rangle = \langle x, iw\rangle$ for all such x. By (7.1) this says exactly $w \in \operatorname{dom} T^*$ with $T^*w = iw$, that is, $w \in \ker(T^* - i)$. Hence

$$\operatorname{ran}(T + i)^\perp = \ker(T^* - i), \qquad \operatorname{ran}(T - i)^\perp = \ker(T^* + i) \tag{7.4}$$

the second identity by the same computation with $-i$ in place of i.

Step 3: (1) $\Rightarrow$ (2). If T is self-adjoint then $T = T^* = \overline{T}$ is closed (proposition 7.2.10). Suppose $T^*w = iw$ for some w. Then, using $T = T^*$ and the reality of $\langle Tw, w\rangle$ from Step 1,

$$i\|w\|^2 = \langle iw, w\rangle = \langle T^*w, w\rangle = \langle Tw, w\rangle \in \mathbb{R}$$

forcing $\|w\|^2 = 0$, so $w = 0$; likewise $\ker(T^* + i) = \{0\}$. This gives (2).

Step 4: (2) $\Rightarrow$ (3). Assume T closed with both deficiency kernels trivial. By Step 1 the ranges $\operatorname{ran}(T \pm i)$ are closed, so $\operatorname{ran}(T + i) = \overline{\operatorname{ran}(T + i)} = \big(\operatorname{ran}(T + i)^\perp\big)^\perp = \ker(T^* - i)^\perp = \{0\}^\perp = \mathbb{H}$, using (7.4); the same argument gives $\operatorname{ran}(T - i) = \mathbb{H}$. This is (3).

Step 5: (3) $\Rightarrow$ (1). Assume $\operatorname{ran}(T \pm i) = \mathbb{H}$. Since T is symmetric, $T \subseteq T^*$, and it remains only to show $\operatorname{dom} T^* \subseteq \operatorname{dom} T$. Let $y \in \operatorname{dom} T^*$. Because $\operatorname{ran}(T + i) = \mathbb{H}$, there is $x \in \operatorname{dom} T$ with $(T + i)x = (T^* + i)y$; here $(T^* + i)y$ makes sense as $y \in \operatorname{dom} T^*$. As $T \subseteq T^*$, also $x \in \operatorname{dom} T^*$ and $(T^* + i)x = (T + i)x = (T^* + i)y$, so

$$(T^* + i)(y - x) = 0$$

that is, $y - x \in \ker(T^* + i) = \operatorname{ran}(T - i)^\perp$ by (7.4). But $\operatorname{ran}(T - i) = \mathbb{H}$ by hypothesis, so its orthogonal complement is $\{0\}$, giving $y = x \in \operatorname{dom} T$. Hence $\operatorname{dom} T^* = \operatorname{dom} T$ and $T = T^*$, as we sought to show.

Deficiency indices. By (7.4) the deficiency space $\ker(T^* \mp i)$ is the orthogonal complement of $\operatorname{ran}(T \pm i)$, whose vanishing (for closed symmetric T) is equivalent to surjectivity of $T \pm i$ by Steps 1 and 4. Thus $n_\pm = 0$ simultaneously characterizes self-adjointness among closed symmetric operators, which is the final assertion.

The criterion converts a domain question, whether T^* overreaches T, into a range question, whether $T \pm i$ hits all of $\mathbb{H}$. Its usefulness is twofold. First, ranges are often computable: verifying $\operatorname{ran}(T \pm i) = \mathbb{H}$ means solving the equations $(T \pm i)x = w$ for every w, which for a differential operator is a boundary-value problem. Second, the deficiency indices (n_+, n_-) organize the failure. When they are equal and finite the symmetric operator has a finite-dimensional family of self-adjoint extensions parametrized by the unitaries between the two deficiency spaces (the von Neumann extension theory), and when one of them is nonzero while the other vanishes no self-adjoint extension exists at all. The endpoint conditions of example 7.2.9 are the visible face of this: the two boundary values give $n_+ = n_- = 1$ there, and the circle of self-adjoint extensions is the circle of admissible matching conditions $f(1) = e^{i\theta} f(0)$.

The equivalence with essential self-adjointness follows by applying the criterion to the closure. Since $\overline{T} = T^{**}$ has the same adjoint as T (both equal T^*, as $(T^{**})^* = \overline{T^*} = T^*$ using proposition 7.2.10 and the closedness of T^*), the deficiency spaces of T and of $\overline{T}$ coincide, and $\overline{T}$ is self-adjoint exactly when $\ker(T^* \mp i) = \{0\}$. This gives the working test used throughout the sequel: a symmetric operator is essentially self-adjoint if and only if $\ker(T^* - i) = \ker(T^* + i) = \{0\}$, equivalently $\operatorname{ran}(T + i)$ and $\operatorname{ran}(T - i)$ are both dense in $\mathbb{H}$. For the momentum operator on the whole line this is the statement that $-i\, d/dx$ on $C_c^\infty(\mathbb{R})$ is essentially self-adjoint, its closure being the self-adjoint operator on $H^1(\mathbb{R})$ of example 7.2.5.

7.3 The Spectral Theorem for Unbounded Self-Adjoint Operators

The spectral theorem for bounded self-adjoint operators (theorem 6.4.6) is not re-derived here. Instead a self-adjoint T is converted into a bounded unitary operator, its Cayley transform, the bounded theorem is applied to that, and the resulting resolution of the identity is transported back to T, yielding the functional calculus.

The first task is to say what the spectrum of an unbounded operator is at all. For a bounded operator the resolvent $(\lambda I - T)^{-1}$ was a bounded inverse in the algebra $B(\mathbb{H})$, and the spectrum was the compact set where no such inverse exists. An unbounded T is defined only on $\operatorname{dom} T$, so $\lambda I - T$ is a map from $\operatorname{dom} T$ into $\mathbb{H}$, and inverting it means finding a bounded operator on all of $\mathbb{H}$ that undoes it. The closedness of T (definition 7.2.4) is exactly the hypothesis that makes such a bounded inverse automatic once $\lambda I - T$ is a bijection, by the closed graph theorem.

Definition 7.3.1: Resolvent Set and Spectrum of a Closed Operator

Let T be a closed, densely-defined operator on $\mathbb{H}$. The *resolvent set* $\rho(T)$ is the set of $\lambda \in \mathbb{C}$ for which $\lambda I - T\colon \operatorname{dom} T \to \mathbb{H}$ is a bijection with bounded inverse; that inverse is the *resolvent*

$$R(\lambda, T) = (\lambda I - T)^{-1} \in B(\mathbb{H})$$

The *spectrum* is the complement $\sigma(T) = \mathbb{C} \setminus \rho(T)$.

For a closed operator the boundedness of the inverse is not an extra demand. If $\lambda I - T$ is a bijection of $\operatorname{dom} T$ onto $\mathbb{H}$, its inverse is a closed operator defined on all of $\mathbb{H}$: the graph of $R(\lambda, T)$ is the graph of $\lambda I - T$ with its two coordinates swapped, and swapping coordinates preserves closedness. A closed operator defined on the whole space is bounded by the closed graph theorem, so $R(\lambda, T)$ is automatically bounded. Thus for closed T one has $\lambda \in \rho(T)$ precisely when $\lambda I - T$ is a bijection $\operatorname{dom} T \to \mathbb{H}$; the bounded-inverse clause in the definition is a consequence, recorded for emphasis. This is the first place the standing assumption of closedness earns its keep.

A concrete unbounded operator makes the spectrum visible.

Example 7.3.2: The Momentum Operator on the Line

Let $\mathbb{H} = L^2(\mathbb{R})$ and let $T = -i\, d/dx$ on the domain $\operatorname{dom} T = \{f \in L^2(\mathbb{R}) \mid f \text{ absolutely continuous}, f' \in L^2(\mathbb{R})\}$, the maximal domain on which the distributional derivative lands back in L^2. This T is self-adjoint, and its spectrum is the whole real line, $\sigma(T) = \mathbb{R}$, with no eigenvalues. For $\lambda \in \mathbb{C}$ with $\operatorname{Im} \lambda \neq 0$ the equation $-if' - \lambda f = g$ is solved by the bounded convolution

$$f(x) = R(\lambda, T)g(x) = \int_{\mathbb{R}} k_\lambda(x - y)\, g(y)\, dy$$

whose kernel k_λ decays exponentially because $\operatorname{Im} \lambda \neq 0$, so $R(\lambda, T)$ is a bounded operator and $\lambda \in \rho(T)$. For real λ the candidate eigenfunction $e^{i\lambda x}$ solves $-if' = \lambda f$ but is not in $L^2(\mathbb{R})$, so λ is not an eigenvalue; nevertheless $\lambda I - T$ fails to be onto (its range is a dense proper subspace, as the plane waves $e^{i\lambda x}$ occupy the "continuous spectrum"), so $\lambda \in \sigma(T)$. The spectrum is real and unbounded, and it consists entirely of continuous spectrum.

The example already exhibits the two structural facts that make the self-adjoint case tractable: the spectrum lies on the real axis, and the resolvent is controlled by the distance from λ to that axis. Both follow from a single inequality, valid for symmetric operators, that bounds $\|(\lambda I - T)x\|$ below by $|\operatorname{Im} \lambda|\, \|x\|$.

Proposition 7.3.3: The Spectrum of a Self-Adjoint Operator is Real

Let T be self-adjoint (definition 7.2.8). Then $\sigma(T) \subseteq \mathbb{R}$, and for every λ with $\operatorname{Im} \lambda \neq 0$,

$$\|R(\lambda, T)\| \le \frac{1}{|\operatorname{Im} \lambda|} \tag{7.5}$$

***Proof*:**
Write $\lambda = a + ib$ with $a, b \in \mathbb{R}$ and $b \neq 0$. For $x \in \operatorname{dom} T$ the vector $(\lambda I - T)x = (aI - T)x + ibx$ has squared norm

$$\|(\lambda I - T)x\|^2 = \|(aI - T)x\|^2 + b^2\|x\|^2 + 2\operatorname{Re}\langle (aI - T)x,\, ibx\rangle$$

The cross term vanishes: since T is symmetric and a is real, $(aI-T)$ is symmetric, so $\langle (aI-T)x, x\rangle$ is real, and $\operatorname{Re}\langle (aI - T)x,\, ibx\rangle = \operatorname{Re}\big(-ib\,\overline{\langle (aI-T)x, x\rangle}\big) = 0$. Hence

$$\|(\lambda I - T)x\|^2 = \|(aI - T)x\|^2 + b^2\|x\|^2 \geq b^2\|x\|^2 \tag{7.6}$$

so $\|(\lambda I - T)x\| \geq |b|\,\|x\| = |\operatorname{Im}\lambda|\,\|x\|$.

This lower bound has three consequences. First, $\lambda I - T$ is injective, since $(\lambda I - T)x = 0$ forces $x = 0$. Second, its range is closed: if $(\lambda I - T)x_n \to y$, the bound makes (x_n) Cauchy, so $x_n \to x$ for some x, and closedness of T (a self-adjoint operator is closed, being an adjoint, definition 7.2.4) gives $x \in \operatorname{dom} T$ with $(\lambda I - T)x = y$, so $y \in \operatorname{ran}(\lambda I - T)$. Third, the range is dense: its orthogonal complement is $\ker(\overline{\lambda} I - T^*) = \ker(\overline{\lambda} I - T)$ (using $T^* = T$), and the lower bound applied to $\overline{\lambda}$ (which also has nonzero imaginary part) shows $\overline{\lambda} I - T$ is injective, so this kernel is $\{0\}$. A closed dense subspace is the whole of $\mathbb{H}$, so $\lambda I - T$ is a bijection onto $\mathbb{H}$, i.e. $\lambda \in \rho(T)$. Every λ with $\operatorname{Im}\lambda \neq 0$ therefore lies in the resolvent set, which is to say $\sigma(T) \subseteq \mathbb{R}$. Finally, for such λ the inverse inherits the reciprocal bound: putting $x = R(\lambda, T)y$ into $\|(\lambda I - T)x\| \geq |\operatorname{Im}\lambda|\,\|x\|$ gives $\|y\| \geq |\operatorname{Im}\lambda|\,\|R(\lambda, T)y\|$ for all y, which is (7.5), completing the proof.

Reality of the spectrum is the unbounded echo of the bounded fact that a self-adjoint operator has real spectrum, proved in the bounded chapter (theorem 6.4.6). The resolvent bound (7.5) is the quantitative refinement: the resolvent blows up no faster than the reciprocal of the distance to the real axis, so $R(\lambda, T)$ is a bounded holomorphic function of λ off the line. It is this bound, evaluated at $\lambda = \pm i$, that will make the two operators $T \pm iI$ boundedly invertible and open the door to the Cayley transform.

The resolvent bound tells us where T is invertible but not what T looks like. To import the bounded spectral theorem we must manufacture a bounded operator out of T that retains all of its spectral information. The device is due to Cayley: the map $z \mapsto (z - i)/(z + i)$ carries the real axis bijectively onto the unit circle minus the point 1, sending ∞ to 1; applying the same fractional linear transformation to the operator T turns a self-adjoint operator (spectrum on $\mathbb{R}$) into a unitary operator (spectrum on the circle). The exceptional point 1 on the circle corresponds to the point at infinity, which is exactly the room an unbounded operator needs.

Theorem 7.3.4: The Cayley Transform

Let T be self-adjoint. The operators $T \pm iI\colon \operatorname{dom} T \to \mathbb{H}$ are bijections onto $\mathbb{H}$, and the *Cayley transform*

$$U = (T - iI)(T + iI)^{-1}$$

is a unitary operator on $\mathbb{H}$ for which 1 is not an eigenvalue (equivalently, $\operatorname{ran}(I - U)$ is dense). Conversely, if U is unitary and 1 is not an eigenvalue of U, then

$$T = i(I + U)(I - U)^{-1}$$

defined on $\operatorname{dom} T = \operatorname{ran}(I - U)$, is self-adjoint and has U as its Cayley transform. The assignment $T \mapsto U$ is a bijection between self-adjoint operators on $\mathbb{H}$ and unitary operators on $\mathbb{H}$ with no eigenvalue 1.

***Proof*:**
Step 1: $T + iI$ and $T - iI$ are bijections onto $\mathbb{H}$. By theorem 7.2.11, a symmetric operator T is self-adjoint if and only if $\operatorname{ran}(T + iI) = \operatorname{ran}(T - iI) = \mathbb{H}$; the self-adjointness of T is thus precisely the surjectivity of $T \pm iI$. Injectivity is the lower bound (7.6) at $\lambda = \mp i$: $\|(T \pm iI)x\| \geq \|x\|$, so $T \pm iI$ is injective with $(T \pm iI)^{-1}$ bounded of norm at most 1. Hence both $T + iI$ and $T - iI$ are bijections of $\operatorname{dom} T$ onto $\mathbb{H}$, and $\pm i \in \rho(T)$ as proposition 7.3.3 already recorded.

Step 2: U is an isometry. For $x \in \operatorname{dom} T$ set $y = (T + iI)x$, so that $x = (T + iI)^{-1}y$ ranges over all of $\mathbb{H}$ as y does, and $Uy = (T - iI)x$. The symmetry of T makes $\langle Tx, x\rangle$ real, so expanding as in (7.6),

$$\|(T + iI)x\|^2 = \|Tx\|^2 + \|x\|^2 = \|(T - iI)x\|^2$$

which says $\|y\| = \|Uy\|$ for every $y \in \mathbb{H}$: U is a linear isometry of $\mathbb{H}$ into itself.

Step 3: U is onto, hence unitary. The range of U is $\operatorname{ran}(T - iI) = \mathbb{H}$ by Step 1, so U is a surjective isometry, i.e. unitary (definition 5.1.23); its inverse is $U^{-1} = (T + iI)(T - iI)^{-1}$.

Step 4: 1 is not an eigenvalue. Compute $I - U$ on a vector $y = (T + iI)x$:

$$(I - U)y = (T + iI)x - (T - iI)x = 2ix \tag{7.7}$$

so $\operatorname{ran}(I - U) = \operatorname{dom} T$, which is dense. If $Uy = y$ then $(I - U)y = 0$ forces $x = 0$ by (7.7), whence $y = (T + iI) \cdot 0 = 0$; so 1 is not an eigenvalue of U. Likewise

$$(I + U)y = (T + iI)x + (T - iI)x = 2Tx \tag{7.8}$$

Comparing (7.8) with (7.7) recovers T: for $y = (T + iI)x$ they read $i(I + U)y = i \cdot 2Tx = T \cdot (2ix) = T(I - U)y$, so $i(I + U) = T(I - U)$ as maps on $\mathbb{H}$, and inverting $I - U$ on its dense range $\operatorname{ran}(I - U) = \operatorname{dom} T$ yields $T = i(I + U)(I - U)^{-1}$.

Step 5: The converse. Let U be unitary with 1 not an eigenvalue, so $I - U$ is injective and $\operatorname{ran}(I - U)$ is dense (its complement is $\ker(I - U^*) = \ker(I - U^{-1}) = \ker(U - I) = \{0\}$, using $U^* = U^{-1}$). Define $T = i(I + U)(I - U)^{-1}$ on $\operatorname{dom} T = \operatorname{ran}(I - U)$. For $x = (I - U)u$ and $x' = (I - U)u'$ in $\operatorname{dom} T$,

$$\langle Tx, x'\rangle - \langle x, Tx'\rangle = i\langle (I + U)u, (I - U)u'\rangle + i\langle (I - U)u, (I + U)u'\rangle$$

Expanding both inner products and cancelling using $\langle Uu, Uu'\rangle = \langle u, u'\rangle$ (unitarity) leaves 0, so T is symmetric. Moreover $T + iI = i(I + U)(I - U)^{-1} + i = i\big[(I + U) + (I - U)\big](I - U)^{-1} =$

$2i(I-U)^{-1}$ and similarly $T - iI = 2iU(I-U)^{-1}$; both are onto $\mathbb{H}$ (since $I-U$ and U are bijections of $\mathbb{H}$ and $\operatorname{dom} T$), so by theorem 7.2.11 T is self-adjoint. Its Cayley transform is $(T-iI)(T+iI)^{-1} = 2iU(I-U)^{-1} \cdot \big(2i(I-U)^{-1}\big)^{-1} = U$, recovering U. Thus $T \mapsto U$ and $U \mapsto T$ are mutually inverse, establishing the claimed bijection, as we sought to show.

The Cayley transform is a discrete analogue of the exponential map: where $e^{i\theta}$ wraps the real line onto the circle in the theory of one-parameter groups, the fractional linear map $\lambda \mapsto (\lambda - i)/(\lambda + i)$ wraps the spectral line onto the circle in one algebraic step, trading the unbounded self-adjoint T for the bounded unitary U with no loss. The single excluded point $z = 1$ is the image of $\lambda = \infty$; it is an eigenvalue of U exactly when T would have to be defined "at infinity," which a genuine operator cannot be, and this is why the correspondence lands on unitaries missing the eigenvalue 1. The formulas (7.7) and (7.8) are worth remembering: $I - U$ recovers the domain and $I + U$ recovers the action of T, so the whole of T is encoded in the two boundary operators $I \pm U$.

Example 7.3.5: The Cayley Transform of Multiplication by x

Let $\mathbb{H} = L^2(\mathbb{R})$ and let T be multiplication by the coordinate, $(Tf)(x) = xf(x)$, on the domain $\operatorname{dom} T = \big\{ f \;\big|\; xf \in L^2(\mathbb{R}) \big\}$. This T is self-adjoint with $\sigma(T) = \mathbb{R}$. Its Cayley transform is again a multiplication operator, now by the scalar function $x \mapsto (x-i)/(x+i)$:

$$(Uf)(x) = \frac{x-i}{x+i}\, f(x)$$

The multiplier $(x-i)/(x+i)$ has modulus 1 for every real x, so U is multiplication by a unimodular function and hence unitary, as the theorem demands. As x runs over $\mathbb{R}$ the value $(x-i)/(x+i)$ traces the unit circle: at $x = 0$ it is -1, and as $x \to \pm\infty$ it approaches 1 without attaining it, so the value 1 is never taken. Consequently $U - I$ is multiplication by a function that vanishes nowhere on $\mathbb{R}$, hence injective, and 1 is not an eigenvalue of U, matching the theorem exactly. The excluded point 1 on the circle is the image of the missing spectral point at infinity.

With the Cayley transform in hand the spectral theorem for unbounded self-adjoint operators is now within reach: apply the bounded spectral theorem to the unitary U, obtaining a projection-valued measure on the unit circle, and push it forward to the real line through the inverse Cayley map $z \mapsto i(1+z)/(1-z)$. The pushed-forward measure integrates the coordinate λ to reconstruct T, exactly as in the bounded case, with the sole new feature that λ is now unbounded, so the integral $\int \lambda\, dE$ produces an unbounded operator with the natural domain that keeps $\int \lambda^2\, d\langle E(\lambda)x, x\rangle$ finite.

Theorem 7.3.6: The Spectral Theorem for Unbounded Self-Adjoint Operators

Let T be self-adjoint on $\mathbb{H}$. There is a unique projection-valued measure E on the Borel σ-algebra of $\mathbb{R}$ (a map assigning orthogonal projections to Borel sets, satisfying the axioms of definition 6.4.1 with $E(\mathbb{R}) = I$) such that

$$T = \int_{\mathbb{R}} \lambda \, dE(\lambda) \tag{7.9}$$

in the sense that

$$\operatorname{dom} T = \left\{ x \in \mathbb{H} \;\middle|\; \int_{\mathbb{R}} \lambda^2 \, d\langle E(\lambda)x, x\rangle < \infty \right\}$$

and $\langle Tx, y\rangle = \int_{\mathbb{R}} \lambda \, d\langle E(\lambda)x, y\rangle$ for all $x \in \operatorname{dom} T$ and $y \in \mathbb{H}$. The support of E is $\sigma(T)$: a real point λ_0 lies in $\sigma(T)$ if and only if $E(\Omega) \neq 0$ for every neighbourhood Ω of λ_0, and $\sigma(T)$ is the smallest closed set of full measure.

***Proof*:**

Step 1: The spectral measure of the Cayley transform. Let $U = (T - iI)(T + iI)^{-1}$ be the Cayley transform (theorem 7.3.4), a unitary operator, hence normal (definition 5.2.8), with spectrum contained in the unit circle $\mathbb{T} = \{z \in \mathbb{C} \mid |z| = 1\}$. The bounded spectral theorem, applied to the normal operator U (theorem 6.4.6), supplies a unique projection-valued measure F on $\mathbb{T}$ with

$$U = \int_{\mathbb{T}} z \, dF(z)$$

and, for every bounded Borel g on $\mathbb{T}$, $g(U) = \int_{\mathbb{T}} g \, dF$ by the Borel functional calculus (theorem 6.3.3). Because 1 is not an eigenvalue of U, the single point $\{1\}$ carries no mass: $F(\{1\})$ is the orthogonal projection onto $\ker(U - I) = \{0\}$, so $F(\{1\}) = 0$. Thus F lives on $\mathbb{T} \setminus \{1\}$.

Step 2: Transport to the line. The Cayley map and its inverse

$$z = \frac{\lambda - i}{\lambda + i}, \qquad \lambda = i\,\frac{1 + z}{1 - z}$$

are mutually inverse homeomorphisms between $\mathbb{R}$ and $\mathbb{T} \setminus \{1\}$. Define a projection-valued measure E on $\mathbb{R}$ by pushing F forward along $\lambda(z) = i(1 + z)/(1 - z)$: for a Borel set $\Omega \subseteq \mathbb{R}$,

$$E(\Omega) = F\big(\{z \in \mathbb{T} \setminus \{1\} \mid \lambda(z) \in \Omega\}\big)$$

Since $\lambda(\cdot)$ is a Borel isomorphism of $\mathbb{T} \setminus \{1\}$ onto $\mathbb{R}$ and $F(\{1\}) = 0$, the assignment E satisfies all the projection-valued-measure axioms ($E(\varnothing) = 0$, $E(\mathbb{R}) = F(\mathbb{T} \setminus \{1\}) = I$, multiplicativity and strong countable additivity transported from F). For any bounded Borel h on $\mathbb{R}$, the change of variables $z \mapsto \lambda(z)$ gives

$$\int_{\mathbb{R}} h(\lambda) \, dE(\lambda) = \int_{\mathbb{T}} h\big(\lambda(z)\big) \, dF(z) = (h \circ \lambda)(U) \tag{7.10}$$

so integration of h against E is the Borel functional calculus of U applied to the bounded Borel function $h \circ \lambda$ on $\mathbb{T}$.

Step 3: The integral recovers T. The coordinate λ is unbounded on $\mathbb{R}$, so its integral must be built as a limit of bounded truncations. For $n \in \mathbb{N}$ let $\lambda_n(\lambda) = \lambda\, \chi_{[-n,n]}(\lambda)$, a bounded Borel function, and set $S_n = \int_{\mathbb{R}} \lambda_n \, dE = \int_{[-n,n]} \lambda \, dE(\lambda) \in B(\mathbb{H})$. Define

$$D = \left\{ x \in \mathbb{H} \;\middle|\; \int_{\mathbb{R}} \lambda^2 \, d\langle E(\lambda)x, x\rangle < \infty \right\}$$

and, for $x \in D$, note $\|S_n x - S_m x\|^2 = \int_{n<|\lambda|\le m} \lambda^2 \, d\langle E(\lambda)x, x\rangle \to 0$ as $n, m \to \infty$ (the tail of a convergent integral), so $S_n x$ converges; call the limit Sx. This S is the operator $\int_{\mathbb{R}} \lambda \, dE$ on $\operatorname{dom} S = D$. It remains to prove $S = T$.

Fix $x \in \operatorname{dom} T$ and put $y = (T + iI)x$, so $x = (T + iI)^{-1} y$. The bounded operator $(T + iI)^{-1}$ is $r(U)$ for the function r continuous on $\mathbb{T} \setminus \{1\}$,

$$r(z) = \frac{1-z}{2i}, \qquad \text{since } (T + iI)^{-1} = \tfrac{1}{2i}(I - U)$$

by (7.7). Since $\lambda(z) + i = 2i/(1 - z)$, the function $r(z) = (1 - z)/(2i)$ equals $1/(\lambda(z) + i)$, so by the transport (7.10) applied to $h(\lambda) = 1/(\lambda + i)$, $(T + iI)^{-1} = r(U) = \int_{\mathbb{R}} \frac{1}{\lambda+i} \, dE(\lambda)$. Likewise (7.8) gives, since $\lambda(z)/(\lambda(z) + i) = (1 + z)/2$,

$$T(T + iI)^{-1} = \tfrac{1}{2}(I + U) = \int_{\mathbb{R}} \frac{\lambda}{\lambda + i} \, dE(\lambda) \tag{7.11}$$

a bounded operator. Now fix $x = (T + iI)^{-1} y \in \operatorname{dom} T$. First, $x \in D$: the scalar measure satisfies $d\langle E(\lambda)x, x\rangle = |1/(\lambda + i)|^2 \, d\langle E(\lambda)y, y\rangle$ (from $x = \int \frac{1}{\lambda+i} \, dE \, y$ and multiplicativity of the bounded calculus), so

$$\int_{\mathbb{R}} \lambda^2 \, d\langle E(\lambda)x, x\rangle = \int_{\mathbb{R}} \frac{\lambda^2}{\lambda^2 + 1} \, d\langle E(\lambda)y, y\rangle \le \|y\|^2 < \infty$$

Thus $Sx = \int_{\mathbb{R}} \lambda \, dE(\lambda) \, x$ is defined. To identify it with Tx, apply the bounded, multiplicative calculus to the two bounded factors $\lambda/(\lambda+i)$ and $(\lambda+i)$ truncated: for each n, $S_n x = \int_{[-n,n]} \lambda \, dE \cdot \int_{\mathbb{R}} \frac{1}{\lambda+i} \, dE \, y = \int_{[-n,n]} \frac{\lambda}{\lambda+i} \, dE \, y$, which converges as $n \to \infty$ to $\int_{\mathbb{R}} \frac{\lambda}{\lambda+i} \, dE \, y = T(T + iI)^{-1} y = Tx$ by (7.11). Hence $Sx = Tx$, and since x ranges over all of $\operatorname{dom} T$ as y ranges over $\mathbb{H}$, this gives $T \subseteq S$. Conversely S is symmetric (each S_n is self-adjoint and $S_n \to S$ strongly on D) and T is self-adjoint, so a self-adjoint operator has no proper symmetric extension: $T \subseteq S$ with both symmetric and T maximal-symmetric forces $S = T$. Thus $T = \int_{\mathbb{R}} \lambda \, dE(\lambda)$ with the stated domain.

Step 4: Uniqueness. Suppose E' is another projection-valued measure on $\mathbb{R}$ with $\int_{\mathbb{R}} \lambda \, dE' = T$ and the corresponding domain. Reversing the transport, define F' on $\mathbb{T} \setminus \{1\}$ by $F'(\Gamma) = E'\big(\{\lambda \in \mathbb{R} \mid z(\lambda) \in \Gamma\}\big)$ where $z(\lambda) = (\lambda - i)/(\lambda + i)$, and set $F'(\{1\}) = 0$. Then $\int_{\mathbb{T}} z \, dF' = \int_{\mathbb{R}} \frac{\lambda-i}{\lambda+i} \, dE' = (T - iI)(T + iI)^{-1} = U$, using the two calculus identities above applied to E'. By the uniqueness in the bounded spectral theorem for the normal operator U (theorem 6.4.6), $F' = F$, and transporting back gives $E' = E$. This proves uniqueness.

Step 5: The support is $\sigma(T)$. By (7.10) the spectrum of U is the support of F on $\mathbb{T}$, and $\lambda(\cdot)$ carries the support of F (minus the point 1) onto the support of E; since $\lambda(z(\lambda)) = \lambda$ conjugates U's spectral data to T's, a real point λ_0 lies in $\sigma(T)$ if and only if $E(\Omega) \ne 0$ for every neighbourhood Ω of λ_0, which is the statement that $\operatorname{supp} E = \sigma(T)$. This completes the proof.

The theorem is the exact analogue of theorem 6.4.6, with two differences forced by unboundedness.

First, the measure lives on all of $\mathbb{R}$ rather than on a compact set, because $\sigma(T)$ need no longer be bounded; multiplication by x (example 7.3.5) and the momentum operator (example 7.3.2) both have $\sigma(T) = \mathbb{R}$. Second, the integral $\int \lambda\, dE$ is genuinely improper: it converges only on the domain D where the "energy" $\int \lambda^2\, d\langle E(\lambda)x, x\rangle$ is finite, and this convergence condition *is* the definition of $\operatorname{dom} T$. The domain is thus not an arbitrary technical choice but is dictated by the spectral measure: x lies in $\operatorname{dom} T$ precisely when its spectral distribution has a finite second moment. The Cayley transform did the entire job of converting the theorem into a bounded statement, and the reality bound (7.5) was what guaranteed $\pm i \in \rho(T)$ so that the transform existed at all.

Example 7.3.7: The Spectral Measure of the Multiplication Operator

Return to $\mathbb{H} = L^2(\mathbb{R})$ with T multiplication by the coordinate, $(Tf)(x) = xf(x)$ (example 7.3.5). Its projection-valued measure is the most transparent possible: for a Borel set $\Omega \subseteq \mathbb{R}$,

$$(E(\Omega)f)(x) = \chi_\Omega(x)\, f(x)$$

multiplication by the indicator of Ω. Each $E(\Omega)$ is the orthogonal projection onto $\{f \mid f = 0 \text{ a.e. off } \Omega\}$, and $E(\mathbb{R}) = I$. The resolution (7.9) reads

$$\langle Tf, f\rangle = \int_{\mathbb{R}} x\, |f(x)|^2\, dx = \int_{\mathbb{R}} \lambda\, d\langle E(\lambda)f, f\rangle$$

since $d\langle E(\lambda)f, f\rangle = |f(\lambda)|^2\, d\lambda$ is the scalar measure with density $|f|^2$. The domain condition $\int_{\mathbb{R}} \lambda^2\, d\langle E(\lambda)f, f\rangle = \int_{\mathbb{R}} x^2 |f(x)|^2\, dx < \infty$ says exactly that $xf \in L^2(\mathbb{R})$, which is the domain of T named at the outset. The spectral theorem here recovers, with no new information, the fact that multiplication operators are already "diagonal"; its force is that *every* self-adjoint operator is unitarily one of these, as the multiplication form below makes explicit.

Once E is available, any Borel function of T, bounded or not, is defined by integrating that function against E, exactly as continuous and bounded Borel functions of a bounded operator were. This is the Borel functional calculus for unbounded operators: it lets one form e^{itT}, $(T - \lambda)^{-1}$, $\chi_\Omega(T)$, and every other spectral gadget of the later sections by a single integral, and it is the calculus that Stone's theorem and the semigroup theory will run on.

Theorem 7.3.8: The Borel Functional Calculus for Unbounded Operators

Let T be self-adjoint with projection-valued measure E on $\mathbb{R}$ (theorem 7.3.6). For each Borel function $f\colon \mathbb{R} \to \mathbb{C}$ the operator

$$f(T) = \int_{\mathbb{R}} f(\lambda)\, dE(\lambda), \qquad \operatorname{dom} f(T) = \left\{ x \in \mathbb{H} \;\middle|\; \int_{\mathbb{R}} |f(\lambda)|^2\, d\langle E(\lambda)x, x\rangle < \infty \right\}$$

is well-defined and satisfies:

1. the map $f \mapsto f(T)$ is a $*$-homomorphism in the sense that $(f+g)(T) \supseteq f(T) + g(T)$, $(fg)(T) \supseteq f(T)g(T)$, and $\overline{f}(T) = f(T)^*$, with equality of domains when f, g are bounded;
2. $f(T) \in B(\mathbb{H})$ if and only if f is (essentially) bounded on $\sigma(T)$, and then $\|f(T)\| = \sup_{\lambda \in \sigma(T)} |f(\lambda)|$ (the essential supremum against E);
3. $f(T)$ is self-adjoint when f is real-valued, and unitary when $|f| = 1$ on $\sigma(T)$; in particular $\chi_\Omega(T) = E(\Omega)$;
4. *(multiplication model)* there is a measure space (M, μ), a unitary $W\colon \mathbb{H} \to L^2(M, \mu)$, and a real-valued measurable $\varphi\colon M \to \mathbb{R}$ with $WTW^{-1} = M_\varphi$ multiplication by φ, and then $Wf(T)W^{-1} = M_{f\circ\varphi}$ for every Borel f.

***Proof*:**
Step 1: Definition and the domain. For bounded Borel f the integral $\int_{\mathbb{R}} f\, dE$ is the bounded operator produced by transporting the bounded calculus of U through (7.10), namely $f(T) = (f \circ \lambda)(U)$, and $\|f(T)\| \le \sup_{\mathbb{R}} |f|$. For unbounded f set $f_n = f\,\chi_{\{|f| \le n\}}$ and, on the domain $D_f = \left\{ x \;\middle|\; \int |f|^2\, d\langle Ex, x\rangle < \infty \right\}$, define $f(T)x = \lim_n f_n(T)x$; the limit exists because $\|f_n(T)x - f_m(T)x\|^2 = \int_{n<|f|\le m} |f|^2\, d\langle E(\lambda)x, x\rangle \to 0$ (the calculation is (7.10) paired with x, using multiplicativity to turn $|f_n - f_m|^2$ into the integrand), and D_f is dense because $\bigcup_n E(\{|f| \le n\})\mathbb{H}$ is dense (its closure is $E(\mathbb{R})\mathbb{H} = \mathbb{H}$). This defines $f(T)$ with the stated domain.

Step 2: The algebraic properties. Each identity in (1) holds for bounded f, g by the corresponding property of the bounded calculus of U (multiplicativity, $*$-preservation, unit) transported through (7.10). For unbounded f, g the operators $f(T) + g(T)$ and $f(T)g(T)$ are defined only where both summands or the composite make sense, so their domains can be strictly smaller than that of $(f+g)(T)$ or $(fg)(T)$; passing to the strong limits $f_n(T), g_n(T)$ and using that a self-adjoint (hence closed) operator contains the closure of these limits yields the inclusions, with equality when the functions are bounded so that all domains are $\mathbb{H}$. The adjoint identity $\overline{f}(T) = f(T)^*$ holds because $\overline{f_n}(T) = f_n(T)^*$ for the bounded truncations and the closed operator $f(T)$ is the closure of $\bigcup_n f_n(T)$, whose adjoint is the intersection, computed to be $\overline{f}(T)$.

Step 3: Boundedness and the norm. If f is bounded on $\sigma(T)$ with $M = \sup_{\sigma(T)} |f|$, then $f = f\,\chi_{\sigma(T)}$ up to an E-null set (since E is supported on $\sigma(T)$, theorem 7.3.6), and $f(T) = (f\chi_{\sigma(T)})(T)$ is bounded of norm $\le M$; conversely if f is unbounded on every set of full measure then $\int |f|^2\, d\langle Ex, x\rangle = \infty$ for suitable x supported spectrally where $|f|$ is large, so $\operatorname{dom} f(T) \ne \mathbb{H}$ and $f(T)$ is unbounded. The norm equals the essential supremum $\operatorname{ess\,sup}_E |f|$ by the corresponding statement for the bounded calculus of U, transported. This is (2), and (3) is the specialization to real-valued f (then $f(T) = f(T)^*$ by the adjoint identity, and

$\operatorname{dom} f(T) = \operatorname{dom} f(T)^*$, so $f(T)$ is self-adjoint), to $|f| = 1$ (then $f(T)^*f(T) = f(T)f(T)^* = I$), and to $f = \chi_\Omega$ (then $f(T) = \int \chi_\Omega \, dE = E(\Omega)$).

Step 4: The multiplication model. Apply the multiplication form of the bounded spectral theorem (theorem 6.4.10) to the normal operator U: there is a unitary $W\colon \mathbb{H} \to L^2(M, \mu)$ and a measurable $\psi\colon M \to \mathbb{T}$ with $WUW^{-1} = M_\psi$. Since 1 is not an eigenvalue of U, $\psi \neq 1$ μ-a.e., so $\varphi := i(1+\psi)/(1-\psi)$ is a well-defined real-valued measurable function on M (real because $|\psi| = 1$ makes $i(1+\psi)/(1-\psi)$ real). Then $WTW^{-1} = i(I + M_\psi)(I - M_\psi)^{-1} = M_\varphi$ by the inverse Cayley formula of theorem 7.3.4 read pointwise, and for any Borel f, transporting $f(T) = (f \circ \lambda)(U)$ through W gives $Wf(T)W^{-1} = (f \circ \lambda)(M_\psi) = M_{f\circ\lambda\circ\psi} = M_{f\circ\varphi}$, since $\lambda \circ \psi = \varphi$. This establishes (4), completing the proof.

The multiplication model is the concrete payoff. It says that every self-adjoint operator, however singular its original description as a differential operator, becomes multiplication by a real function φ once the space is re-coordinatized by the unitary W, and that any Borel function of the operator is then just multiplication by $f \circ \varphi$. Boundedness of $f(T)$ becomes the elementary question of whether $f \circ \varphi$ is essentially bounded, and the operator e^{itT} that generates a dynamics is multiplication by the unimodular function $e^{it\varphi}$. This diagonalization is what makes the spectral theorem a working tool rather than an existence statement: the resolvent $(\lambda - T)^{-1} = M_{1/(\lambda-\varphi)}$, the projection $\chi_\Omega(T) = M_{\chi_{\varphi^{-1}(\Omega)}}$, and the unitary group $e^{itT} = M_{e^{it\varphi}}$ are all read off from a single real function on a measure space.

Example 7.3.9: Functions of the Laplacian

Let $\mathbb{H} = L^2(\mathbb{R}^n)$ and let $T = -\Delta$ be the (positive) Laplacian, self-adjoint on the domain $\{f \in L^2 \mid \Delta f \in L^2\}$ (the distributional Laplacian landing in L^2). Although the Fourier transform is the fastest diagonalizer, the spectral theorem alone already fixes the shape: $\sigma(-\Delta) = [0, \infty)$, and the projection-valued measure E is supported on $[0, \infty)$, so $-\Delta$ is unitarily multiplication by a nonnegative φ. Two Borel functions of it govern the fundamental evolution equations. Taking $f(\lambda) = e^{-t\lambda}$ for $t > 0$, which is bounded by 1 on $[0, \infty)$, gives the bounded operator

$$e^{-t(-\Delta)} = \int_{[0,\infty)} e^{-t\lambda} \, dE(\lambda)$$

the *heat semigroup*, whose norm is $\sup_{\lambda\geq 0} e^{-t\lambda} = 1$ (attained at $\lambda = 0$), consistent with (2). Taking $f(\lambda) = e^{-it\lambda}$, which has modulus 1 on the spectrum, gives the *unitary* Schrödinger group $e^{it\Delta} = \int e^{-it\lambda} \, dE(\lambda)$ of (3). The heat operator smooths (its multiplier decays for large λ, damping high spectral modes) while the Schrödinger operator rotates (its multiplier has modulus 1), and this single distinction, bounded-decaying versus unimodular multiplier, separates parabolic from dispersive dynamics.

7.4 Stone's Theorem

The functional calculus of section 7.3 produces from a self-adjoint operator A the family $U(t) = e^{itA}$, a group of unitaries read here as a dynamics. Stone's theorem is the converse: every such group arises this way, from a self-adjoint operator uniquely determined as its infinitesimal generator.

Definition 7.4.1: Strongly Continuous One-Parameter Unitary Group

A *strongly continuous one-parameter unitary group* on a complex Hilbert space $\mathbb{H}$ is a map $U : \mathbb{R} \to B(\mathbb{H})$ such that

1. each $U(t)$ is unitary (definition 5.1.23);
2. $U(s+t) = U(s)U(t)$ for all $s, t \in \mathbb{R}$, and $U(0) = I$;
3. for every $x \in \mathbb{H}$ the orbit $t \mapsto U(t)x$ is continuous from $\mathbb{R}$ to $\mathbb{H}$.

The first two conditions make U a group homomorphism from $(\mathbb{R}, +)$ into the unitary group of $\mathbb{H}$; in particular $U(t)^{-1} = U(-t) = U(t)^*$. The third condition, *strong* continuity, is the correct regularity requirement here. Norm continuity of $t \mapsto U(t)$ would be far too strong: it forces the generator to be bounded, excluding every operator the chapter exists to treat. Strong continuity asks only that the group move each vector continuously, which is what a solution of an evolution equation does, and it is automatically upgraded from a single point: because the $U(t)$ are uniformly bounded (each has norm 1) and satisfy the group law, continuity of every orbit at $t = 0$ already gives continuity everywhere, since $U(t+h)x - U(t)x = U(t)(U(h)x - x)$ and $\|U(t)\| = 1$.

The motivating example is the translation group, which will also fix the reading of the generator as a differential operator.

Example 7.4.2: The Translation Group on $L^2(\mathbb{R})$

On $\mathbb{H} = L^2(\mathbb{R})$ define, for each $t \in \mathbb{R}$, the translation

$$(U(t)f)(x) = f(x-t)$$

Each $U(t)$ is unitary: translation is a bijection of $\mathbb{R}$ preserving Lebesgue measure, so $\|U(t)f\|_2 = \|f\|_2$, and its inverse is $U(-t)$. The group law $U(s+t) = U(s)U(t)$ is the identity $f(x-s-t) = (U(s)f)(x-t)$, and $U(0) = I$. For strong continuity, fix $f \in L^2(\mathbb{R})$. Continuous functions of compact support are dense in $L^2(\mathbb{R})$, and such a g is uniformly continuous, so $\|U(t)g - g\|_2 \to 0$ as $t \to 0$ by dominated convergence (the integrand is bounded by an integrable dominating function once $|t| \leq 1$, since the supports stay inside a fixed compact set). For general f, given $\varepsilon > 0$ pick such a g with $\|f - g\|_2 < \varepsilon$; then

$$\|U(t)f - f\|_2 \leq \|U(t)(f-g)\|_2 + \|U(t)g - g\|_2 + \|g - f\|_2 < 2\varepsilon + \|U(t)g - g\|_2$$

using $\|U(t)\| = 1$, and the middle term tends to 0. So U is a strongly continuous one-parameter unitary group. Its generator, computed below, is $A = i\, d/dx$ on the natural domain, and the group is the flow of the transport equation $\partial_t u = -\partial_x u$.

The generator is the object that recovers A from the group $U(t) = e^{itA}$ by differentiating at $t = 0$. Since $e^{it\lambda} = 1 + it\lambda + o(t)$, one expects $(U(t)x - x)/(it) \to Ax$, but only for those x on which the limit exists. That set of x, not prescribed in advance, is exactly the domain of the generator.

Definition 7.4.3: Generator of a Unitary Group

Let U be a strongly continuous one-parameter unitary group on $\mathbb{H}$. Its *generator* is the operator A with domain

$$\operatorname{dom} A = \left\{ x \in \mathbb{H} \;\middle|\; \lim_{t\to 0} \frac{U(t)x - x}{it} \text{ exists in } \mathbb{H} \right\}$$

and $Ax = \lim_{t\to 0}(U(t)x - x)/(it)$ for $x \in \operatorname{dom} A$.

The factor i in the denominator is a convention that makes A self-adjoint rather than skew-adjoint: with it, $U(t) = e^{itA}$ pairs a real spectral parameter λ with the phase $e^{it\lambda}$, matching the functional calculus of theorem 7.3.8. The domain is genuinely a proper subspace in the unbounded case, but it is never small: the following lemma produces a dense supply of vectors in $\operatorname{dom} A$ by averaging the orbit against smooth weights, the standard device for regularizing a merely continuous group.

Lemma 7.4.4: Density of the Generator's Domain

Let U be a strongly continuous one-parameter unitary group with generator A. Then $\operatorname{dom} A$ is dense in $\mathbb{H}$, and A is symmetric (definition 7.2.8).

***Proof*:**
Step 1: A dense set of smoothed vectors lies in the domain. For $x \in \mathbb{H}$ and $\varphi \in C_c^\infty(\mathbb{R})$ (smooth, compactly supported) define the averaged vector

$$x_\varphi = \int_{\mathbb{R}} \varphi(s)\, U(s)x \, ds$$

a Bochner integral of the continuous, compactly supported $\mathbb{H}$-valued function $s \mapsto \varphi(s)U(s)x$. Apply the group law to $U(t)x_\varphi$ and substitute $s \mapsto s - t$ inside the integral:

$$U(t)x_\varphi = \int_{\mathbb{R}} \varphi(s)\, U(s+t)x \, ds = \int_{\mathbb{R}} \varphi(s-t)\, U(s)x \, ds$$

Hence

$$\frac{U(t)x_\varphi - x_\varphi}{it} = \int_{\mathbb{R}} \frac{\varphi(s-t) - \varphi(s)}{it}\, U(s)x \, ds$$

As $t \to 0$ the difference quotient $(\varphi(s-t) - \varphi(s))/t$ converges uniformly to $-\varphi'(s)$, with all supports inside a fixed compact interval, so the integrand converges in the norm of $\mathbb{H}$ uniformly in s and the integral converges. Therefore $x_\varphi \in \operatorname{dom} A$ and

$$Ax_\varphi = i \int_{\mathbb{R}} \varphi'(s)\, U(s)x \, ds$$

Choosing $\varphi = \varphi_n$ an approximate identity (nonnegative, integral 1, supports shrinking to 0) gives $x_{\varphi_n} = \int \varphi_n(s)U(s)x\, ds \to U(0)x = x$ by strong continuity, since $\|x_{\varphi_n} - x\| \le \int \varphi_n(s)\|U(s)x - x\|\, ds \to 0$. So every $x \in \mathbb{H}$ is a limit of vectors $x_{\varphi_n} \in \operatorname{dom} A$, proving $\operatorname{dom} A$ dense.

Step 2: A is symmetric. Fix $x, y \in \operatorname{dom} A$. Because each $U(t)$ is unitary, $\langle U(t)x, U(t)y\rangle = \langle x, y\rangle$ for all t, so

$$0 = \frac{d}{dt}\Big|_{t=0} \langle U(t)x, U(t)y\rangle$$

Computed as a limit of difference quotients, using $\langle U(t)x, U(t)y\rangle = \langle x, y\rangle$ and the definition of A,

$$0 = \left\langle \lim_{t\to 0} \frac{U(t)x - x}{t}, y \right\rangle + \left\langle x, \lim_{t\to 0} \frac{U(t)y - y}{t} \right\rangle = \langle iAx, y\rangle + \langle x, iAy\rangle$$

where the two limits exist precisely because $x, y \in \operatorname{dom} A$. Thus $i\langle Ax, y\rangle - i\langle x, Ay\rangle = 0$, that is $\langle Ax, y\rangle = \langle x, Ay\rangle$ for all $x, y \in \operatorname{dom} A$. So $A \subseteq A^*$ (definition 7.2.6), which is symmetry, completing the proof.

The lemma has done the analytic work: the generator of any unitary group is a densely-defined symmetric operator. To promote symmetry to self-adjointness one direction of Stone's theorem needs the deficiency criterion of theorem 7.2.11, and to invert the construction the other direction needs the exponential e^{itA} to be defined and to depend continuously on t, which the functional calculus supplies. Both are assembled in the theorem.

Before the theorem, one more preparation. In the converse direction A is given self-adjoint and $U(t) := e^{itA}$ is defined through theorem 7.3.8 as $\int_{\mathbb{R}} e^{it\lambda}\, dE(\lambda)$ against the spectral measure E of A. The functions $e_t(\lambda) = e^{it\lambda}$ are bounded of modulus 1, so each $U(t)$ is unitary, and their pointwise products and limits control the group law and the strong continuity through the calculus.

Theorem 7.4.5: Stone's Theorem

Let $\mathbb{H}$ be a complex Hilbert space. There is a bijection between strongly continuous one-parameter unitary groups on $\mathbb{H}$ and self-adjoint operators on $\mathbb{H}$, given in the two directions as follows.

1. If A is self-adjoint, then $U(t) := e^{itA} = \int_{\mathbb{R}} e^{it\lambda}\, dE(\lambda)$, defined through the functional calculus of theorem 7.3.8 against the spectral measure E of A, is a strongly continuous one-parameter unitary group.

2. Conversely, every strongly continuous one-parameter unitary group U is of this form for a unique self-adjoint operator A, namely its generator (definition 7.4.3),

$$Ax = \lim_{t\to 0} \frac{U(t)x - x}{it}$$

on its natural domain.

The two constructions are mutually inverse: the generator of e^{itA} is A, and the exponential of the generator of U is U.

***Proof*:**
Step 1: a self-adjoint A yields a group (part 1). Let E be the spectral measure of A (theorem 7.3.6) and $e_t(\lambda) = e^{it\lambda}$. Each e_t is a bounded Borel function, so $U(t) = e_t(A) = \int_{\mathbb{R}} e^{it\lambda}\, dE(\lambda)$ is a bounded operator by theorem 7.3.8. The functional calculus is a $*$-homomorphism on bounded Borel functions, and $\overline{e_t} = e_{-t}$ with $e_t \cdot e_{-t} = 1$, so

$$U(t)^* U(t) = e_{-t}(A) e_t(A) = (e_{-t} e_t)(A) = 1(A) = I$$

and likewise $U(t)U(t)^* = I$, so each $U(t)$ is unitary. The pointwise identity $e_{s+t} = e_s e_t$ gives $U(s+t) = U(s)U(t)$ through the same homomorphism property, and $e_0 = 1$ gives $U(0) = I$. It remains to check strong continuity. Fix $x \in \mathbb{H}$ and let $\mu_x(\Omega) = \langle E(\Omega)x, x\rangle$ be the associated finite Borel measure on $\mathbb{R}$, with $\mu_x(\mathbb{R}) = \|x\|^2$. The functional calculus gives, for the difference $U(t)x - x = (e_t - 1)(A)x$,

$$\|U(t)x - x\|^2 = \int_{\mathbb{R}} |e^{it\lambda} - 1|^2 \, d\mu_x(\lambda)$$

As $t \to 0$ the integrand $|e^{it\lambda} - 1|^2$ tends to 0 pointwise in λ and is bounded by the μ_x-integrable constant 4, so dominated convergence gives $\|U(t)x - x\| \to 0$. Strong continuity at $t = 0$, hence everywhere by the group law, follows. This proves part 1.

Step 2: the generator of e^{itA} is A (half of mutual inversion). Let $U(t) = e^{itA}$ as in Step 1 and let B denote its generator (definition 7.4.3). For $x \in \operatorname{dom} A$, that is x with $\int_{\mathbb{R}} \lambda^2 \, d\mu_x(\lambda) < \infty$, estimate the difference quotient against $Ax = \int \lambda \, dE(\lambda) \, x$:

$$\Big\| \frac{U(t)x - x}{it} - Ax \Big\|^2 = \int_{\mathbb{R}} \Big| \frac{e^{it\lambda} - 1}{it} - \lambda \Big|^2 d\mu_x(\lambda)$$

The integrand tends to 0 pointwise as $t \to 0$; moreover $|(e^{it\lambda} - 1)/(it)| \le |\lambda|$ for all t (the chord of $e^{i\theta}$ is no longer than its arc), so the integrand is bounded by $(2|\lambda|)^2 = 4\lambda^2$, which is μ_x-integrable exactly because $x \in \operatorname{dom} A$. Dominated convergence gives the limit 0, so $x \in \operatorname{dom} B$ and $Bx = Ax$; thus $A \subseteq B$. But A is self-adjoint, hence maximally symmetric, and B is symmetric (lemma 7.4.4) with $A \subseteq B \subseteq B^* \subseteq A^* = A$, forcing $B = A$. So the generator of e^{itA} is exactly A.

Step 3: the generator is self-adjoint, and $U = e^{itA}$ (part 2). Now let U be an arbitrary strongly continuous one-parameter unitary group and A its generator. By lemma 7.4.4, A is densely defined and symmetric, so theorem 7.2.11 reduces self-adjointness to showing $\operatorname{ran}(A \pm i) = \mathbb{H}$, equivalently $\ker(A^* \mp i) = 0$. Suppose $y \in \operatorname{dom} A^*$ with $A^* y = iy$; the goal is $y = 0$ (the case $A^* y = -iy$ is identical). For any $x \in \operatorname{dom} A$ the orbit $t \mapsto U(t)x$ is differentiable with derivative

$$\frac{d}{dt} U(t)x = iA\, U(t)x = i\, U(t)Ax \tag{7.12}$$

the first equality because $U(t)x \in \operatorname{dom} A$ with $AU(t)x = U(t)Ax$ (the group commutes with its generator, since $U(t)$ commutes with each $U(h)$ and hence with the difference quotient defining A), and both expressions agreeing by that commutation. Consider the scalar function $g(t) = \langle U(t)x, y\rangle$. Using (7.12) and $A^* y = iy$,

$$g'(t) = \langle iA\, U(t)x, y\rangle = i\langle AU(t)x, y\rangle = i\langle U(t)x, A^* y\rangle = i\langle U(t)x, iy\rangle = \langle U(t)x, y\rangle = g(t)$$

So g solves $g' = g$, giving $g(t) = g(0)e^t$. But $|g(t)| = |\langle U(t)x, y\rangle| \le \|U(t)x\| \, \|y\| = \|x\| \, \|y\|$ is bounded in t, while $g(0)e^t$ is unbounded as $t \to +\infty$ unless $g(0) = 0$. Hence $\langle x, y\rangle = g(0) = 0$ for every x in the dense set $\operatorname{dom} A$, so $y = 0$. This kills both deficiency spaces, and theorem 7.2.11 makes A self-adjoint.

It remains to identify U with e^{itA}. Both are strongly continuous unitary groups; write $V(t) = e^{itA}$ for the group of Step 1, whose generator is A by Step 2, and fix $x \in \operatorname{dom} A$. The vector $w(t) = U(t)x - V(t)x$ is differentiable in t, and by (7.12) applied to each group (both have generator A, and $U(t)x, V(t)x \in \operatorname{dom} A$),

$$\frac{d}{dt} w(t) = iA\, U(t)x - iA\, V(t)x = iA\, w(t)$$

Its squared norm has derivative

$$\frac{d}{dt}\|w(t)\|^2 = \langle iAw(t), w(t)\rangle + \langle w(t), iAw(t)\rangle = i\big(\langle Aw(t), w(t)\rangle - \langle w(t), Aw(t)\rangle\big) = 0$$

using the symmetry of A on $\operatorname{dom} A$. So $\|w(t)\|^2$ is constant; since $w(0) = x - x = 0$, it is identically 0, giving $U(t)x = V(t)x$ for all t and all $x \in \operatorname{dom} A$. As $\operatorname{dom} A$ is dense and $U(t), V(t)$ are bounded (norm 1), $U(t) = V(t) = e^{itA}$ for every t. This proves part 2 and, together with Step 2, the mutual inversion.

Step 4: uniqueness. If A_1, A_2 are self-adjoint with $e^{itA_1} = e^{itA_2}$ for all t, then by Step 2 both equal the generator of the common group, so $A_1 = A_2$. The two constructions are therefore inverse bijections, completing the proof.

Stone's theorem is the exact analogue, in infinite dimensions, of writing a one-parameter group of unitary matrices as e^{itH} for a Hermitian H. The content beyond the matrix case is entirely in the domain bookkeeping: the generator A is unbounded, defined only where the difference quotient converges, and the whole force of the theorem is that this a-priori-partial operator turns out to be self-adjoint, so the spectral calculus applies to it and reconstitutes the group. The self-adjointness in Step 3 is where symmetry is not enough; the boundedness of the orbit $\langle U(t)x, y\rangle$ against a putative eigenvector of A^* at $\pm i$ is exactly what a merely symmetric operator could fail to provide, and it is the unitarity of U that supplies it. In physical terms, A is the Hamiltonian (up to sign and units), the group $U(t) = e^{itA}$ is the time evolution, and the theorem says that the reversible dynamics on a Hilbert space are in bijection with the self-adjoint observables generating them.

The theorem also settles the translation group, whose generator was announced but not yet computed.

Example 7.4.6: The Momentum Operator as Generator of Translation

Return to the translation group $(U(t)f)(x) = f(x-t)$ on $L^2(\mathbb{R})$ (example 7.4.2). Its generator A is computed directly from definition 7.4.3: for f in the domain,

$$(Af)(x) = \lim_{t\to 0} \frac{f(x-t) - f(x)}{it} = \frac{1}{i}\lim_{t\to 0}\frac{f(x-t)-f(x)}{t} = -\frac{1}{i}f'(x) = if'(x)$$

so $A = i\,d/dx$ acts as i times the derivative. The domain is not all of $L^2(\mathbb{R})$: it is exactly the set of $f \in L^2(\mathbb{R})$ for which the limit converges in L^2, which is the Sobolev space $\operatorname{dom} A = \{f \in L^2(\mathbb{R}) \mid f' \in L^2(\mathbb{R})\}$ where f' is the weak derivative (section 4.1), that is $H^1(\mathbb{R})$. On this domain $A = i\,d/dx$ is self-adjoint, by Stone's theorem applied to the unitary group U, without a separate integration-by-parts verification. Up to sign this A is the momentum operator of quantum mechanics, $P = -i\,d/dx = -A$: Stone's theorem identifies it as the self-adjoint generator of spatial translation, and $U(t) = e^{itA} = e^{-t\,d/dx}$ is the unitary that translates a wavefunction by t.

The identification $A = i\,d/dx$ closes the loop with the chapter's opening: the differential operator that is unbounded and discontinuous, and to which the bounded theory did not apply, is recovered here as the generator of a bounded, well-behaved group of unitaries. Its self-adjointness was earned through the group rather than assumed, and the same route, from a group of symmetries to its self-adjoint generator, is how the position, momentum, and energy observables of a quantum system acquire the spectral calculus that makes their measurement statistics meaningful.

7.5 C_0-Semigroups and the Hille-Yosida Theorem

This section takes up one-directional dynamics on a Banach space X: which operators arise as the infinitesimal generator of a C_0-semigroup. The Hille-Yosida theorem answers this completely for the contractive case, reconstructing the semigroup from the resolvent of the generator by an approximation due to Yosida.

Definition 7.5.1: C_0-Semigroup

A *C_0-semigroup* (or *strongly continuous semigroup*) on a Banach space X is a family $(T(t))_{t\geq 0}$ of bounded operators $T(t) \in B(X)$ such that

1. $T(0) = I$;
2. $T(s+t) = T(s)T(t)$ for all $s, t \geq 0$ (the semigroup law);
3. for each $x \in X$ the orbit $t \mapsto T(t)x$ is continuous from $[0, \infty)$ to X (strong continuity).

The semigroup is a *contraction semigroup* if $\|T(t)\| \leq 1$ for all $t \geq 0$.

The three axioms encode a deterministic, time-homogeneous, one-directional dynamics: the state at time $s+t$ is obtained from the state at time s by evolving for a further time t, and the evolution depends only on the elapsed time, not on the absolute clock. Strong continuity is the minimal regularity that makes the orbit a genuine trajectory in X while still admitting the unbounded generators of analysis; requiring norm continuity of $t \mapsto T(t)$ instead would force the generator to be bounded and exclude every differential operator. The two stronger conditions suppressed here, that T extend to all of $\mathbb{R}$ and preserve norms, are exactly what distinguishes the unitary groups of section 7.4 from the present setting.

The semigroup law forces the norms to grow at most exponentially. This bound is the first structural fact and the reference point against which the resolvent is later defined.

Proposition 7.5.2: Exponential Bound

Let $(T(t))_{t\geq 0}$ be a C_0-semigroup on X. There are constants $M \geq 1$ and $\omega \in \mathbb{R}$ with

$$\|T(t)\| \leq Me^{\omega t} \qquad (t \geq 0) \tag{7.13}$$

***Proof*:**
Step 1: Local boundedness. The claim is that $\sup_{0\leq t\leq 1} \|T(t)\| = M < \infty$. Suppose not; then there are $t_n \in [0, 1]$ with $\|T(t_n)\| \to \infty$. For each fixed $x \in X$ the orbit $t \mapsto T(t)x$ is continuous on the compact interval $[0, 1]$, hence bounded, so $\sup_n \|T(t_n)x\| < \infty$. The family $\{T(t_n)\}$ is therefore pointwise bounded, and by the Banach-Steinhaus uniform boundedness principle it is uniformly bounded, $\sup_n \|T(t_n)\| < \infty$, contradicting the choice of t_n. Thus $M := \sup_{0\leq t\leq 1} \|T(t)\|$ is finite, and $M \geq \|T(0)\| = 1$.

Step 2: Exponential growth from the semigroup law. Fix $t \geq 0$ and write $t = n + r$ with $n \in \mathbb{N}$ (or $n = 0$) and $r \in [0, 1)$. The semigroup law gives $T(t) = T(1)^n T(r)$, so

$$\|T(t)\| \leq \|T(1)\|^n \|T(r)\| \leq M^{n+1} = M \cdot M^n \leq M \cdot M^t$$

using $\|T(1)\| \leq M$, $\|T(r)\| \leq M$, and $n \leq t$ with $M \geq 1$. Setting $\omega = \log M$ gives $M^t = e^{\omega t}$, hence (7.13), as we sought to show.

The pair (M, ω) is not unique; the infimum of the admissible ω is the growth bound of the semigroup. When ω may be taken to be 0 and $M = 1$ the semigroup is a contraction semigroup, the case the Hille-Yosida theorem below characterizes. The general case reduces to it by the rescaling $S(t) = e^{-\omega t}T(t)$, which is again a C_0-semigroup with a smaller growth bound; only a bounded shift of the generator separates the two.

The prototype is the heat semigroup, where the abstract axioms become the concrete statement that heat diffuses and the total is conserved.

Example 7.5.3: The Heat Semigroup

On $X = L^2(\mathbb{R})$ define, for $t > 0$,

$$(T(t)f)(x) = \frac{1}{\sqrt{4\pi t}} \int_{\mathbb{R}} e^{-|x-y|^2/(4t)} f(y)\, dy$$

and $T(0) = I$. The kernel $g_t(x) = (4\pi t)^{-1/2} e^{-|x|^2/(4t)}$ is the Gaussian of variance $2t$, so $T(t)f = g_t * f$ is convolution with g_t. Young's inequality gives $\|T(t)f\|_2 \leq \|g_t\|_1 \|f\|_2 = \|f\|_2$, since $\int_{\mathbb{R}} g_t = 1$; thus each $T(t)$ is a contraction. The Gaussians satisfy the convolution identity $g_s * g_t = g_{s+t}$ (the variances add), which is exactly the semigroup law $T(s)T(t) = T(s+t)$. Strong continuity $\|T(t)f - f\|_2 \to 0$ as $t \to 0^+$ is the statement that $(g_t)_{t>0}$ is an approximate identity on L^2 (section 2.6). The orbit $u(t, \cdot) = T(t)f$ solves the heat equation $\partial_t u = \partial_{xx} u$ with initial datum f: this is the reason the generator, computed below on suitable f, is the second-derivative operator ∂_{xx}.

The whole theory of a C_0-semigroup is encoded in the rate of change of its orbits at time zero. That rate is a densely defined operator, generally unbounded, and it plays the role for the semigroup that a differential equation's right-hand side plays for its flow.

Definition 7.5.4: Generator of a C_0-Semigroup

Let $(T(t))_{t \geq 0}$ be a C_0-semigroup on X. Its *generator* is the operator A with domain

$$\operatorname{dom} A = \left\{ x \in X \;\middle|\; \lim_{t \to 0^+} \frac{T(t)x - x}{t} \text{ exists in } X \right\}$$

defined on that domain by

$$Ax = \lim_{t \to 0^+} \frac{T(t)x - x}{t}$$

The generator is the right derivative of the orbit at the origin, and by time homogeneity it controls the derivative at every later time. For a differential-equation reader, A is the "vector field" whose flow is $(T(t))$: the orbit $u(t) = T(t)x$ is the solution of the abstract Cauchy problem $u'(t) = Au(t)$, $u(0) = x$, whenever $x \in \operatorname{dom} A$. The content of the next results is that this domain is large (dense) and that A is closed, so that despite being unbounded it is a well-behaved object to which the resolvent machinery applies.

For the heat semigroup the generator is the operator anticipated by the heat equation.

Example 7.5.5: Generator of the Heat Semigroup

For the heat semigroup of example 7.5.3 on $L^2(\mathbb{R})$, the difference quotient $(T(t)f - f)/t$ converges in L^2 precisely when $f'' \in L^2$ (interpreting the derivative weakly), and its limit is f''. Thus the generator is

$$Af = f'', \qquad \operatorname{dom} A = \left\{ f \in L^2(\mathbb{R}) \;\middle|\; f'' \in L^2(\mathbb{R}) \right\}$$

the second-derivative operator on the Sobolev space $\left\{ f \in L^2 \;\middle|\; f, f' \text{ absolutely continuous}, f'' \in L^2 \right\}$. This domain is dense (it contains the Schwartz functions), and A is unbounded. To witness this, fix a nonzero Schwartz bump φ and set $f_k(x) = e^{ikx}\varphi(x)$ for $k \in \mathbb{N}$. Modulation preserves the L^2 norm, so $\|f_k\|_2 = \|\varphi\|_2$ is fixed, while $f_k'' = e^{ikx}(\varphi'' + 2ik\varphi' - k^2\varphi)$ has $\|f_k''\|_2 \geq k^2\|\varphi\|_2 - 2k\|\varphi'\|_2 - \|\varphi''\|_2 \to \infty$. Thus $\|Af_k\|_2/\|f_k\|_2 \to \infty$ on a family of fixed norm, so $\sup_{\|f\|_2=1} \|Af\|_2 = \infty$: the second derivative amplifies high frequencies without bound, which on the Fourier side is the unboundedness of the multiplier ξ^2. The abstract Cauchy problem $u' = Au$ is the heat equation, recovering the interpretation of example 7.5.3.

The differentiation properties of the orbits are collected next; they are the computational engine behind density, closedness, and the resolvent formula. The key identity is that integrating the orbit lands one back in the domain of A, so that A can be studied through the everywhere-defined averaging operators $x \mapsto \int_0^t T(s)x\,ds$.

Lemma 7.5.6: Orbit Calculus

Let $(T(t))_{t\geq 0}$ be a C_0-semigroup on X with generator A.

1. For every $x \in X$ and $t \geq 0$ the vector $\int_0^t T(s)x\,ds$ lies in $\operatorname{dom} A$, and
$$A\int_0^t T(s)x\,ds = T(t)x - x \tag{7.14}$$
2. If $x \in \operatorname{dom} A$ then $T(t)x \in \operatorname{dom} A$ for every $t \geq 0$, the orbit $t \mapsto T(t)x$ is continuously differentiable, and
$$\frac{d}{dt}T(t)x = AT(t)x = T(t)Ax \tag{7.15}$$

***Proof*:**

The integral $\int_0^t T(s)x\,ds$ is the Bochner integral of the continuous X-valued function $s \mapsto T(s)x$ (theorem 7.1.9); continuity makes it Bochner integrable on $[0, t]$ by theorem 7.1.6.

Part (1). Fix x and t, and let $h > 0$. Since each $T(h)$ is bounded, it commutes with the Bochner integral (a bounded operator is closed, so proposition 7.1.8 applies, or directly by approximation by simple functions), giving

$$\frac{T(h) - I}{h}\int_0^t T(s)x\,ds = \frac{1}{h}\int_0^t \big(T(s+h)x - T(s)x\big)\,ds$$

Substituting $\sigma = s + h$ in the first term and splitting the intervals,

$$\frac{1}{h}\int_0^t T(s+h)x\,ds - \frac{1}{h}\int_0^t T(s)x\,ds = \frac{1}{h}\int_t^{t+h} T(\sigma)x\,d\sigma - \frac{1}{h}\int_0^h T(s)x\,ds$$

As $h \to 0^+$ each average converges to the value of the continuous integrand at the endpoint, by the fundamental theorem of calculus for Bochner integrals (theorem 7.1.9): the first tends to $T(t)x$ and the second to $T(0)x = x$. Hence $\int_0^t T(s)x\,ds \in \operatorname{dom} A$ and (7.14) holds.

Part (2). Let $x \in \operatorname{dom} A$. For $h > 0$, the semigroup law and boundedness of $T(t)$ give

$$\frac{T(h) - I}{h} T(t)x = T(t)\frac{T(h) - I}{h} x \to T(t)Ax \quad (h \to 0^+)$$

the limit existing because $x \in \operatorname{dom} A$ and $T(t)$ is continuous. Therefore $T(t)x \in \operatorname{dom} A$ with $AT(t)x = T(t)Ax$, and the right derivative of $t \mapsto T(t)x$ is $T(t)Ax$. To identify this as a two-sided derivative, consider for $t > 0$ and $0 < h < t$ the left difference quotient

$$\frac{T(t)x - T(t-h)x}{h} = T(t-h)\frac{T(h)x - x}{h}$$

The factor $(T(h)x - x)/h \to Ax$, while $T(t-h) \to T(t)$ strongly, and the two convergences combine on the bounded family $\{T(t-h)\}_{0 \le h \le t/2}$ (bounded by proposition 7.5.2): writing $T(t-h)y_h - T(t)Ax = T(t-h)(y_h - Ax) + (T(t-h) - T(t))Ax$ with $y_h = (T(h)x - x)/h$, the first term is bounded in norm by $(\sup \|T(t-h)\|)\|y_h - Ax\| \to 0$ and the second tends to 0 by strong continuity. Thus the left quotient also converges to $T(t)Ax$, so $t \mapsto T(t)x$ is differentiable with derivative $T(t)Ax = AT(t)x$, which is continuous in t; this is (7.15) and shows the orbit is C^1, completing the proof.

Density and closedness now follow by reading off consequences of the orbit calculus.

Proposition 7.5.7: The Generator is Densely Defined and Closed

The generator A of a C_0-semigroup $(T(t))_{t \ge 0}$ on X is densely defined and closed.

***Proof*:**

Density. For $x \in X$ set $x_t = \frac{1}{t}\int_0^t T(s)x\,ds$ for $t > 0$. By lemma 7.5.6(1), $x_t \in \operatorname{dom} A$. As $t \to 0^+$ the average of the continuous function $s \mapsto T(s)x$ over $[0, t]$ converges to its value $T(0)x = x$ at the endpoint (theorem 7.1.9). Hence every $x \in X$ is a limit of elements $x_t \in \operatorname{dom} A$, so $\operatorname{dom} A$ is dense.

Closedness. Let $x_n \in \operatorname{dom} A$ with $x_n \to x$ and $Ax_n \to y$ in X. By (7.15) the orbit of each x_n is C^1 with $\frac{d}{ds}T(s)x_n = T(s)Ax_n$, so the fundamental theorem of calculus for Bochner integrals (theorem 7.1.9) gives

$$T(t)x_n - x_n = \int_0^t T(s)Ax_n\,ds$$

Fix $t > 0$. On the compact interval $[0, t]$ the operators $T(s)$ are uniformly bounded by $C = \sup_{0 \le s \le t} \|T(s)\| < \infty$ (proposition 7.5.2), so $\|T(s)Ax_n - T(s)y\| \le C\|Ax_n - y\| \to 0$ uniformly in s. Passing to the limit under the integral (uniform convergence of the integrand on a finite interval) and using $x_n \to x$, $T(t)x_n \to T(t)x$,

$$T(t)x - x = \int_0^t T(s)y\,ds$$

Dividing by t and letting $t \to 0^+$, the right side converges to $T(0)y = y$ by theorem 7.1.9, while the left side is the difference quotient defining Ax. Hence $x \in \operatorname{dom} A$ and $Ax = y$, so A is closed, as we sought to show.

That A is closed and densely defined is what admits it to the resolvent theory of section 7.2: its spectrum and resolvent are defined exactly as for the closed operators studied there. The next result computes the resolvent explicitly and reveals the analytic identity organizing the whole subject. The resolvent of the generator is the Laplace transform of the semigroup.

The motivation is the scalar identity $\int_0^\infty e^{-\lambda t} e^{at}\,dt = (\lambda - a)^{-1}$, valid for $\operatorname{Re}\lambda > a$. Reading e^{at} as the semigroup $T(t)$ and $(\lambda - a)^{-1}$ as the resolvent $R(\lambda, A)$ suggests that the resolvent is the operator Laplace transform of the semigroup, and this is exactly the theorem.

Theorem 7.5.8: The Resolvent as Laplace Transform

Let $(T(t))_{t\geq 0}$ be a C_0-semigroup on X with generator A and bound $\|T(t)\| \leq Me^{\omega t}$ as in proposition 7.5.2. Then every λ with $\operatorname{Re}\lambda > \omega$ lies in the resolvent set $\rho(A)$, the improper Bochner integral

$$R(\lambda)x = \int_0^\infty e^{-\lambda t} T(t)x\,dt \qquad (x \in X) \tag{7.16}$$

converges, and $R(\lambda) = R(\lambda, A) = (\lambda I - A)^{-1}$. Moreover $\|R(\lambda, A)\| \leq M/(\operatorname{Re}\lambda - \omega)$.

***Proof*:**
Step 1: Convergence and the norm bound. Fix λ with $\operatorname{Re}\lambda > \omega$ and $x \in X$. The integrand $t \mapsto e^{-\lambda t}T(t)x$ is continuous, hence strongly measurable, and

$$\|e^{-\lambda t}T(t)x\| \leq e^{-(\operatorname{Re}\lambda)t}Me^{\omega t}\|x\| = M\|x\|e^{-(\operatorname{Re}\lambda - \omega)t}$$

whose integral over $[0, \infty)$ is $M\|x\|/(\operatorname{Re}\lambda - \omega) < \infty$. By the Bochner integrability criterion (theorem 7.1.6) the integral (7.16) converges and $\|R(\lambda)x\| \leq M\|x\|/(\operatorname{Re}\lambda - \omega)$, giving the norm bound on the bounded operator $R(\lambda)$.

Step 2: $R(\lambda)$ maps into $\operatorname{dom} A$ *and* $(\lambda I - A)R(\lambda) = I$. For $h > 0$, commuting the bounded operator $(T(h) - I)/h$ past the Bochner integral and shifting as in lemma 7.5.6,

$$\frac{T(h) - I}{h}R(\lambda)x = \frac{1}{h}\int_0^\infty e^{-\lambda t}\big(T(t+h)x - T(t)x\big)\,dt = \frac{e^{\lambda h} - 1}{h}R(\lambda)x - \frac{e^{\lambda h}}{h}\int_0^h e^{-\lambda t}T(t)x\,dt$$

where the substitution $\sigma = t+h$ in the first term produces the factor $e^{\lambda h}$ and the boundary integral over $[0, h]$. Letting $h \to 0^+$: the first term tends to $\lambda R(\lambda)x$, and the average $\frac{1}{h}\int_0^h e^{-\lambda t}T(t)x\,dt \to x$ by theorem 7.1.9, so the second tends to x. Hence $R(\lambda)x \in \operatorname{dom} A$ and

$$AR(\lambda)x = \lambda R(\lambda)x - x$$

that is, $(\lambda I - A)R(\lambda)x = x$ for all $x \in X$.

Step 3: $R(\lambda)(\lambda I - A) = I$ on $\operatorname{dom} A$. Let $x \in \operatorname{dom} A$. Since A is closed and commutes with the semigroup on its domain (lemma 7.5.6(2)), A commutes with the integral defining $R(\lambda)$ by proposition 7.1.8: the functions $t \mapsto e^{-\lambda t}T(t)x$ and $t \mapsto e^{-\lambda t}T(t)Ax = A(e^{-\lambda t}T(t)x)$ are both Bochner integrable, so $R(\lambda)x \in \operatorname{dom} A$ and $AR(\lambda)x = R(\lambda)Ax$. Therefore

$$R(\lambda)(\lambda I - A)x = \lambda R(\lambda)x - R(\lambda)Ax = \lambda R(\lambda)x - AR(\lambda)x = x$$

the last equality by Step 2. Steps 2 and 3 exhibit $R(\lambda)$ as a two-sided inverse of $\lambda I - A$, so $\lambda \in \rho(A)$ and $R(\lambda) = R(\lambda, A)$, completing the proof.

Two consequences of (7.16) organize what follows. First, the resolvent bound $\|R(\lambda, A)\| \le M/(\operatorname{Re}\lambda - \omega)$ is the operator reflection of the exponential bound on the semigroup; for a contraction semigroup ($M = 1$, $\omega = 0$) it reads $\|R(\lambda, A)\| \le 1/\lambda$ for $\lambda > 0$, precisely the inequality that will characterize generators. Second, the theorem is not yet a construction: it computes the resolvent from a semigroup already in hand. The Hille-Yosida theorem runs the implication the other way, building the semigroup from an operator whose resolvent satisfies that bound, and it is the substantive result of the section.

The reconstruction rests on approximating the unbounded generator A by bounded operators whose exponentials can be formed directly. The device is the Yosida approximation.

Definition 7.5.9: Yosida Approximation

Let A be a densely defined closed operator on X with $(0, \infty) \subseteq \rho(A)$. For $\lambda > 0$ the *Yosida approximation* of A is the bounded operator

$$A_\lambda = \lambda A R(\lambda, A) = \lambda^2 R(\lambda, A) - \lambda I$$

where the second expression uses $AR(\lambda, A) = \lambda R(\lambda, A) - I$.

The Yosida approximation replaces the unbounded A by the bounded operator $\lambda A R(\lambda, A)$, which one should read as A composed with the "mollifier" $\lambda R(\lambda, A)$. The second expression $A_\lambda = \lambda^2 R(\lambda, A) - \lambda I$ shows A_λ is bounded, so its exponential $e^{tA_\lambda} = \sum_{n \ge 0} (tA_\lambda)^n / n!$ converges in operator norm and defines a genuine semigroup for each λ. The plan is to show these approximating semigroups converge strongly as $\lambda \to \infty$ and that the limit is the semigroup generated by A. The next lemma records the two facts that make the plan work: the mollifier tends to the identity, and A_λ tends to A.

Lemma 7.5.10: Convergence of the Yosida Approximation

Let A be densely defined and closed with $(0, \infty) \subseteq \rho(A)$ and $\|R(\lambda, A)\| \le 1/\lambda$ for all $\lambda > 0$. Then:

1. $\|\lambda R(\lambda, A)\| \le 1$ for all $\lambda > 0$, and $\lambda R(\lambda, A)x \to x$ as $\lambda \to \infty$, for every $x \in X$;
2. $A_\lambda x \to Ax$ as $\lambda \to \infty$, for every $x \in \operatorname{dom} A$.

***Proof*:**
Part (1). The bound $\|\lambda R(\lambda, A)\| \le \lambda \cdot (1/\lambda) = 1$ is immediate. For $x \in \operatorname{dom} A$, the resolvent identity $AR(\lambda, A) = \lambda R(\lambda, A) - I$ rearranges to $\lambda R(\lambda, A)x - x = R(\lambda, A)Ax$, whence

$$\|\lambda R(\lambda, A)x - x\| = \|R(\lambda, A)Ax\| \le \frac{1}{\lambda}\|Ax\| \to 0 \quad (\lambda \to \infty)$$

Thus $\lambda R(\lambda, A)x \to x$ on the dense subspace $\operatorname{dom} A$. Since the operators $\lambda R(\lambda, A)$ are uniformly bounded by 1, the convergence extends to all $x \in X$: given $x \in X$ and $\varepsilon > 0$, choose $y \in \operatorname{dom} A$ with $\|x - y\| < \varepsilon$; then $\|\lambda R(\lambda, A)x - x\| \le \|\lambda R(\lambda, A)(x - y)\| + \|\lambda R(\lambda, A)y - y\| + \|y - x\| \le 2\varepsilon + \|\lambda R(\lambda, A)y - y\|$, and the middle term tends to 0.

Part (2). For $x \in \operatorname{dom} A$, the operators A and $R(\lambda, A)$ commute on $\operatorname{dom} A$, so $A_\lambda x =$

$\lambda AR(\lambda, A)x = \lambda R(\lambda, A)Ax$. By Part (1) applied to the vector $Ax \in X$, $\lambda R(\lambda, A)Ax \to Ax$ as $\lambda \to \infty$. Hence $A_\lambda x \to Ax$, completing the proof.

Everything is in place for the theorem. It characterizes exactly which operators generate a C_0-contraction semigroup, and its content is that the resolvent conditions read off from theorem 7.5.8 are not merely necessary but sufficient. The proof constructs the semigroup from the resolvent via the Yosida approximation.

Theorem 7.5.11: Hille-Yosida

Let A be an operator on a Banach space X. Then A is the generator of a C_0-contraction semigroup $(T(t))_{t\geq 0}$ on X (that is, $\|T(t)\| \leq 1$ for all $t \geq 0$) if and only if

1. A is densely defined and closed;
2. $(0, \infty) \subseteq \rho(A)$; and
3. $\|R(\lambda, A)\| \leq 1/\lambda$ for every $\lambda > 0$.

When these hold the semigroup is unique, and $T(t)x = \lim_{\lambda\to\infty} e^{tA_\lambda}x$ for every $x \in X$, uniformly for t in compact intervals, where A_λ is the Yosida approximation of definition 7.5.9.

***Proof*:**
Necessity. Suppose A generates a C_0-contraction semigroup. Then A is densely defined and closed by proposition 7.5.7. With $M = 1$ and $\omega = 0$ in proposition 7.5.2, theorem 7.5.8 gives $(0, \infty) \subseteq \rho(A)$ and $\|R(\lambda, A)\| \leq 1/\lambda$ for $\lambda > 0$. This is (1), (2), (3).

Sufficiency. Assume (1), (2), (3). The construction proceeds in four steps: form the bounded-operator semigroups e^{tA_λ}, bound them uniformly, prove they converge strongly, and verify the limit is a C_0-contraction semigroup with generator A.

Step 1: The approximating semigroups are contractions. Fix $\lambda > 0$. Since $A_\lambda = \lambda^2 R(\lambda, A) - \lambda I$ is bounded, $e^{tA_\lambda} = e^{-\lambda t}e^{t\lambda^2 R(\lambda,A)}$ is defined by its norm-convergent power series. Using $\|\lambda R(\lambda, A)\| \leq 1$ from lemma 7.5.10(1),

$$\|e^{tA_\lambda}\| \leq e^{-\lambda t}\sum_{n=0}^{\infty}\frac{(t\lambda^2)^n}{n!}\|R(\lambda, A)\|^n \leq e^{-\lambda t}\sum_{n=0}^{\infty}\frac{(t\lambda^2)^n}{n!}\left(\frac{1}{\lambda}\right)^n = e^{-\lambda t}e^{t\lambda} = 1$$

so each e^{tA_λ} is a contraction, for every $t \geq 0$ and $\lambda > 0$.

Step 2: The approximations commute. For fixed $\lambda, \mu > 0$ the operators A_λ and A_μ are polynomials in the commuting resolvents $R(\lambda, A)$ and $R(\mu, A)$ (which commute by the resolvent identity), hence commute with each other and with each e^{sA_μ}.

Step 3: Strong convergence of e^{tA_λ}. Fix $x \in \operatorname{dom} A$ and $t \geq 0$, and let $\lambda, \mu > 0$. The operators e^{sA_λ} and $e^{(t-s)A_\mu}$ commute (Step 2), so the telescoping identity

$$e^{tA_\lambda}x - e^{tA_\mu}x = \int_0^t \frac{d}{ds}\left(e^{sA_\lambda}e^{(t-s)A_\mu}x\right) ds = \int_0^t e^{sA_\lambda}e^{(t-s)A_\mu}(A_\lambda - A_\mu)x\, ds$$

holds, the integrand being continuous in s. Since e^{sA_λ} and $e^{(t-s)A_\mu}$ are contractions (Step 1),

$$\|e^{tA_\lambda}x - e^{tA_\mu}x\| \leq \int_0^t \|(A_\lambda - A_\mu)x\|\, ds = t\|A_\lambda x - A_\mu x\|$$

By lemma 7.5.10(2) the sequence $(A_\lambda x)$ is Cauchy as $\lambda \to \infty$ (it converges to Ax), so $(e^{tA_\lambda}x)$ is Cauchy in X, uniformly for t in any compact interval $[0,t_0]$. Define $T(t)x = \lim_{\lambda\to\infty} e^{tA_\lambda}x$ for $x \in \operatorname{dom} A$. Each $T(t)$ is a contraction on $\operatorname{dom} A$, and since $\operatorname{dom} A$ is dense (proposition 7.5.7) and the e^{tA_λ} are uniformly bounded by 1, the same density argument as in lemma 7.5.10(1) extends $T(t)$ to a contraction on all of X with $e^{tA_\lambda}x \to T(t)x$ for every $x \in X$, uniformly on compact t-intervals.

Step 4: The limit is a C_0-contraction semigroup generated by A. The semigroup properties pass to the limit: $T(0)x = \lim e^0 x = x$, and for the semigroup law, each $e^{(s+t)A_\lambda} = e^{sA_\lambda}e^{tA_\lambda}$, so letting $\lambda \to \infty$ and using uniform boundedness gives $T(s+t)x = T(s)T(t)x$. Strong continuity holds because $e^{tA_\lambda}x \to T(t)x$ uniformly on compact intervals and each $t \mapsto e^{tA_\lambda}x$ is continuous, so the uniform limit $t \mapsto T(t)x$ is continuous. Thus $(T(t))$ is a C_0-contraction semigroup; let B denote its generator.

It remains to identify B with A. For $x \in \operatorname{dom} A$, the orbit calculus applied to each approximating semigroup gives $e^{tA_\lambda}x - x = \int_0^t e^{sA_\lambda}A_\lambda x\, ds$. The integrand converges to $T(s)Ax$ uniformly in $s \in [0,t]$: indeed $\|e^{sA_\lambda}A_\lambda x - T(s)Ax\| \le \|e^{sA_\lambda}(A_\lambda x - Ax)\| + \|(e^{sA_\lambda} - T(s))Ax\| \le \|A_\lambda x - Ax\| + \|(e^{sA_\lambda} - T(s))Ax\|$, and both terms tend to 0 uniformly on $[0,t]$ by lemma 7.5.10(2) and Step 3. Passing to the limit,

$$T(t)x - x = \int_0^t T(s)Ax\, ds$$

Dividing by t and letting $t \to 0^+$, the right side tends to $T(0)Ax = Ax$ by theorem 7.1.9, so $x \in \operatorname{dom} B$ and $Bx = Ax$. Hence $A \subseteq B$.

For the reverse inclusion, note that $1 \in \rho(A)$ by (2) and $1 \in \rho(B)$ since B generates a contraction semigroup (necessity applied to B). Now $I - A$ maps $\operatorname{dom} A$ bijectively onto X, and $I - B$ maps $\operatorname{dom} B$ bijectively onto X; since $A \subseteq B$, the operator $I - B$ restricted to $\operatorname{dom} A$ already surjects onto X. If $x \in \operatorname{dom} B$, pick $y \in \operatorname{dom} A$ with $(I-A)y = (I-B)x$; then $(I-B)y = (I-A)y = (I-B)x$ because $A \subseteq B$, and injectivity of $I - B$ forces $y = x$, so $x = y \in \operatorname{dom} A$. Thus $\operatorname{dom} B \subseteq \operatorname{dom} A$ and $B = A$.

Uniqueness. Suppose $(S(t))$ is another C_0-contraction semigroup with generator A. Fix $x \in \operatorname{dom} A$ and $t > 0$, and consider the X-valued function $s \mapsto S(t-s)T(s)x$ on $[0,t]$, where $(T(t))$ is the semigroup constructed above. By lemma 7.5.6(2) it is differentiable with

$$\frac{d}{ds}S(t-s)T(s)x = -S(t-s)AT(s)x + S(t-s)AT(s)x = 0$$

the two terms cancelling because $T(s)x \in \operatorname{dom} A$ and both semigroups have generator A. A function with zero derivative on $[0,t]$ is constant, so evaluating at $s = 0$ and $s = t$ gives $S(t)x = T(t)x$ for all $x \in \operatorname{dom} A$, and by density $S(t) = T(t)$. This establishes uniqueness and completes the proof.

The theorem is the exact analogue for irreversible dynamics of Stone's theorem for the reversible case. Stone's theorem realizes every strongly continuous unitary group as e^{itA} for a self-adjoint A (theorem 7.4.5); Hille-Yosida realizes every C_0-contraction semigroup as, informally, e^{tA} for an operator A satisfying the resolvent bound, with the exponential made precise by the Yosida limit $e^{tA} = \lim_\lambda e^{tA_\lambda}$. On a Hilbert space the two theorems meet: a self-adjoint $A \le 0$ satisfies the Hille-Yosida conditions (its spectrum lies in $(-\infty, 0]$, so $(0,\infty) \subseteq \rho(A)$ with the required bound), and the contraction semigroup it generates is the $e^{tA} = \int_{(-\infty,0]} e^{t\lambda}\, dE(\lambda)$ of the functional calculus of theorem 7.3.8. The heat semigroup of example 7.5.3 is this case with $A = \partial_{xx} \le 0$.

The general (non-contractive) case follows by the rescaling noted after proposition 7.5.2: A generates a C_0-semigroup with $\|T(t)\| \leq Me^{\omega t}$ if and only if $A - \omega I$ generates a bounded (M-bounded) semigroup, and the sharp characterization of that case requires bounding all powers $\|R(\lambda, A)^n\| \leq M/(\lambda - \omega)^n$ rather than the resolvent alone. This is the Feller-Miyadera-Phillips theorem, whose proof is the same Yosida construction with the contraction bound of Step 1 replaced by the power bound; the contraction case above carries all the ideas, and the general statement is developed where it is needed, in the evolution-equation volumes.

One refinement of the generation theory is indispensable for parabolic problems and is recorded here for orientation, its development deferred to the volumes that use it. A C_0-semigroup is *analytic* when the orbit map $t \mapsto T(t)$ extends holomorphically from the positive real axis to a sector of the complex plane. Such semigroups are exactly those whose generators have resolvent bounds on a sector, and they carry the smoothing that makes parabolic equations regularize their data instantly.

Proposition 7.5.12: Analytic Semigroups and Sectorial Generators

Let A be a densely defined closed operator on a Banach space X, and for $\theta \in (0, \pi/2]$ let $\Sigma_\theta = \{\lambda \in \mathbb{C} \setminus \{0\} \mid |\arg \lambda| < \pi/2 + \theta\}$. Suppose $\Sigma_\theta \subseteq \rho(A)$ and there is a constant C with

$$\|R(\lambda, A)\| \leq \frac{C}{|\lambda|} \qquad (\lambda \in \Sigma_\theta)$$

(such an A is called *sectorial*). Then A generates a C_0-semigroup $(T(t))_{t \geq 0}$ that extends to a holomorphic map $z \mapsto T(z)$ on the open sector $\{z \in \mathbb{C} \setminus \{0\} \mid |\arg z| < \theta\}$, bounded on each subsector $|\arg z| \leq \theta' < \theta$, with $T(z)X \subseteq \operatorname{dom} A$ for every z in the sector. The semigroup is given by the Dunford integral $T(z) = \frac{1}{2\pi i} \int_\Gamma e^{z\lambda} R(\lambda, A)\, d\lambda$ over a suitable contour Γ in Σ_θ.

The sectorial resolvent bound is what turns the Laplace-transform representation of theorem 7.5.8 into a contour integral that may be differentiated in the time variable arbitrarily often, so an analytic semigroup maps X into $\operatorname{dom} A$ (indeed into $\bigcap_n \operatorname{dom} A^n$) for every $t > 0$. This is the abstract form of parabolic smoothing: the solution $T(t)f$ of $u' = Au$ is infinitely differentiable in t and lies in the domain of every power of A for $t > 0$, however rough the datum f. The Laplacian on L^p and the elliptic operators of the theory of linear parabolic equations are the governing examples, and the contour-integral construction, the perturbation theory, and the maximal-regularity estimates that accompany it belong to *Linear Elliptic and Parabolic PDE* [4], where the heat and reaction-diffusion equations are solved by exactly this machinery.